CS3591-COMPUTER NETWORKS

UNIT I

INTRODUCTION AND APPLICATION LAYER

Data Communication – Networks – Network Types – Protocol Layering – TCP/IP Protocol suite – OSI Model – Introduction to Sockets – Application Layer protocols: HTTP – FTP – Email protocols (SMTP – POP3 – IMAP – MIME) – DNS – SNMP

UNIT II TRANSPORT LAYER

Introduction – Transport-Layer Protocols: UDP – TCP: Connection Management – Flow control – Congestion Control – Congestion avoidance (DECbit, RED) – SCTP – Quality of Service

UNIT III NETWORK LAYER

Switching : Packet Switching – Internet protocol – IPV4 – IP Addressing – Subnetting – IPV6, ARP, RARP, ICMP, DHCP

UNIT IV ROUTING

Routing and protocols: Unicast routing – Distance Vector Routing – RIP – Link State Routing – OSPF – Path-vector routing – BGP – Multicast Routing: DVMRP – PIM.

UNIT V DATA LINK AND PHYSICAL LAYERS

Data Link Layer – Framing – Flow control – Error control – Data-Link Layer Protocols – HDLC – PPP – Media Access Control – Ethernet Basics – CSMA/CD – Virtual LAN – Wireless LAN (802.11) – Physical Layer: Data and Signals – Performance – Transmission media- Switching – Circuit Switching.

INTRODUCTION

DATA COMMUNICATIONS : Sharing Information

Local

Remote



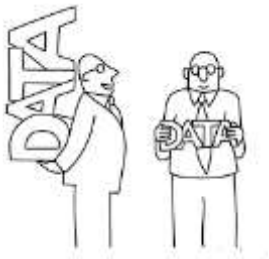




INTRODUCTION

DATA COMMUNICATION

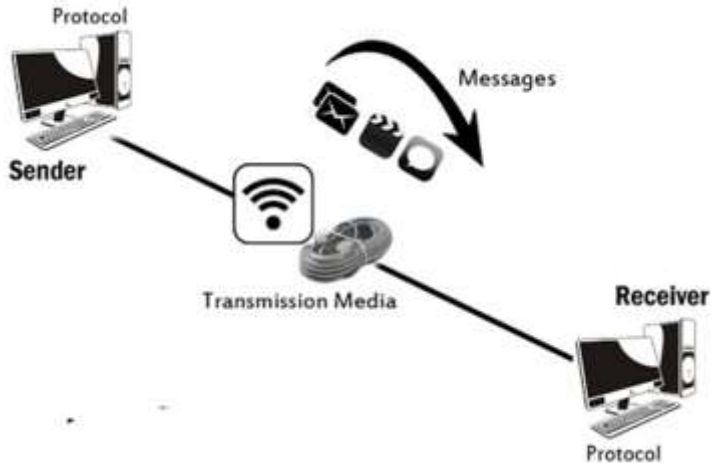
DATA: Information presented in whatever form is agreed upon by the parties creating and using the data.





INTRODUCTION

DATA COMMUNICATION: exchange of data between two devices via some form of transmission medium.



INTRODUCTION

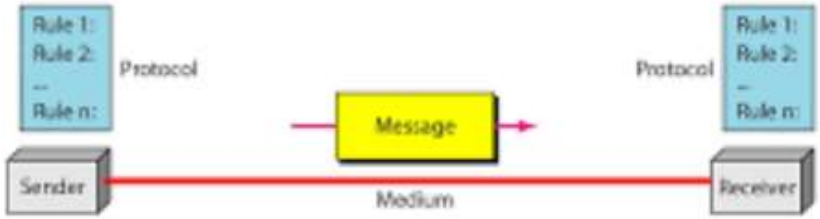
The effectiveness of a data communications system depends on four fundamental characteristics:

- Deliv 
- Accur 
- timeli  and
- jitter. 

- **Delivery.** The system must deliver data to the correct destination. Data must be received by the intended device
- **Accuracy.** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
- **Timeliness.** The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, with out delay.
- **Jitter** refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

INTRODUCTION

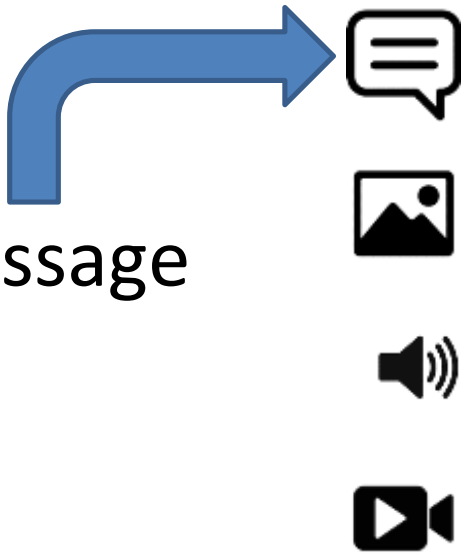
A data communications system has five components.



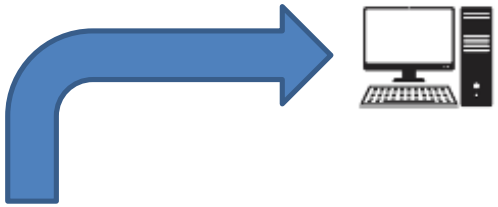
- Message
- Sender
- Receiver
- Transmission Medium
- Protocol

INTRODUCTION

- Message



INTRODUCTION

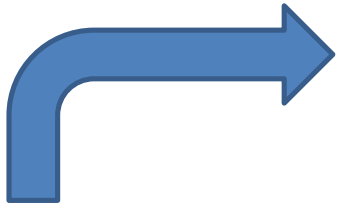




- Sender & Receiver



INTRODUCTION







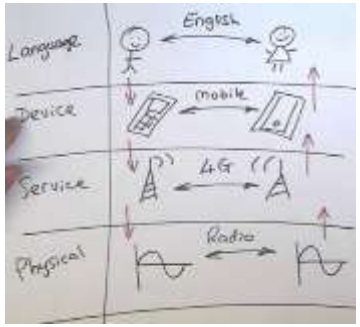
- Transmission Medi





INTRODUCTION





- Protocol

INTRODUCTION

Data Representation

- Text
- Numbers
- Images
- Audio
- Video

INTRODUCTION

Data Representation

- Text  1010100

Letter	ASCII Code	Binary	Letter	ASCII Code	Binary
a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	100	01100100	D	068	01000100
e	101	01100101	E	069	01000101
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111
h	104	01101000	H	072	01001000
i	105	01101001	I	073	01001001
j	106	01101010	J	074	01001010
k	107	01101011	K	075	01001011
l	108	01101100	L	076	01001100
m	109	01101101	M	077	01001101
n	110	01101110	N	078	01001110
o	111	01101111	O	079	01001111
p	112	01110000	P	080	01010000
q	113	01110001	Q	081	01010001
r	114	01110010	R	082	01010010

INTRODUCTION

Data Representation

- Numbers → 1010100

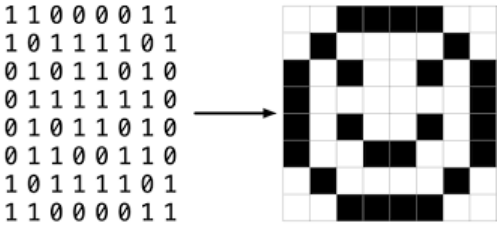
$(160)_{10}$

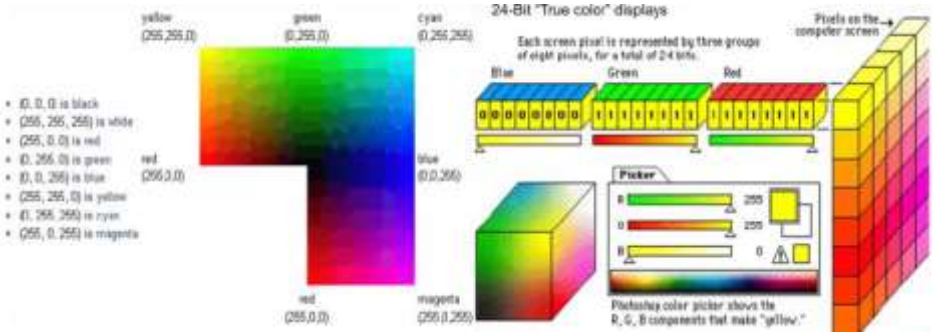
2	160	
2	80	0
2	40	0
2	20	0
2	10	0
2	5	0
2	2	1
	1	0

INTRODUCTION

Data Representation

- Images





INTRODUCTION

Data Representation

- Audio





INTRODUCTION

Data Representation

- Video







INTRODUCTION

Data Flow

- Simplex
- Half Duplex
- Full Duplex

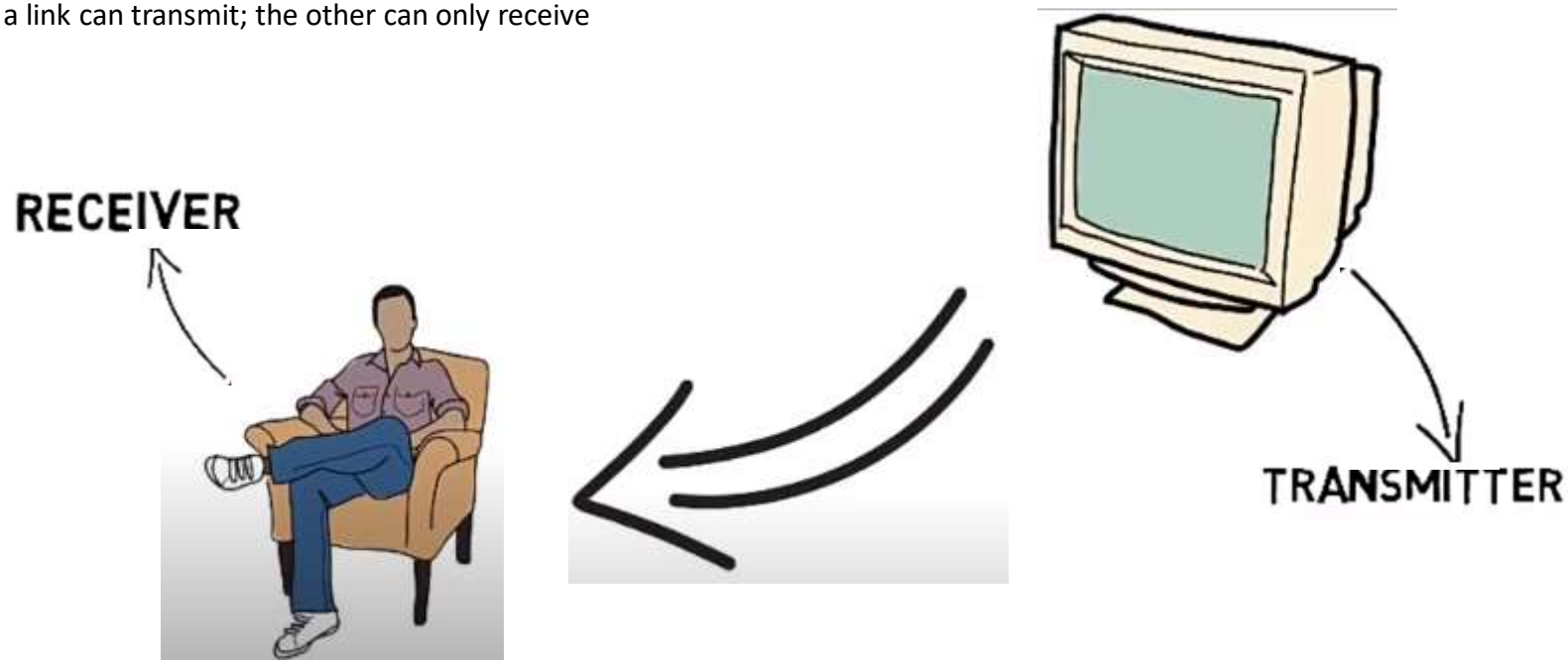
INTRODUCTION

Simplex



INTRODUCTION

- Simplex In **simplex mode,** the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive



INTRODUCTION

- Half Duplex



In **half-duplex mode,** each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa

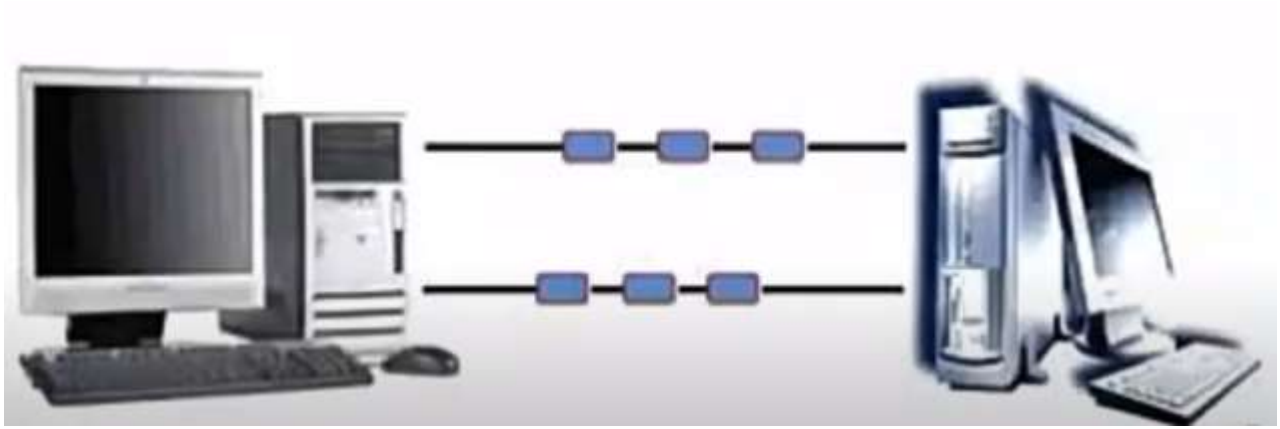






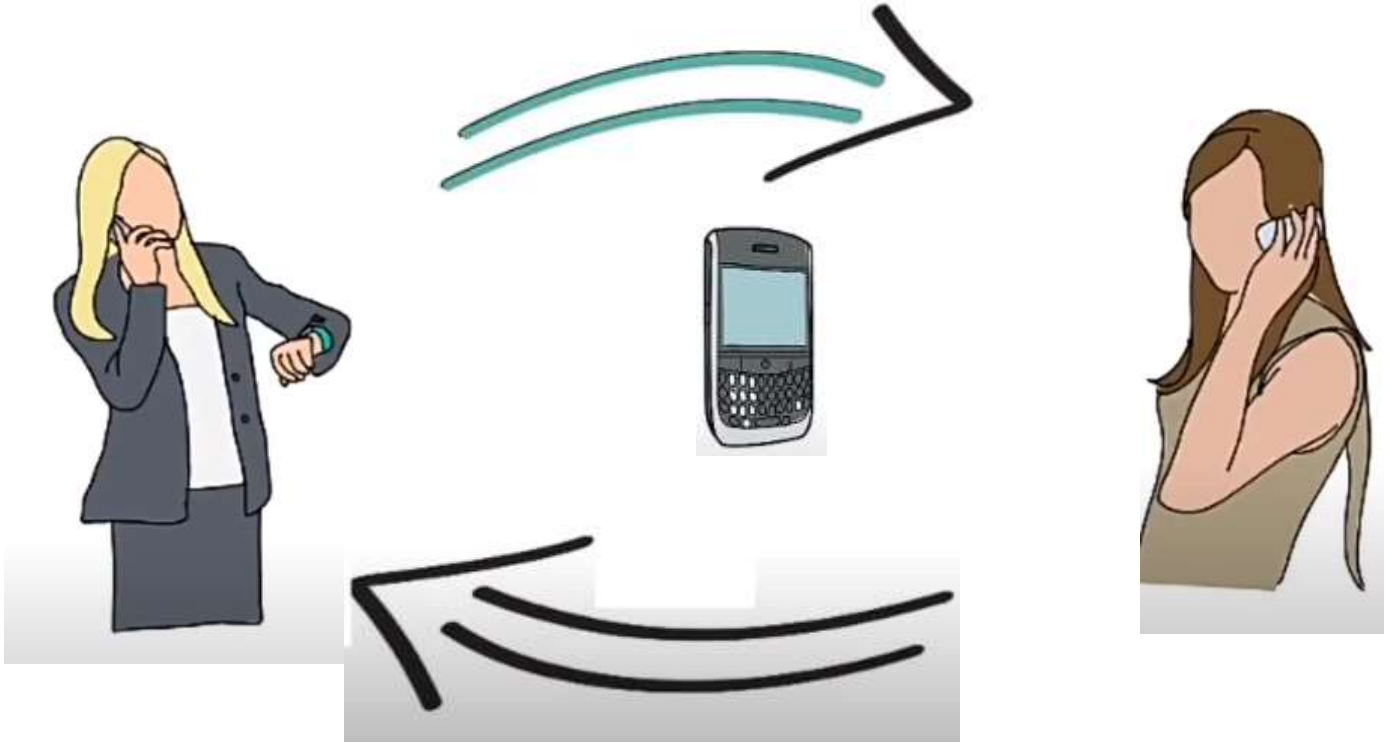
INTRODUCTION

Full Duplex



In **full-duplex mode** (also called *duplex*), both stations can transmit and receive simultaneously

Full Duplex



NETWORKS

- A network is a set of devices (often referred to as nodes) connected by communication links.
- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.



NETWORKS

Network Criteria

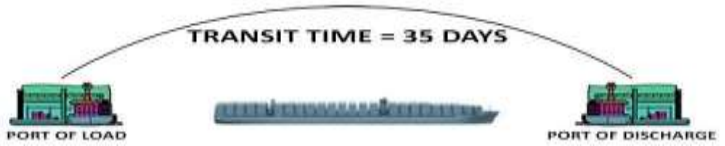
- Performance
- Reliability
- Security

NETWORKS

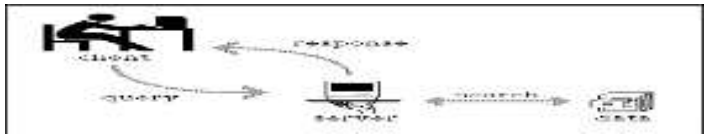
Network Criteria

- Performance

Transit time



Response time



Metrics

Throughput



Delay.



NETWORKS

Network Criteria

- Reliability

  Frequency of failure

NETWORKS

Network Criteria

- Security

  Protecting data

  Recovery from breaches and data losses

Networks

Networks Attributes

Types of connection

Physical Topology

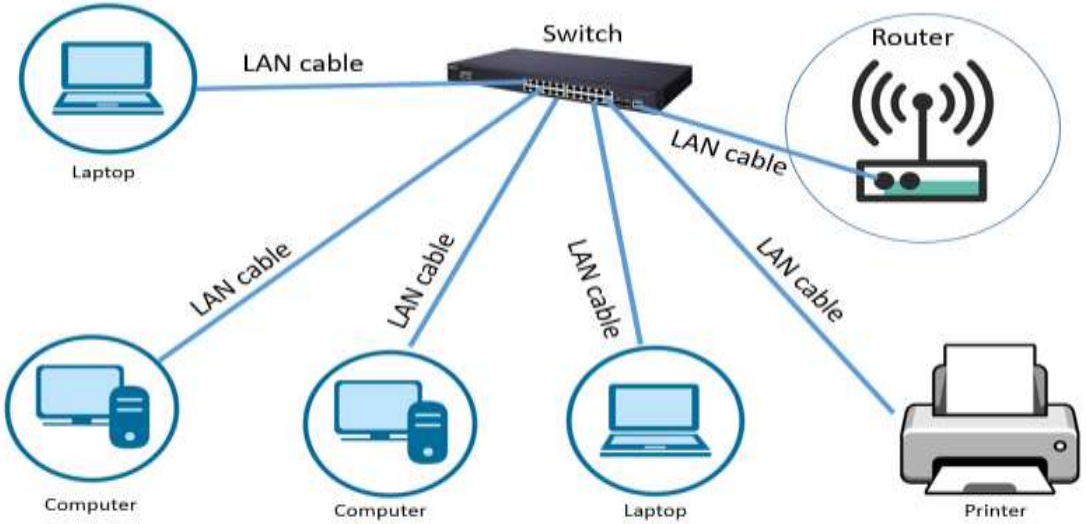
NETWORKS MODELS

Categories of Networks

- LAN
- WAN
- MAN
- Interconnection of networks: Internetworks

LAN:

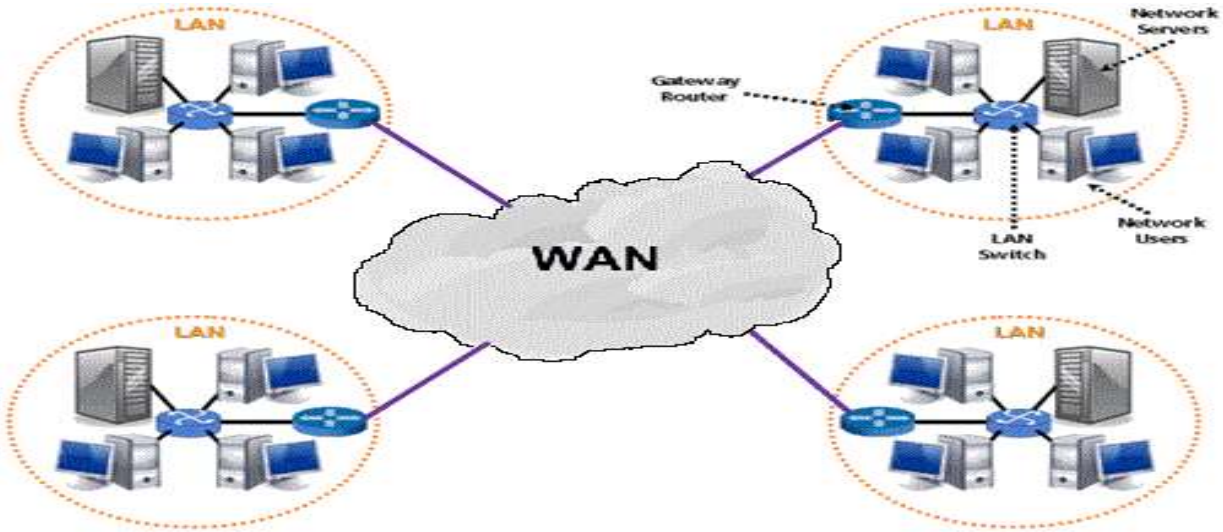
- LAN connects network devices over short distance.
- LAN's are controlled and managed by a single person or organization.
- LAN allows sharing of resources in the network like printers, software etc.
- The IEEE standard LAN are: Bus,ring,Ethernet (IEEE 802.4,IEEE 802.5,IEEE 802.3)



Local Area Network

WAN: (Wide Area Network)

- WAN provides long distance transmission of data,voice,image and video
- WAN is a data communication system spanning states, countries or the whole world.
- Private owned WAN is called Enterprise Network
- All the host are connected by each other through a communication subnet.
- Subnet consist of transmission line and switching element.



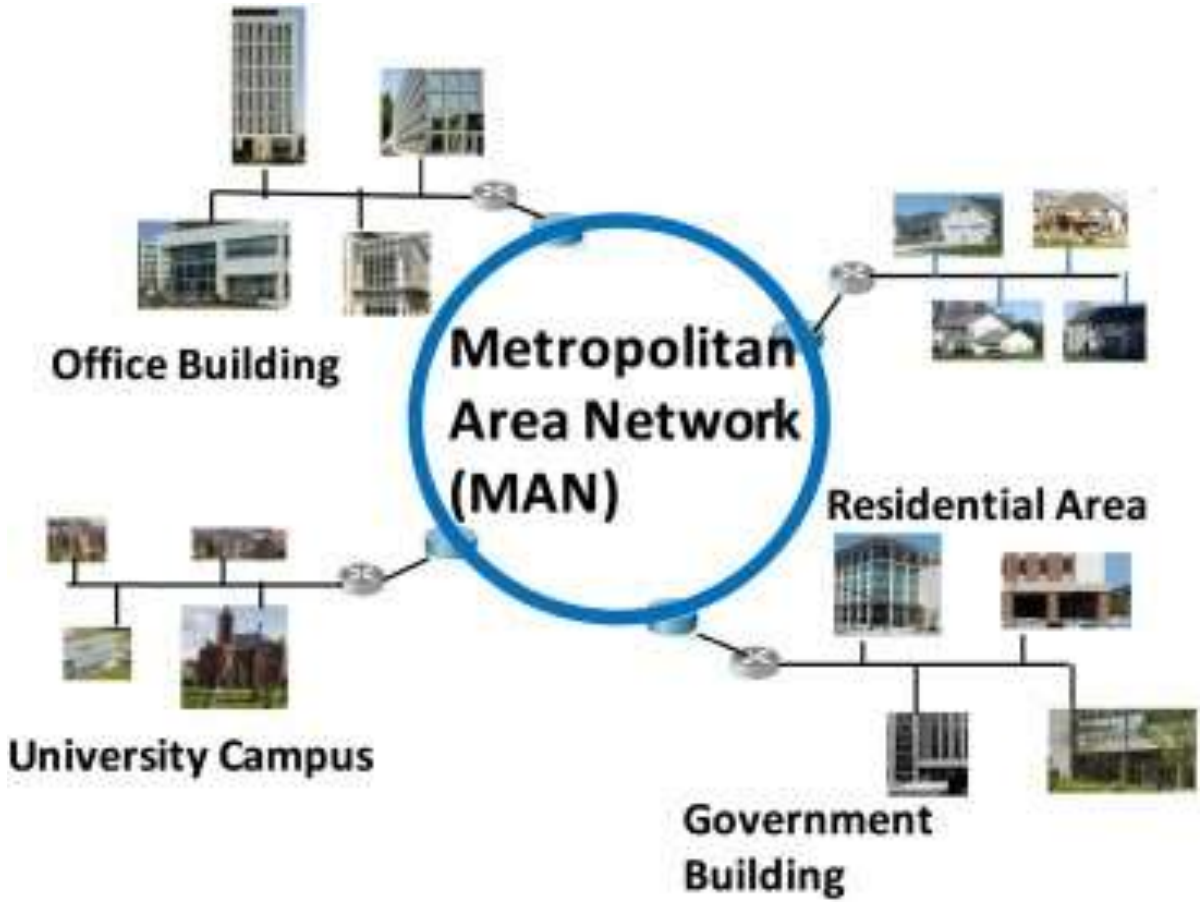
WAN - Wide Area Network

MAN (Metropolitan Area Network)

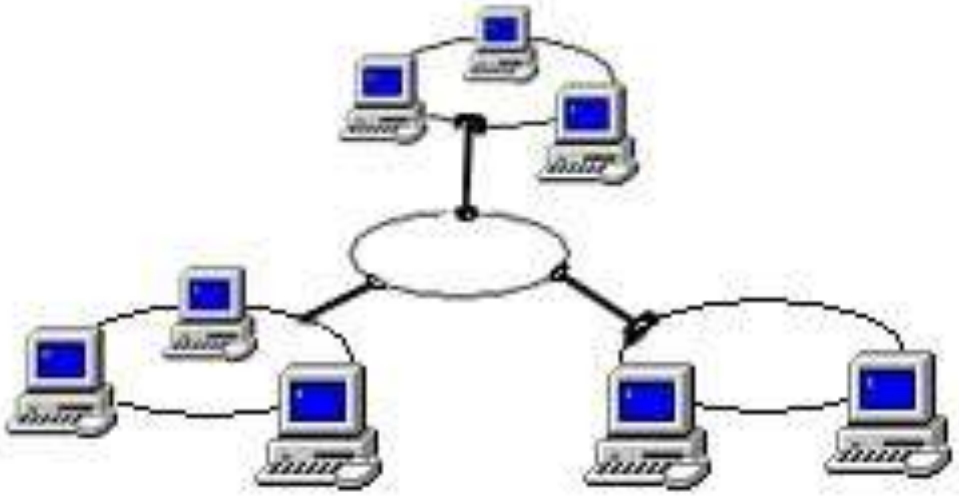
- Designed to extend over entire city.
- It can be a single n/w such as cable teleivision n/w, or it may be connecting no of LAN into a Large n/w.

Example:

- Organization can use MAN to connect LAN all of its office throughout a city.


Office Building
Metropolitan Area Network (MAN)
Residential Area
University Campus
Government Building

- **Internetworking**
- is **the practice of interconnecting multiple computer networks**.
- The Internet is **a vast network that connects computers all over the world**. Through the Internet, people can share information and communicate from anywhere with an Internet connection



The Internet

A Network of Networks

Protocol Layering

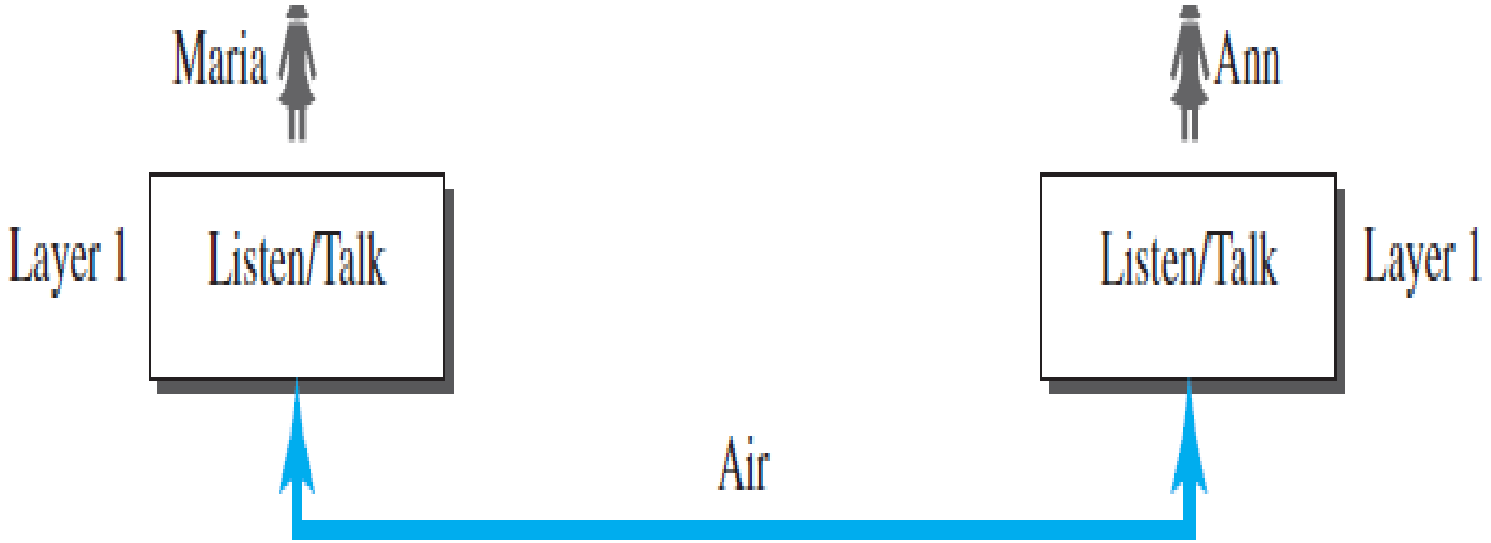
- In data communication and networking, a protocol defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.
- When communication is simple, we may need only one simple protocol; when the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or **protocol layering.**

Scenarios

- Let us develop two simple scenarios to better understand the need for protocol layering.

First Scenario

- In the first scenario, communication is so simple that it can occur in only one layer.
- Imagine Maria and Ann are neighbours with a lot of common ideas.
- Communication between Maria and Ann takes place in one layer, face to face, in the same language, as shown in this figure



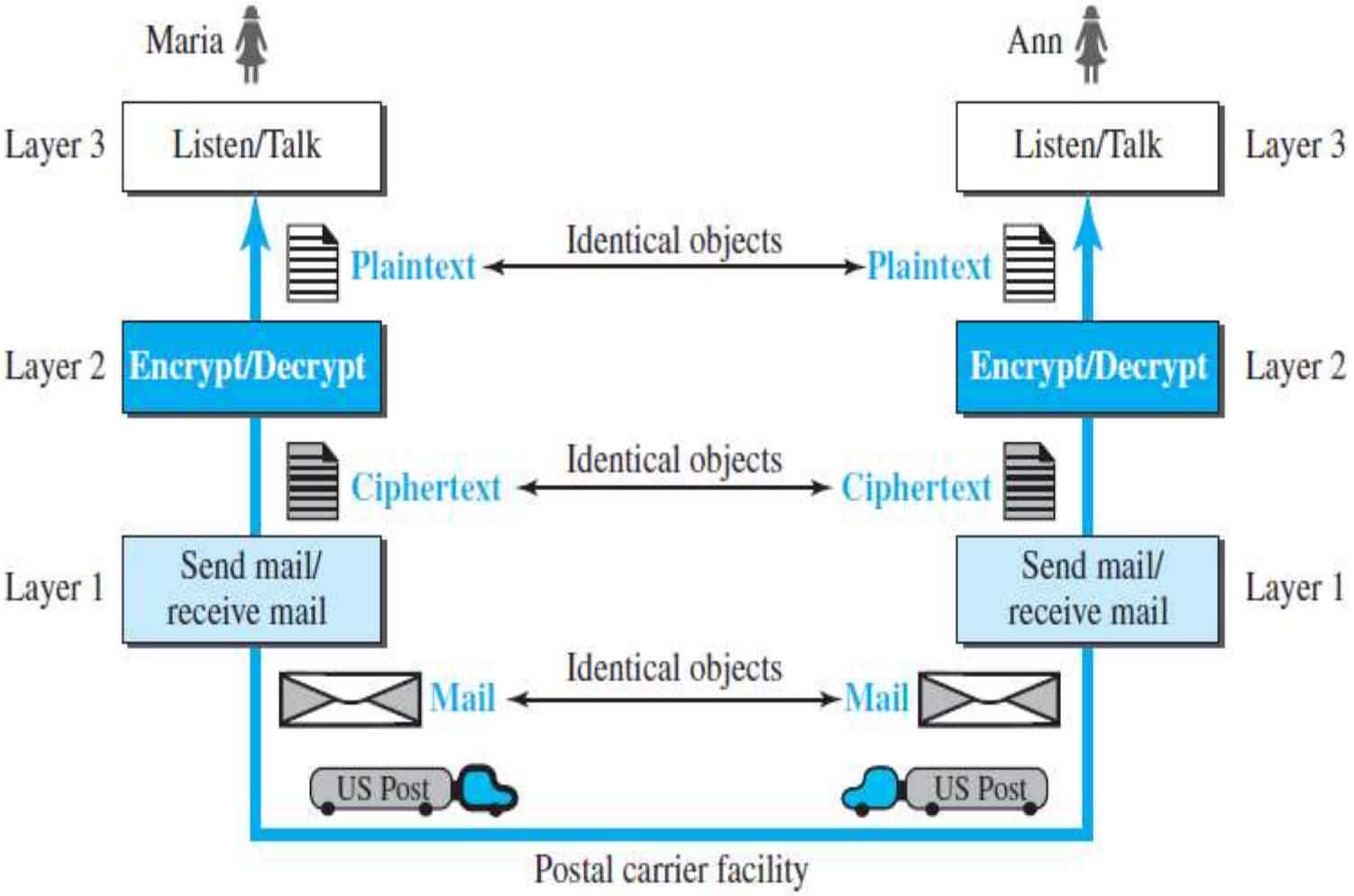
Even in this simple scenario, we can see that a set of rules needs to be followed.

- First, Maria and Ann should greet each other when they meet.
- Second, should confine their vocabulary to the level of their friendship.
- Third, each party knows that she should refrain from speaking when the other party is speaking.
- Fourth, each party knows that the conversation should be a dialog, not a monolog.
- Fifth, they should exchange some nice words when they leave.

Second Scenario

- In the second scenario, we Imagine that Ann is offered a higher-level position in her company, but needs to move to another branch located in a city very far from Maria.
- The two friends still want to continue their communication and exchange ideas because they have come up with an innovative project to start a new business when they both retire.
- They decide to continue their conversation using regular mail through the post office. However, they do not want their ideas to be revealed by other people if the letters are intercepted.

- They agree on an encryption/decryption technique. The sender of the letter encrypts it to make it unreadable by an intruder; the receiver of the letter decrypts it to get the original letter. (Fig-Three layer protocol)



- Protocol layering enables us to divide a complex task into several smaller and simpler tasks. For example, in above figure, if Maria and Ann decide that the encryption/decryption done by the machine is not enough to protect their secrecy, they would have to change the whole machine.
- In the present situation, they need to change only the second layer machine; the other two can remain the same. This is referred to as *modularity*.
- Modularity in this case means independent layers.

Principles of Protocol Layering

Let us discuss two principles of protocol layering.

First Principle

- The first principle if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.
- For example, the third layer task is to listen (in one direction) and *talk* (in the other direction).
- The second layer needs to be able to encrypt and decrypt. The first layer needs to send and receive mail.

Second Principle

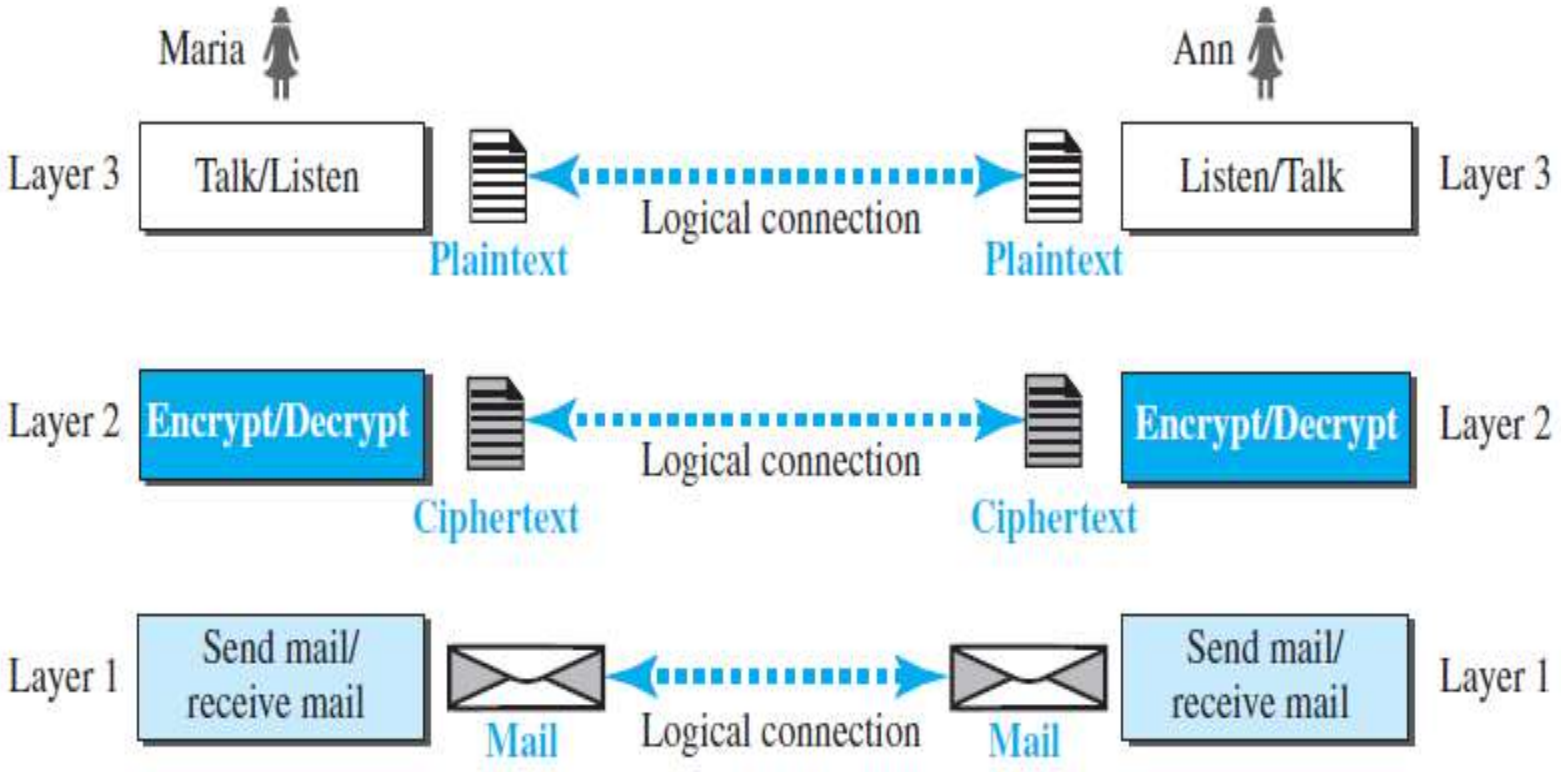
- The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.
- For example, the object under layer 3 at both sites should be a plaintext letter.
- The object under layer 2 at both sites should be a cipher text letter.
- The object under layer 1 at both sites should be a mail.

Logical Connections

- After following the above two principles, we can think about logical connection between each layer as shown in Figure .

- This means that we have layer-to-layer communication. Maria and Ann can think that there is a logical (imaginary) connection at each layer through which they can send the object created from that layer

Figure 2.3 *Logical connection between peer layers*



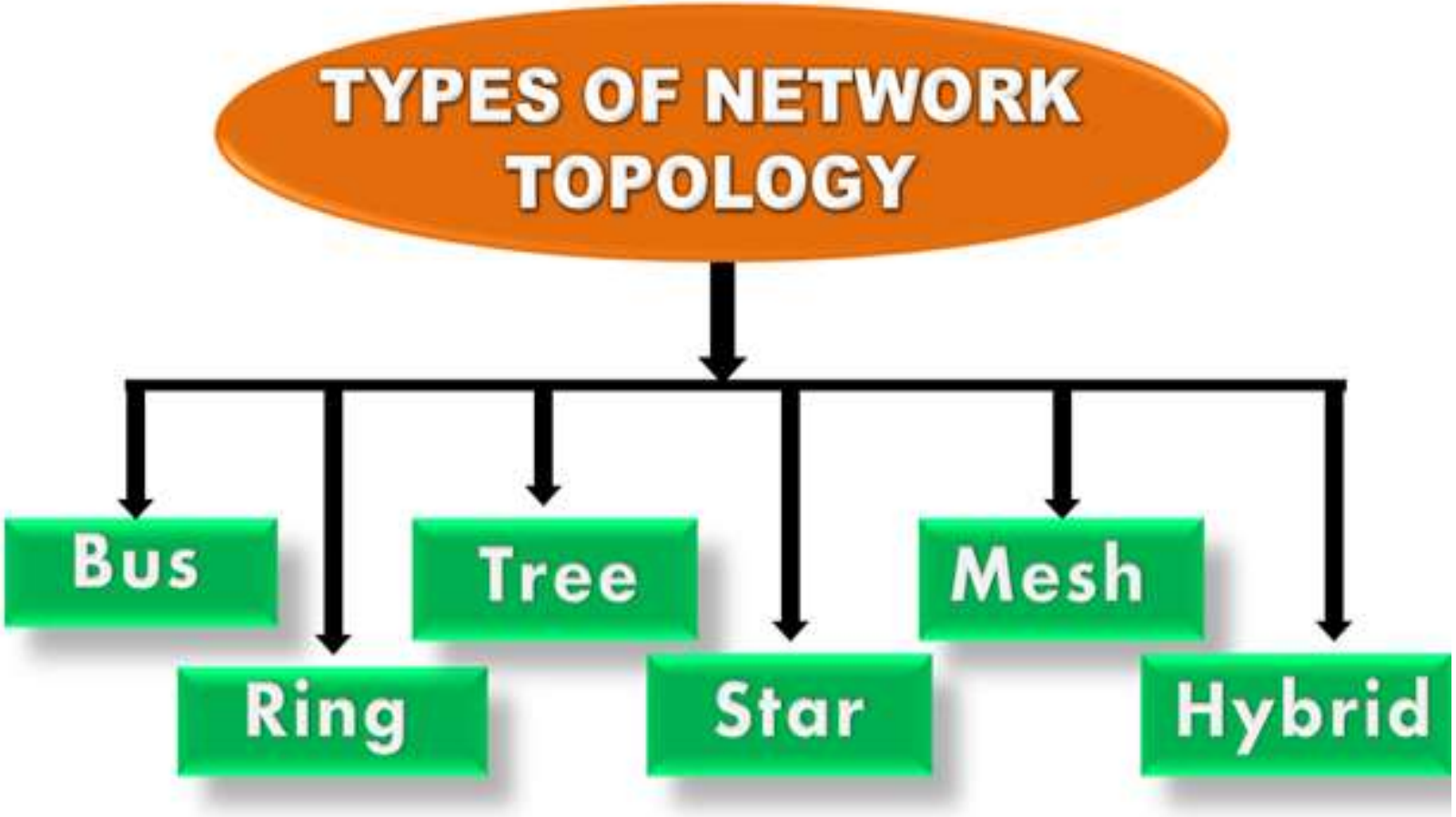
Topology

- Topology defines the structure of the network of how all the components are interconnected to each other.

There are two types of topology:

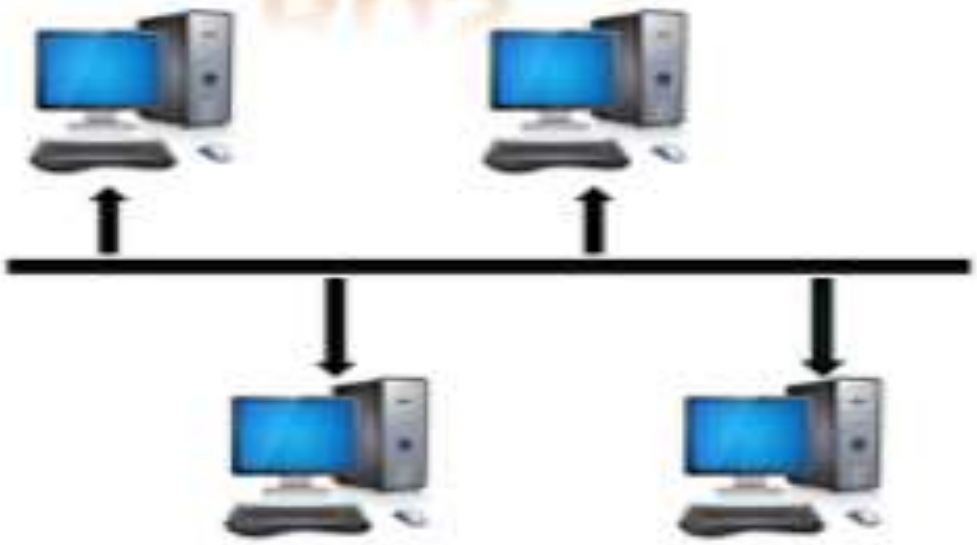
- physical Toplogy
- logical topology.

Physical topology is the geometric representation of all the nodes in a network.


TYPES OF NETWORK TOPOLOGY
Bus
Ring
Tree
Star
Mesh
Hybrid

BUS

NETWORK BACKBONE



BUS Topology:

- The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- When a node wants to send a message over the network, it puts a message over the network. All the stations available in the network will receive the message whether it has been addressed or not.
- The bus topology is mainly used in 802.3 (ethernet) and 802.4 standard networks.

The most common access method of the bus topologies is **CSMA** (Carrier Sense Multiple Access)

Advantages of Bus topology:

Low cost cable

Limited Failure

Moderate data transmission speed

Drawbacks:

Extensive cabling

Difficult trouble shooting

Reconfiguration is difficult


RING

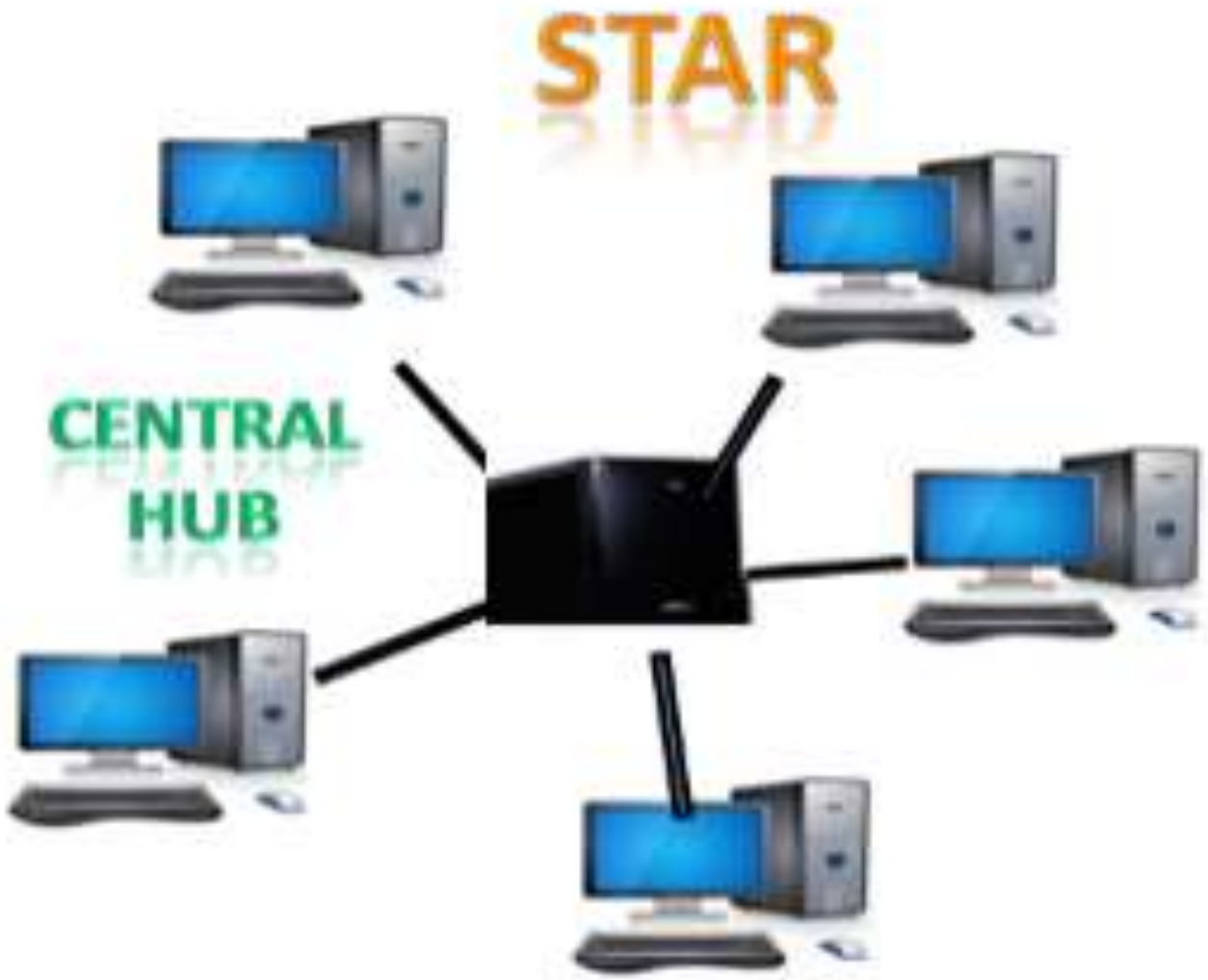
- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.
- The data flows in one direction, i.e., it is unidirectional.
- The data in a ring topology flow in a clockwise direction.
- The most common access method of the ring topology is **token passing**

Advantages of Ring Topology

- The data transmission is high-speed.
- Cheap to install and expand.
- It is less costly than a star topology.

Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- Less secure.


STAR
CENTRAL
HUB

- Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
- The central computer is known as a **server**, and the peripheral devices attached to the server are known as **clients**.
- Coaxial cable or RJ-45 cables are used to connect the computers.
- Hubs or Switches are mainly used as connection devices in a **physical star topology**.
- Star topology is the most popular topology in network implementation.

Advantages of Star Topology

- It is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.

Disadvantages of Star Topology

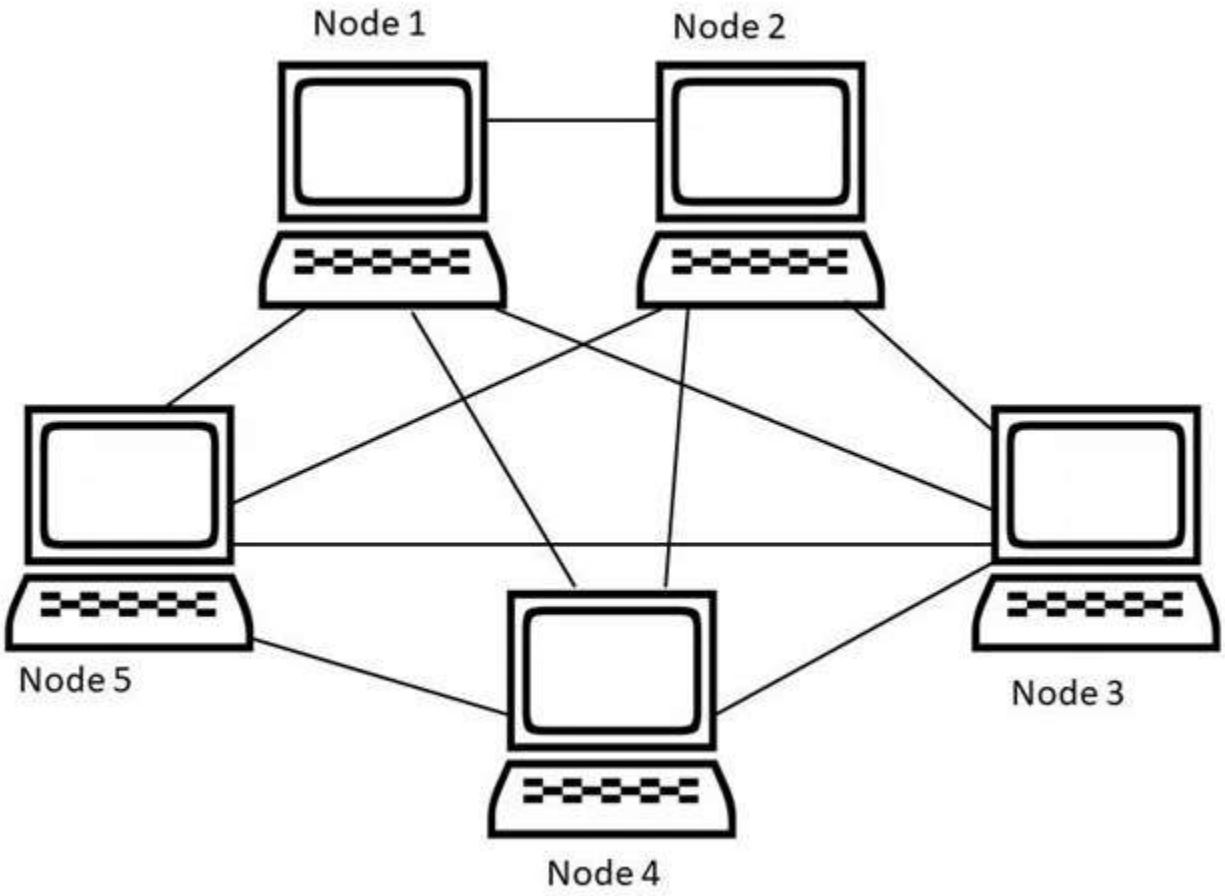
- If the hub fails the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.

TREE



- Tree topology combination of bus topology and star topology.
- A tree topology is a type of structure in which all the computers are connected with each other in hierarchical fashion.
- The top-most node in tree topology is known as a root node, and all other nodes are the descendants of the root node.
- There is only one path exists between two nodes for the data transmission. Thus, it forms a parent-child hierarchy.

- MESH TOPOLOGY



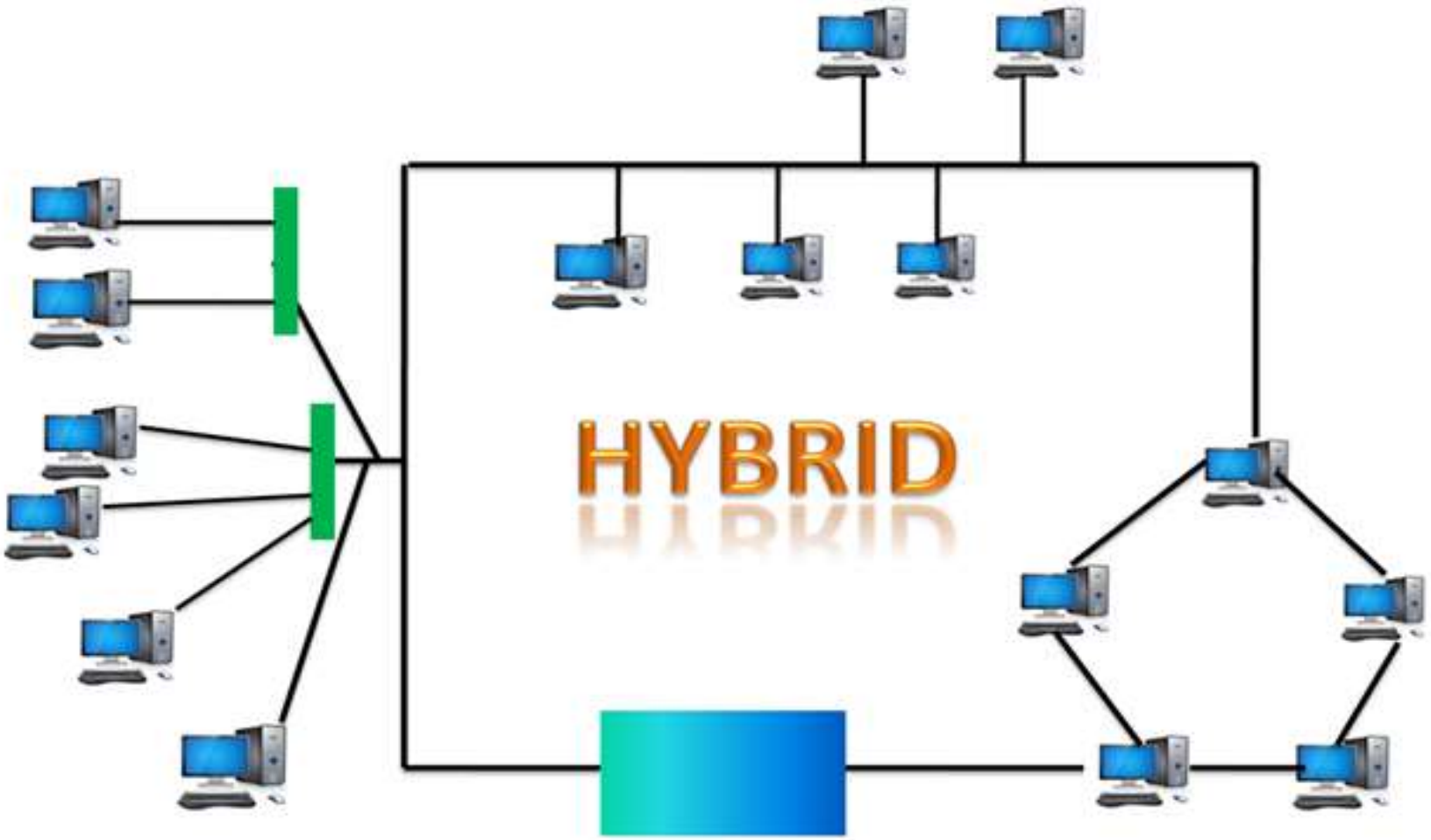
- Mesh technology is an arrangement of the network in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- It does not contain the switch, hub or any central computer which acts as a central point of communication.
- The Internet is an example of the mesh topology.
- Mesh topology can be formed by using the formula: **Number of cables = (n*(n-1))/2;**
- Where n is the number of nodes that represents the network.

Advantages of Mesh Topology

- Communication is very fast between the nodes.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy**.**

Disadvantages of Mesh Topology

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.


HYBRID

- The combination of various different topologies is known as **Hybrid topology**.
- A Hybrid topology is a connection between different links and nodes to transfer the data.
- When two or more different topologies are combined together is termed as Hybrid topology and if similar topologies are connected with each other will not result in Hybrid topology.

The OSI Model

- An ISO (International standard Organization) that covers all aspects of network communications is the **Open System Interconnection (OSI) model**.
- **An open system is a model that allows any two different systems to communicate regardless of their underlying architecture (hardware or software).**
- The OSI model is not a protocol; it is model for understanding and designing a network architecture that is flexible, robust and interoperable.

- The OSI model is a layered framework for the design of network systems that allows for communication across all types of computer systems.
- The OSI model is built of seven ordered layers:

1. (layer 1) physical layer
2. (layer 2) data link
3. (layer 3) network layer
4. (layer 4) transport layer
5. (layer 5) session layer
6. (layer 6) presentation layer
7. (layer 7) application layer

Figure 3.1 *The OSI model*

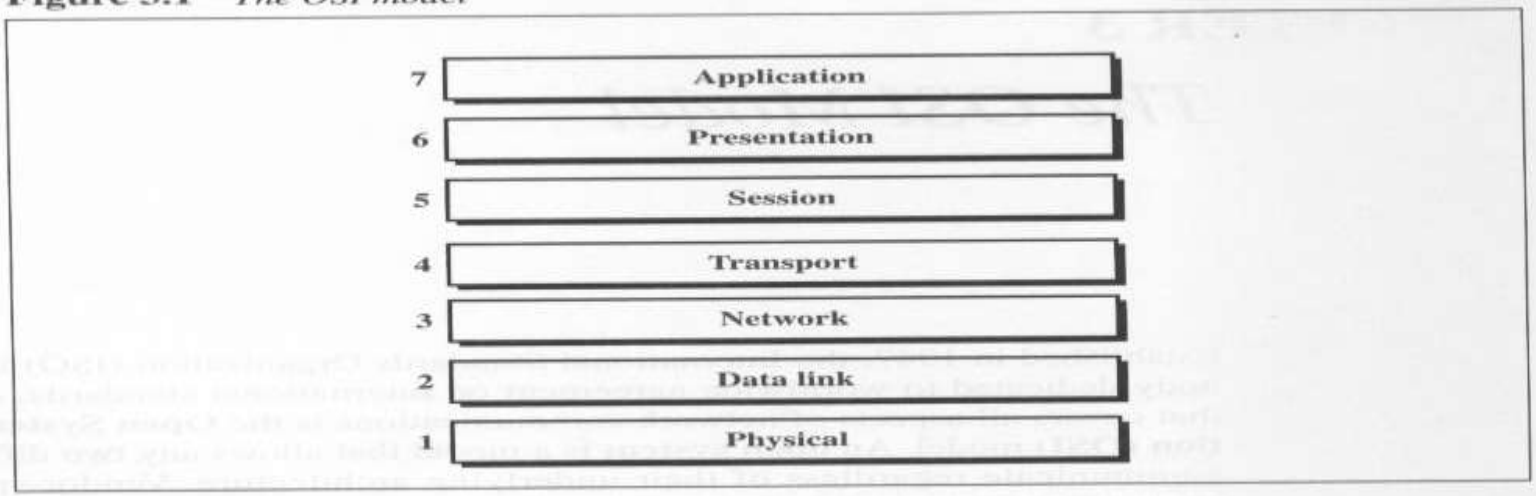
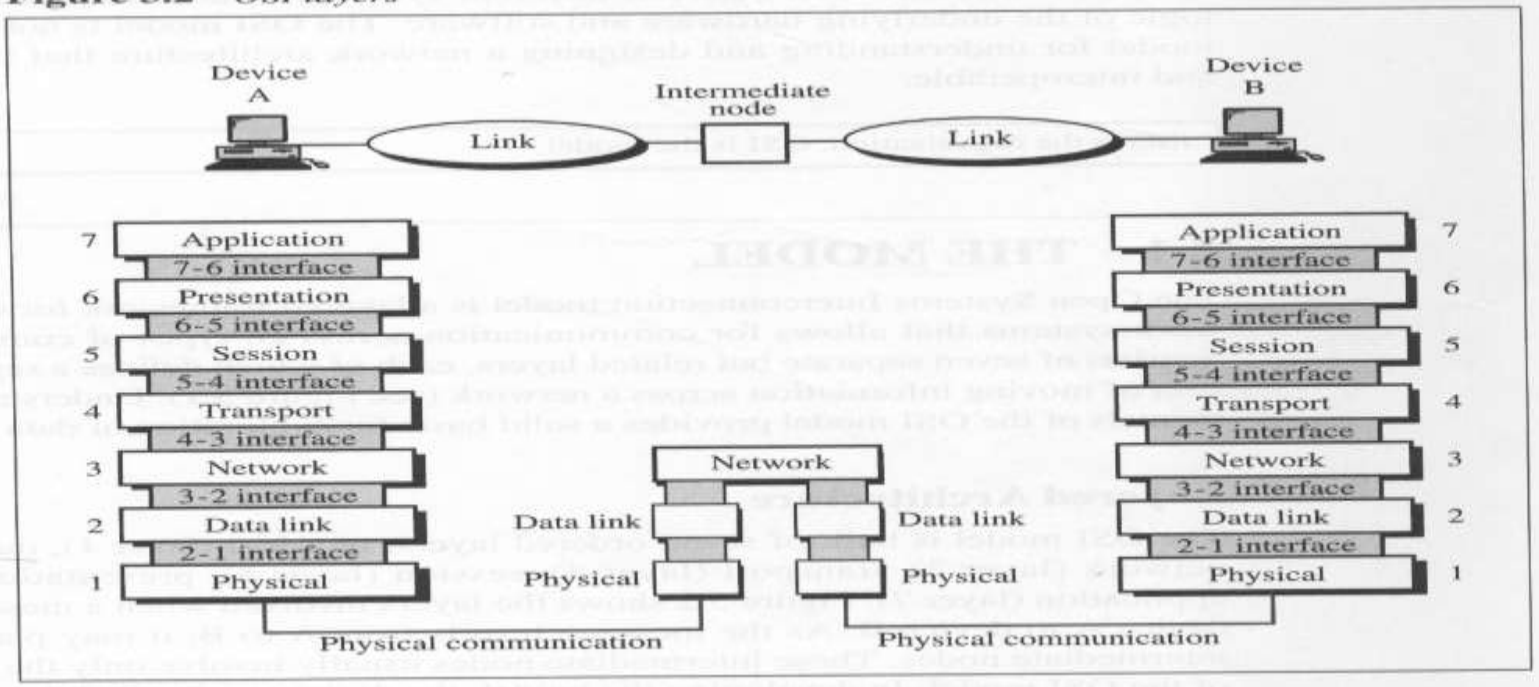


Figure 3.2 *OSI layers*

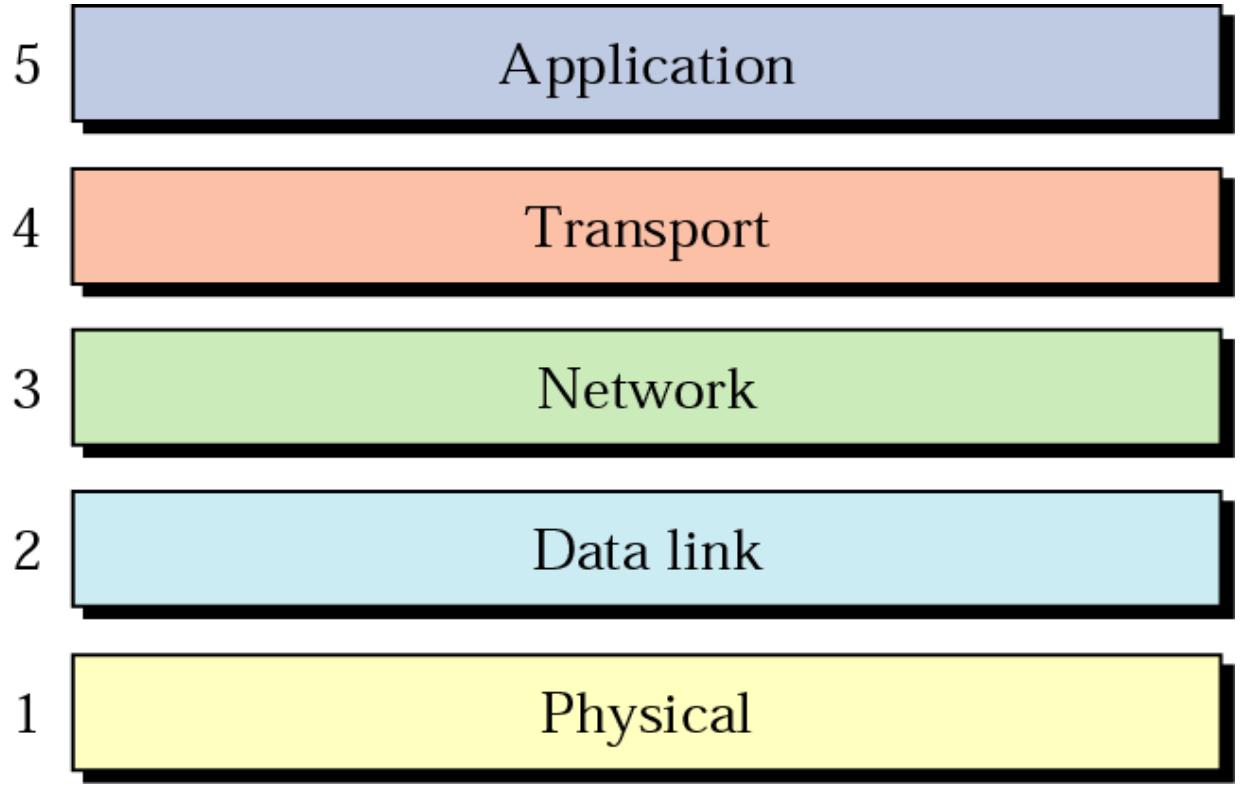


- **P**lease Do Not tell Secret Password Anytime
- Please Do Not Throw Sandwich Pizza Away

Peer-to-Peer Process

- Within a single machine, each layer calls upon services of the layer just below it.
- Layer 3, for example, uses the services provided by layer 2 and provides services for layer 4.
- Between machines, layer x on one machine communicates with layer x on another machine, by using a protocol (**this is Peer-to-Peer Process**).
- Communication between machines is therefore a peer-to-peer process using protocols appropriate to a given layer.

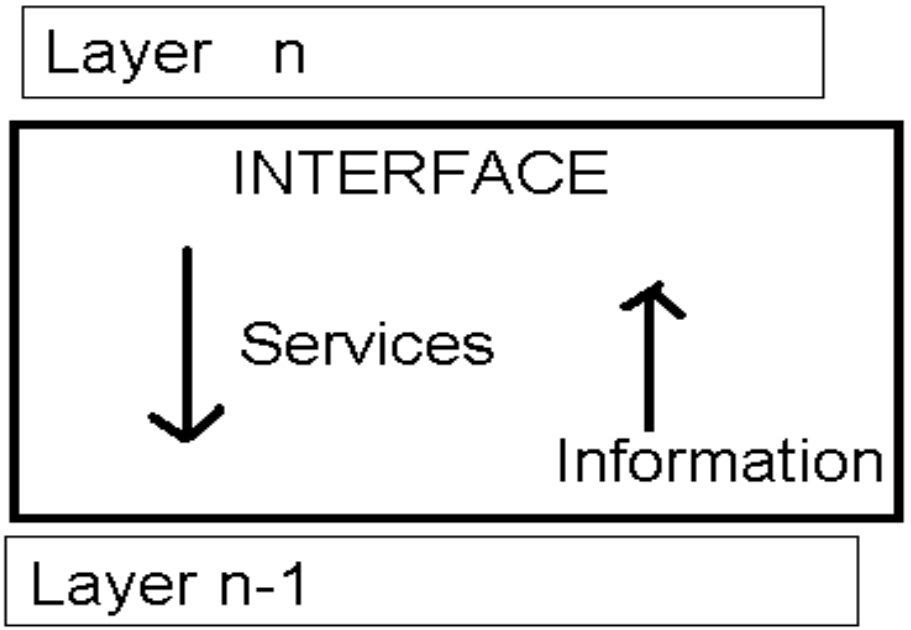
Internet Layers (TCP/IP)



Interfaces Between Layers

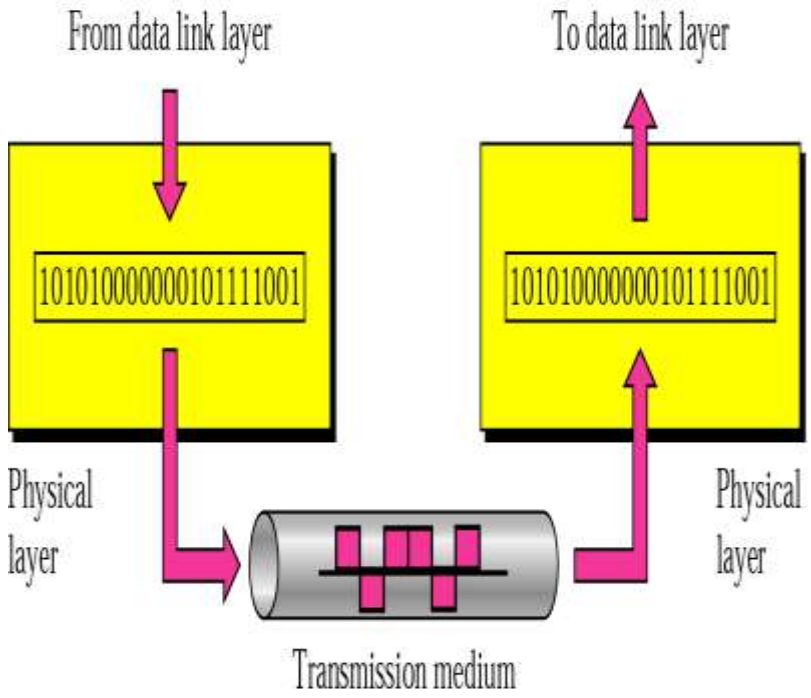
- There is an **interface** between each pair of adjacent layers.

This interface defines what information and services a layer must provide for the layer above it.



Physical Layer

- The physical layer coordinates the functions required to transmit a bit stream over a physical medium. It also defines the procedures and functions that physical devices and interfaces have to perform for transmission occur.



The physical layer is responsible for transmitting individual bits from one node to the next.

Physical layer

The physical layer is concerned with the following:

- **Physical characteristics of interfaces and media**: The physical layer defines the characteristics of the interface between devices and the transmission media, including its type.
- **Representation of the bits**: the physical layer data consist of a stream of bits without any interpretation. To be transmitted, bits must be encoded into signals – electrical or optical-. The physical layer defines the type of **encoding**.
- **Data rate**: The physical layer defines the transmission rate, the number of bits sent each second.

Physical Layer

- **Line configuration:** the physical layer is concerned with the connection of devices to the medium.
- **Physical topology**
- **Transmission Mode**

Example of physical layer:

Hub

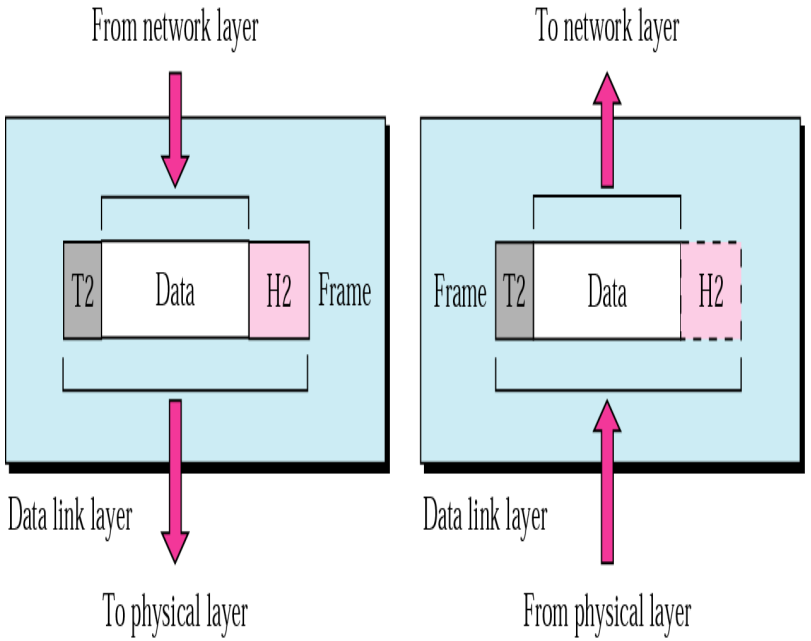
Repeater

Modem

various type of cables

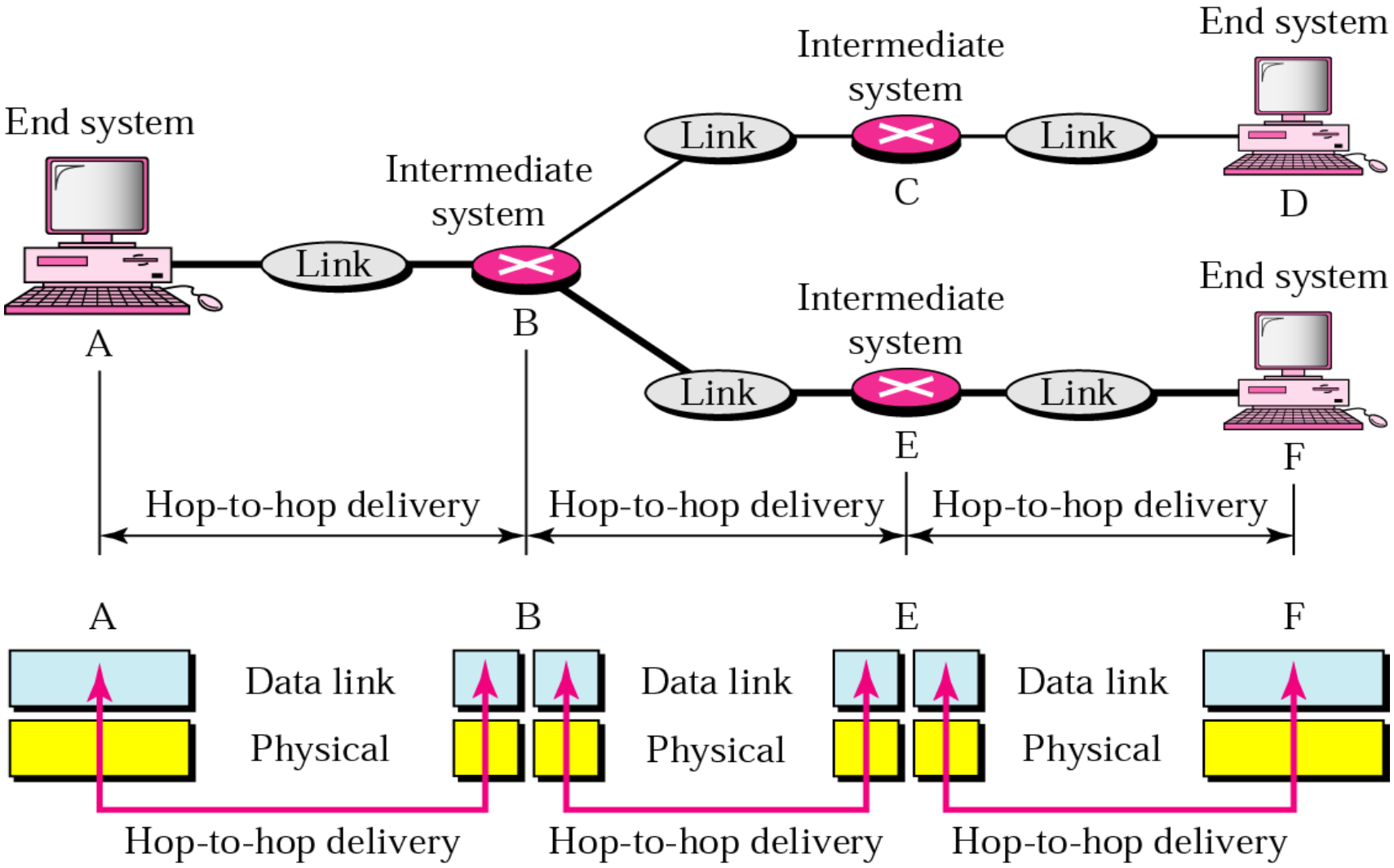
Data Link Layer

- Responsible for node-to-node delivery. It makes the physical layer appear error free to the upper layer (network layer).



The data link layer is responsible for transmitting frames from one node to the next.

Node-to-node delivery

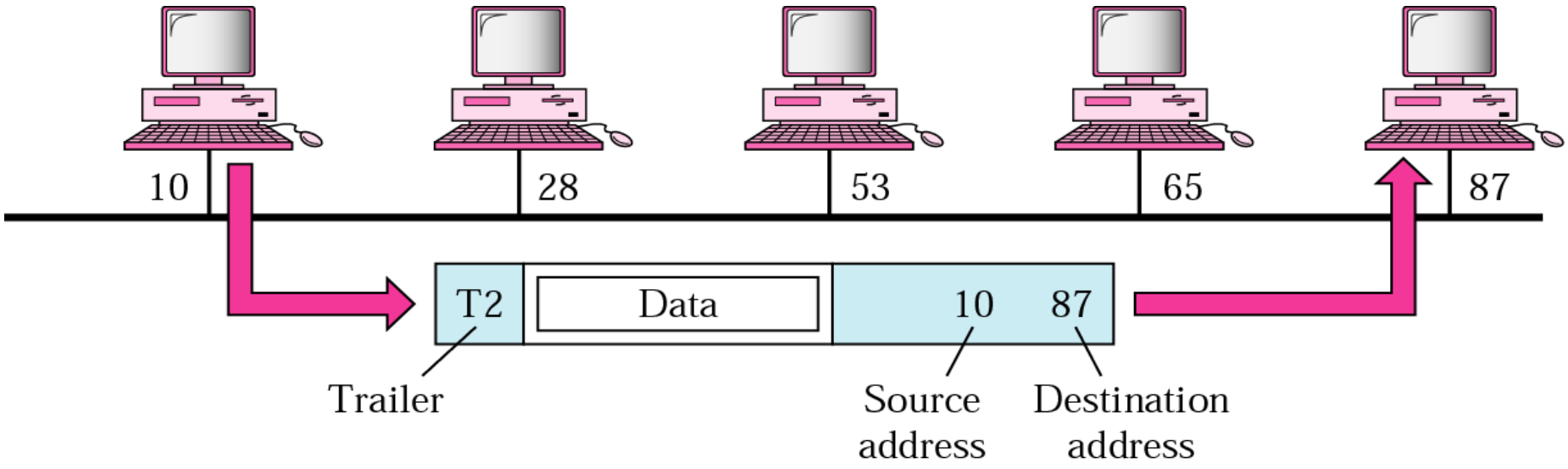


Functions of the data link layer:

- **Framing.** The data link layer divides the stream of bits received from the network layer into data units called **frames**.

- **Physical addressing**. If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the physical address of the sender (source address) and/or receiver (destination address) of the frame.

- **Flow Control**. If the rate at which the data are absorbed by the receiver is less than the rate produced in the sender, the data link layer imposes a flow control mechanism to prevent flow control.


10
28
53
65
87
T2
Data
10
87
Trailer
Source address
Destination address

- **Error control**. The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. Error control is normally achieved through a trailer to the end of the frame.

- **Access Control.** When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any time.

Example of Data link layer:

Switch

Bridges

Network Layer

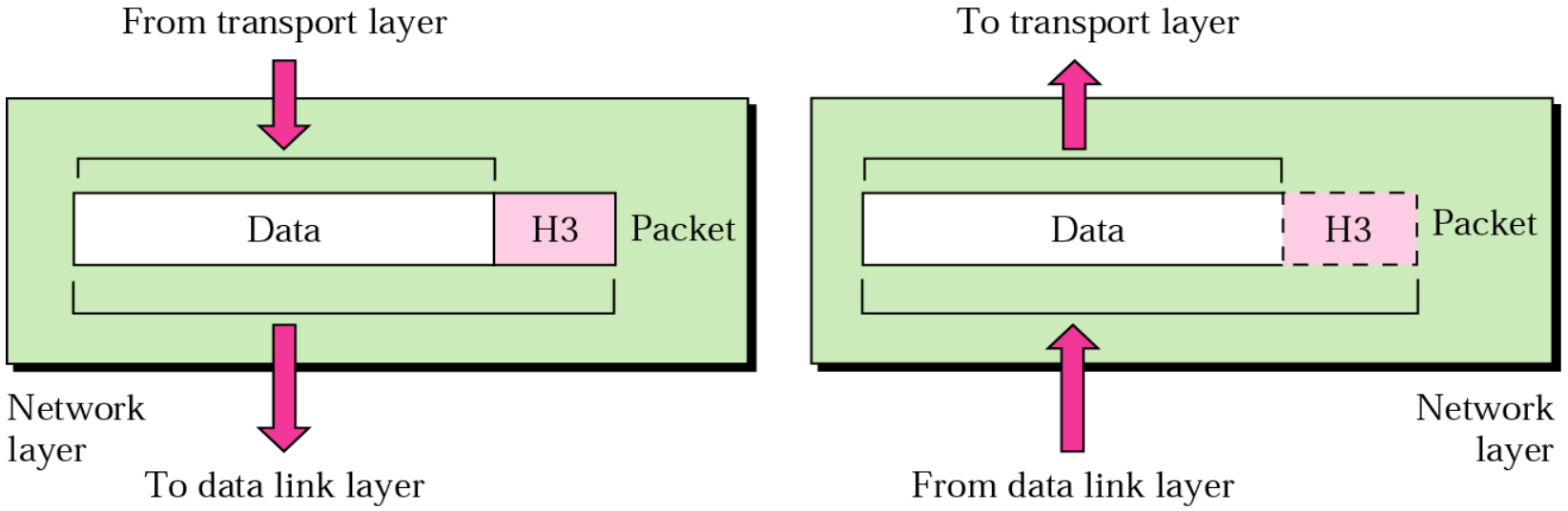
The Network layer is responsible for the source-to-destination delivery of a packet possible across multiple networks.

If two systems are connected to the same link, there is usually no need for a network layer. However, if the two systems are attached to different networks, there is often a need for the network layer to accomplish source-to-destination delivery.

Network Layer

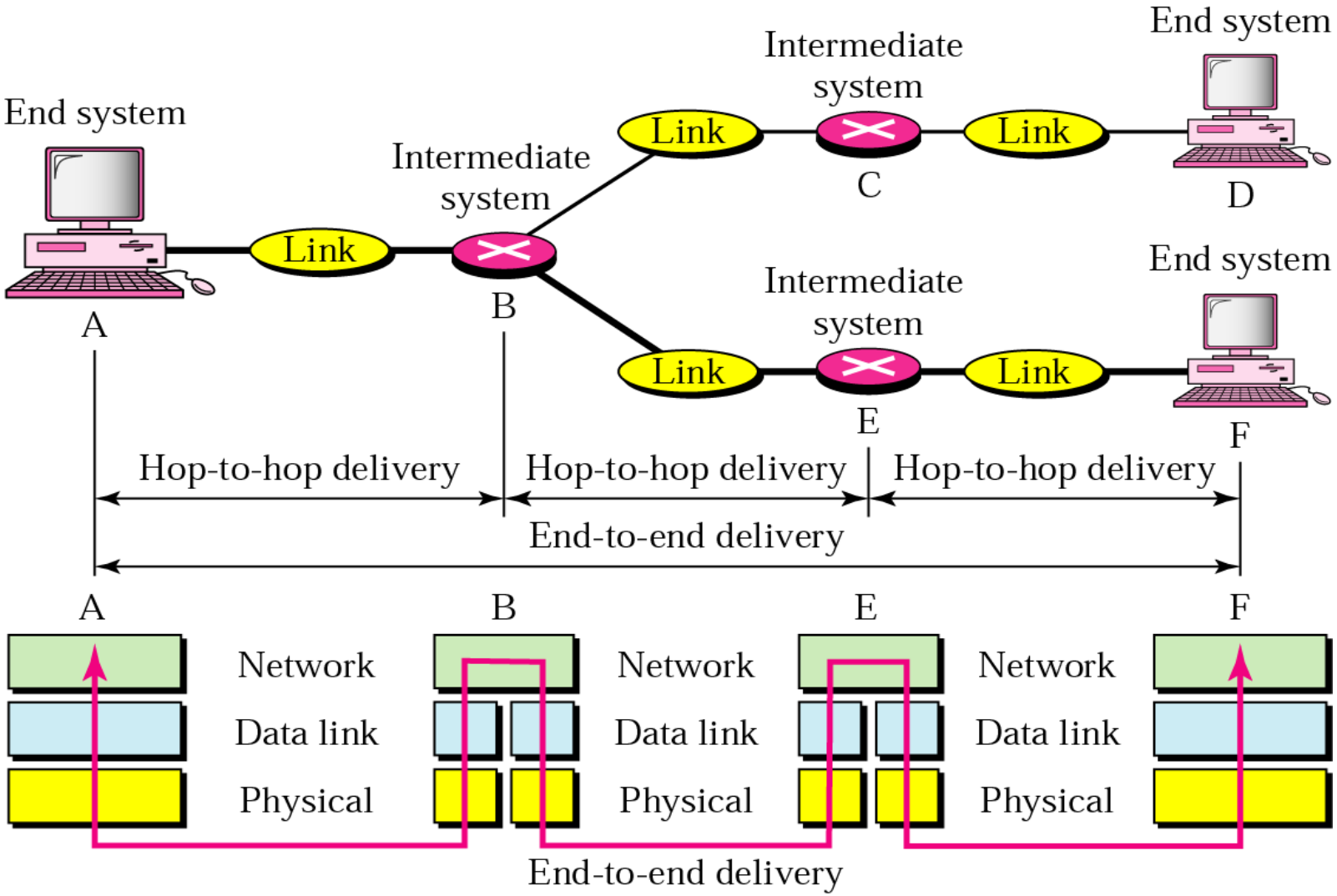
Functions:

- Logical addressing.
- Routing



The network layer is responsible for the delivery of packets from the original source to the final destination.

Source-to-destination delivery

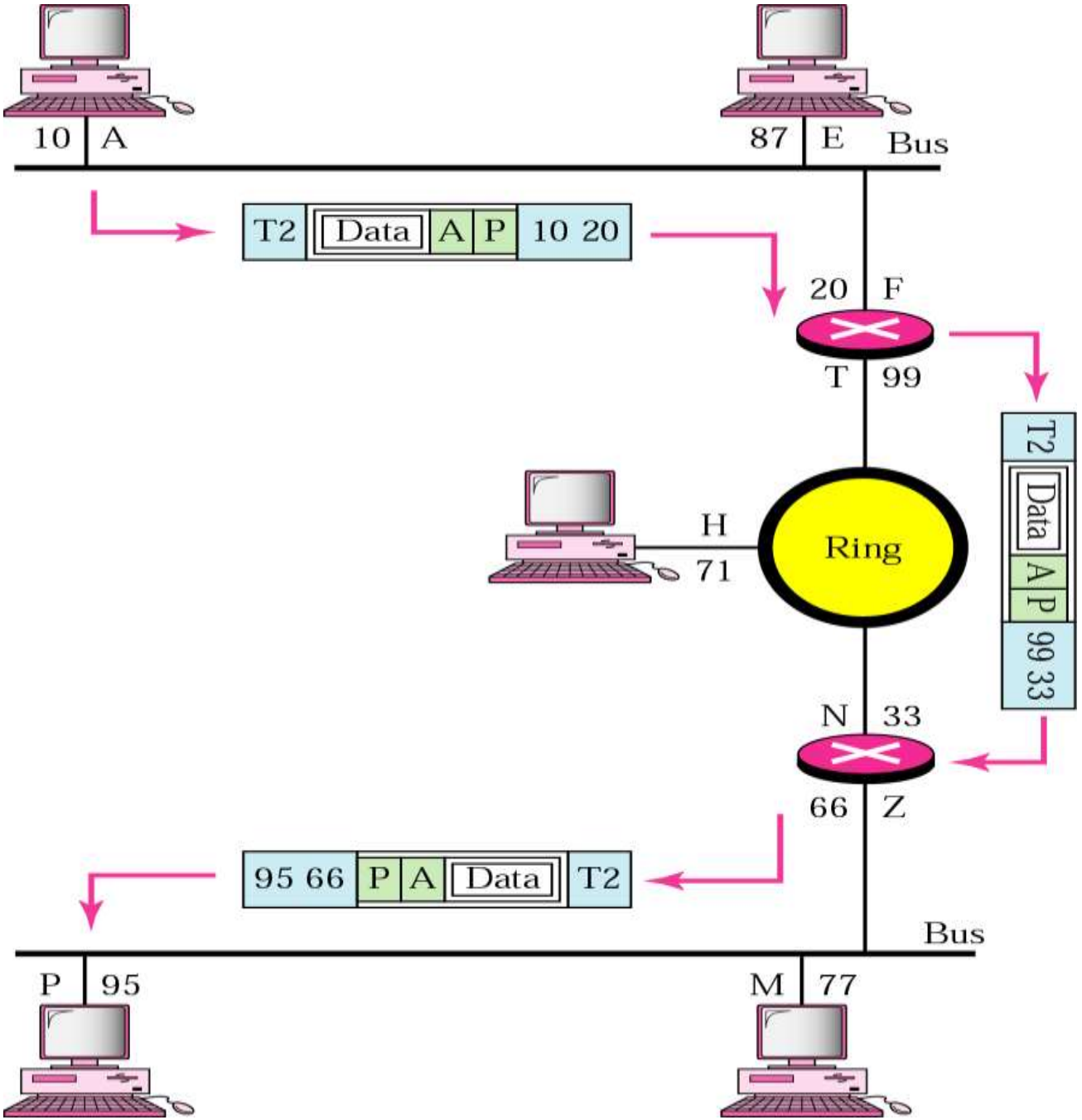


- **Logical addressing**. The physical addressing implemented by the data link layer handles the addressing problem locally.
- The network layer adds a header to the packet coming from the upper layer, the header includes the logical address of the sender and receiver.

- **Routing**. Route or switch the packets to their final destination.

Example:

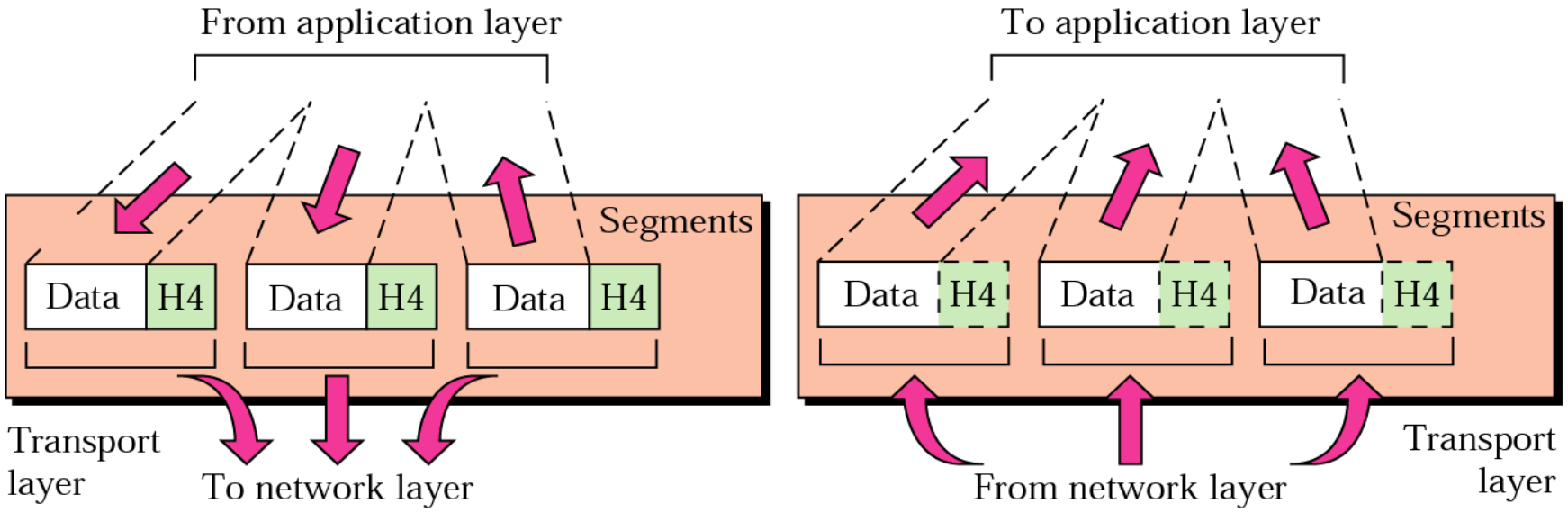
Router


10 A
87 E
Bus
T2 Data A P 10 20
20 F
T 99
T2 Data A P 99 33
H 71
Ring
N 33
66 Z
95 66 P A Data T2
Bus
P 95
M 77

Transport Layer

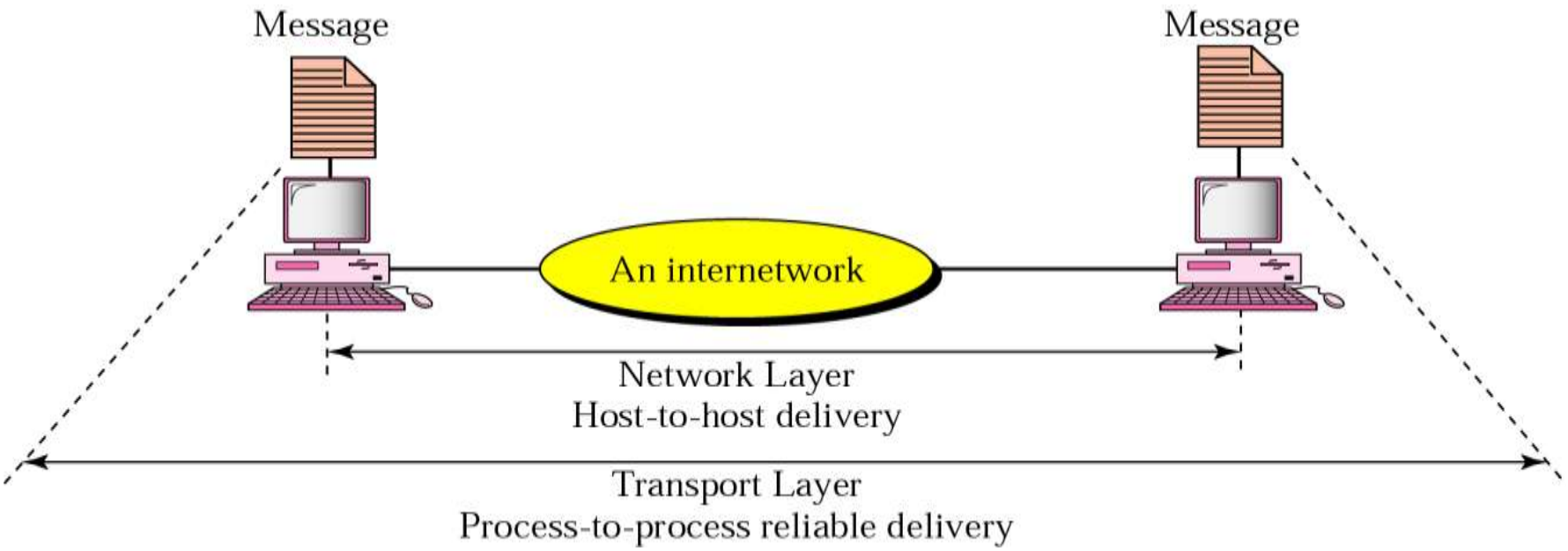
- The transport layer is responsible for process-to-process delivery of the entire message.
- The network layer oversees host-to-destination delivery of individual packets, it does not recognize any relationship between those packets.
- The transport layer ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the process-to-process level.

Transport layer



The transport layer is responsible for delivery of a message from one process to another.

Reliable process-to-process delivery of a message



Functions of the transport layer

- **Port addressing**:
- Computer often run several processes (running programs) at the same time. Process-to-process delivery means delivery from a specific process on one computer to a specific process on the other.
- The transport layer header include a type of address called port address.
- The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.

Functions of the transport layer

- **Segmentation and reassembly**: a message is divided into transmittable segments, each having a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arrival at the destination.
- **Connection control**: The transport layer can be either connectionless or connection-oriented.
- A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine.
- A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.

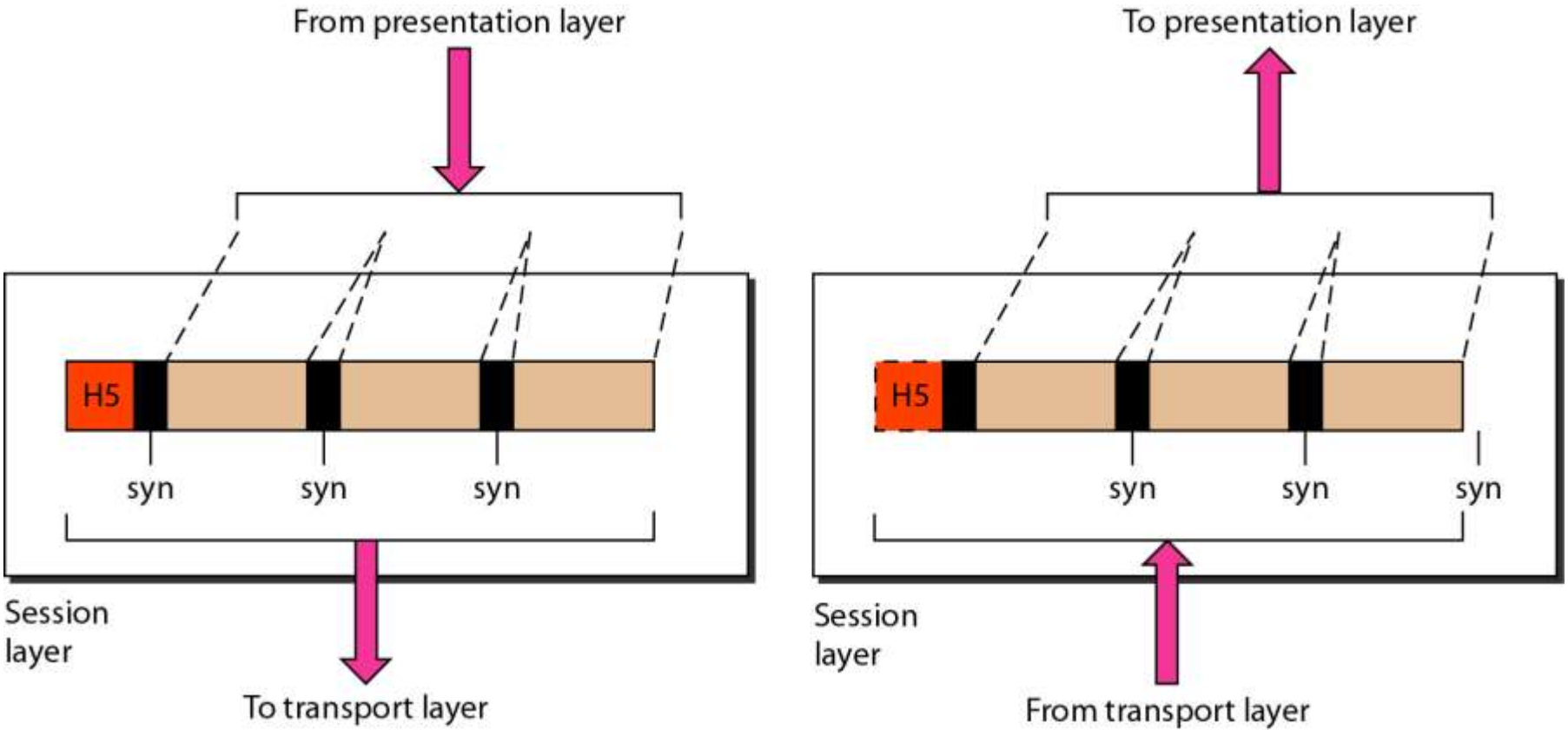
Functions of the transport layer

- **Flow control**: the transport layer performs a flow control end to end. The data link layer performs flow control across a single link.
- **Error control**: the transport layer performs error control end to end. The data link layer performs control across a single link.
- Some examples port numbers

S.No.	Name of the Protocol	Port no
1	FTP	20 and 21
2	SMTP	25
3	HTTP	80
4	DNS	53

Session layer

- The session layer is responsible for dialog control and synchronization.



Function of Session Layer:

Dialog control:

The session layer allow two system to enter into dialog. It allows the communication between two processes to take place either in half duplex or full duplex.

Synchronization:

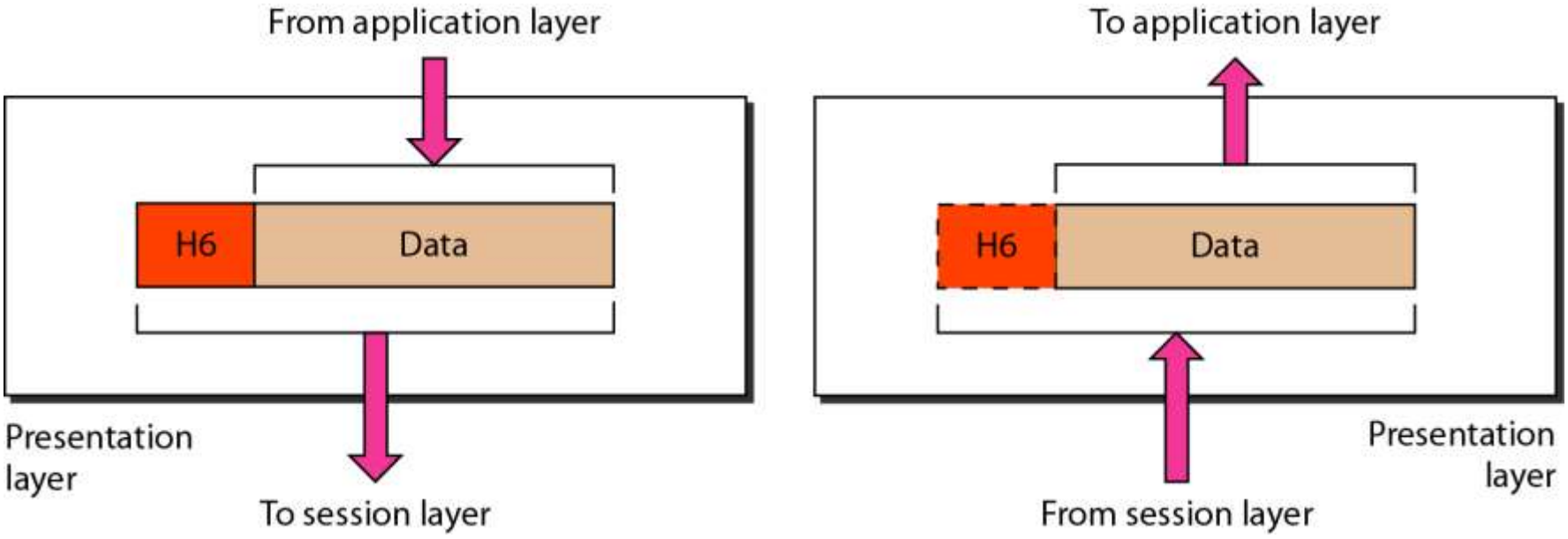
Session layer allows a process to add check points (Synchronization points) into a stream of data.

For example if a system is sending a file of 2000 pages, it is advisable to insert check point after every 100 pages.

In this case if any crashes are happens during the transmission of page 523, retransmission begin at page 501. There is no need to send starting pages from the file.

Presentation layer

- The presentation layer is responsible for translation, compression, and encryption.



Function:

Translation:

Different computers use different encoding scheme the layer is responsible for interoperability between these different encoding methods.

Encryption:

It means that the sender transforms the original information to another form and sends the resulting message out over the network.

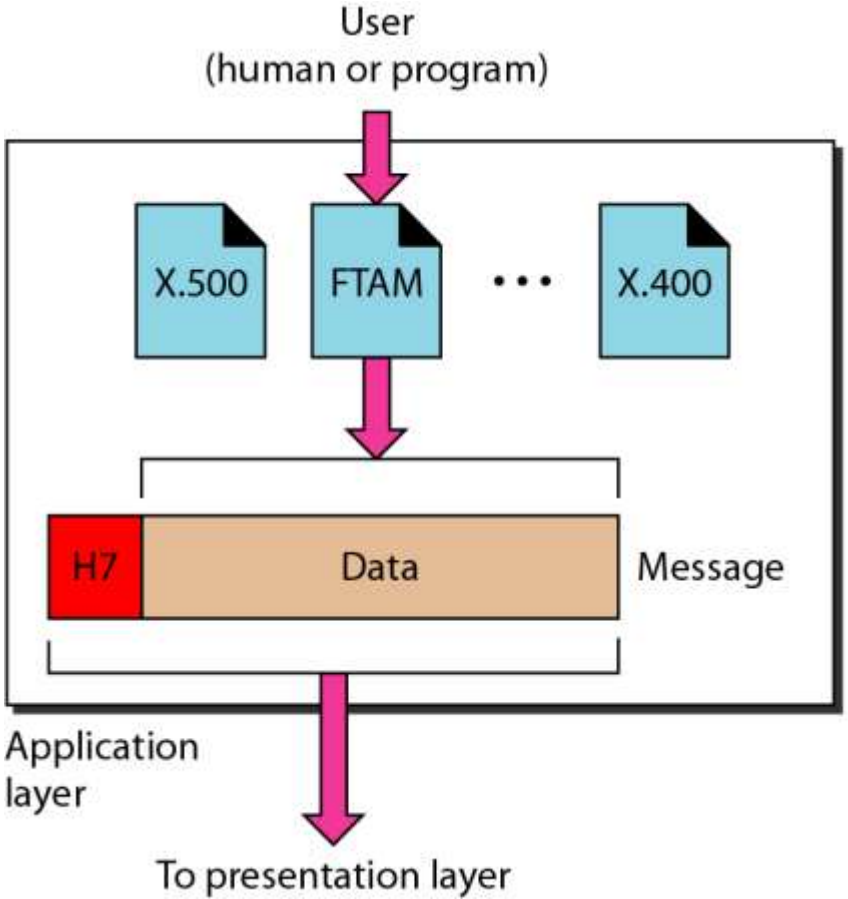
Decryption is reverse of encryption.

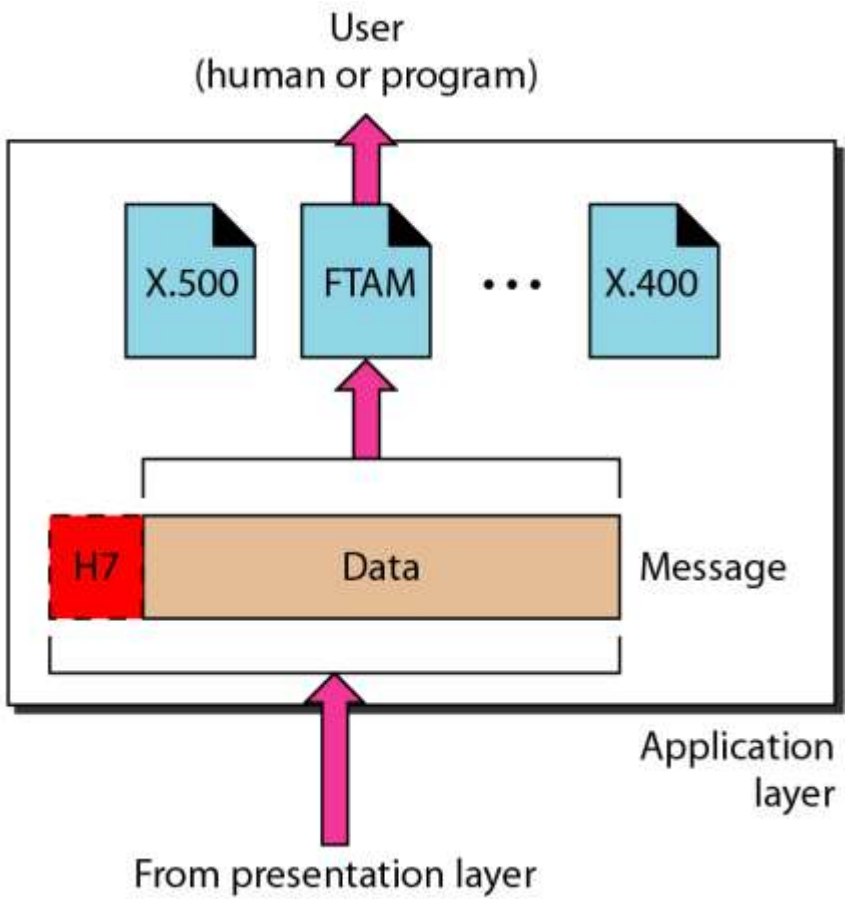
Compression:

Data compression reduces the no of bits to be transmitted.

Application layer

- The application layer is responsible for providing services to the user.





Application layer services:

X.500 Directory services:

The application provide distributed database sources and access for global information about various objects and services.

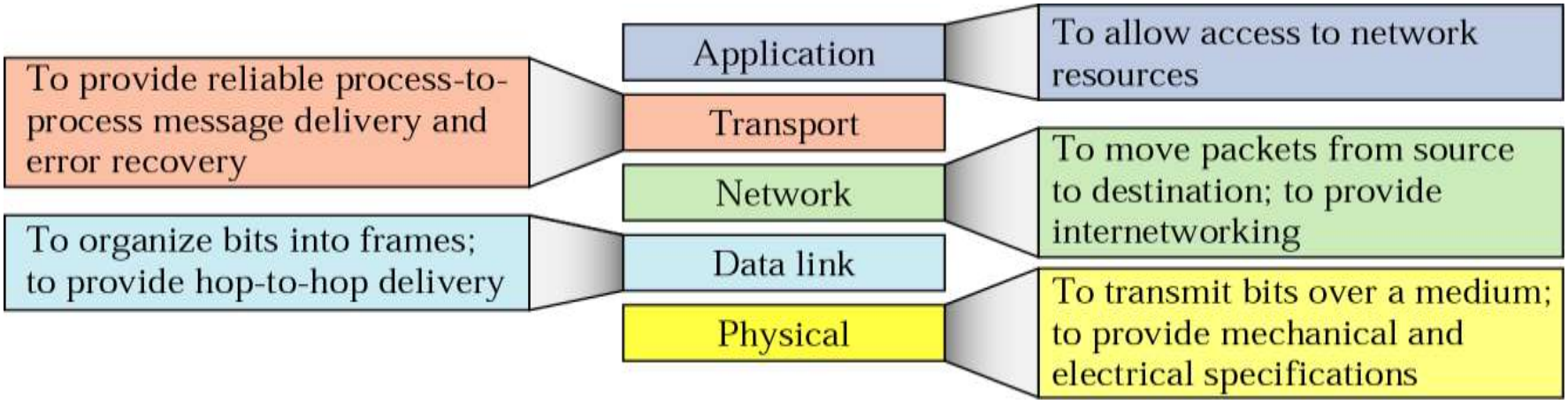
X.400 message handling services:

The application provides the basic for e-mail forwarding and storage.(SMTP)

FTAM:(FILE TRANSFER ACCESS MANAGEMENT)

The application allows user to access files in a remote computer , to retrieve file from a remote computer and to manage or control file in a remote computer.

Summary of duties

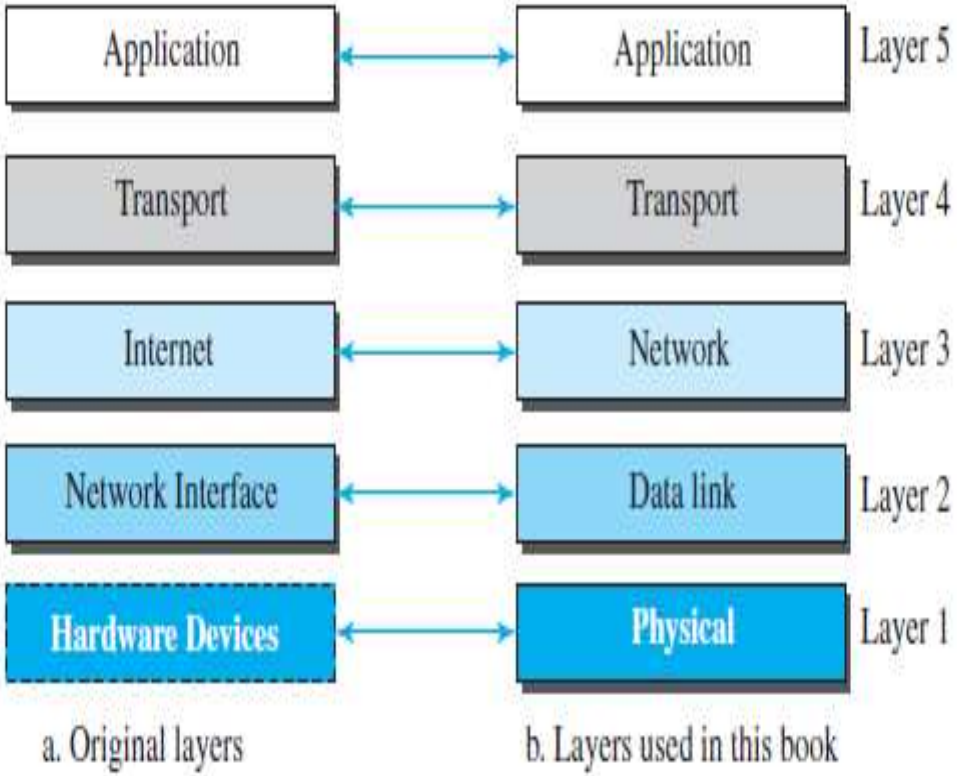


TCP/IP protocol suite

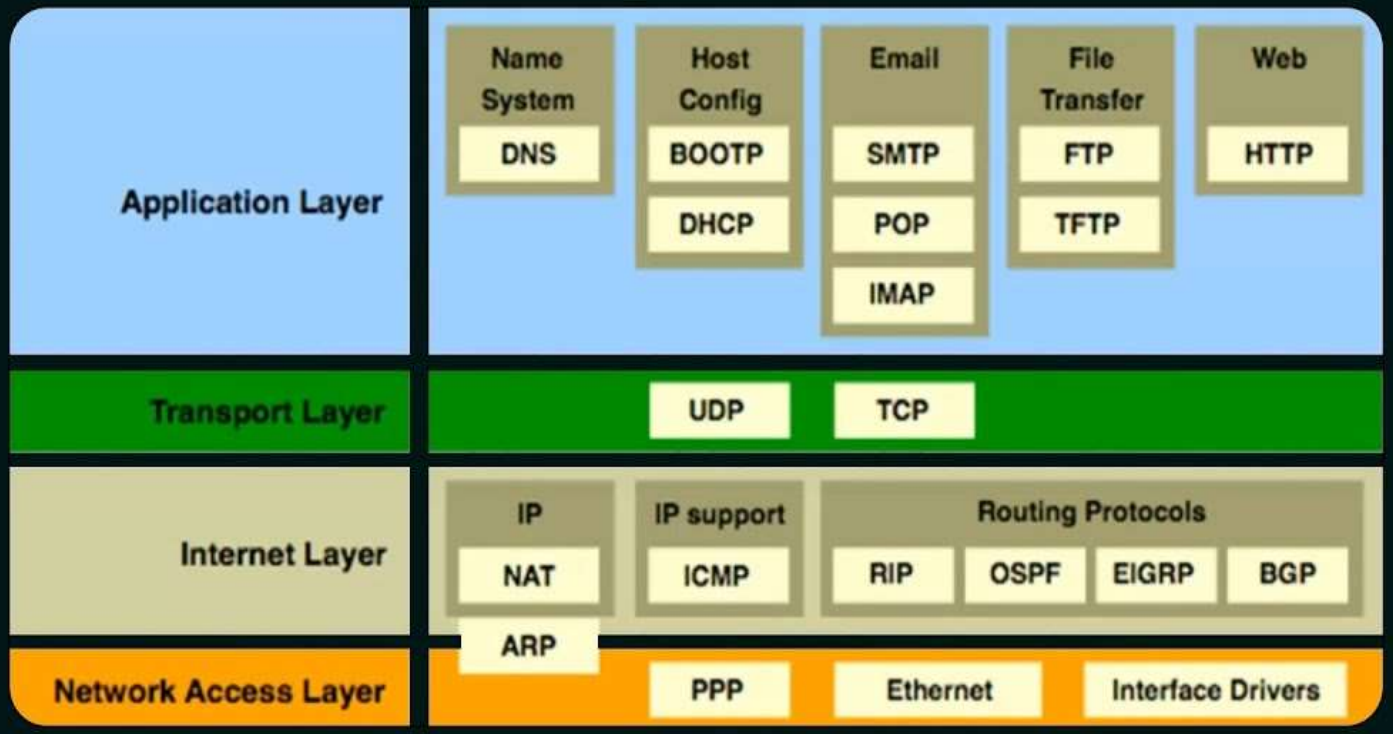
(Transmission Control Protocol/Internet Protocol)

- TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today.
- It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.
- The term *hierarchical* means that each upper level protocol is supported by the services provided by one or more lower level protocols.
- TCP/IP is as a five-layer model.

Figure 2.4 *Layers in the TCP/IP protocol suite*



TCP/IP Protocol Suite

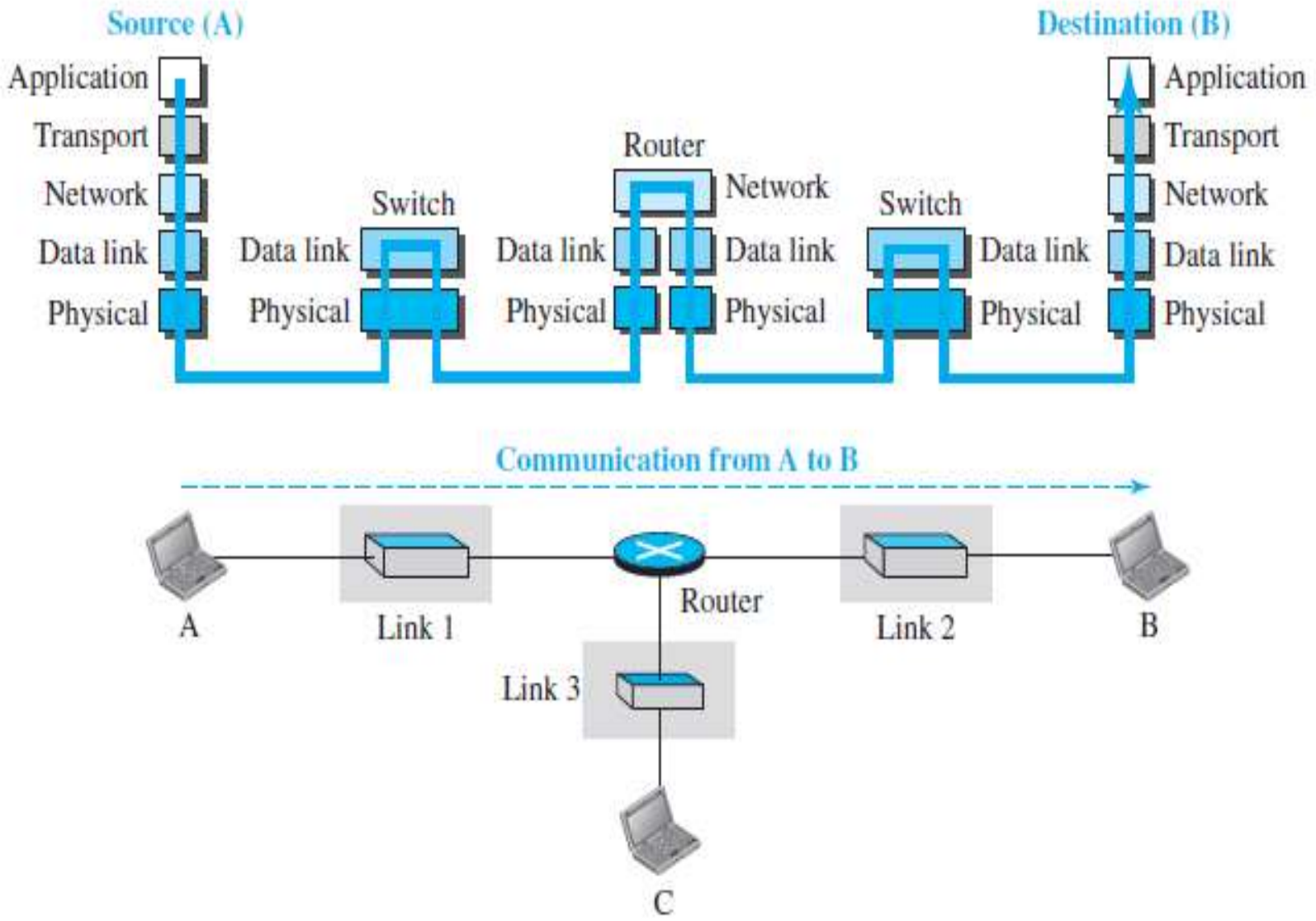


19 Computer Networks

Layered Architecture

- To show how the layers in the TCP/IP protocol suite are involved in communication between two hosts,
- we imagine that we want to use the suite in a small internet made up of three LANs (links), each with a link-layer switch.
- We also assume that the links are connected by one router, as shown in Figure 2.5.
- Let us assume that computer A communicates with computer B. As the figure shows, we have five communicating devices in this communication: source host (computer A), the link-layer switch in link 1, the router, the link-layer switch in link 2, and the destination host (computer B).

Figure 2.5 *Communication through an internet*



- Each device is involved with a set of layers depending on the role of the device in the internet.
- The two hosts are involved in all five layers; the source host needs to create a message in the application layer and send it down the layers so that it is physically sent to the destination host.
- The destination host receive the Message at the physical layer and then deliver it through the other layers to the application layer.
- The router is involved in only three layers; there is no transport or application layer in a router as long as the router is used only for routing.

Layers in the TCP/IP Protocol Suite:

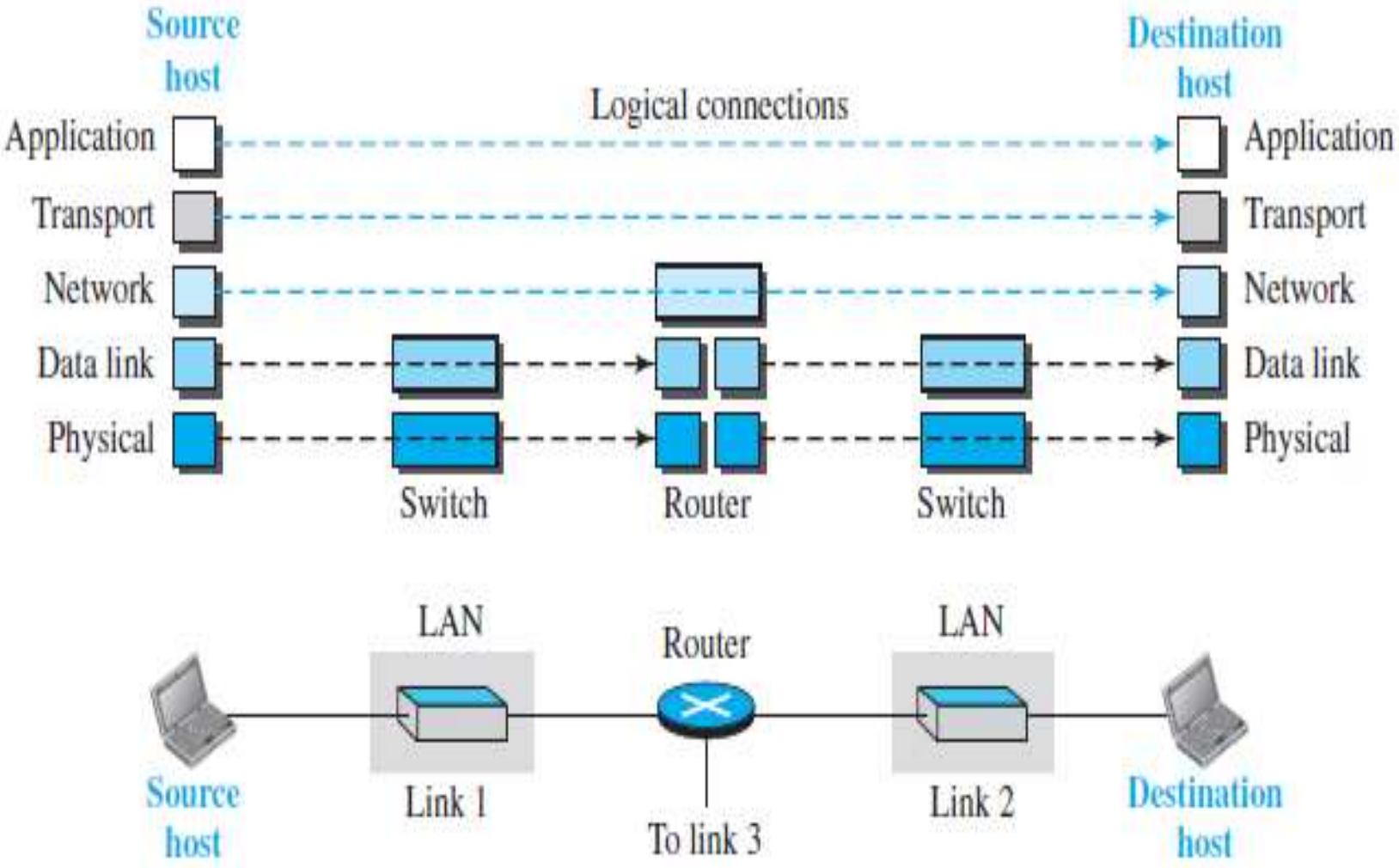
- Using logical connections makes it easier for think about the duty of each layer.
- As the figure shows, the duty of the application, transport, and network layers is end-to-end.
- The duty of the data-link and physical layers is hop-to-hop, in which a hop is a host or router.

Description of each layer:

Physical layer:

- physical layer is responsible for carrying individual bits in a frame across the link.

Figure 2.6 *Logical connections between layers of the TCP/IP protocol suite*



- Although the physical layer is the lowest level in the TCP/IP protocol suite,
- The communication between two devices at the physical layer is still a logical communication because there is another, hidden layer, the transmission media, under the physical layer. Two devices are connected by a transmission medium (cable or air).
- Datalink Layer:
- The routers are responsible for choosing the *best* links.
- when the next link to travel is determined by the router, the data-link layer is responsible for taking the datagram and moving it across the link. The link can be a wired LAN with a link-layer switch, a wireless LAN, a wired WAN, or a wireless WAN.

- The data-link layer takes a datagram and encapsulates it in a packet called a *frame*.

Network Layer

- The communication at the network layer is source host-to- Destination host.
- Since there can be several routers from the source to the destination, the routers are responsible for choosing the best route for each packet.
- **The network layer in the Internet includes the main protocol, Internet Protocol (IP)**, that defines the format of the packet, called a datagram at the network layer.
- IP also defines the format and the structure of addresses.

- IP is also responsible for routing a packet from its source to its destination, which is achieved by each router forwarding the datagram to the next router in its path.
- **IP is a connectionless protocol** that provides no flow control, no error control, and no congestion control services.

Transport Layer

- The logical connection at the transport layer is also end-to-end.
- The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet (called a *segment* or a *user datagram* in different protocols) and sends it, through the logical (imaginary) connection, to the transport layer at the destination host.

- Transport-layer protocols in the Internet, each designed for some specific task. The main protocol, TCP.
- **T**ransmission **C**ontrol **P**rotocol (TCP)
- connection-oriented protocol that first establishes a logical connection between transport layers at two hosts before transferring data.
- It creates a logical pipe between two TCPs for transferring a stream of bytes. TCP provides flow control , error control ,and congestion control to reduce the loss of segments due to congestion in the network.
- The other common protocol, User Datagram Protocol (UDP

UDP:

- is a connectionless protocol
- That transmits user datagrams without first creating a logical connection. In UDP, each user datagram is an independent entity without being related to the previous or the next one (the meaning of the term *connectionless*).
- UDP is a simple protocol that does not provide flow, error, or congestion control.

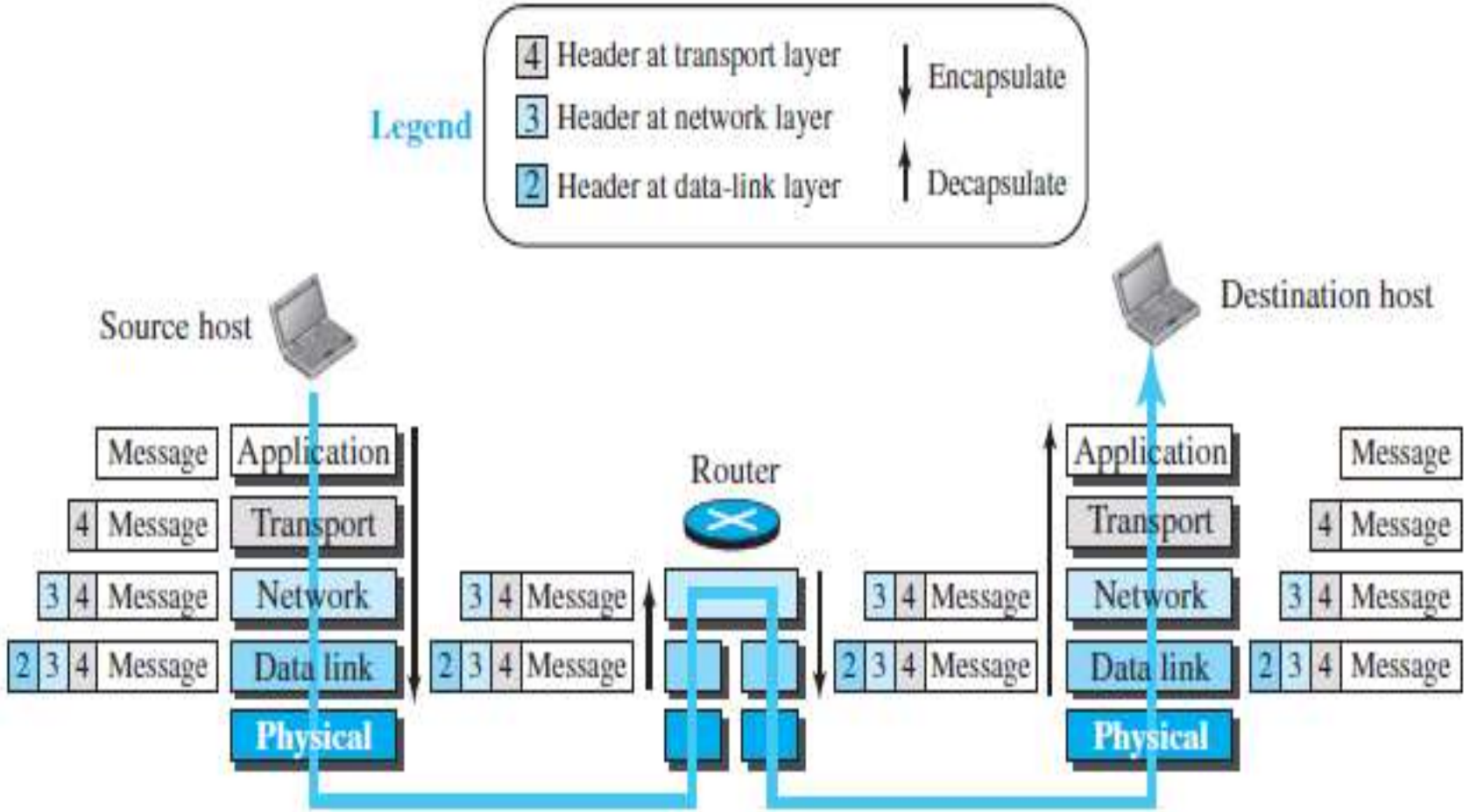
Application Layer:

- logical connection between the two application layers is end to- end.
- The application layer in the Internet includes many predefined protocols

- The Hypertext Transfer Protocol (HTTP) is a vehicle for accessing the World Wide Web (WWW).
- The Simple Mail Transfer Protocol (SMTP) is the main protocol used in electronic mail (e-mail) service.
- The File Transfer Protocol (FTP) is used for transferring files from one host to another.
- The Terminal Network (TELNET) used for accessing a site remotely.
- The Simple Network Management Protocol (SNMP) is used by an administrator to manage the Internet at global and local levels.
- The Domain Name System (DNS) is used by other protocols to find the network-layer address of a computer

Encapsulation and Decapsulation:

Figure 2.8 *Encapsulation/Decapsulation*



- At the source, we have only encapsulation.
- **1.** At the application layer, the data to be exchanged is referred to as a *message*. A message normally does not contain any header or trailer, the message is passed to the transport layer.
- **2.** The transport layer takes the message as the payload, It adds the transport layer header to the payload, which contains the identifiers of the source and destination application programs that want to communicate .The result of the transport-layer packet, which is called the *segment* (in TCP) and the *user datagram* (in UDP). The transport layer then passes the packet to the network layer.

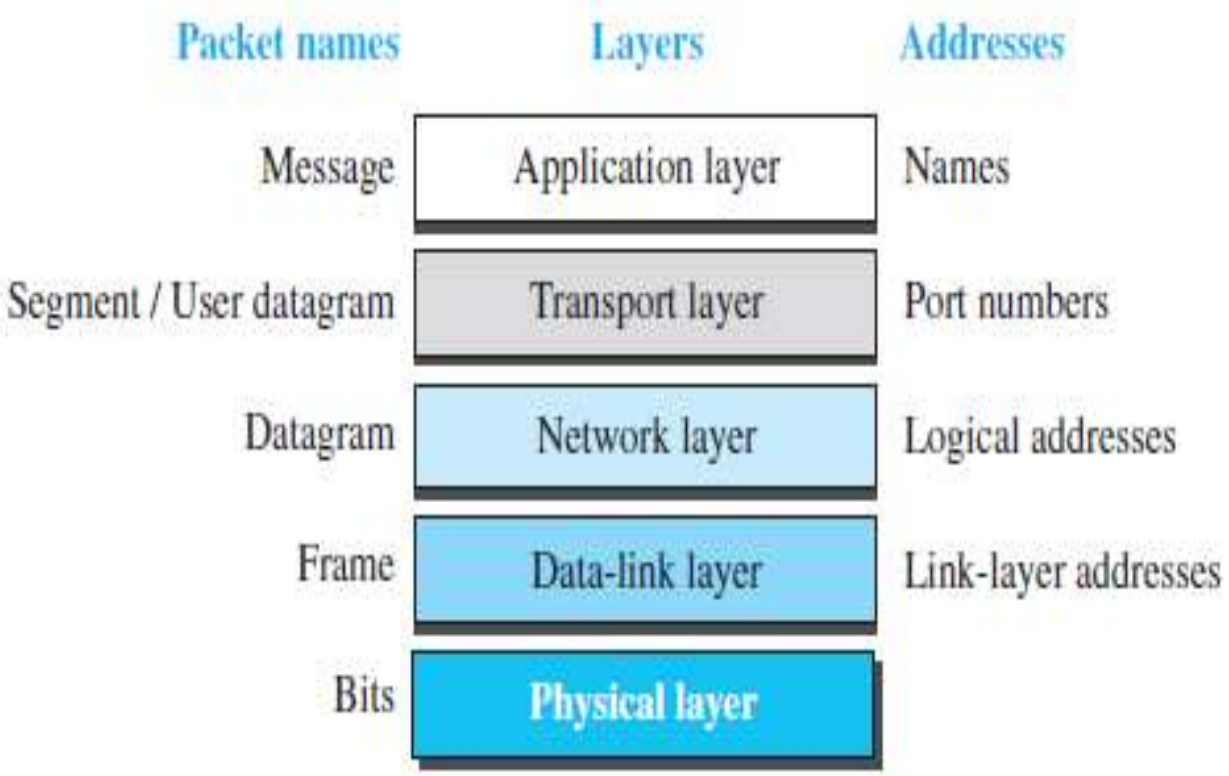
3. The network layer takes the transport-layer packet as data or payload and adds its own header to the payload. The header contains the addresses of the source and destination.

- The result is the network-layer packet, called a *datagram*. The network layer then passes the packet to the data-link layer.

4. The data-link layer takes the network-layer packet as data or payload and adds its own header, which contains the link-layer addresses of the host or the next hop (the router). The result is the link-layer packet, which is called a *frame*. The frame is passed to the physical layer for transmission.

Addressing:

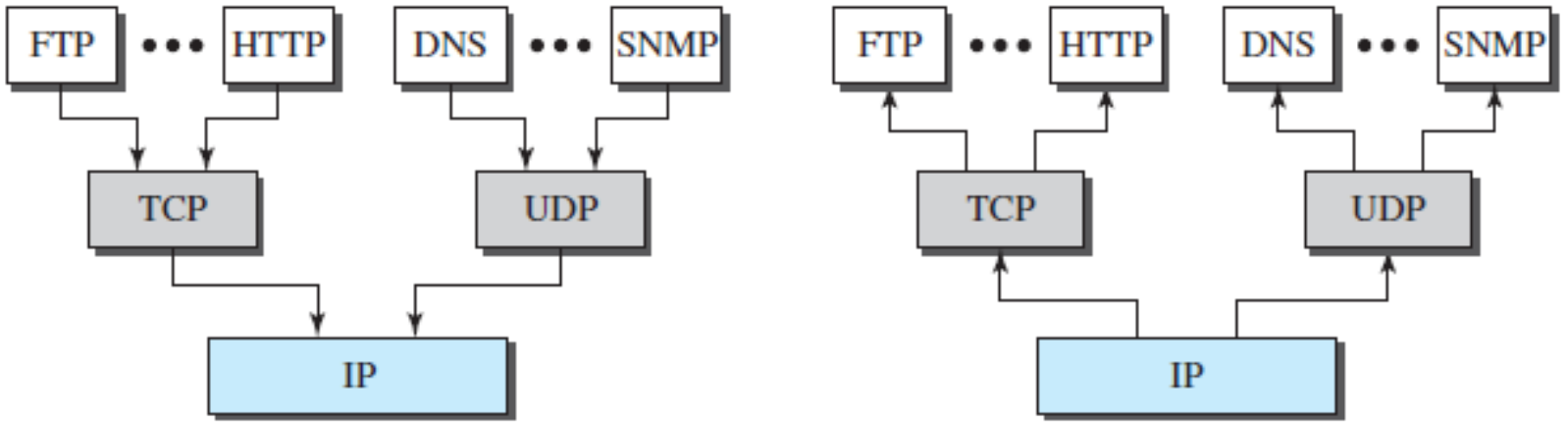
Figure 2.9 *Addressing in the TCP/IP protocol suite*



Multiplexing and Demultiplexing

- Multiplexing means that a protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time);
- Demultiplexing means that a protocol can decapsulate and deliver a packet to several next-higher layer protocols (one at a time).

Figure 2.10 *Multiplexing and demultiplexing*



a. Multiplexing at source

b. Demultiplexing at destination

In General:

Multiplexing is the process in which multiple Data Streams, coming from different Sources, are combined and Transmitted over a Single Data Channel or Data Stream.

At the Receiving End, the Composite Signal is separated by an equipment called Demultiplexer (DEMUX). Demultiplexer performs the reverse process of Multiplexing and routes the separated signals to their corresponding Receivers or Destinations.

SOCKETS

- A **socket** is one endpoint of a **two way** communication link between two programs running on the network.
- The socket mechanism provides a means of inter-process communication (IPC) by establishing named contact points between which the communication take place.
- Each socket has a specific address. This address is composed of an IP address and a port number.

Figure 10.7 *A socket address*



- socket defines the end-point of the communication, we can say that a socket is identified by a pair of socket addresses, a local and a remote

Server Site

- The server needs a local (server) and a remote (client) socket address for communication

Local Socket Address

socket address is provided by the operating system

Remote Socket Address

The remote socket address for a server is the socket address of the client that makes the connection.

Client Site

Local Socket Address

socket address is provided by the operating system

Remote Socket Address

Application Layer Protocols

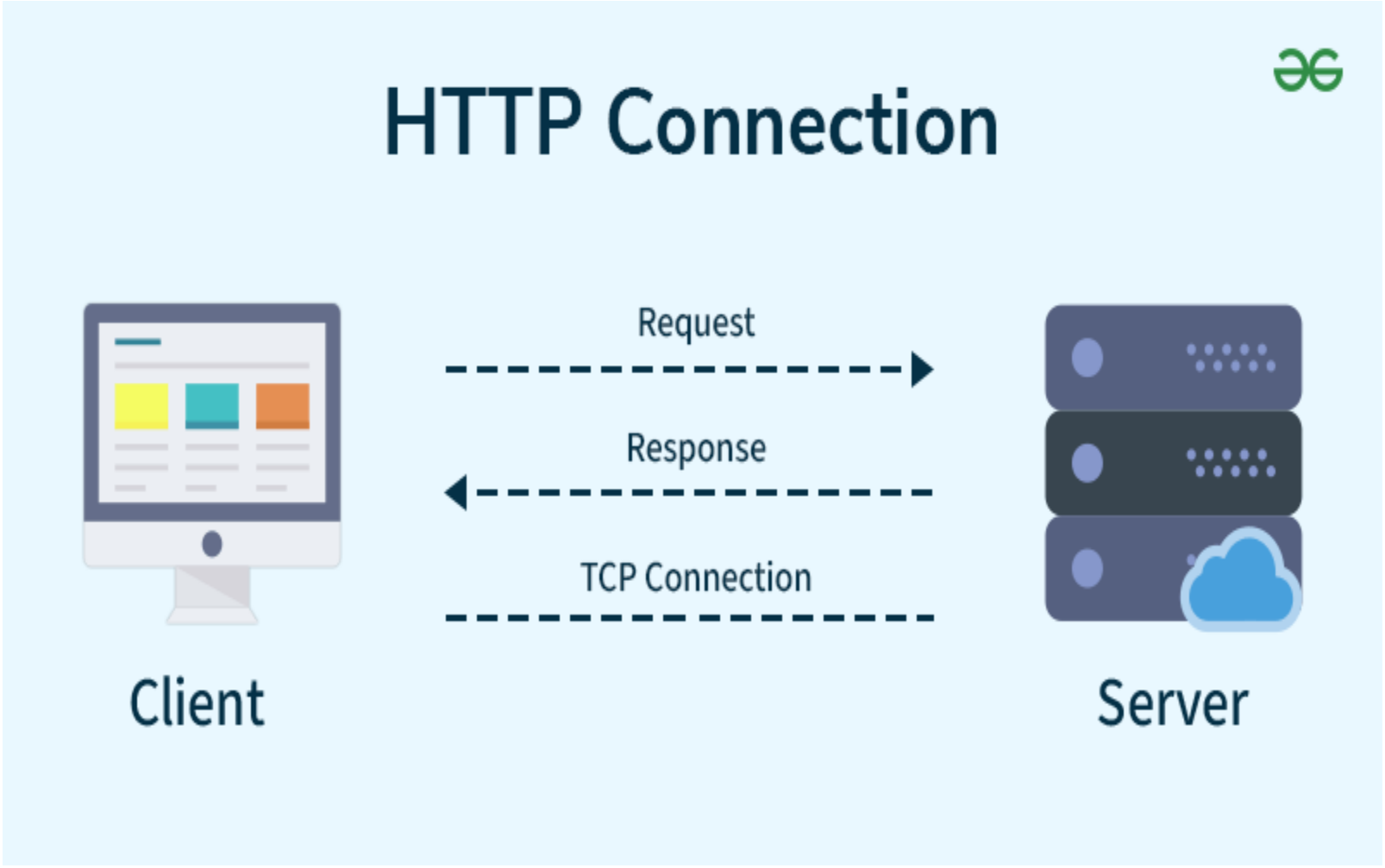
HTTP: Hypertext Transfer Protocol

- Define how the client- server programs can be written to retrieve web pages from the Web.
- It is a protocol used to access the data on the World Wide Web (WWW).
- An HTTP client sends a request; an HTTP server returns a response.
- The server uses the port number 80; the client uses a temporary port number. HTTPS use the port number of 443.

- HTTP is a text-oriented protocol. It contains embedded URL known as links.

- When hypertext is clicked, browser opens a new connection, retrieves file from the server and displays the file.

- HTTP is a connectionless protocol. HTTP client initiates a request and waits for a response from the server. When the server receives the request, the server processes the request and sends back the response to the HTTP client after which the client disconnects the connection.

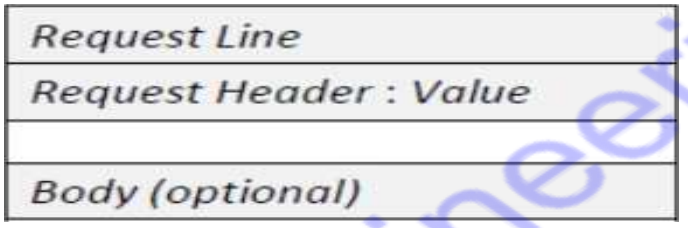

HTTP Connection
Request
Response
TCP Connection
Client
Server

Request Message:

- The request message is sent by the client that consists of a request line, headers, and sometimes a body.

Request Line

- There are three fields in this request line - Method, URL and Version.
- The Method field defines the request types.
- The URL field defines the address and name of the corresponding web page.
- The Version field gives the version of the protocol; the most current version of HTTP is 1.1.
- Some of the Method types are:



www.eduengineering.net

Method	*Action*
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
PUT	Sends a document from the client to the server
POST	Sends some information from the client to the server
TRACE	Echoes the incoming request
DELETE	Removes the web page
CONNECT	Reserved
OPTIONS	Inquires about available options

GET:

Retrieve data from the server.

It does not modify data on the server and can be repeated without side effects.

Example: GET /users

POST:

Submit data to the server to create a new resource or trigger a process. **Example**: Post/Users (Create a new user)

PUT:

Update or replace an existing resource.

Repeating a PUT request with the same data should have the same effect as doing it once.

Example: PUT /users/123 (to update the user with ID 123)

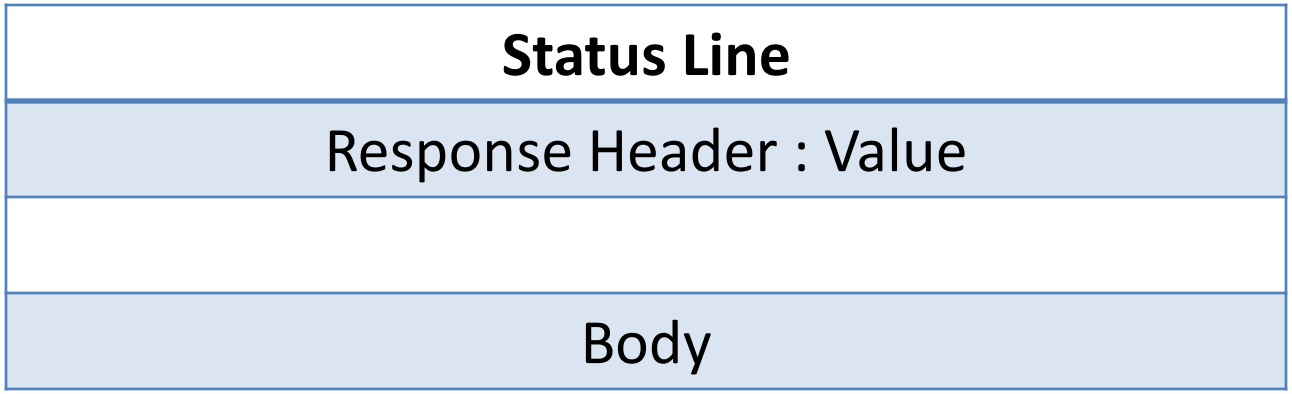
Delete:

Remove a resource from the server.

Example: Delete/Users/123

Response Message:

- The response message is sent by the server to the client that consists of a status line, headers, and sometimes a body.



Http Cookies

- An HTTP cookie (also called web cookie, Internet cookie, browser cookie, or simply cookie) is a small piece of data sent from a website and stored on the user's computer by the user's web browser while the user is browsing.
- HTTP is stateless , Cookies are used to add State

HTTP CONNECTIONS

- HTTP Clients and Servers exchange multiple messages over the same TCP connection.

Non-Persistent Connections

- In a non-persistent connection, one TCP connection is made for each request/response.
- Only one object can be sent over a single TCP connection
- The client opens a TCP connection and sends a request.
- The server sends the response and closes the connection.

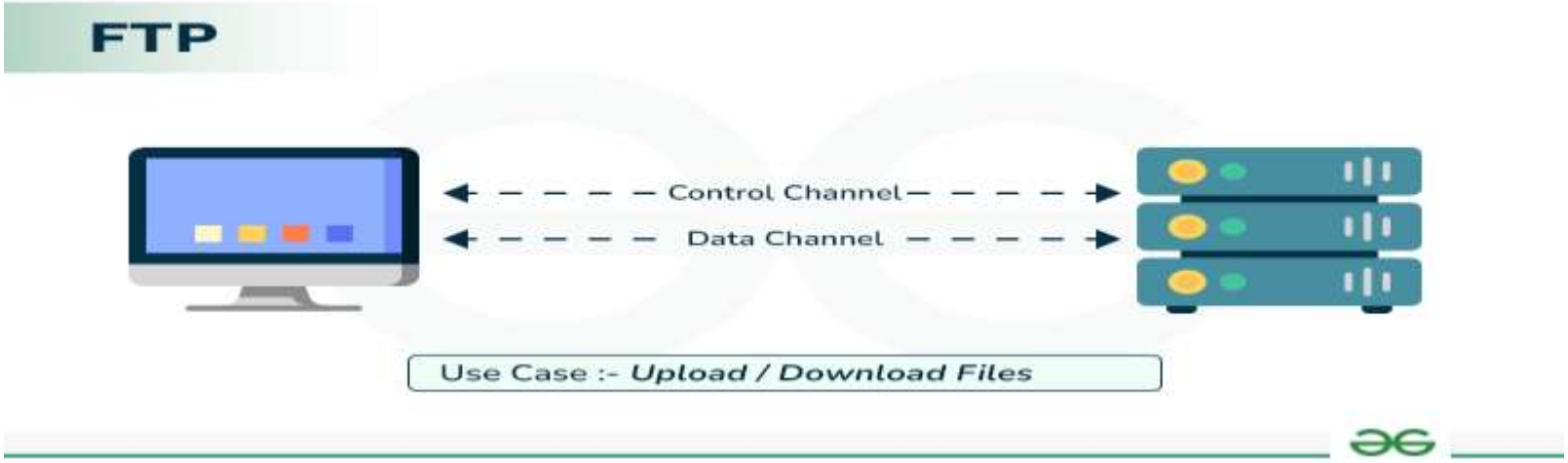
Persistent Connections

- HTTP specifies a persistent connection by default.
- Multiple objects can be sent over a single TCP connection.
- In a persistent connection, the server leaves the connection open for more requests after sending a response.
- The server can close the connection at the request of a client or if a time-out has been reached.

HTTP SECURITY

- HTTP does not provide security.
- However HTTP can be run over the Secure Socket Layer (SSL).
- In this case, HTTP is referred to as HTTPS.
- HTTPS provides confidentiality, client and server authentication, and data integrity.

File Transfer Protocol (FTP) in Application Layer

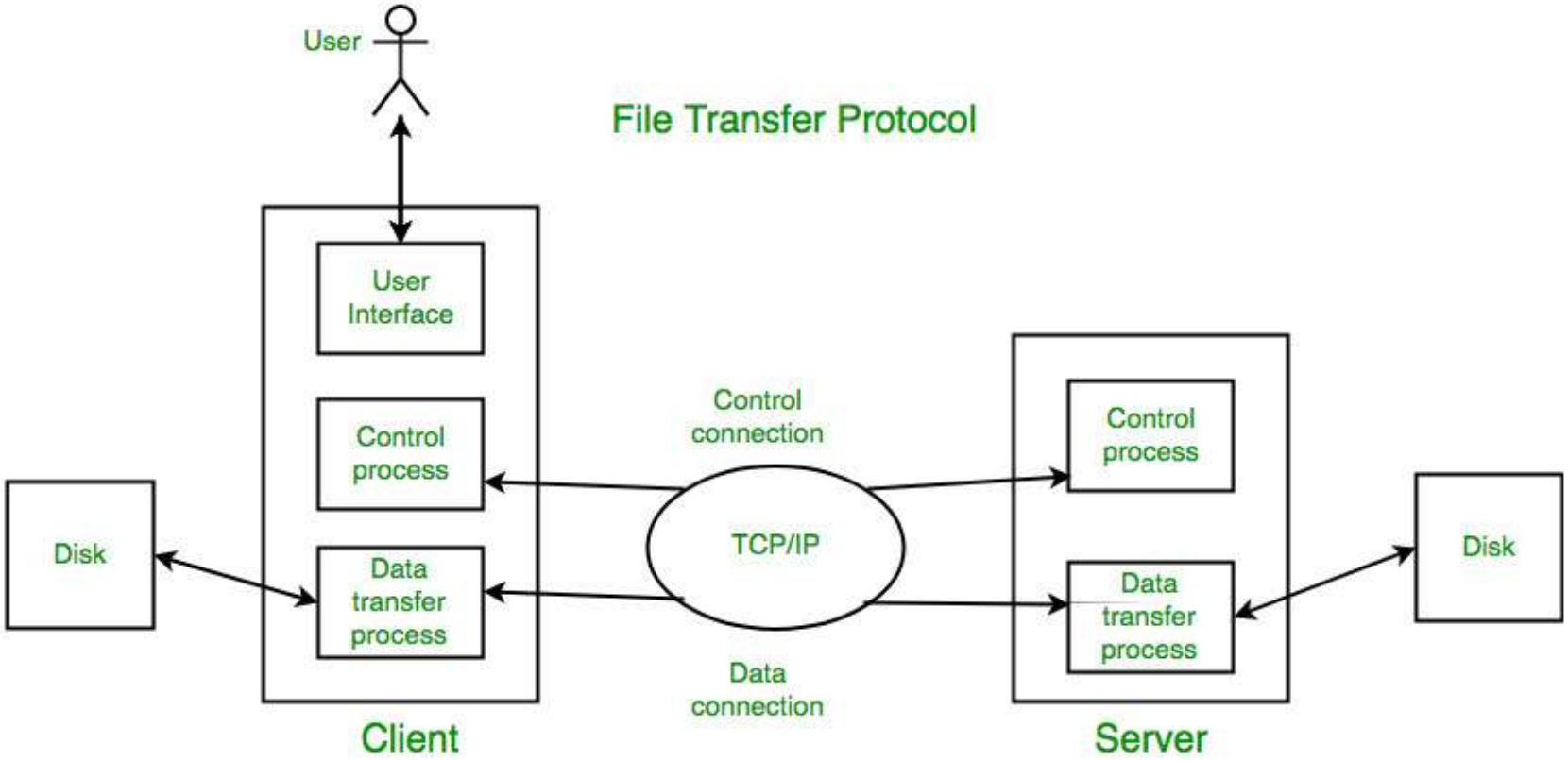


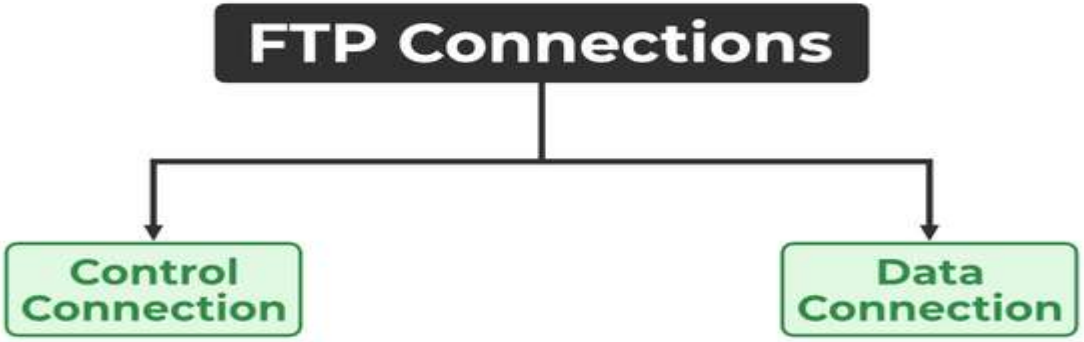
Located in the application layer of the OSI model, FTP is a basic system that helps in transferring files between a client and a server.

When u uploading a files, the files are transferred from a personal Computer to Server

Types of FTP

- **There are different ways through which a server and a client do a file transfer using FTP. Some of them are mentioned below:**
- **Anonymous FTP:** Anonymous FTP is enabled on some sites whose files are available for public access. A user can access these files without having any username or password.
- **Password Protected FTP:** This type of FTP is similar to the previous one, but the change in it is the use of username and password.
- **FTP Secure (FTPS):** It is also called as FTP Secure Sockets Layer (FTP SSL). It is a more secure version of FTP data transfer. Whenever FTP connection is established, Transport Layer Security (TLS) is enabled.
- **Secure FTP (SFTP):** SFTP is not a FTP Protocol, but it is a subset of Secure Shell Protocol, as it works on port 22.


User
File Transfer Protocol
User Interface
Control process
Data transfer process
Disk
Client
Control connection
TCP/IP
Data connection
Control process
Data transfer process
Disk
Server



Control Connection

For sending control information like user identification, password, commands to change the remote directory, commands to retrieve and store files, etc., FTP makes use of a control connection. The control connection is initiated on port number 21.

Data connection

For sending the actual file, FTP makes use of a data connection. A data connection is initiated on port number 20.

E-MAIL PROTOCOLS

SMTP (Simple Mail Transfer Protocol):

Used primarily for sending emails from the sender's client to the mail server and between mail servers.

SMTP manages email transmission but does not handle receiving or storing emails. It works over TCP/IP and is connection-oriented.

POP3 (Post Office Protocol version 3):

Used for receiving emails by downloading them from a mail server to a client device. POP3 typically downloads and removes emails from the server, allowing offline access but limiting synchronization between multiple devices.

IMAP (Internet Message Access Protocol):

Used for receiving and managing emails directly on the mail server, allowing multiple clients to access and synchronize the same mailbox concurrently through various devices such as tablet, mobile phone, laptop….

Architectural Components of Email protocol :

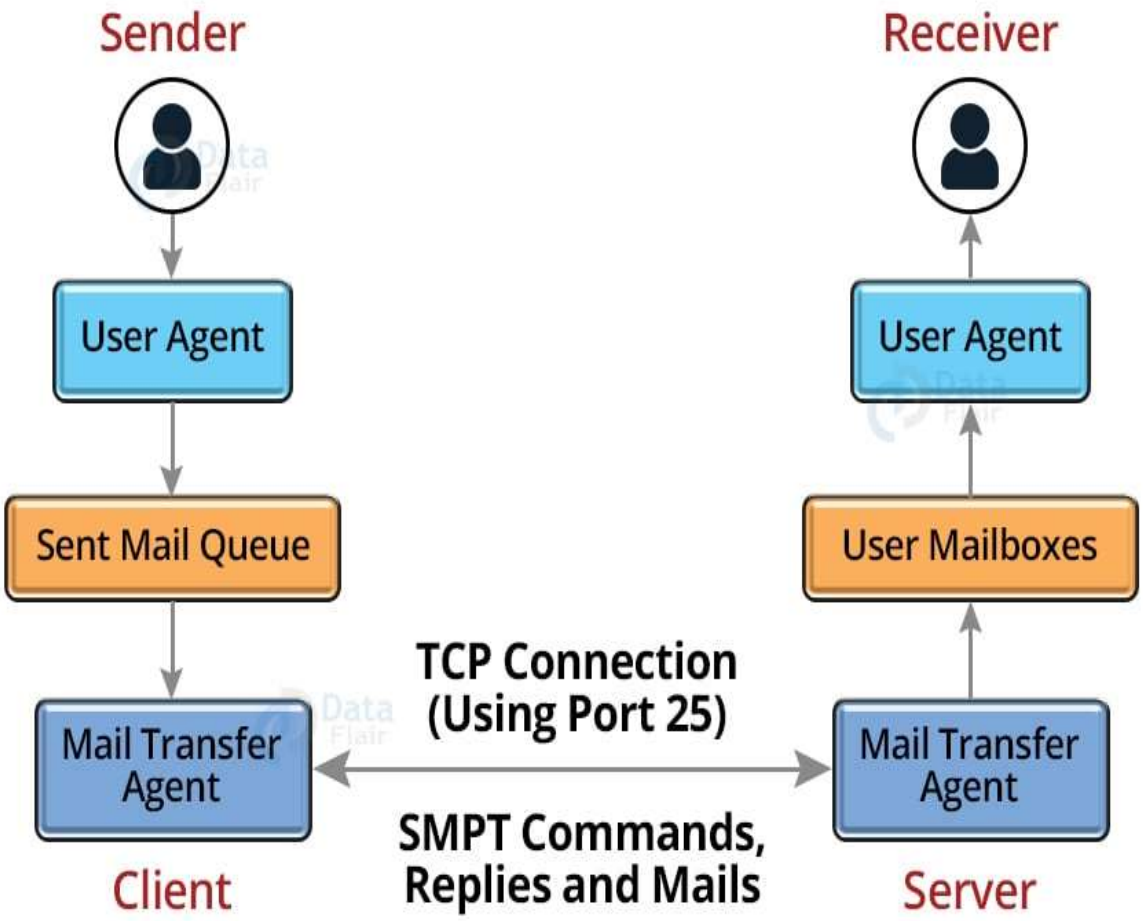
- **User Agent** (UA): This is the email client used by the user to compose, send, receive, and read emails. It acts as the interface between the user and the email service.

  Ex. Gmail ,yahoo

- **Message Transfer Agent (MTA):** This component handles the transfer of email messages between servers using protocols such as SMTP (Simple Mail Transfer Protocol). It routes and delivers emails across the network.

- **Message Access Agent (MAA):** This component enables users to access and retrieve their emails from the mail server using protocols like POP3 (Post Office Protocol version 3) or IMAP (Internet Message Access Protocol).

Working of SMTP

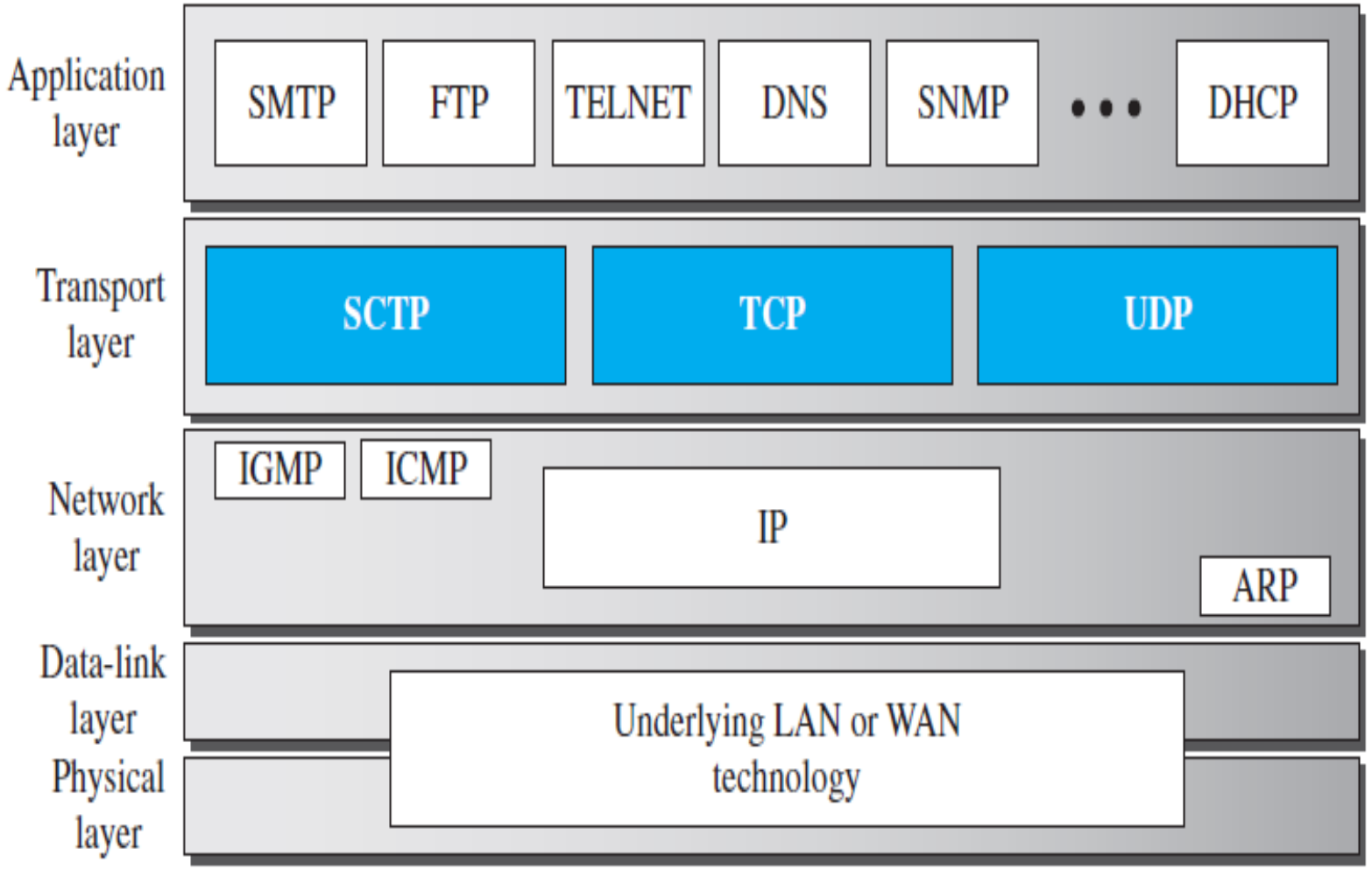


Unit-2 Transport Layer

The transport layer is the fourth layer of the OSI model and is the core of the Internet model.

- It responds to service requests from the session layer and issues service requests to the network Layer.
- The transport layer provides transparent transfer of data between hosts.
- It provides end-to-end control and information transfer with the quality of service needed by the application program.
- It is the first true end-to-end layer, implemented in all End Systems (ES).

Figure 24.1 *Position of transport-layer protocols in the TCP/IP protocol suite*



TRANSPORT LAYER FUNCTIONS / SERVICES

- The transport layer is located between the network layer and the application layer.
- The transport layer is responsible for providing services to the application layer; it receives services from the network layer.
- The services that can be provided by the transport layer are

1. Process-to-Process Communication

2. Addressing : Port Numbers

3. Encapsulation and Decapsulation

4. Multiplexing and Demultiplexing

5. Flow Control

6. Error Control

7. Congestion Control

Congestion Control

- Congestion in a network may occur if the *load* on the network (the number of packets sent to the network) is greater than the *capacity* of the network (the number of packets a network can handle).
- Control the congestion and keep the load below the capacity.
- Congestion Control refers to techniques and mechanisms that can either prevent congestion, before it happens, or remove congestion, after it has happened
- Congestion control mechanisms are divided into two categories

1. Open loop - prevent the congestion before it happens.
2. Closed loop - remove the congestion after it happens.

USER DATAGRAM PROTOCOL

- The **User Datagram Protocol (UDP)** is a connectionless, unreliable transport protocol.
- UDP is a very simple protocol using a minimum of overhead. If a process wants to send a small message and does not care much about reliability, it can use UDP.
- Sending a small message using UDP takes much less interaction between the sender and receiver than using TCP.

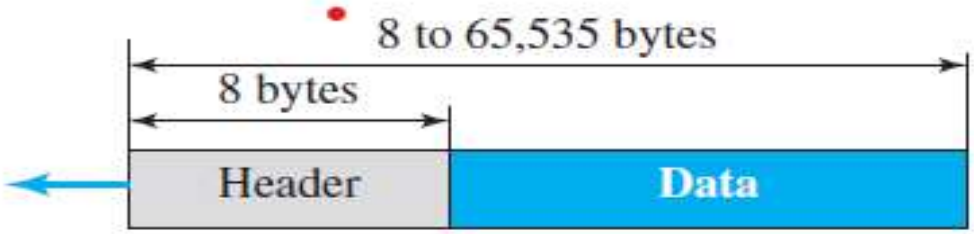
User Datagram

UDP packets, called ***user datagrams,*** have a fixed-size header of 8 bytes made of four fields, each of 2 bytes (16 bits). Figure shows the format of a user datagram

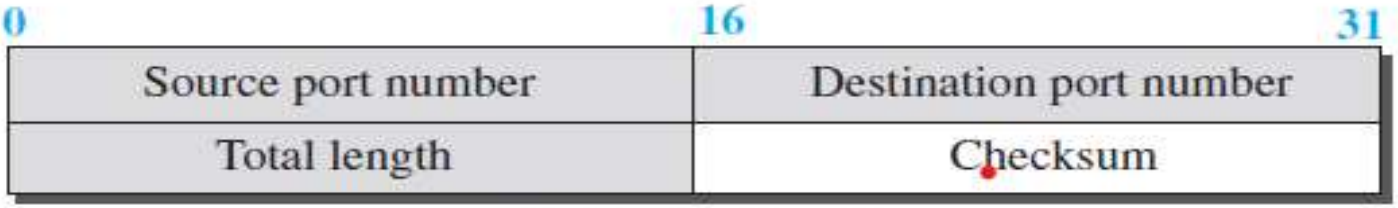
.

The first two fields define the source and destination port numbers

User datagram packet format



a. UDP user datagram



b. Header format

UDP Services:

Process-to-Process Communication

UDP provides process-to-process communication using **socket addresses,** a combination of IP addresses and port numbers.

Connectionless Services

UDP provides a *connectionless service.* This means that each user datagram sent by UDP is an independent datagram. There is no relationship between the different user datagrams even if they are coming from the same source process and going to the same destination program.

Flow Control :UDP is a very simple protocol. There is no *flow control*

Error Control :There is no *error control* mechanism in UDP except for the checksum. This means that the sender does not know if a message has been lost or duplicated.

Checksum

UDP checksum calculation includes three sections:

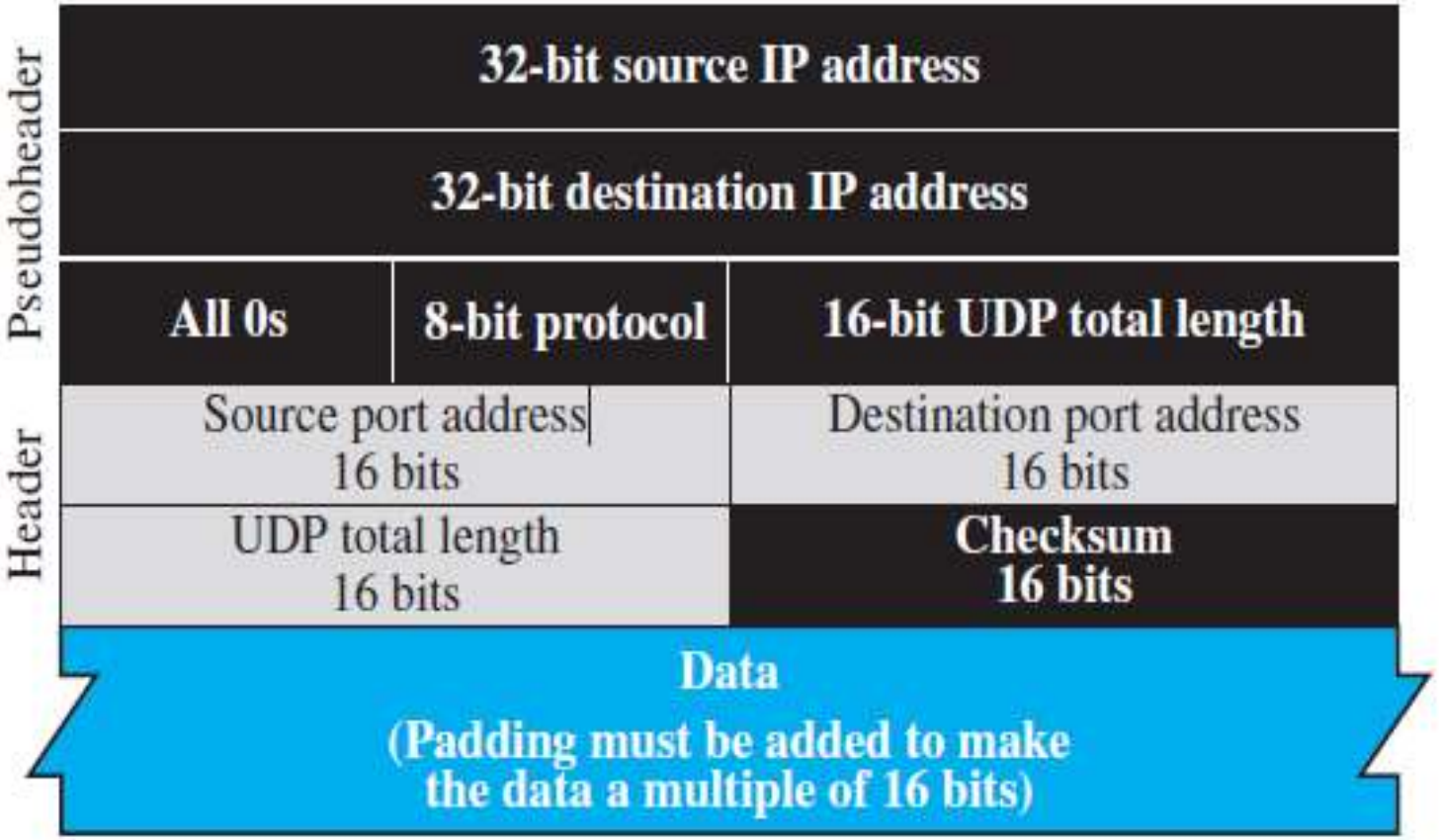
a pseudoheader

UDP header,

and the data coming from the application layer.

The *pseudoheader* is the part of the header of the IP packet

Pseudoheader for checksum calculation



Optional Inclusion of Checksum

- The sender of a UDP packet can choose not to calculate the checksum. In this case, the checksum field is filled with all 0s before being sent.
- In the situation where the sender decides to calculate the checksum, but it happens that the result is all 0s, the checksum is changed to all 1s before the packet is sent.

Typical Applications Of UDP:

- UDP is suitable for a process that requires simple request-response communication with little concern for flow and error control.
- UDP is suitable for a process with internal flow- and error-control mechanisms. For example, the Trivial File Transfer Protocol (TFTP) process includes flow and error control. It can easily use UDP.
- UDP is used for some route updating protocols such as Routing Information Protocol (RIP)
- UDP is normally used for interactive real-time applications

TRANSMISSION CONTROL PROTOCOL

- **Transmission Control Protocol (TCP)** is a connection-oriented, reliable protocol.
- TCP explicitly defines connection establishment, data transfer, and connection teardown phases to provide a connection-oriented service.
- TCP uses a combination of GBN (Go back-N) and SR (Selective Repeat) protocols to provide reliability.

TCP Services:

Process-to-Process Communication

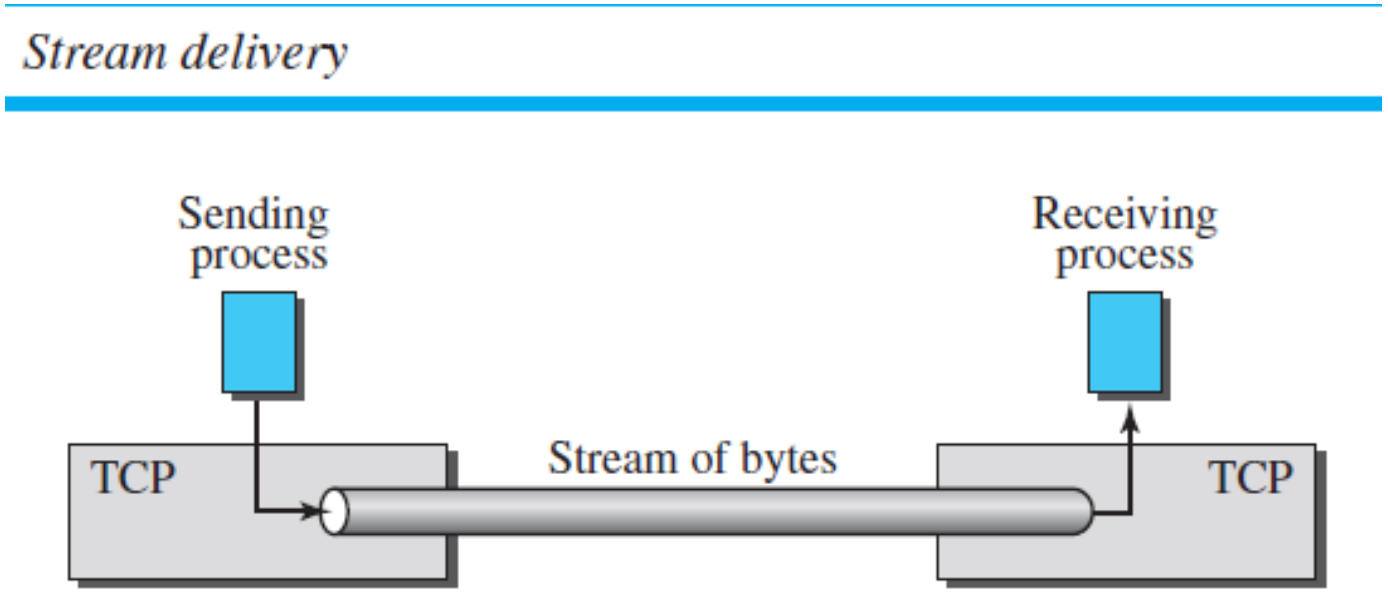
As with UDP, TCP provides process-to-process communication using port numbers.

Stream Delivery Service

TCP, unlike UDP, is a stream-oriented protocol. In UDP, a process sends messages with predefined boundaries to UDP for delivery.

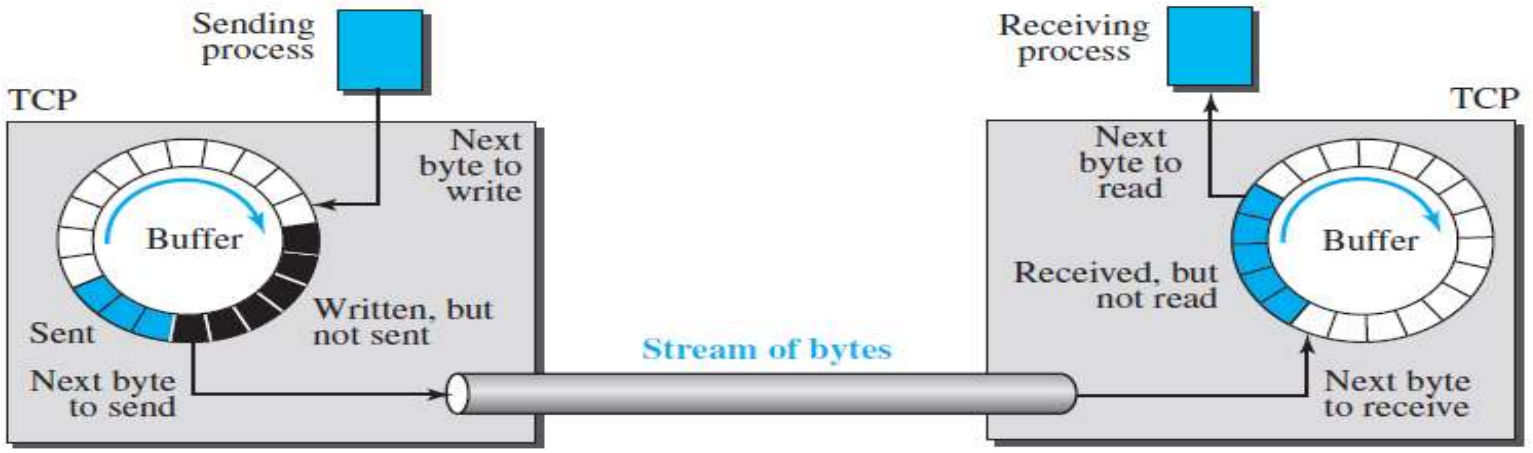
TCP, on the other hand, allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as a stream of bytes.

TCP creates an environment in which the two processes seem to be connected by an imaginary “tube” that carries their bytes across the Internet. This imaginary environment is depicted in Figure . The sending process produces (writes to) the stream and the receiving process consumes (reads from) it.



Sending and Receiving Buffers

Figure 24.5 *Sending and receiving buffers*



The figure shows the movement of the data in one direction. At the sender, the buffer has three types of chambers.

The white section contains empty chambers that can be filled by the sending process (producer).

The colored area holds bytes that have been sent but not yet acknowledged.
The TCP sender keeps these bytes in the buffer until it receives an acknowledgment.
The shaded area contains bytes to be sent by the sending TCP.

The operation of the buffer at the receiver is simpler. The circular buffer is divided into two areas (shown as white and colored).

The white area contains empty chambers to be filled by bytes received from the network.

The colored sections contain received bytes that can be read by the receiving process. When a byte is read by the receiving process, the chamber is recycled and added to the pool of empty chambers.

Full-Duplex Communication

TCP offers *full-duplex service,* where data can flow in both directions at the same time. Each TCP endpoint then has its own sending and receiving buffer, and segments move in both directions.

Multiplexing and Demultiplexing

Like UDP, TCP performs multiplexing at the sender and demultiplexing at the receiver.

However, since TCP is a connection-oriented protocol, a connection needs to be established for each pair of processes.

Connection-Oriented Service

TCP is a connection-oriented protocol. When a process at site A wants to send to and receive data from another process at site B, the following three phases occur:

1. The two TCP's establish a logical connection between them.

2. Data are exchanged in both directions.

3. The connection is terminated.

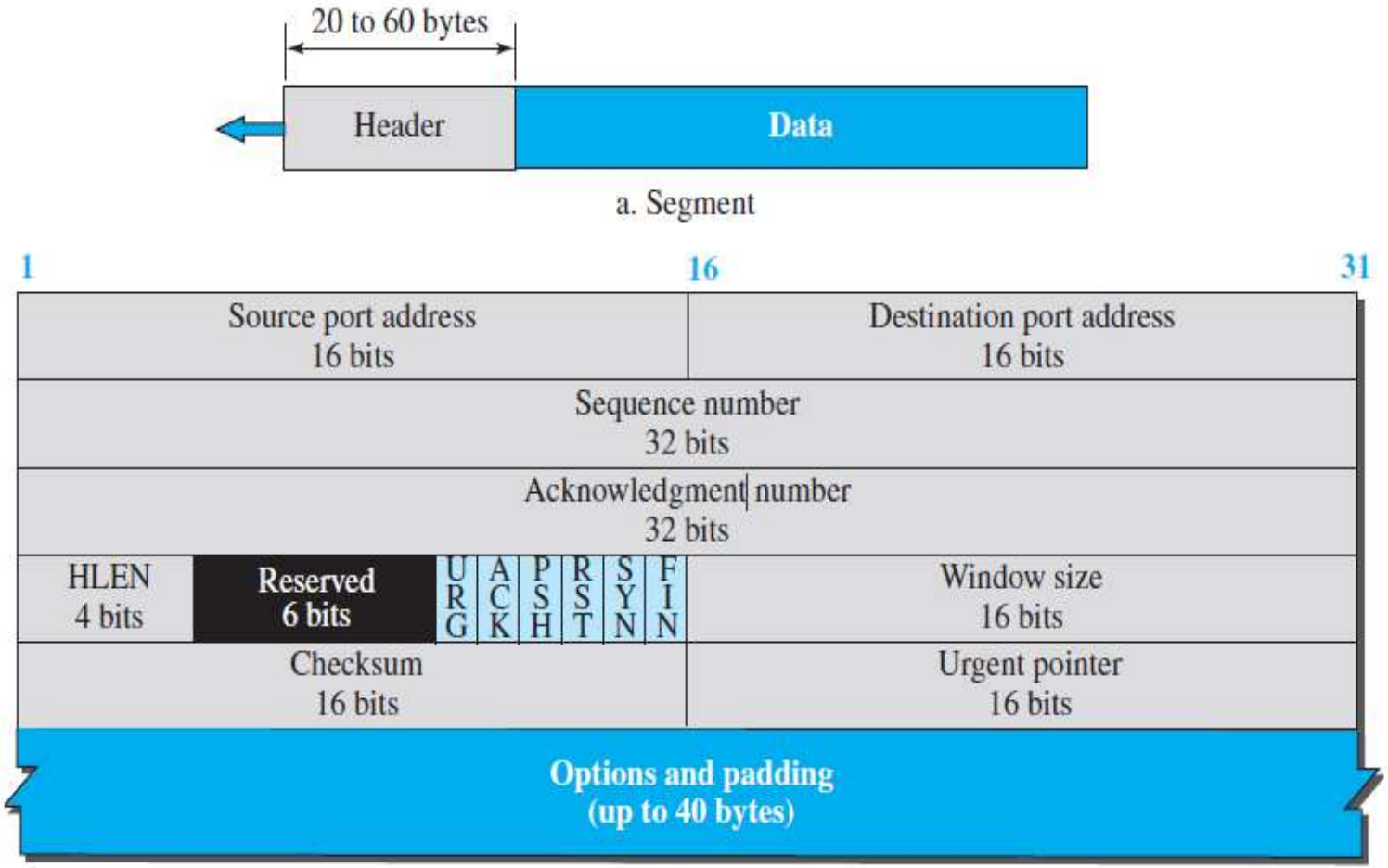
Reliable Service

TCP is a reliable transport protocol. It uses an acknowledgment mechanism to check the safe and sound arrival of data

Segments

- Buffering handles the disparity between the speed of the producing and consuming processes, we need one more step before we can send data.
- The network layer, as a service provider for TCP, needs to send data in packets, not as a stream of bytes. At the transport layer, TCP groups a number of bytes together into a packet called a *segment*.
- TCP adds a header to each segment and delivers the segment to the network layer for transmission. The segments are encapsulated in an IP datagram and transmitted. This entire operation is transparent to the receiving process.

Figure 24.7 *TCP segment format*



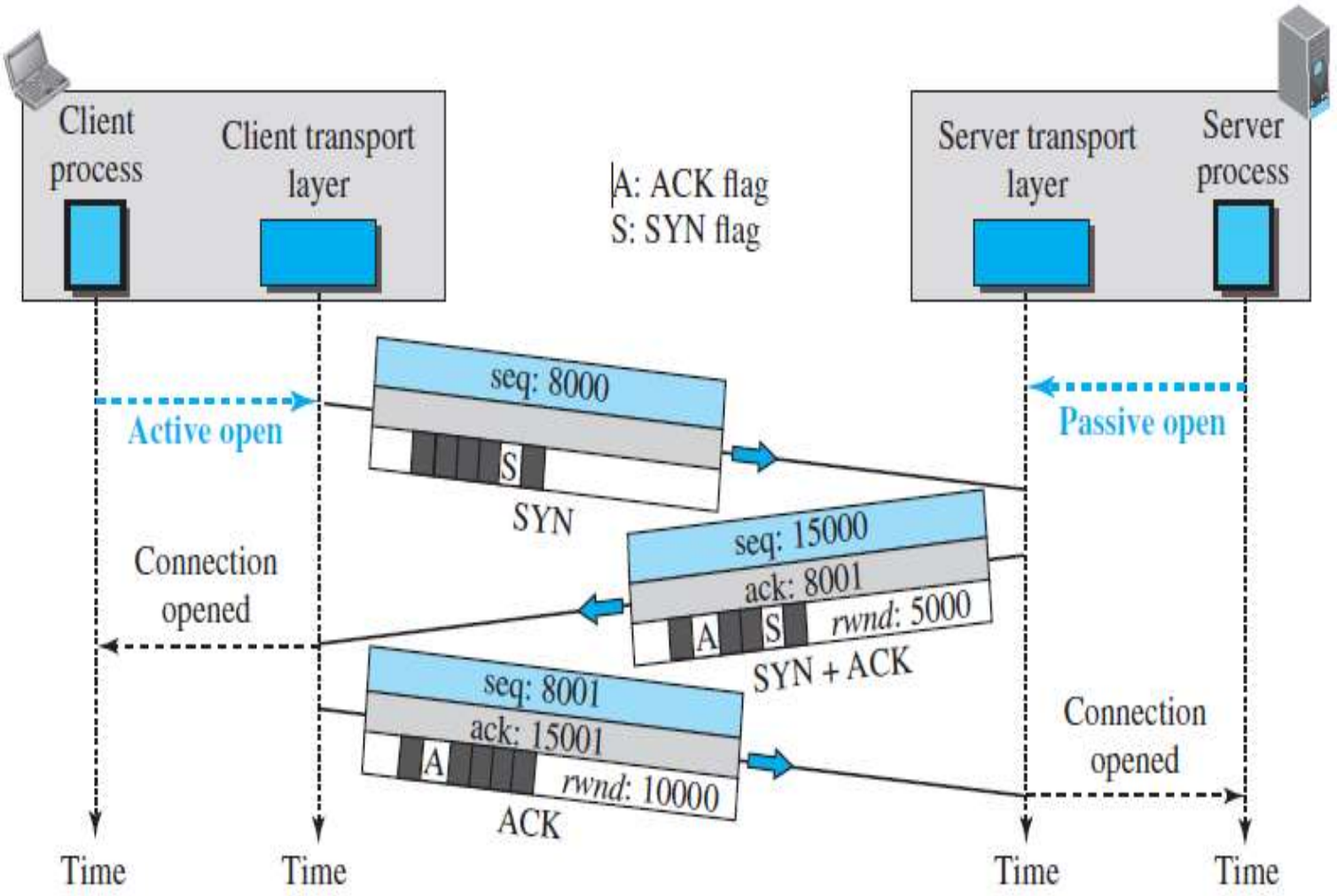
A TCP Connection:

Connection Establishment-Three-Way Handshaking

The connection establishment in TCP is called ***three-way handshaking.***

In our example, an application program, called the *client,* wants to make a connection with another application program, called the *server,* using TCP as the transport-layer protocol.

Figure 24.10 *Connection establishment using three-way handshaking*



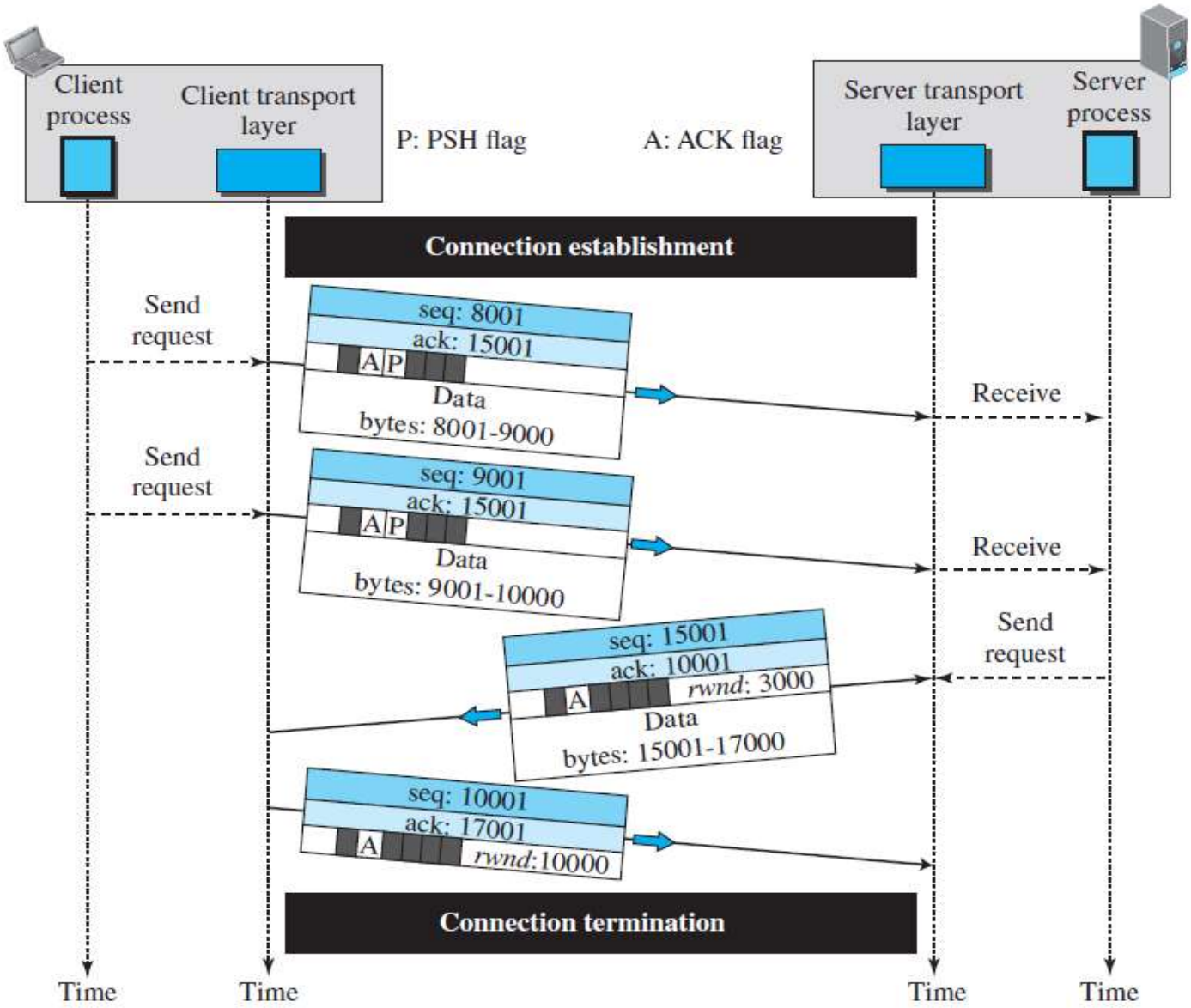
- The client sends the first segment, a SYN segment, in which only the SYN flag is set.
- This segment is for synchronization of sequence numbers. The client in our example chooses a random number as the first sequence number and sends this number to the server. This sequence number is called the *initial sequence number (ISN).*
- **A SYN segment cannot carry data, but it consumes one sequence number**
- The server sends the second segment, a SYN + ACK segment with two flag bits set as: SYN and ACK. This segment has a dual purpose. First, it is a SYN segment for communication in the other direction.
- The server uses this segment to initialize a sequence number.
- The server also acknowledges the receipt of the SYN segment from the client by setting the ACK flag and displaying the next sequence number it expects to receive from the client.

- The client sends the third segment. This is just an ACK segment. It acknowledges the receipt of the second segment with the ACK flag and acknowledgment number field.
- **An ACK segment, if carrying no data, consumes no sequence number.**

Data Transfer :

- After connection is established, bidirectional data transfer can take place. The client and server can send data and acknowledgments in both directions.
- After a connection is established, the client sends 2,000 bytes of data in two segments. The server then sends 2,000 bytes in one segment. The client sends one more segment.
- The first three segments carry both data and acknowledgment, but the last segment carries only an acknowledgment because there is no more data to be sent.

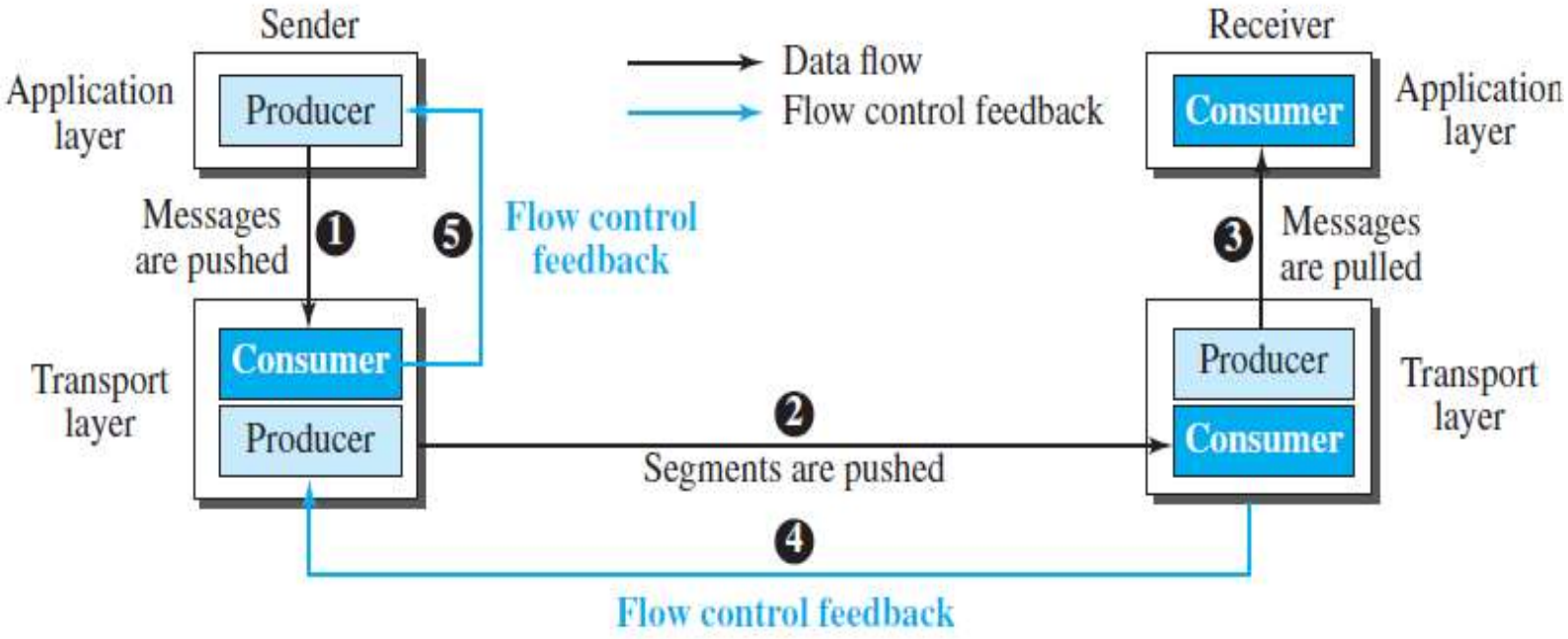
Figure 24.11 *Data transfer*



Flow Control:

- *Flow control* balances the rate a producer creates data with the rate a consumer can use the data. TCP separates flow control from error control.

Figure 24.19 *Data flow and flow control feedbacks in TCP*



- The figure shows that data travel from the sending process down to the sending TCP, from the sending TCP to the receiving TCP, and from the receiving TCP up to the receiving process (paths 1, 2, and 3).
- Flow control feedbacks, however, are traveling from the receiving TCP to the sending TCP and from the sending TCP up to the sending process (paths 4 and 5).
- Most implementations of TCP do not provide flow control feedback from the receiving process to the receiving TCP; they let the receiving process pull data from the receiving TCP whenever it is ready to do so.

- In other words, the receiving TCP controls the sending TCP; the sending TCP controls the sending process.
- Flow control feedback from the sending TCP to the sending process (path 5) is achieved through simple rejection of data by the sending TCP when its window is full.
- This means that our discussion of flow control concentrates on the feedback sent from the receiving TCP to the sending TCP (path 4).

Error Control

TCP is a reliable transport-layer protocol. This means that an application program that delivers a stream of data to TCP relies on TCP to deliver the entire stream to the application program on the other end in order, without error, and without any part lost or duplicated

- TCP provides reliability using error control. Error control includes mechanisms for detecting and resending corrupted segments, resending lost segments, storing out-of order segments until missing segments arrive, and detecting and discarding duplicated segments. Error control in TCP is achieved through the use of three simple tools: checksum, acknowledgment, and time-out.

Checksum

- Each segment includes a checksum field, which is used to check for a corrupted segment.
- If a segment is corrupted, as detected by an invalid checksum, the segment is discarded by the destination TCP and is considered as lost. TCP uses a 16-bit checksum that is mandatory in every segment.

TCP Congestion Control

Congestion Window

- TCP congestion control refers to the mechanisms used in the **Transmission Control Protocol (TCP)** to prevent network congestion. It ensures that data is transmitted across a network efficiently without overloading

Congestion Window (cwnd):

- The **congestion window** is a key mechanism that controls the number of packets a sender can send without receiving an acknowledgment (ACK). The size of this window is adjusted based on network conditions.

Actual window size =**minimum (*rwnd*, *cwnd*)**

rwnd-Receive window size

Congestion Detection

- TCP sender can detect the possible existence of congestion in the network.
- The TCP sender uses the occurrence of two events as signs of congestion in the network: time-out and receiving three duplicate ACKs.
- *time-out*- If a TCP sender does not receive an ACK for a segment or a group of segments before the time-out occurs, it assumes that the corresponding segment or segments are lost and the loss is due to congestion.
- Another event is the receiving of three duplicate ACKs . Recall that when a TCP receiver sends a one duplicate ACK, it is the sign that a segment has been delayed, but sending three duplicate ACKs is the sign of a missing segment, which can be due to congestion in the network.

- time-out, is the sign of a strong congestion; the receiving of three duplicate ACKs is the sign of a weak congestion in the network

Congestion Policies

TCP's general policy for handling congestion is based on three algorithms:

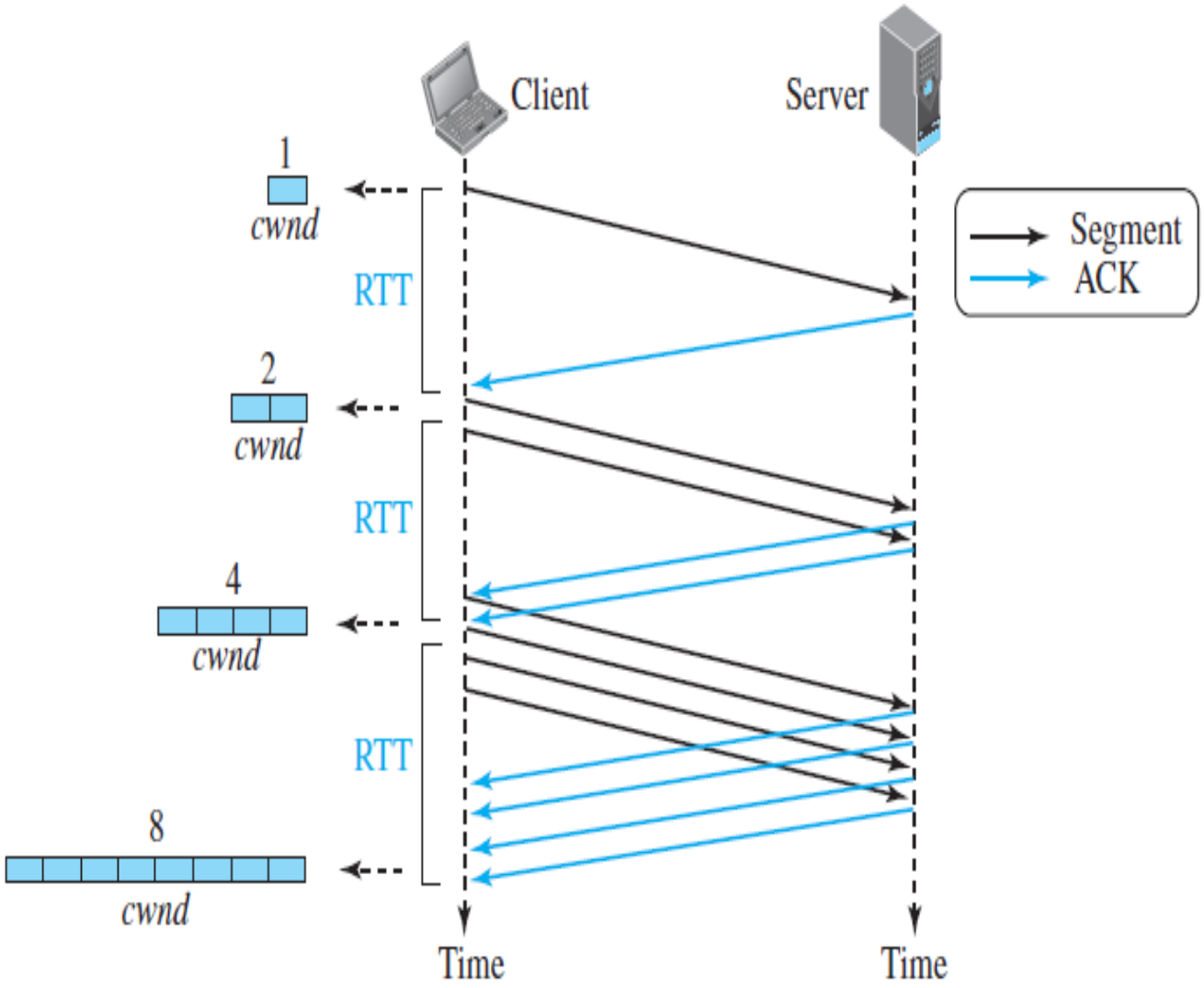
- Slow start
- Congestion avoidance
- Fast recovery.

Slow Start: Exponential Increase

Initially TCP starts with a small **congestion window** (typically 1 MSS, or **Maximum Segment Size**), which represents the amount of unacknowledged data that can be sent into the network

For every **ACK** received, the congestion window doubles during each round-trip time (**RTT**)

igure 24.29 *Slow start, exponential increase*



Start → *cwnd* = **1** → $\mathbf{2^0}$

After 1 RTT → *cwnd* = *cwnd* + **1** = **1** + **1** = **2**→ $\mathbf{2^1}$

After 2 RTT → *cwnd* = *cwnd* + **2** = **2** + **2** = **4**→ $\mathbf{2^2}$

After 3 RTT → *cwnd* = *cwnd* + **4** = **4** + **4** = **8**→ $\mathbf{2^3}$

Congestion Avoidance: Additive Increase

- If we continue with the slow-start algorithm, the size of the congestion window increases exponentially. To avoid congestion before it happens, we must slow down this exponential growth. TCP defines another algorithm called ***congestion avoidance,*** which increases the *cwnd* additively instead of exponentially.
- When the size of the congestion window reaches the slow-start threshold in the case where *cwnd* = *i*, the slow-start phase stops and the additive phase begins.

UNIT-3 IP ADDRESS

IP address is a unique address that is used to identify computers or nodes on the internet.

It is generally expressed in a set of numbers for example 192.155.12.1. Here each number in the set is from 0 to 255 range.

we can say that a full IP address ranges from 0.0.0.0 to 255.255.255.255.

Types of IP Address

1. **IPv4:** Internet Protocol version 4. It consists of 4 numbers separated by the dots. Each number can be from 0-255 in decimal numbers.

But computers do not understand decimal numbers, they instead change them to binary numbers which are only 0 and 1. Therefore, in binary, this (0-255) range can be written as (00000000 – 11111111).

Since each number N can be represented by a group of 8-digit binary digits. So, a whole IPv4 binary address can be represented by 32-bits of binary digits

2.IPv6: But, there is a problem with the IPv4 address. With IPv4, we can connect only the above number of 4 billion devices uniquely, and apparently, there are much more devices in the world to be connected to the internet.

- So, gradually we are making our way to **IPv6 Address** which is a 128-bit IP address. In human-friendly form, IPv6 is written as a group of 8 hexadecimal numbers separated with colons(:).

- IPV6 can be written as

2011:0bd9:75c5:0000:0000:6b3e:0170:8394

The maximum limit of IPv4 addresses is 4,294,967,296 (approximately 4.3 billion) unique addresses.

Convert Binary to IP Address

Given the binary IP address:
11000000.10101000.00000001.00001010
Solution:
Break into octets (8 bits each):
11000000 | 10101000 | 00000001 | 00001010
Convert each octet from Binary to Decimal:
11000000 = 192
10101000 = 168
00000001 = 1
00001010 = 10
Final IP address: 192.168.1.10

Classful IP address Ranges

The IP address classful ranges are:

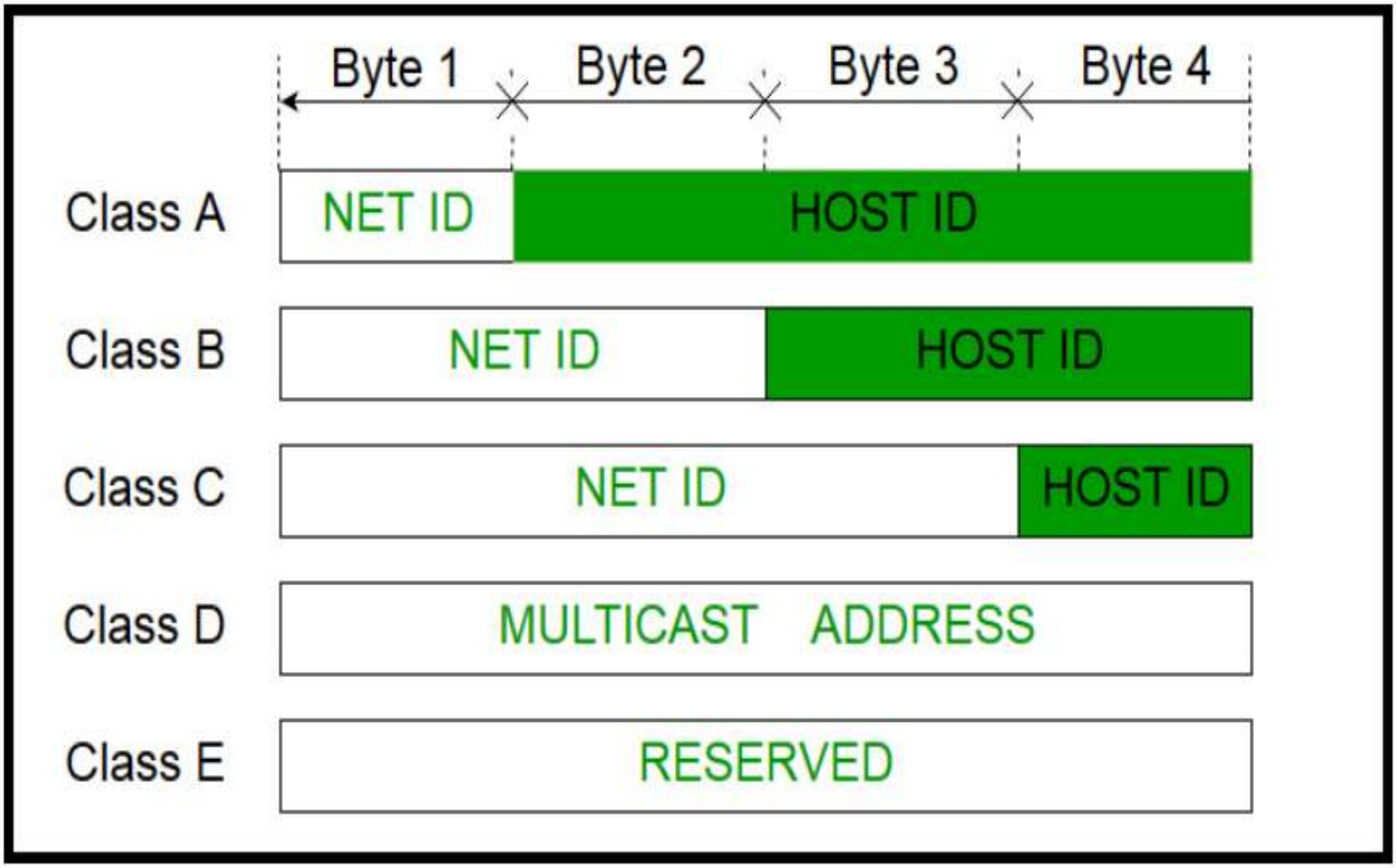
Class A: 0.0.0.0 to 127.255.255.255

Class B: 128.0.0.0 to 191.255.255.255

Class C: 192.0.0.0 to 223.255.255.255

Class D: 224.0.0.0 to 239.255.255.255

Class E: 240.0.0.0 to 255.255.255.255


Byte 1
Byte 2
Byte 3
Byte 4
Class A
NET ID
HOST ID
Class B
NET ID
HOST ID
Class C
NET ID
HOST ID
Class D
MULTICAST ADDRESS
Class E
RESERVED

Find the Class, Network Id, Host Id and Network Address of following ip address

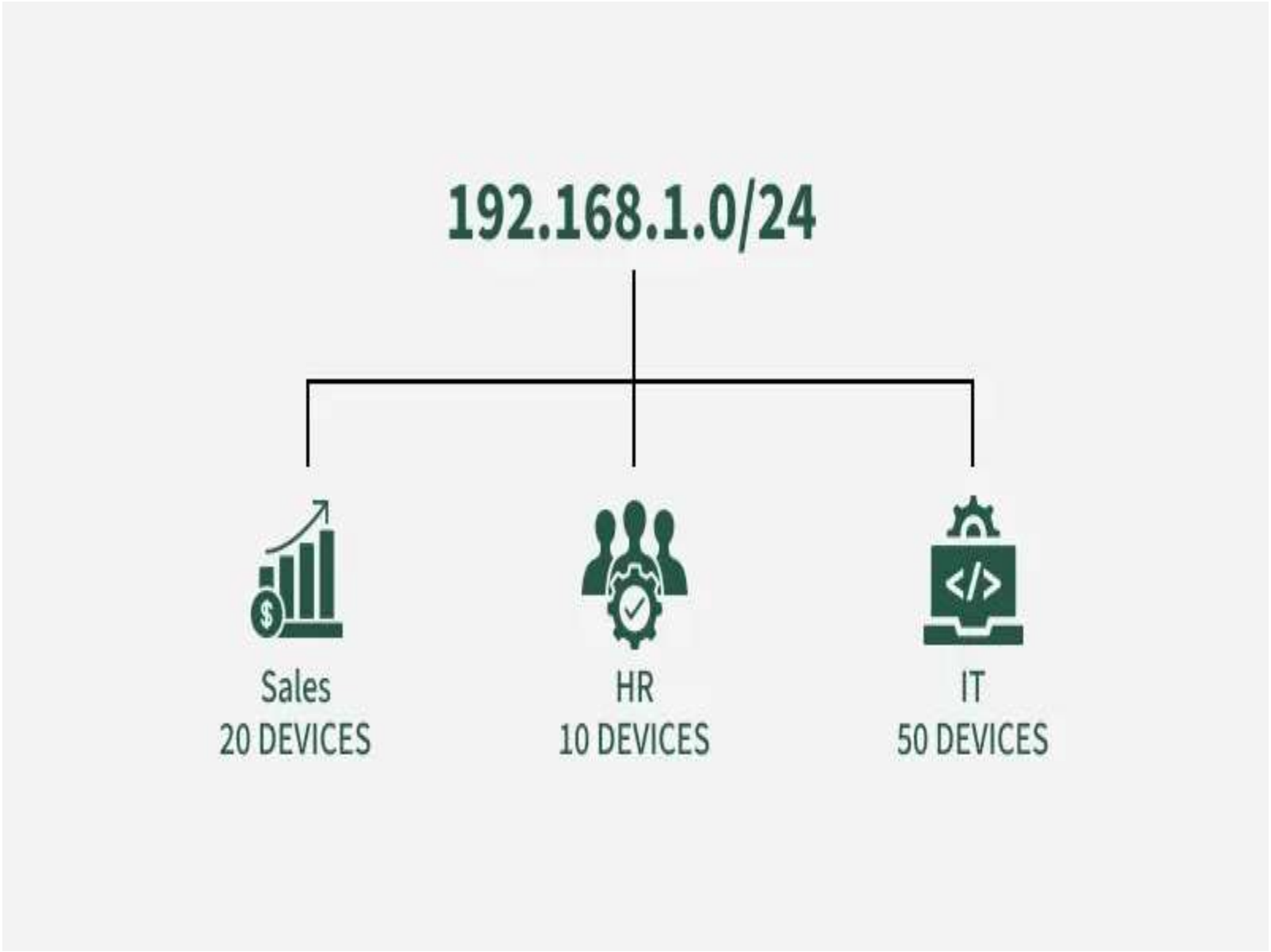
1. 4.23.145.90
2. 135.7.97.56
3. 227.34.78.7
4. 246.7.3.8
5. 129.6.8.4
6. 141.5.7.5
7. 198.76.9.23
8. 223.56.78.90
9. 275.89.90.56
10. 190.09.56.78

CIDR

- Classless Inter-Domain Routing
- It is the Standard routing and address allocation system used by all major ISPs and modern network equipment
- It prevents the wastage of IP address by allowing precise network sizing
- Addresses are written in CIDR notation is like
- 192.168.1.0/24 its indicates first 24 bits are network parts and remaining 8 bits are host parts

SUBNET

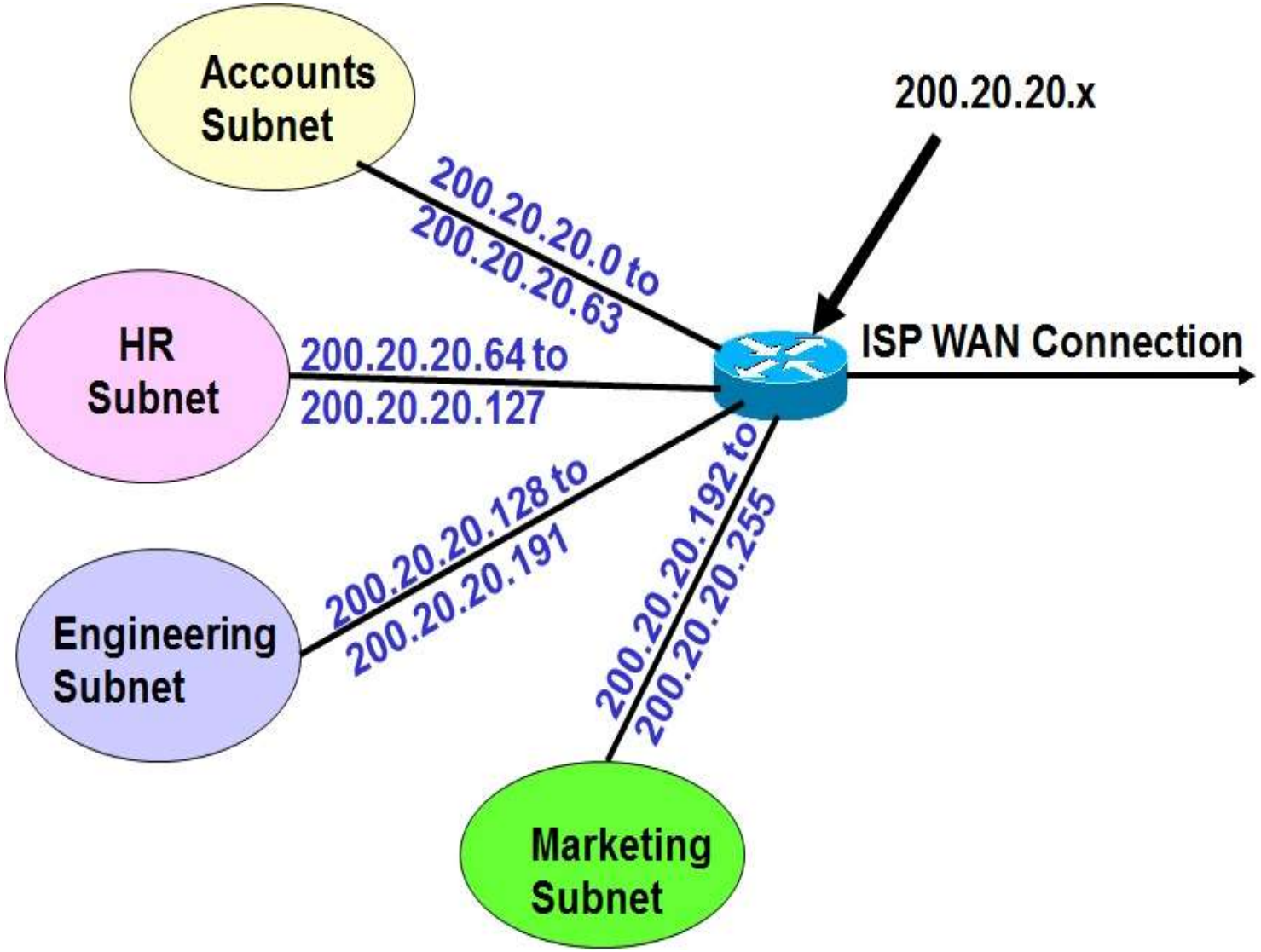
- Subnetting is necessary to divide a large network into smaller, more manageable sub-networks (subnets), which improves efficiency, performance, and security.
- Subnets help reduce IP address waste by allocating just the number of addresses needed for different groups or departments, minimizing unused addresses.


192.168.1.0/24
Sales
20 DEVICES
HR
10 DEVICES
IT
50 DEVICES

- Without subnetting, all departments share the same network, and all **256 IP addresses** are available to everyone, which leads to:
- **IP Waste**: Only **80 devices** are needed (20 + 10 + 50), but all 256 addresses are allocated, wasting **176 addresses**.
- **With Subnetting**, we split the network into three subnets, allocating just enough IP addresses for each department:
- **Sales**: 192.168.1.0/27 → 32 IPs (for 20 devices, 12 spare)
- **HR**: 192.168.1.32/28 → 16 IPs (for 10 devices, 6 spare)
- **IT**: 192.168.1.48/26 → 64 IPs (for 50 devices, 14 spare)

By subnetting, we:

- **Save IP addresses (Efficiency)**: Only 112 addresses are used (80 + some spare), leaving 144 unused for future growth.
- **Keep networks faster (Better Performance)**: Data within each department stays in its subnet. For example, HR traffic stays in HR, reducing network congestion for Sales and IT.
- **Protect sensitive data (Improved Security)**: Each department is isolated. If someone in Sales tries to access HR systems, subnet restrictions block that access.


Accounts Subnet
200.20.20.0 to 200.20.20.63
200.20.20.x
HR Subnet
200.20.20.64 to 200.20.20.127
ISP WAN Connection
Engineering Subnet
200.20.20.128 to 200.20.20.191
200.20.20.192 to 200.20.20.255
Marketing Subnet

Masking in a subnet:

- Masking is the process of getting the physical network address from an IP address
- Masking can be done whether we have subnetting or not
- If we have subnetting masking getting sub network address from an IP address
- It helps computers and devices determine which part of an IP address refers to the network they are present, and which part refers to their specific location or address within that network.

Without Subnetting



With Subnet



Subnet the given Network based on the Host Requirements

Problem: Subnet the IP address **194.5.6.0** into at least 30 Host in Each Subnet

Steps:

1) Identify the Class of the ip address and Default Subnet Mask
2) Convert the Default Subnet Mask into Binary
3) Note the Number of Hosts Required per subnet and find the Subnet Generator(SG) and Octet Position
4) Generate the New Subnet Mask
5) Use the Subnet generator to generate the network Ranges (Subnets) in the appropriate Octet Position

1. Class C IP address

Default Subnet Mask 255.255.255.0

Binary Conversion of Default Subnet Mask

2. **1111111.11111111.11111111.0000000**

3.No. of Host Approximately Required per subnet =30

Subnet Generator = ?

Octet Position=4

Binary value of 30 is (11110) it consist of 5 bits then reserve the 5 bits in 4th octet Right to left , Replace the remaining 0s (zeros) in 4th octet as 1

1 1 1 1 1 1 1.1 1 1 1 1 1 1 1.1 1 1 1 1 1 1 1.1 1 1 0 0 0 0 0

The value of first one(1) in 4th octet position is 32

Now we found 32 is Subnet Generator(SG)

Our new Subnet Mask is **255.255.255.224 or /27**

Subnet generator to generate the Network Ranges

1. **194.5.6.0 to 194.5.6.31**
2. **194.5.6.32 to 194.5.6.63**
3. **194.5.6.64 to 194.5.6.95**
4. **194.5.6.96 to 194.5.6.127**
5. **194.5.6.128 to 194.5.6.159**
6. **194.5.6.160 to 194.5.6.191**
7. **194.5.6.192 to 194.5.6.223**
8. **194.5.6.224 - Broadcast Address**

What is the maximum number of valid subnets and usable hosts per subnet that you can get from the network 192.168.186.0/25?

Subnet:_____ and Host usable:_____

Solution:

Default subnet mask for this ip addres 192.168.186.0/25 is

255.255.255.128

11111111.11111111.11111111.10000000

Number of subnet = 2^n where n is number of ones in 4th octet is 1 therefore 2^1 = **2 subnet**

Host Range = 2^n where n is number of zeroes in 4th octet is 7 therefore $2^7 = 128$

Valid and usable host is $2^n - 2$

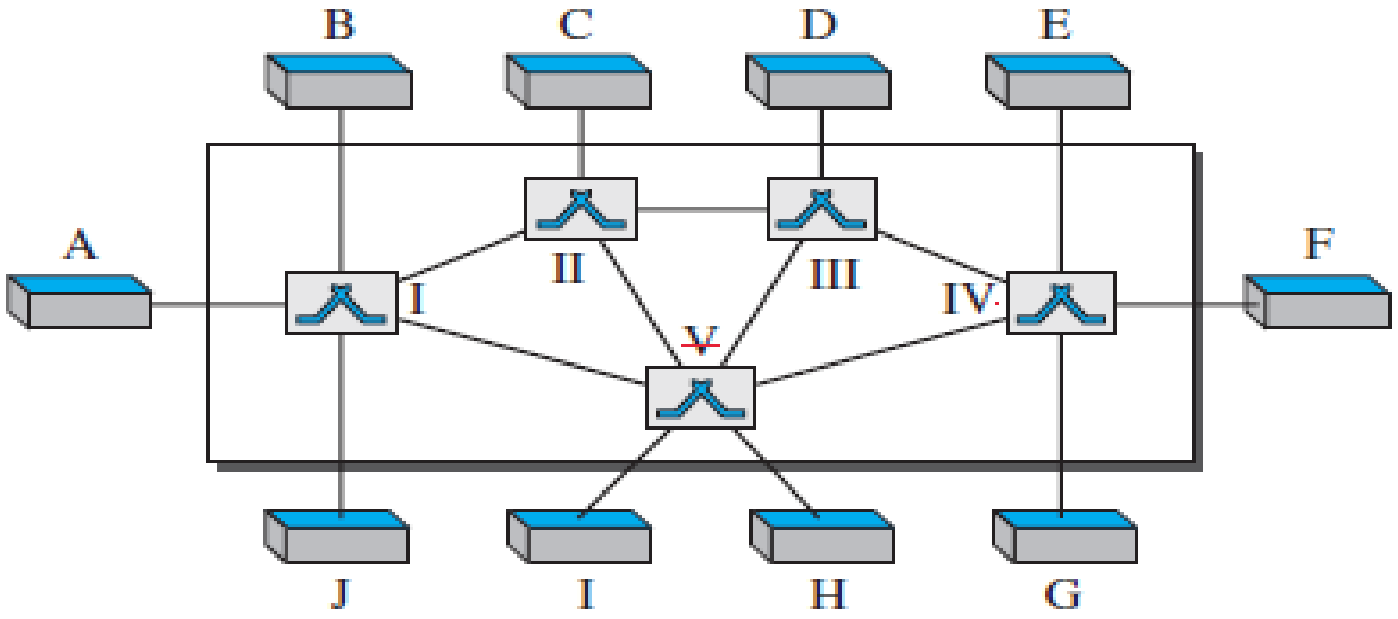
Hence 128-2= **126 usable host**

UNIT-3
Switching

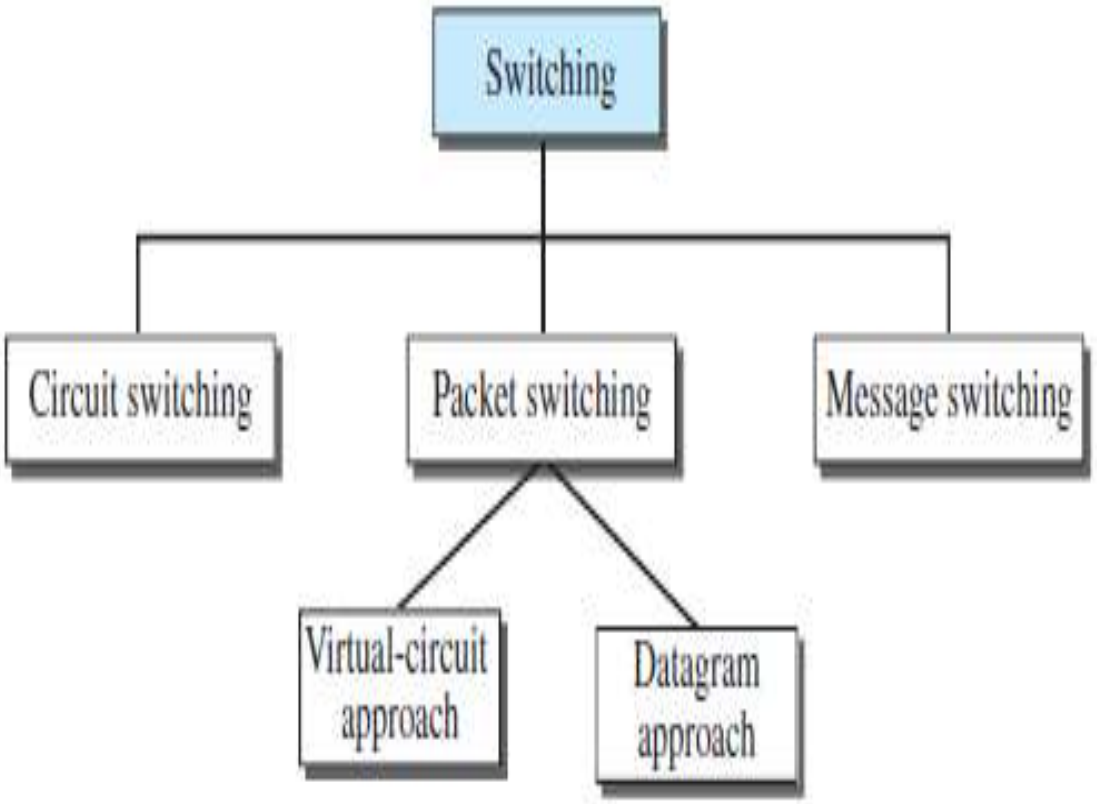
- Switch is a mechanism to allows us to interconnect multiple links to form a larger network to communicate efficiently.
- A switched network consist of a series of interlinked node is called switch.
- A switch is Multi input and Multi Ouput(MIMO) device and its primary job is to receive incoming packets on one of its link and to transmit them on to some other output link until they reach its destination. The function otherwise called as Switching or forwarding

- The **end systems** (communicating devices) are labeled A, B, C, D, and so on, and the
- switches are labeled I, II, III, IV, and V. Each switch is connected to multiple links.

Switched network



Three Methods of Switching



Switching and TCP/IP Layers

- Switching can happen at several layers of the TCP/IP protocol suite.

Switching at Physical Layer

- At the physical layer, we can have only circuit switching. There are no packets exchanged at the physical layer. The switches at the physical layer allow signals to travel in one path or another.

Switching at Data-Link Layer

- At the data-link layer, we can have packet switching. However, the term *packet* in this case means *frames* or *cells*.
- Packet switching at the data-link layer is normally done using a virtual-circuit approach.

Switching at Network Layer

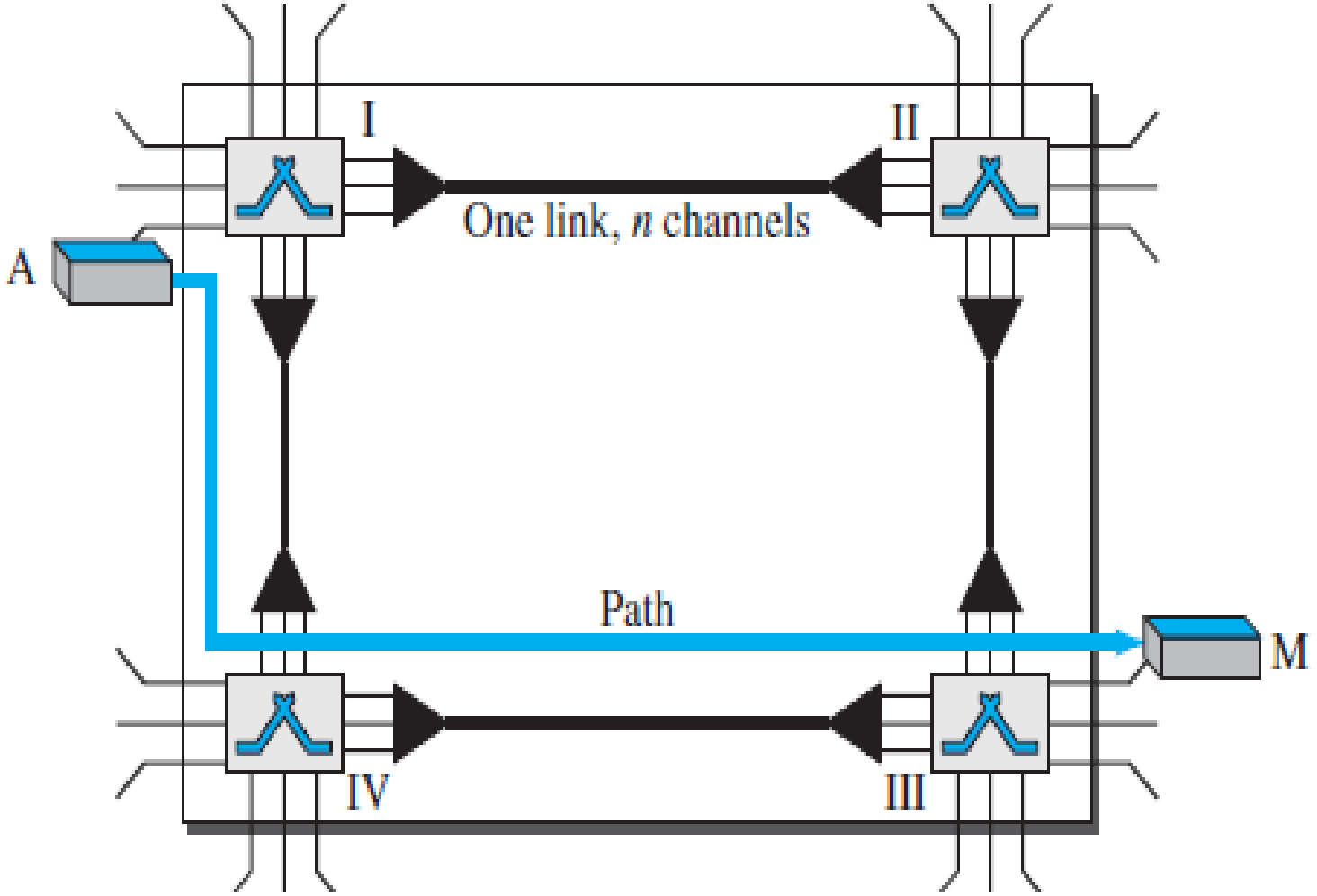
- At the network layer, we can have packet switching. In this case, either a virtual-circuit approach or a datagram approach can be used.
- Currently the Internet uses a datagram approach,

Switching at Application Layer

- At the application layer, we can have only message switching. The communication at the application layer occurs by exchanging messages.
- Conceptually, we can say that communication using e-mail is a kind of message-switched communication

UNIT-5
Circuit Switching

- Circuit switching creates a direct physical connection between two devices such as phones (or) computers, which is mainly designed for voice communication.
- It is less well suited for data and other non-voice transmission.
- A circuit switched network is made up of a set of switches connected by physical links, in which each link is divided into ‘n’ channels by using FDM or TDM
- **In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.**


I
II
One link, n channels
A
Path
M
IV
III

Three phases:

1.Setup Phase

2.Data transfer phase

3.Tear down phase

Setup phase:

- Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established.
- The end systems are normally connected through dedicated lines to the switches, so
- connection setup means creating dedicated channels between the switches

- when system A needs to connect to system M, it sends a setup request that includes the address of system M, to switch I.
- Switch I finds a channel between itself and switch IV that can be dedicated for this purpose. Switch I then sends the request to switch IV, which finds a dedicated channel between itself and switch III. Switch III informs system M.
- In the next step to making a connection, an acknowledgment from system M needs to be sent in the opposite direction to system A. Only after system A receives this acknowledgment is the connection established.

- **Data-Transfer Phase:**
- After the establishment of the dedicated circuit (channels), the two parties can transfer data.
- **Teardown Phase(Connection Termination)**
- When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

UNIT-3
Packet Switching

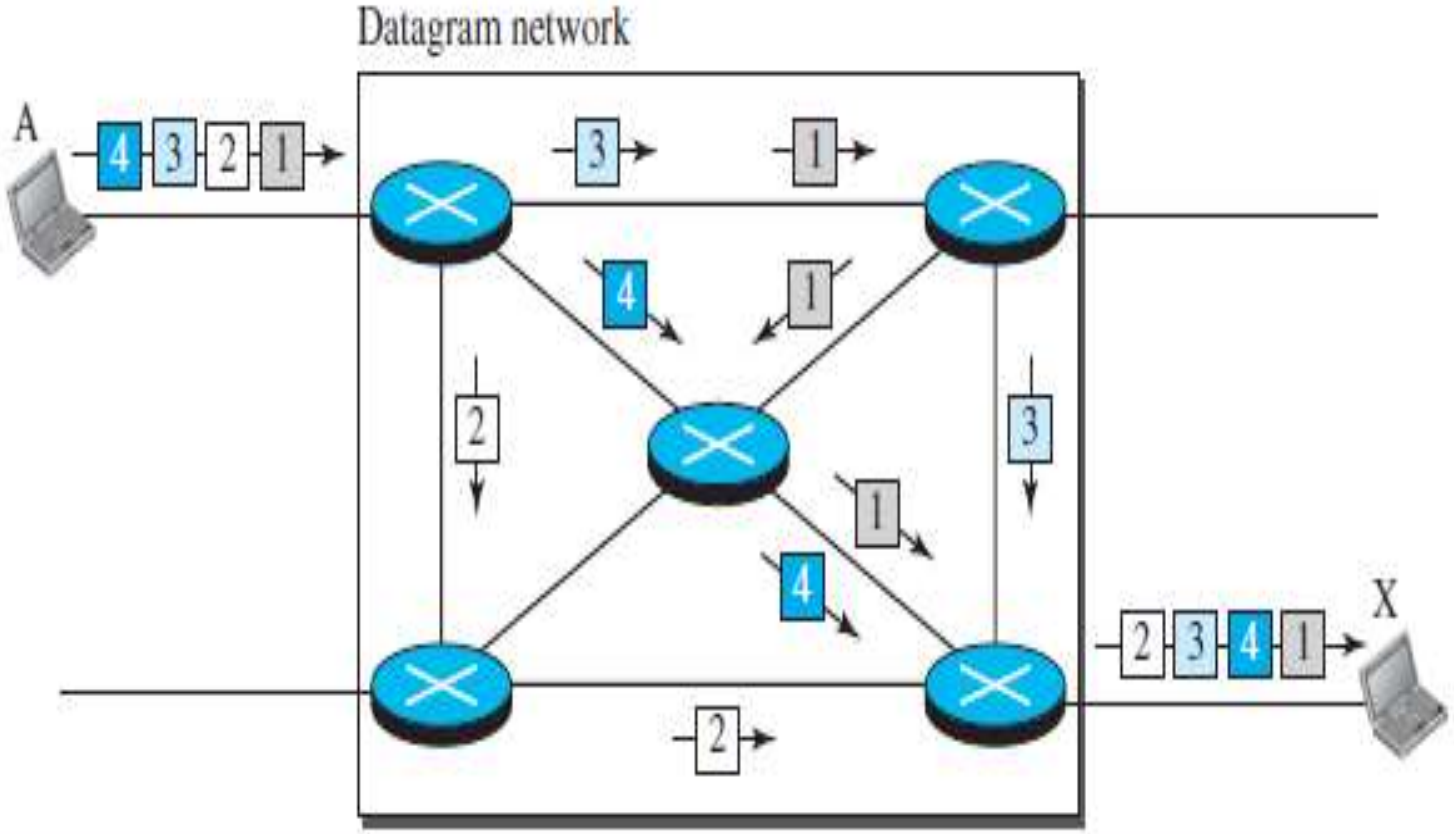
- In Data communication Send messages from one system to another. If the message is going to pass through a **packet-switched network**, it needs to be divided into packets of fixed or variable size according to their governing protocol.
- In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet. Resources are allocated on demand.
- The allocation is done on a first come, first-served basis.

- When a switch receives a packet, no matter what the source or destination is, the packet must wait if there are other packets being processed.
- We can have two types of packet-switched networks:
- **Datagram networks**
- **Virtual-Circuit networks**

Datagram Networks:

- In a **datagram network**, each packet is treated independently of all others.
- Packets in this approach are referred to as ***datagrams***.
- Datagram switching is normally done at the network layer

Figure 8.7 *A datagram network with four switches (routers)*



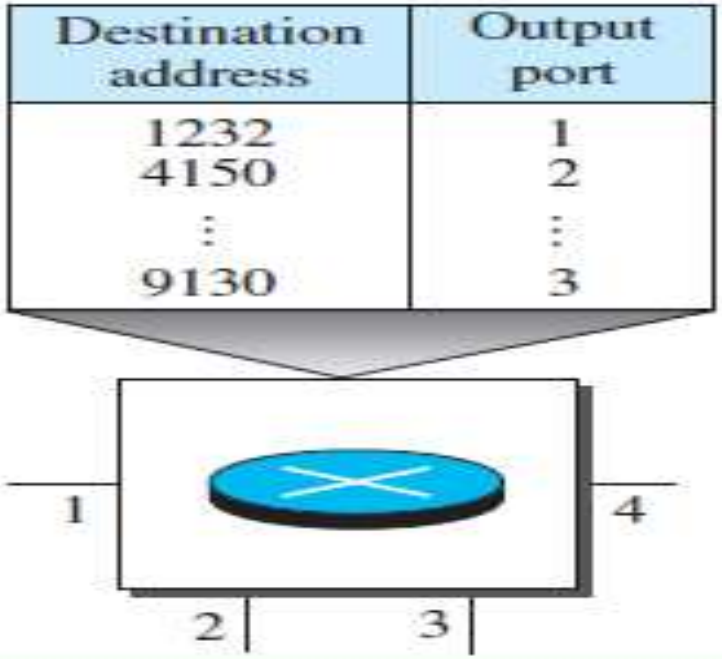
- In this above diagram shows how the datagram approach is used to deliver four packets from station A to station X. The switches in a datagram network are traditionally referred to as routers.
- In this example, all four packets (or datagrams) belong to the same message, but may travel different paths to reach their destination.
- The datagram networks are sometimes referred to as *connectionless networks*. The term *connectionless* here means that the switch (packet switch) does not keep information about the connection state. There are no setup and teardown phases.

Routing Table

- If there are no setup or teardown phases, how are the packets routed to their destinations in a datagram network? In this type of network, each switch (or packet switch) has a routing table which is based on the destination address.
- The routing tables are dynamic and are updated periodically.
- The destination addresses and the corresponding forwarding output ports are recorded in the tables.

- Every packet in a datagram network carries a header that contains, among other information,
- The destination address of the packet. When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded.

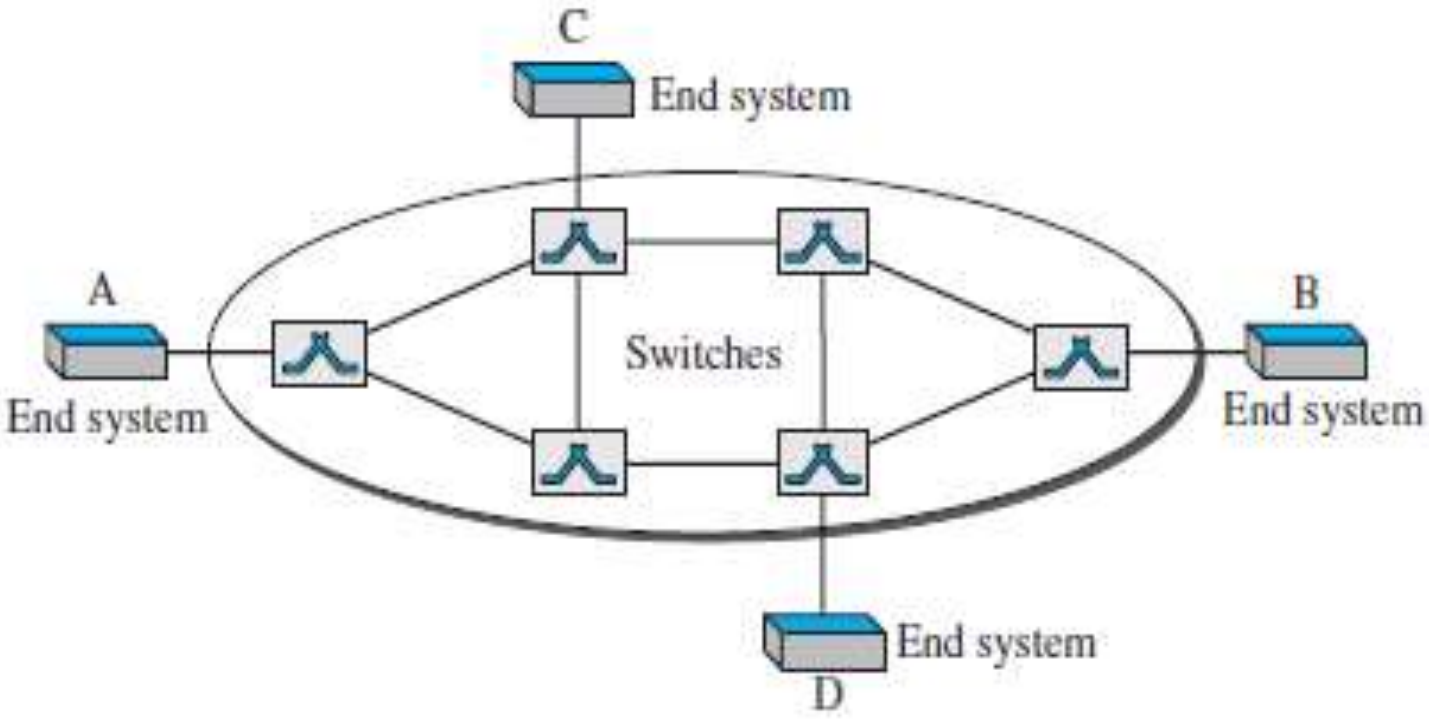
Routing table in a datagram network



Virtual-Circuit Networks(Connection oriented service)

- A **virtual-circuit network** is a cross between a circuit-switched network and a datagram network.
- A virtual-circuit network is normally implemented in the data-link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer. But this may change in the future.
- In this below diagram a source or destination can be a computer, packet switch, bridge, or any other device that connects other networks.

- Virtual Circuit network (one of the route is established, all the packets between a pair of communicating parties follow this same route through the network. During the connection establishment phase one of available path is selected. This path is called virtual circuit)



Addressing

- In a virtual-circuit network, two types of addressing are involved:
- global
- local(virtual-circuit identifier).

Global Addressing

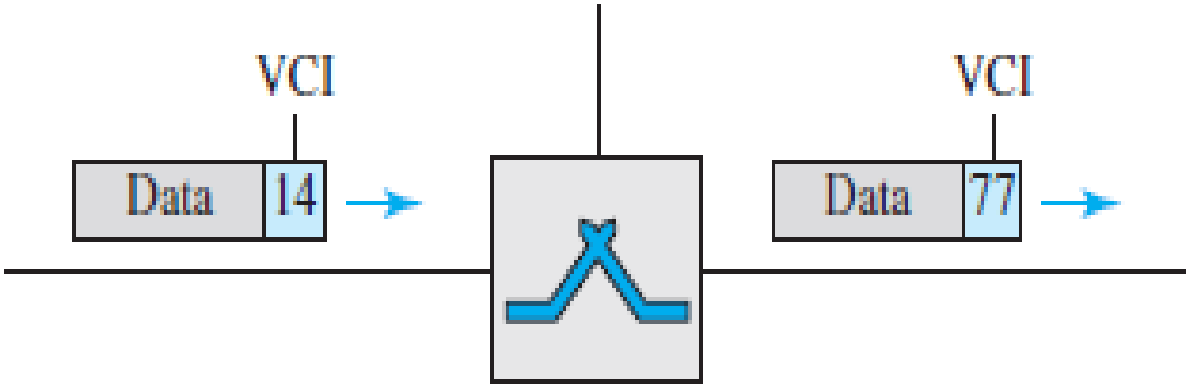
- A source or a destination needs to have a global address—an address that can be unique in the scope of the network or if the network is part of an international network.
- Using this global address to create virtual circuit identifier(VCI)

Virtual-Circuit Identifier

- The identifier that is actually used for data transfer is called the ***virtual-circuit identifier*** **(VCI)** or the ***label*****.**

- A VCI, unlike a global address, is a small number that has only switch scope; it is used by a frame between two switches.
- When a frame arrives at a switch, it has a VCI; when it leaves, it has a different VCI.

Virtual-circuit identifier



Three phases

1.Setup phase

- In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required:
- Setup request
- Acknowledgment.

2.Data transfer phase

3.Teardown

In this phase, source A, after sending all frames to B, sends a special frame called a *teardown request*. Destination B responds with a teardown confirmation frame. All switches delete the corresponding entry from their tables.

Figure 8.14 *Setup request in a virtual-circuit network*

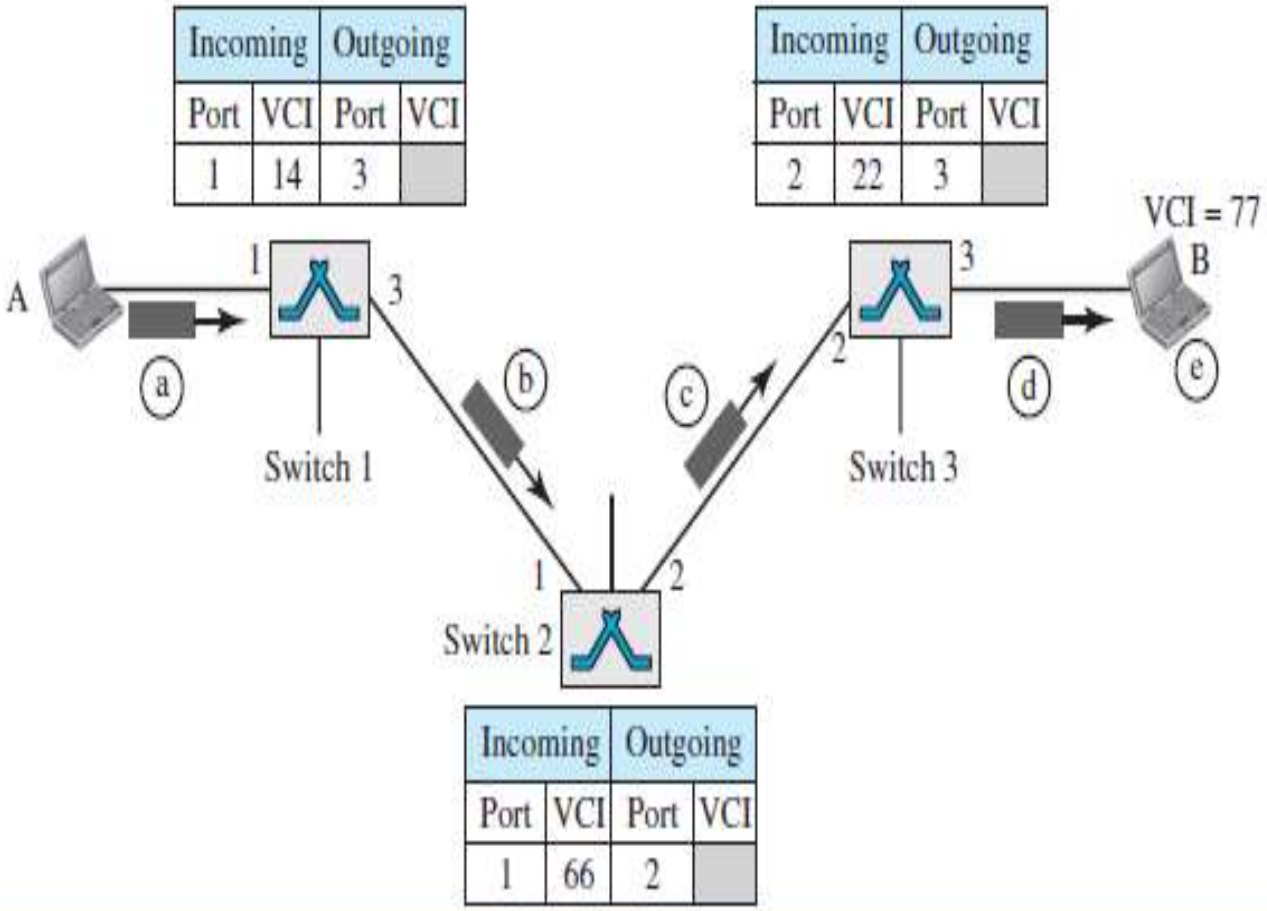


Figure 8.15 *Setup acknowledgment in a virtual-circuit network*

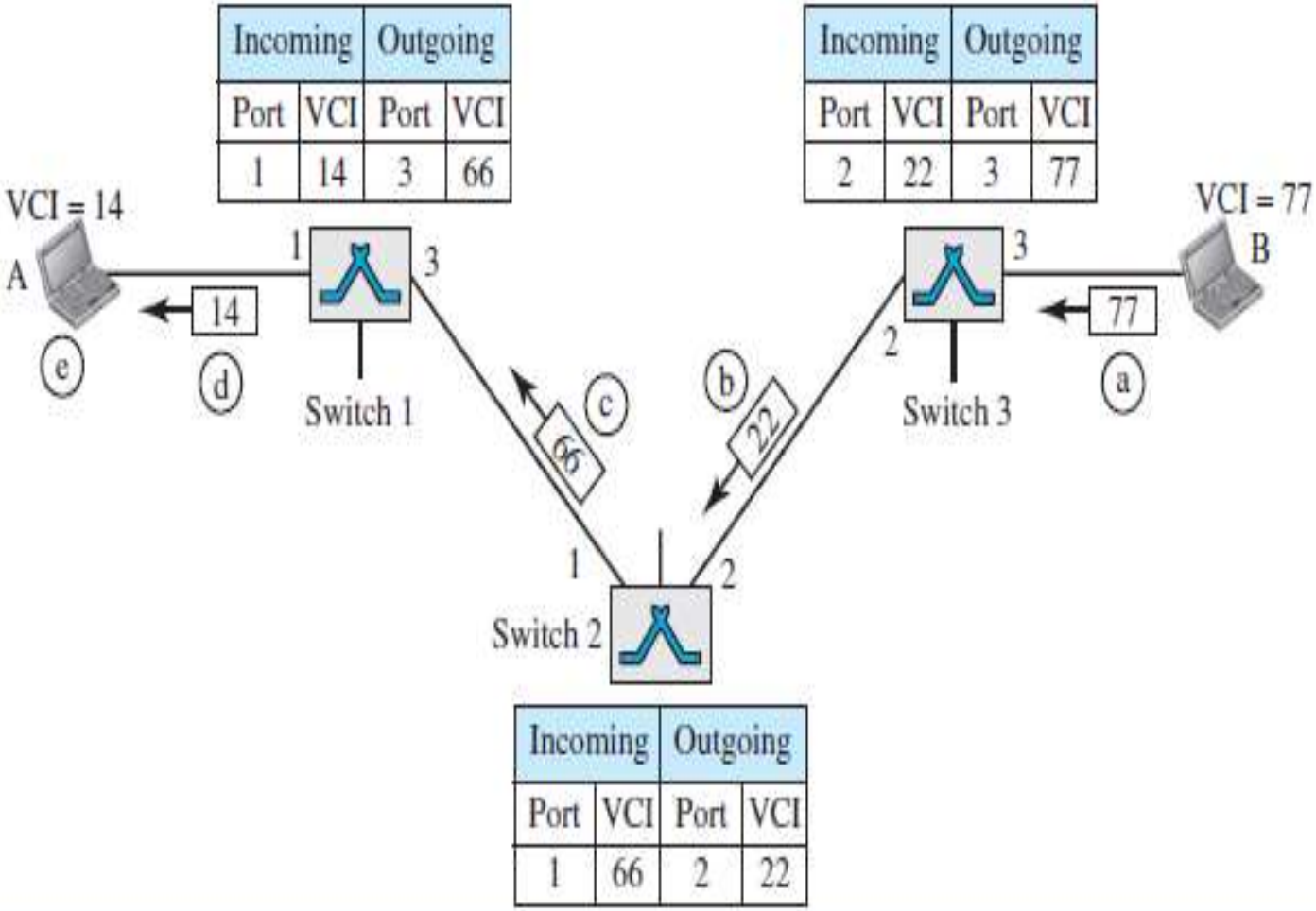


Figure 8.12 *Switch and tables in a virtual-circuit network*

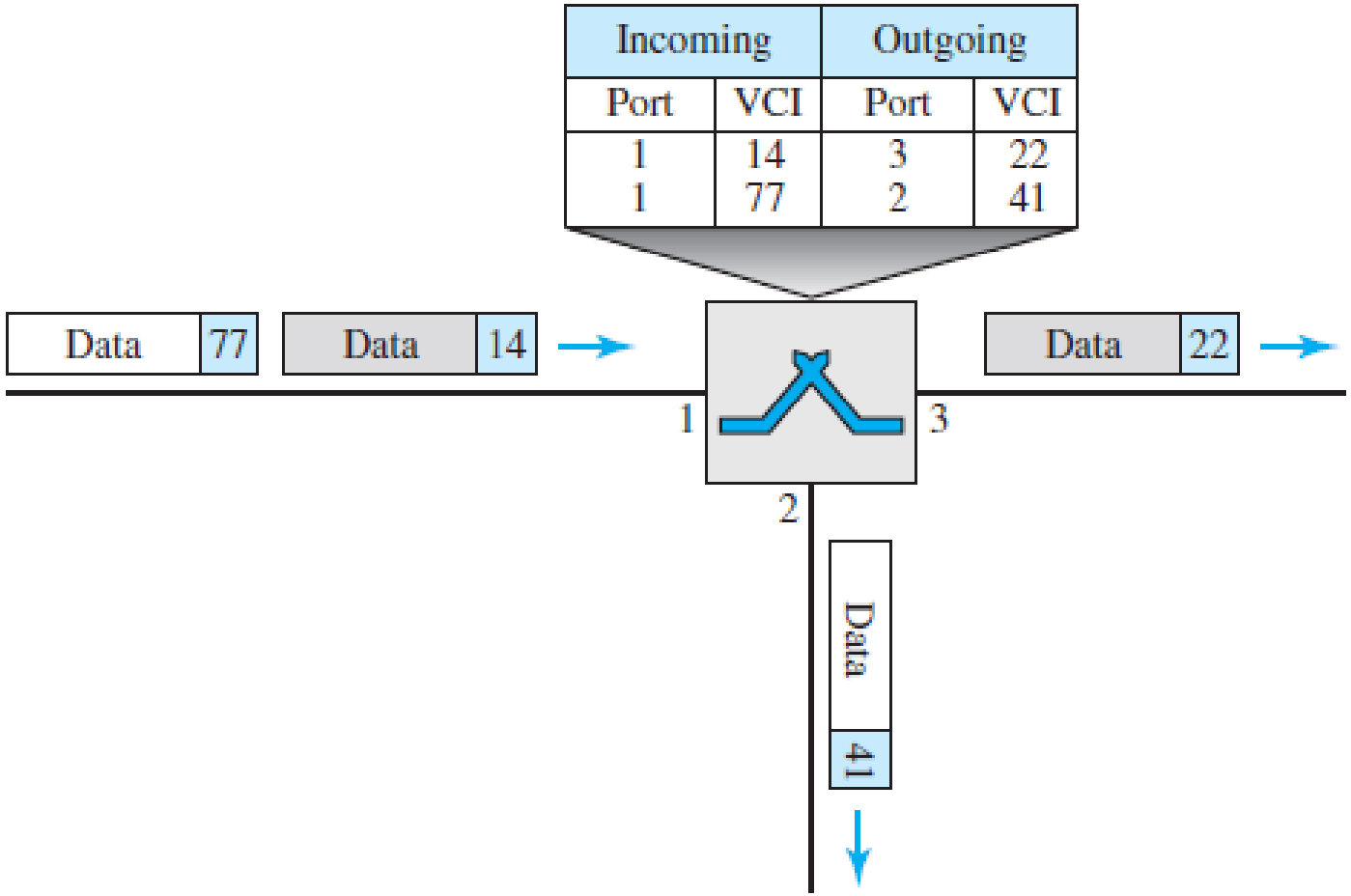
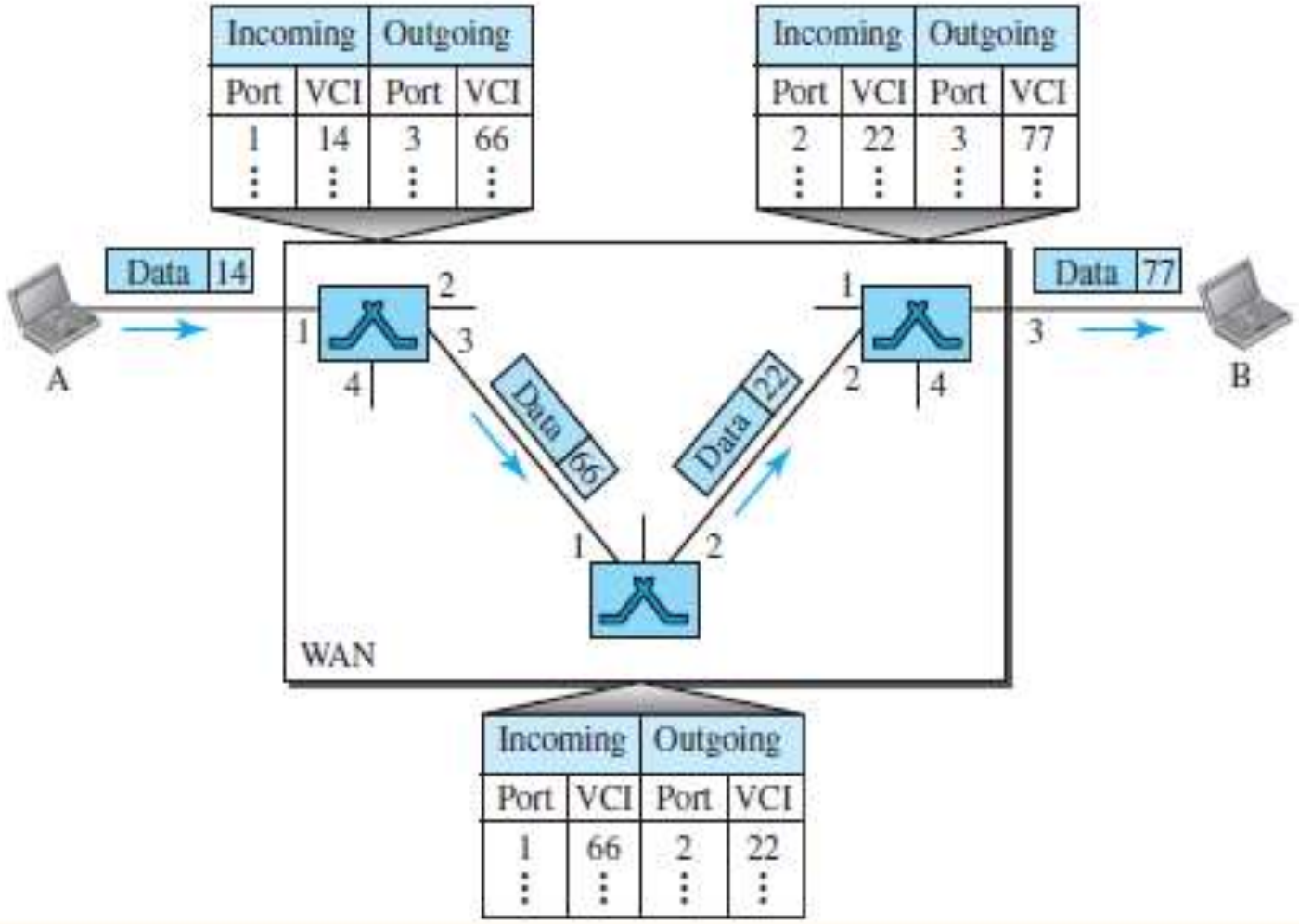


Figure 8.13 *Source-to-destination data transfer in a virtual-circuit network*

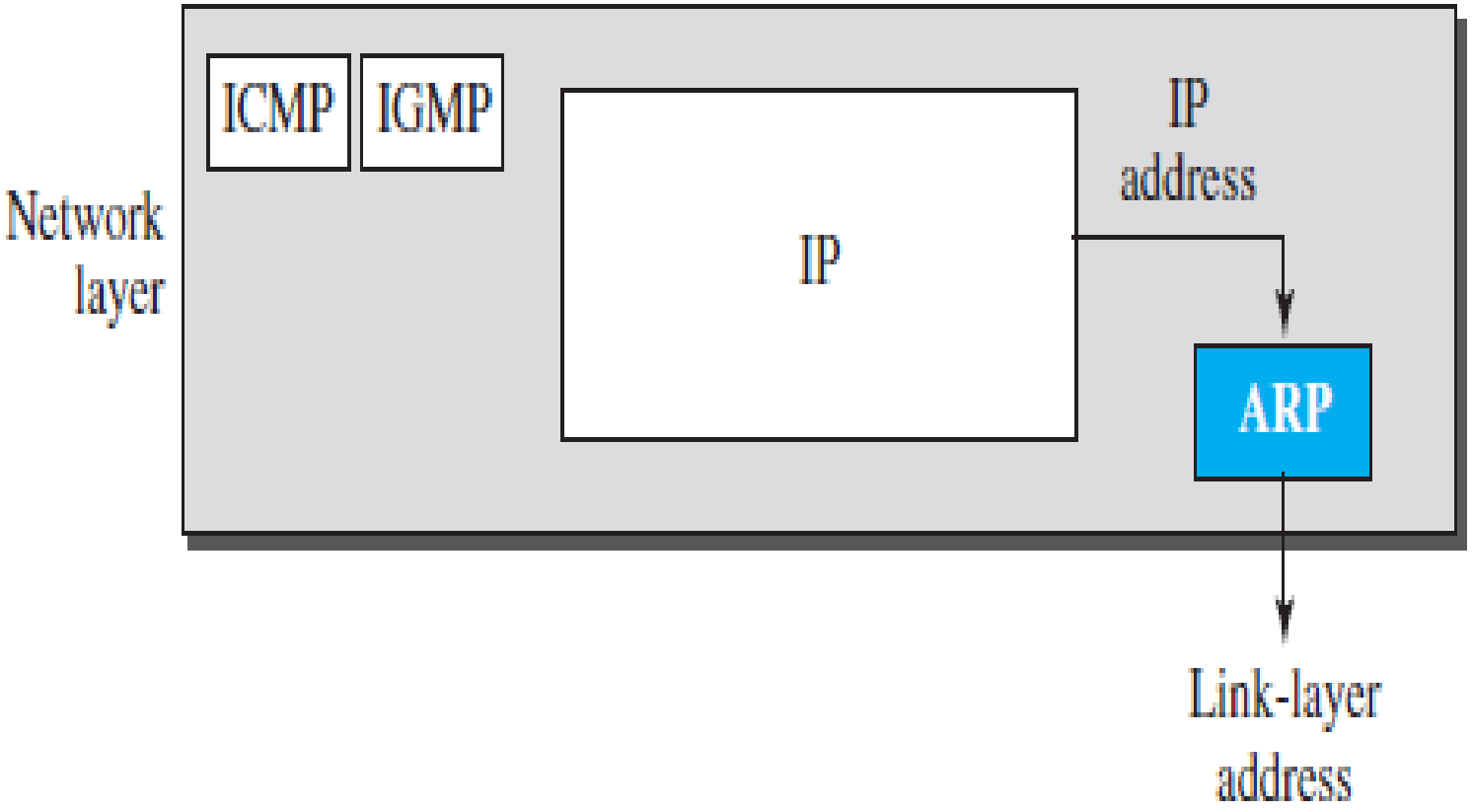


UNIT-3

ARP

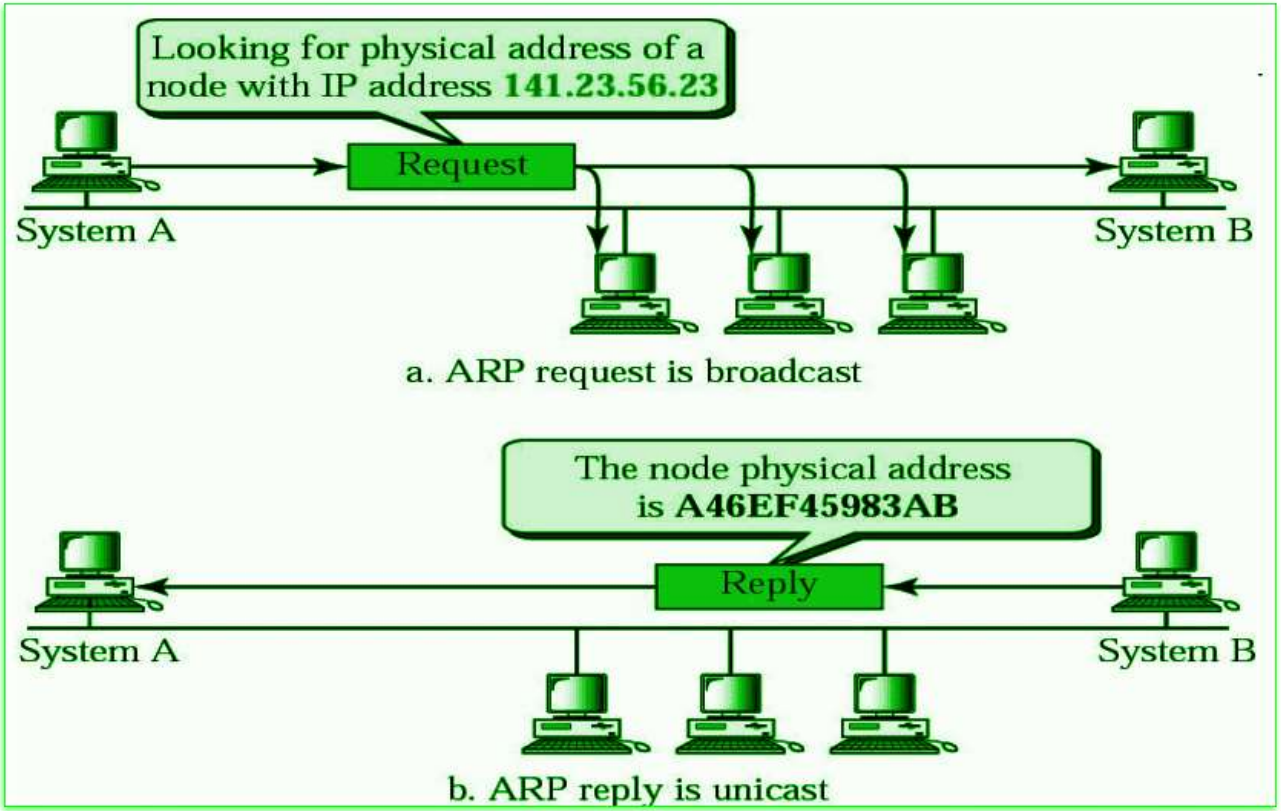
- ARP –Address Resolution Protocol, is used to map IP addresses to the link layer address (MAC address, Local address) addresses used by the data link protocol.
- The ARP protocol operates between the network layer and the data link layer in the Open System Interconnection (OSI) model.
- ARP accepts an IP address from the IP protocol, maps the address to the corresponding link-layer address, and passes it to the data-link layer.

Position of ARP in TCP/IP Protocol Suite



ARP request and Reply operation:

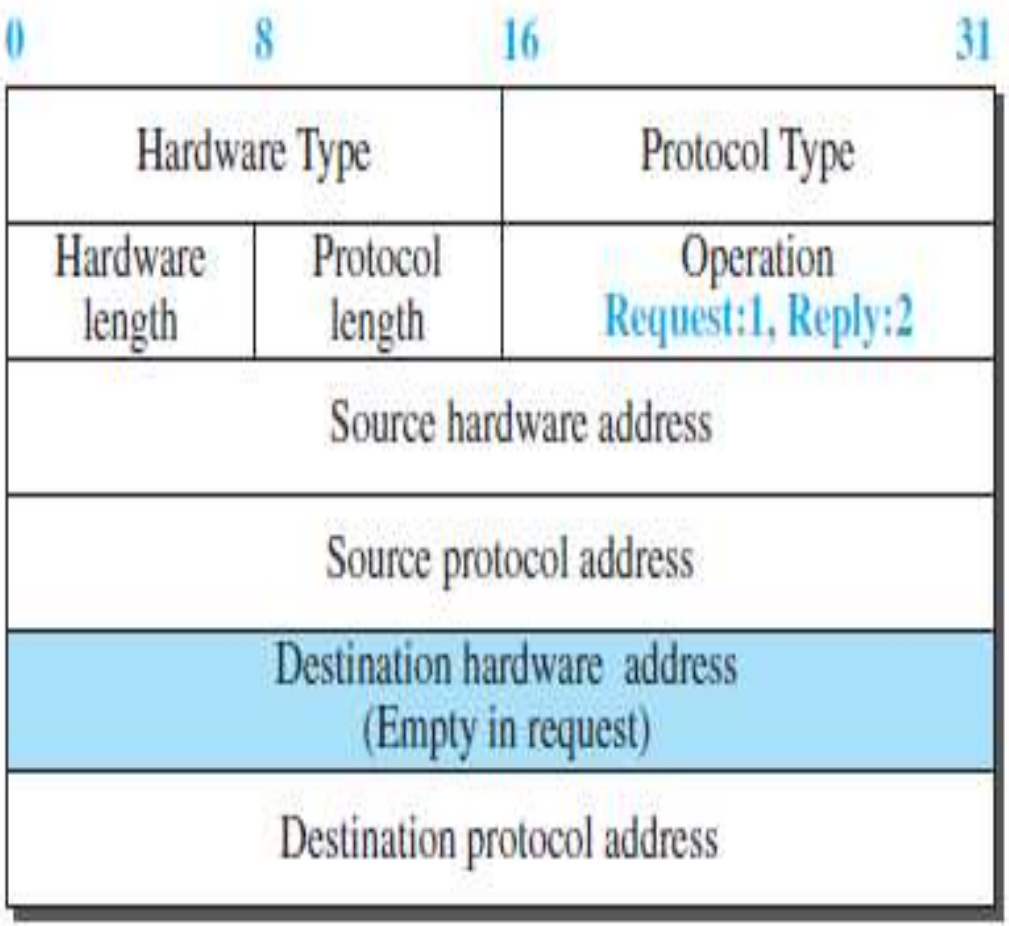
- host or a router needs to find the link-layer address of another host or router in its network, it sends an ARP request packet.
- The packet includes the link-layer and IP addresses of the sender and the IP address of the receiver. Because the sender does not know the link-layer address of the receiver,
- the query is broadcast over the link using the link-layer broadcast address


Looking for physical address of a node with IP address 141.23.56.23
Request
System A
System B
a. ARP request is broadcast
The node physical address is A46EF45983AB
Reply
System A
System B
b. ARP reply is unicast

- Every host or router on the network receives and processes the ARP request packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet.

- The ARP response packet contains the recipient's IP and link-layer addresses. The packet is unicast directly to the node that sent the request packet.

ARP packet Format:



Hardware: LAN or WAN protocol

Protocol: Network-layer protocol

RARP

Reverse Address Resolution Protocol

- At the beginning of the Internet era, a protocol called Reverse Address Resolution Protocol (RARP) was designed to provide the IP address for a booted computer
- RARP networking protocol that is used to map a physical (MAC) address to an Internet Protocol (IP) address.
- It is the reverse of the more commonly used Address Resolution Protocol (ARP), which maps an IP address to a MAC address.
- Uses

RARP is used to convert the Ethernet address to an IP address. It is available for the LAN technologies like FDDI, token ring LANs, etc.

Unit-3 ICMP-Internet Control Message Protocol

- IP provides unreliable and connectionless datagram delivery. It was designed this way to make efficient use of network resources.
- The IP protocol is a best-effort delivery service that delivers a datagram from its original source to its final destination.

However, it has two deficiencies:

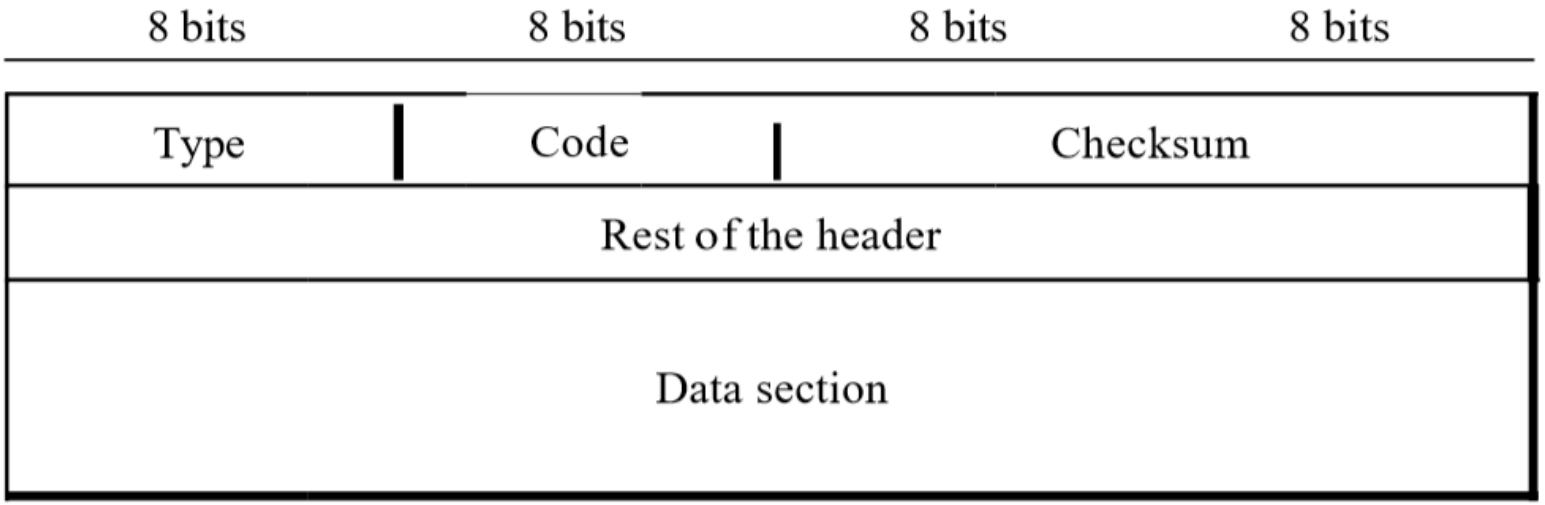
- lack of error control and
- lack of assistance mechanisms.

The Internet Control Message Protocol (ICMP) has been designed to compensate for the above two deficiencies. It is a companion to the IP protoco1

Types of Messages

ICMP messages are divided into two broad categories:

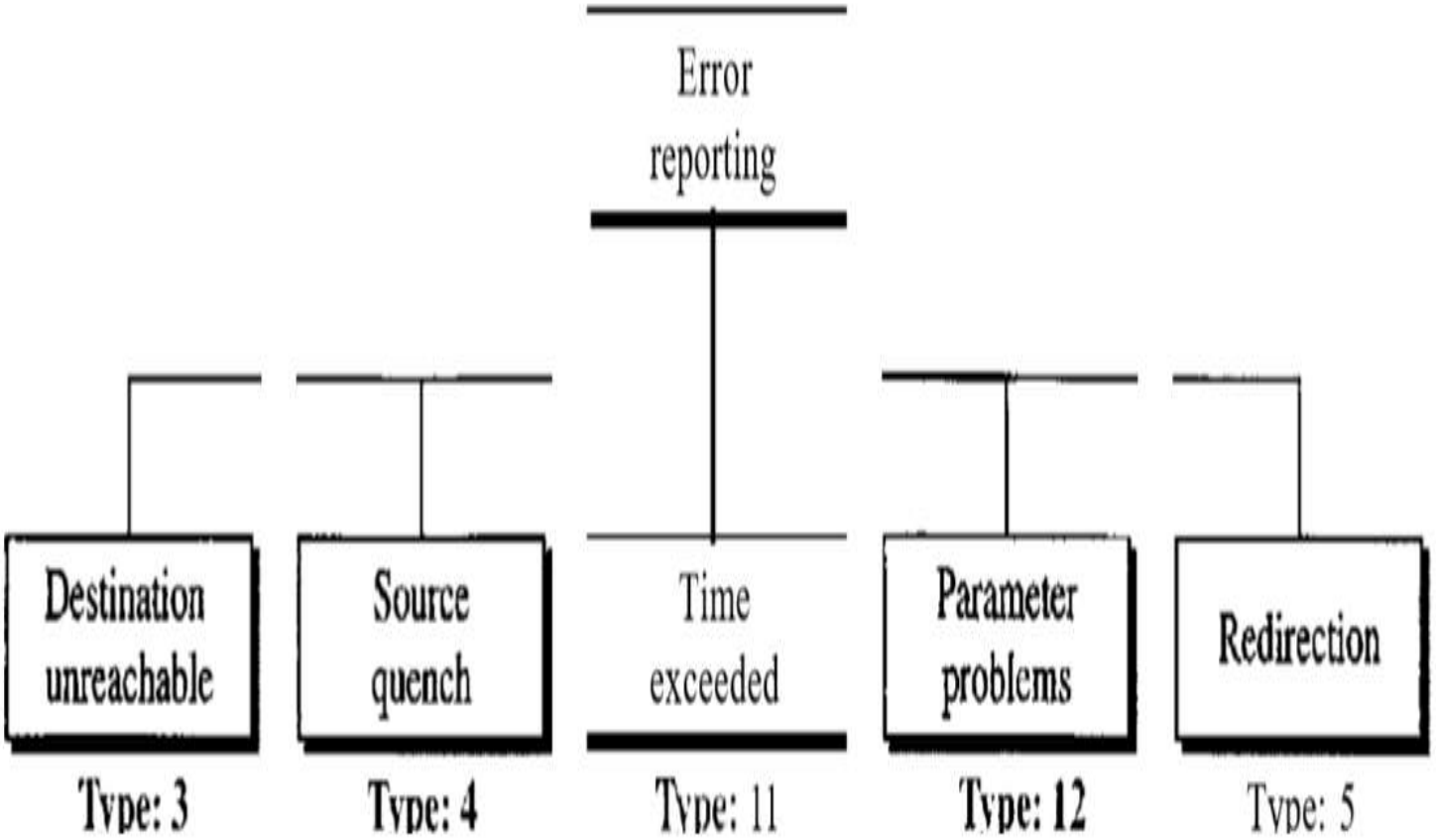
- Error-reporting messages and
- Query messages.
- The error-reporting messages report problems that a router or a host (destination) may encounter when it processes an IP packet.
- The query messages, which occur in pairs, help a host or a network manager get specific information from a router or another host. For example, nodes can discover their neighbors. Also, hosts can discover and learn about routers on their network, and routers can help a node redirect its messages



ICMP Message Format

Error Reporting

- One of the main responsibilities of ICMP is to report errors
- ICMP does not correct errors-it simply reports them.
- Error messages are always sent to the original source because the only information available in the datagram about the route is the source and destination IP addresses.
- ICMP uses the source IP address to send the error message to the source (originator) of the datagram.


Error reporting
Destination unreachable
Source quench
Time exceeded
Parameter problems
Redirection
Type: 3
Type: 4
Type: 11
Type: 12
Type: 5

Destination Unreachable :

- When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host.

Source Quench :

- When a router or host discards a datagram due to congestion, it sends a source-quench message to the sender of the datagram.
- This message has two purposes. First, it informs the source that the datagram has been discarded.
- Second, it warns the source that there is congestion somewhere in the path and that the source should slow down (quench) the sending process.

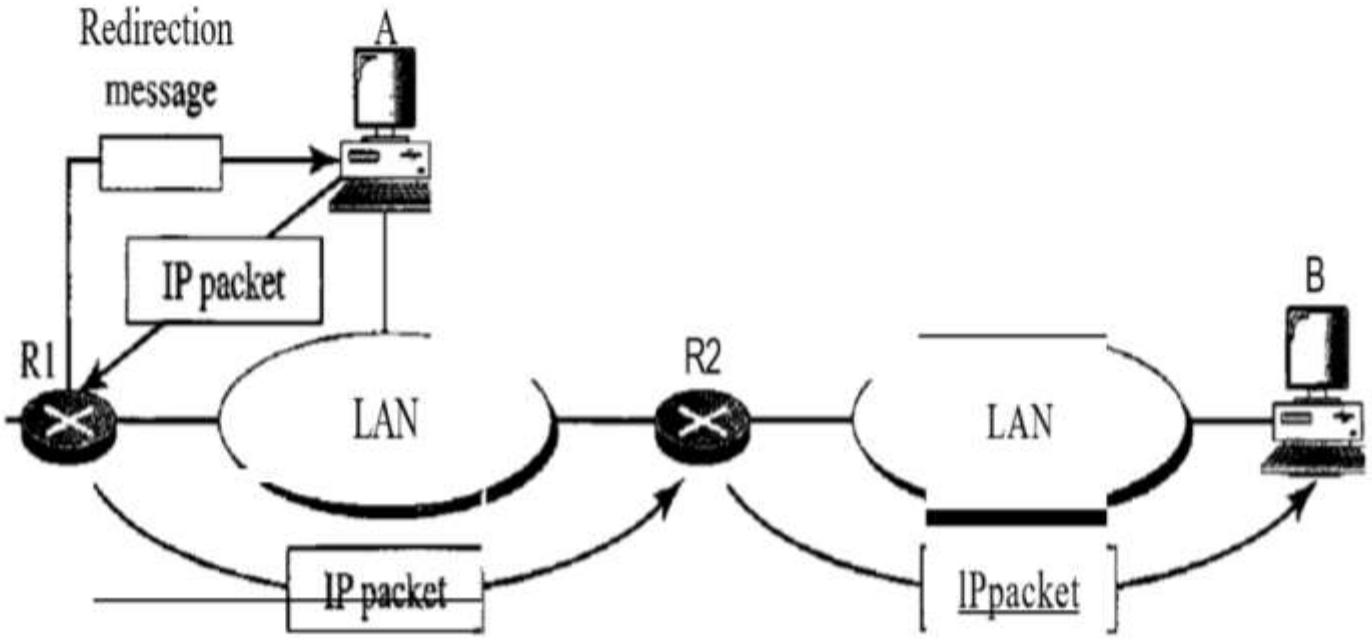
Time Exceeded :

- When a datagram visits a router, the value of this field is decremented by 1.
- When the time-to-live value reaches 0, after decrementing, the router discards the datagram. However, when the datagram is discarded, a time-exceeded message must be sent by the router to the original source.

Parameter Problem

- If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source.

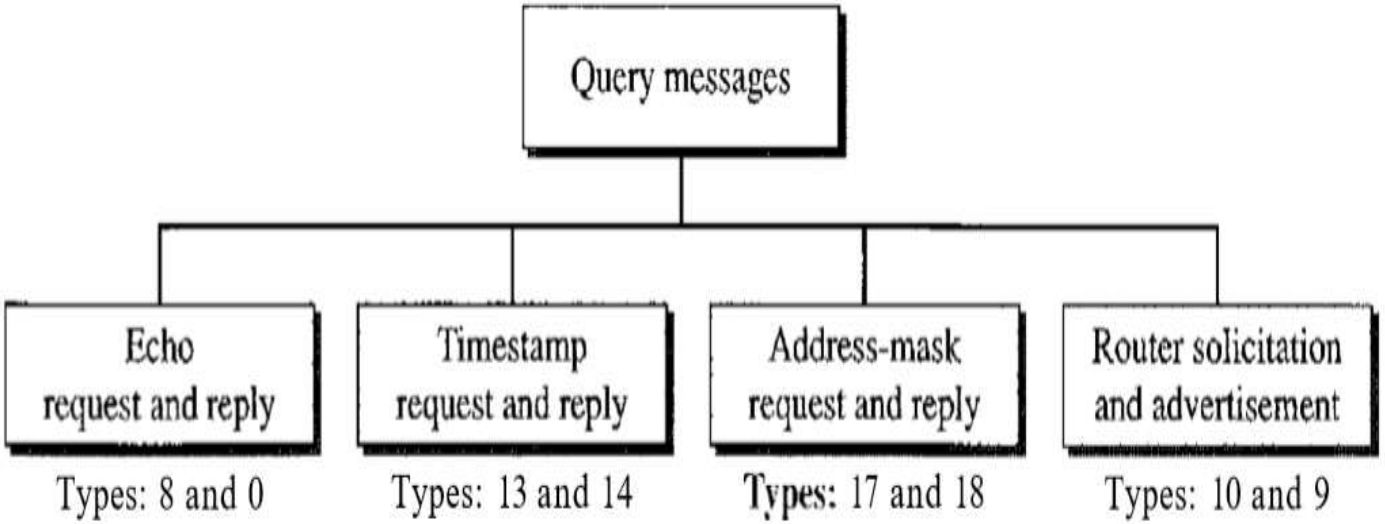
Redirection



- Host A wants to send a datagram to host B.
- Router R2 is obviously the most efficient routing choice, but host A did not choose router R2.
- The datagram goes to R1 instead. Router R1, after consulting its table, finds that the packet should have gone to R2. It sends the packet to R2 and, at the same time, sends a redirection message to host A.
- Host A's routing table can now be updated.

Query Messages:

In addition to error reporting, ICMP can diagnose some network problems. This is accomplished through the query messages, a group of four different pairs of messages



Echo Request and Reply :

The echo-request and echo-reply messages are designed for diagnostic purposes

The combination of echo-request and echo-reply messages determines whether two systems (hosts or routers) can communicate with each other.

Today, most systems provide a version of the ping command that can create a series (instead of just one) of echo-request and echo-reply messages

Timestamp Request and Reply

- Two machines (hosts or routers) can use the timestamp request and timestamp reply messages to determine the round-trip time needed for an IP datagram to travel between them.
- It can also be used to synchronize the clocks in two machines

Address-Mask Request and Reply:

- Host may know its IP address as 159.31.17.24, but it may not know that the corresponding mask.
- To obtain its mask, a host sends an address-mask-request message to a router on the LAN. If the host knows the address of the router, it sends the request directly to the router. If it does not know, it broadcasts the message.

- The router receiving the address-mask-request message responds with an address-mask-reply message.

Router Solicitation and Advertisement

- Host must know if the routers are alive and functioning. The router-solicitation and router-advertisement messages can help in this situation.
- A host can broadcast (or multicast) a router-solicitation message. The router or routers that receive the solicitation message broadcast their routing information using the router-advertisement message. A router can also periodically send router-advertisement messages even if no host has solicited.

Unit-3- DHCP
Dynamic Host Configuration Protocol

- The Dynamic Host Configuration Protocol (DHCP) is a client/server protocol designed to provide the four pieces of information for a diskless computer or a computer that is booted for the first time.

➢ **The IP address of the computer**

➢ **The subnet mask of the computer**

➢ **The IP address of a Gateway / router**

➢ **The IP address of a name server**

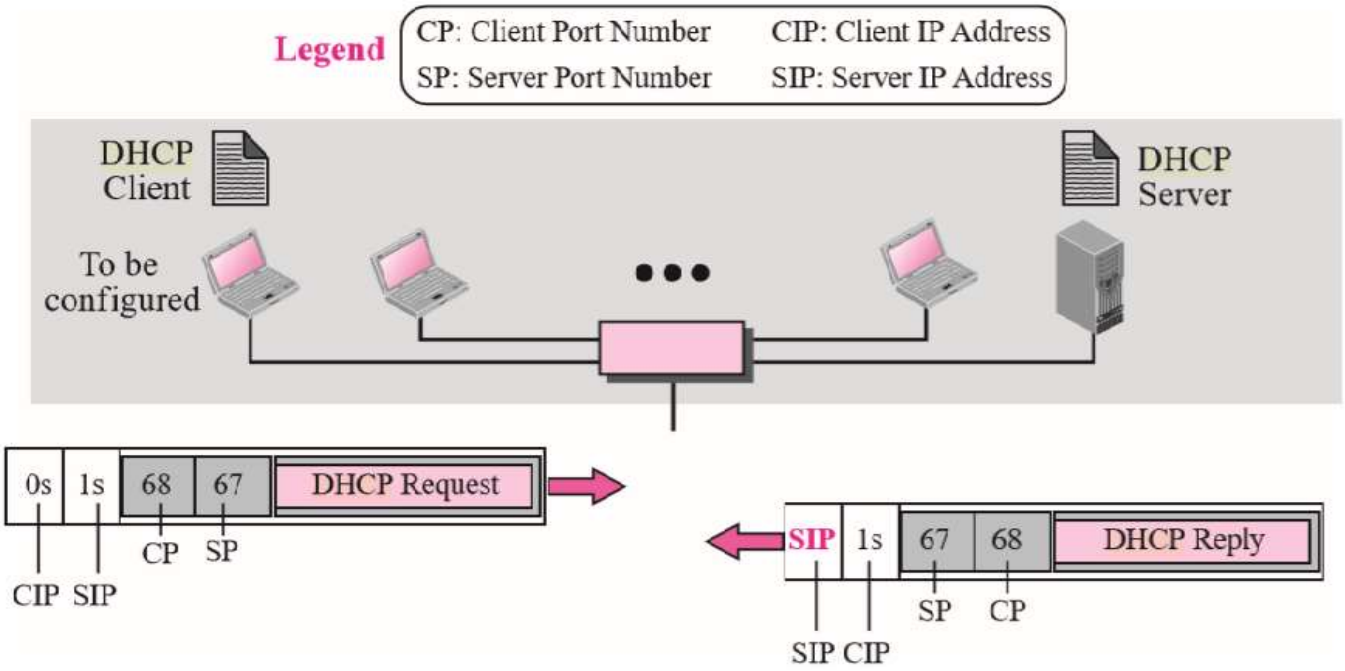
- DHCP is a successor to BOOTP -Bootstrap protocol (Static Configuration Protocol)

DHCP Operation

The DHCP client and server can either be on the same network or on different networks.

Same Network

Client and server on the same network



- The DHCP server issues a passive open command on UDP port number 67 and waits for a client.

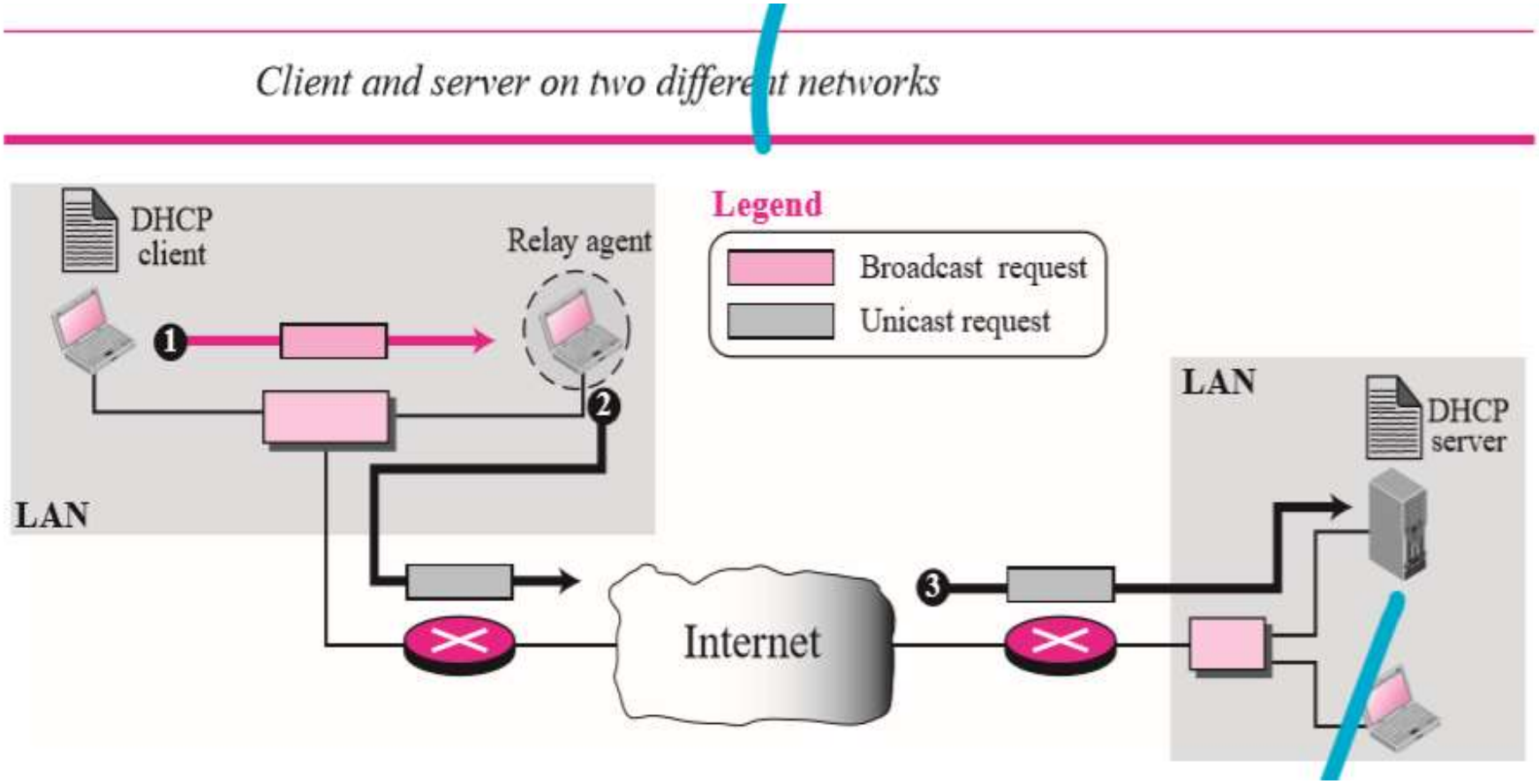
From DHCP Client side:

- A booted client issues an active open command on port number 68 .
- the Destination port number 67(Default UDP port no for DHCP Server)
- Source/Client port number 68.
- The client uses all 0s as the Client/Source IP address and all 1s as the Destination/Server IP address.

From DHCP Server Side:

- The server responds with either a broadcast or a unicast message using UDP source port number 67 and destination port number 68. The response can be unicast because the server knows the IP address of the client. It also knows the physical address of the client.

Different Network



The DHCP request is broadcast because the client does not know the IP address of the server.

A broadcast IP datagram cannot pass through any router. A router receiving such a packet discards it.

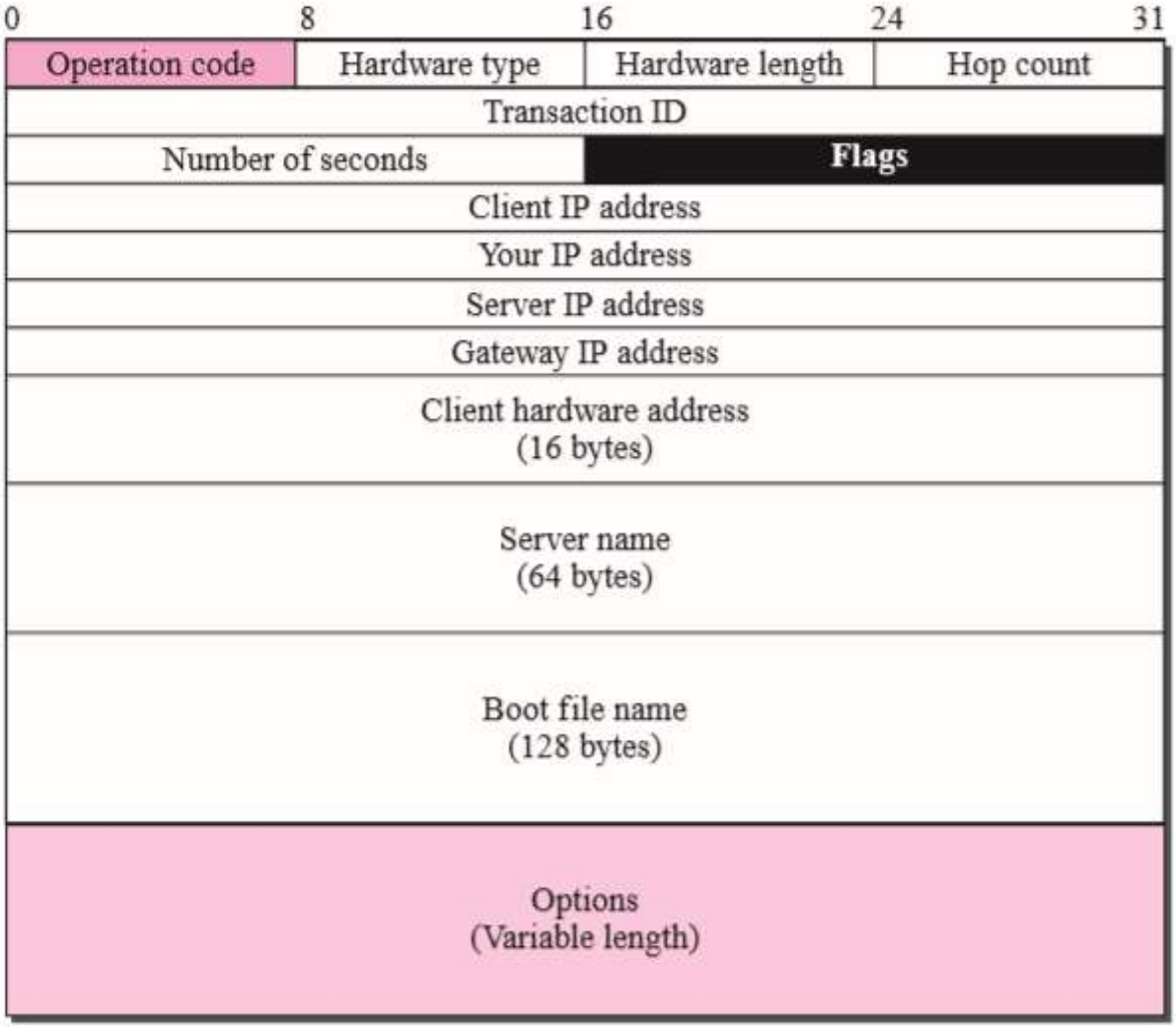
To solve the problem, there is a need for an intermediary. One of the hosts can be used as a relay.

The host in this case is called a relay agent. The relay agent knows the unicast address of a DHCP server and listens for broadcast messages on port 67.

When it receives this type of packet, it encapsulates the message in a unicast datagram and sends the request to the DHCP server.

The DHCP server knows the message comes from a relay agent because one of the fields in the request message defines the IP address of the relay agent. The relay agent, after receiving the reply, sends it to the DHCP client.

DHCP packet format

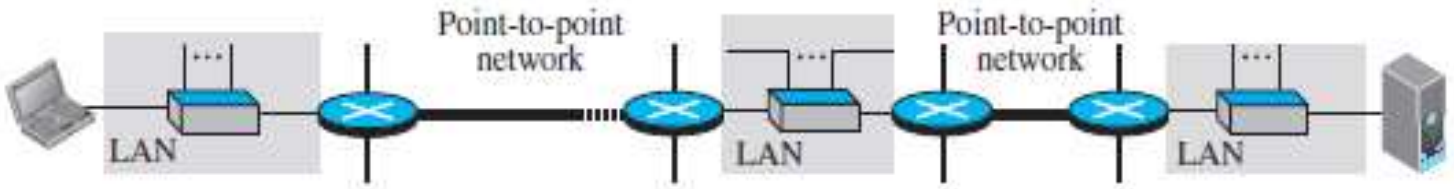


UNIT-5
Introduction to Data Link Layer

- The data-link layer is located between the physical and the network layers.
- The data link layer provides services to the network layer; it receives services from the physical layer
- The first service provided by the data-link layer is **framing**.
- The data-link layer at each node needs to encapsulate the datagram (packet received from the network layer) in a **frame** before sending it to the next node.
- The node also needs to decapsulate the datagram from the frame received on the logical channel.

- A packet at the data-link layer is normally called a *frame*.

Figure 9.2 *Nodes and Links*



a. A small part of the Internet



b. Nodes and links

Services

- The data-link layer is located between the physical and the network layers.
- The data link layer provides services to the network layer; it receives services from the physical layer.
- The duty scope of the data-link layer is node-to-node.
- When a packet is travelling in the Internet, the data-link layer of a node (host or router) is responsible for delivering a datagram to the next node in the path.
- For this purpose, the data-link layer of the sending node needs to encapsulate the datagram received from the network in a frame, and the data-link layer of the receiving node needs to decapsulate the datagram from the frame.

Framing:

Framing is a process in which the data link layer receive the packets or datagram from the network layer and divided into smaller frames for transmission.

Flow control:

its refers to a set of procedures used to restrict the amount of data that the sender can send before receiving an acknowledgement from the receiver.

1. Stop and wait flow control(Send one frames at a time)

2. Sliding window flow control(Send several frames at a time).

Error control:

At the sending node, a frame in a data-link layer needs to be changed to bits, transformed to electromagnetic signals, and transmitted through the transmission media.

At the receiving node, electromagnetic signals are received, transformed to bits, and put together to create a frame.

When a signals are transmitted from one place to another, transmission error may occur, it was due to electrical interference from natural sources such as lightning, earth quake as well as man made sources as motors, generator failures.

To maintain a quality of signals transmitted, error control mechanism has to be implemented, The error needs first to be detected. After detection, it needs to be either corrected at the receiver node or discarded and retransmitted by the sending node.

Congestion Control

- Although a link may be congested with frames, which may result in frame loss, most data-link-layer protocols do not directly use a congestion control to alleviate congestion, although some wide-area networks do.
- It refers to the techniques and mechanism that can either prevent congestion before it happens or remove congestion after it has happened.
- It also maintain the load below the maximum capacity.

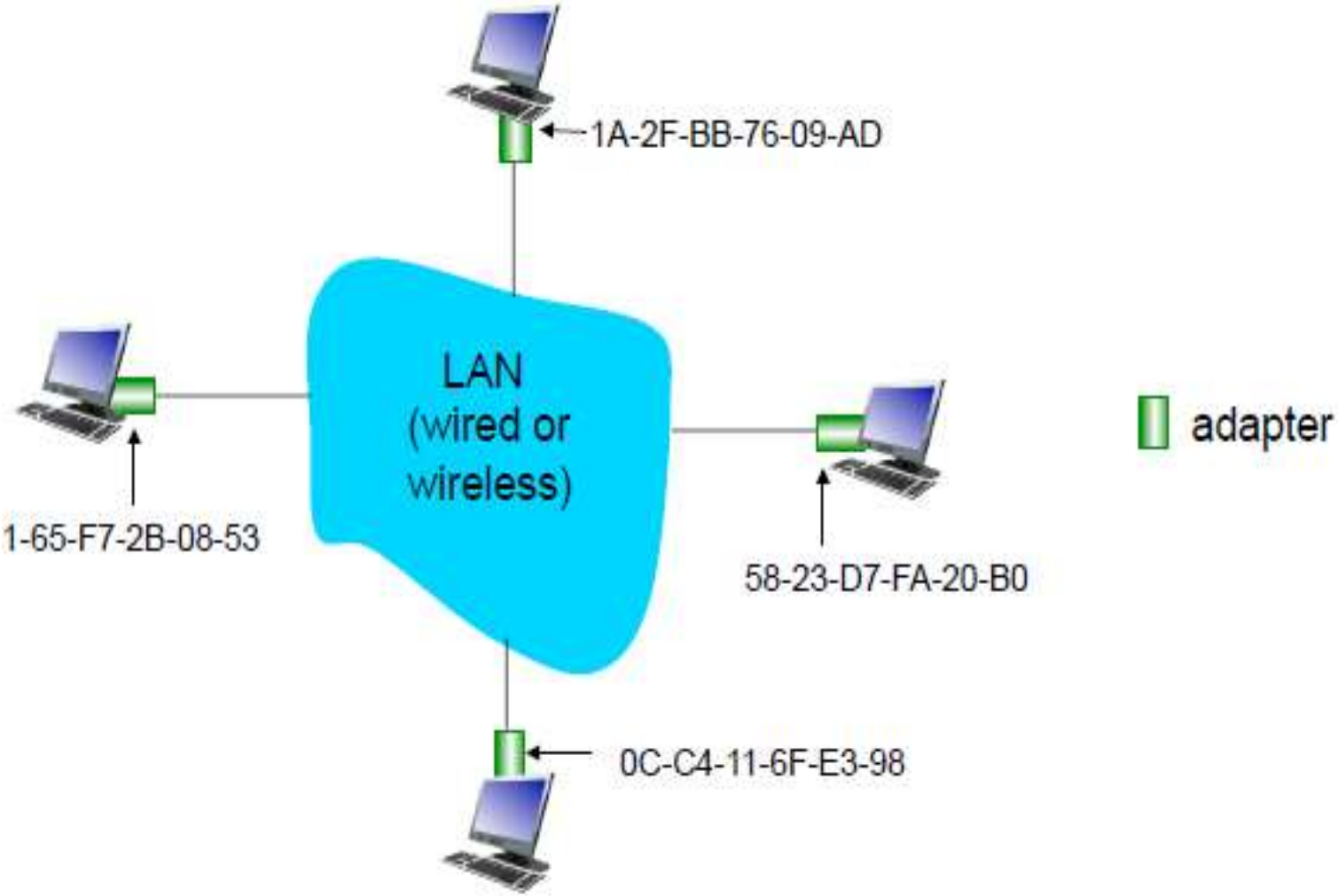
Types of Link:

1. Point to point link(Dedicated between two devices)
2. Broadcast link(Link is shared between several pairs)

LINK-LAYER ADDRESSING

- The link-layer addresses of the two nodes. A *link-layer address* is sometimes called a *link address*, sometimes a *physical address*, and sometimes a *MAC address*.
- 48 bit MAC address (for most LANs) burned in NIC ROM, also sometimes software settable e.g.: 1A-2F-BB-76-09-AD (Hexadecimal notation)

each adapter on LAN has unique *LAN* address



- Manufacturer buys portion of MAC address space (to assure uniqueness)
- MAC address: like Aadhar card Number
- IP address: like postal address
- MAC flat address → portability
- can move LAN card from one LAN to another
- IP hierarchical address *not* portable address depends on IP subnet to which node is attached

Three Types of addresses

Some link-layer protocols define three types of addresses:

- unicast
- multicast
- broadcast

Unicast Address

- Each host or each interface of a router is assigned a unicast address.
- Unicasting means one-to-one communication.

Example of unicast address: **4A:30:10:21:10:1A**

This is a unicast address because A in binary is 1010 (even).

Multicast Address

- Some link-layer protocols define multicast addresses. Multicasting means one-to-many communication.

Example of multicast address:

47:20:1B:2E:08:EE

This is a multicast address because 7 in binary is 0111 (odd).

Broadcast Address

- Some link-layer protocols define a broadcast address. Broadcasting means one-to-all communication.

Example:**FF:FF:FF:FF:FF:FF**

This is a broadcast address because all digits are Fs in hexadecimal.

UNIT-IV Unicast Routing

- In an internet, the goal of the network layer is to deliver a datagram from its source to its destination or destinations.
- If a datagram is destined for only one destination (one-to-one delivery), we have *unicast routing*.
- If the datagram is destined for several destinations (one-to-many delivery), we have *multicast routing*.

Routing Table:

- Routing table is maintained by a host or router, which contains an information a router needs to route packets from source to destination.
- The information that includes network address, the cost and the address of the next hop and so on.

Types of Routing table:

- Static routing table(non adaptive routing table)
- Dynamic routing table(Adaptive routing table)
- The **static Routing table** contains the information entered manually by the network administrator. When the table is created it can not be updated automatically when there is a change in the internet.

In **dynamic routing table** updated automatically by dynamic routing protocols such as RIP,OSPF.

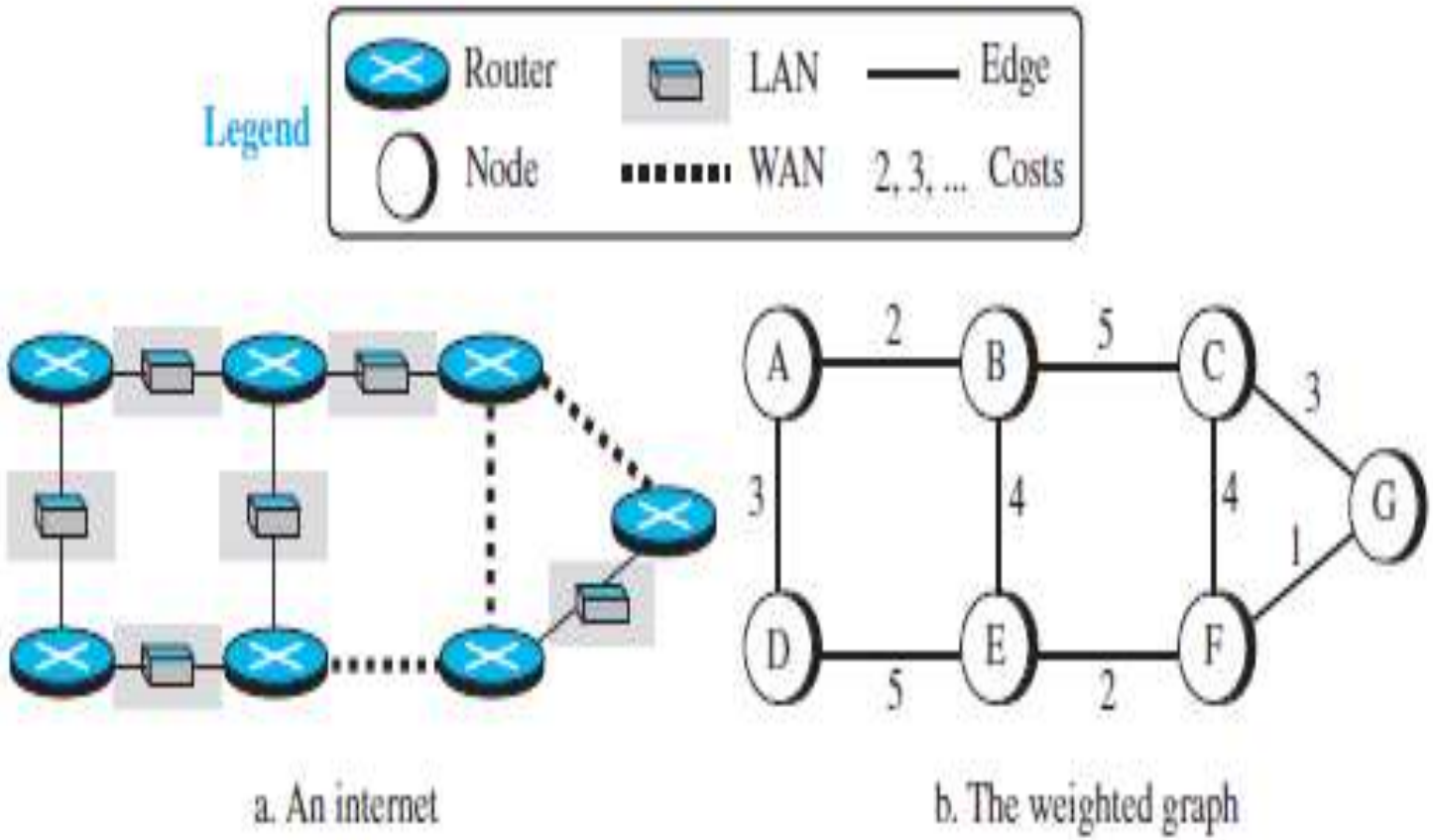
It will change the routing decision to reflect changes in the internet topology and also network traffic.

Ex: Distance Vector Routing (DVR), Link state Routing

An Internet as a Graph

- To find the best route, an internet can be modelled as a *graph*.
- A graph in computer science is a set of *nodes* and *edges* (lines) that connect the nodes.
- To model an internet as a graph, we can think of each router as a node and each network between a pair of routers as an edge.
- The basic problem of routing is to find the lowest-cost path between any two nodes, where the cost of a path equals the sum of the costs of all the edges that make up the path

Figure 20.1 *An internet and its graphical representation*



a. An internet

b. The weighted graph

Routing domain:

Interdomain:

Routing between different autonomous system.

Intradomain:

Routing within a single autonomous system.

Example: Within a college campus, office and any organization

Autonomous System:

A group of networks and routers under the authority of a single administration is called as **autonomous system**.

INTERDOMAIN ROUTING PROTOCOL:

EGP(Exterior Gateway protocol)

BGP(Border Gateway protocol)

INTRADOMAIN ROUTING PROTOCOL:

1.Distance vector algorithm –Routing information protocol(RIP)

2.Link state algorithm-Open Shortest Path First(OSPF)protocol

Unit-IV - Distance Vector Routing Algorithm

- **Distance-Vector (RIP)(Intradomain Routing protocol)**
- The idea behind the distance-vector algorithm is suggested by its name Each node constructs a one-dimensional array (a vector) containing the “distances” (costs) to all other node
- Distributes that vector to its immediate neighbors.
- Distance vector routing periodically share it knowledge about the entire network to its neighbors.
- The starting assumption for distance-vector routing is that each node knows the cost of the link to each of its directly connected neighbors.
- These costs may be provided when the router is configured by a network manager. A link that is down is assigned an infinite cost.

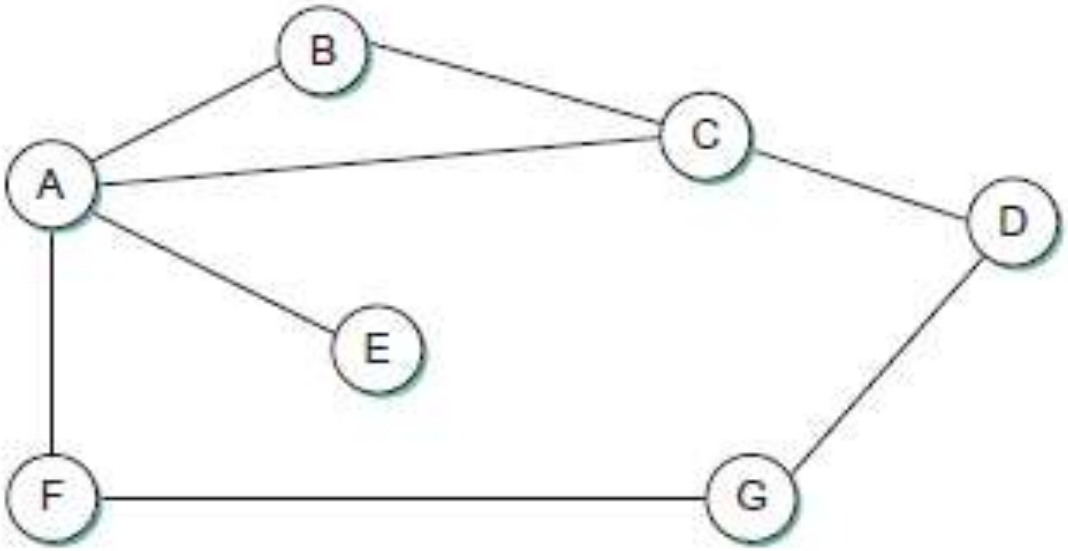


FIGURE 3.29 Distance-vector routing: an example network.

Three keys to understand how this algorithm works:

1. Knowledge about the whole network.
2. Routing only to neighbors
3. Information sharing at regular intervals

Each row represent list of distances from one node to all other node.

Initially each node sets a cost of 1 to its directly connected neighbors and infinite to all other nodes which is shown in table

Table 3.10 Initial Distances Stored at Each Node (Global View)

Information Stored at Node	Distance to Reach Node						
	A	B	C	D	E	F	G
A	0	1	1	∞	1	1	∞
B	1	0	1	∞	∞	∞	∞
C	1	1	0	1	∞	∞	∞
D	∞	∞	1	0	∞	∞	1
E	1	∞	∞	∞	0	∞	∞
F	1	∞	∞	∞	∞	0	1
G	∞	∞	∞	1	∞	1	0

- Next step every node sends message to its directly connected neighbors containing its personal list of distances.
- For example, node F tells node A that it can reach node G at a cost of 1; A also knows it can reach F at a cost of 1, so it adds these costs to get the cost of reaching G by means of F. This total cost of 2 is less than the current cost of infinity,

Table 3.11 Initial Routing Table at Node A

Destination	Cost	NextHop
B	1	B
C	1	C
D	∞	—
E	1	E
F	1	F
G	∞	—

Final routing table at Node A

Destination	Cost	Next hop
B	01	B
C	01	C
D	02	C
E	01	E
F	01	F
G	02	F

Table 3.13 Final Distances Stored at Each Node (Global View)

Information Stored at Node	Distance to Reach Node						
	A	B	C	D	E	F	G
A	0	1	1	2	1	1	2
B	1	0	1	2	2	2	3
C	1	1	0	1	2	2	2
D	2	2	1	0	3	2	1
E	1	2	2	3	0	2	3
F	1	2	2	2	2	0	1
G	2	3	2	1	3	1	0

Routing Information Protocol(RIP)

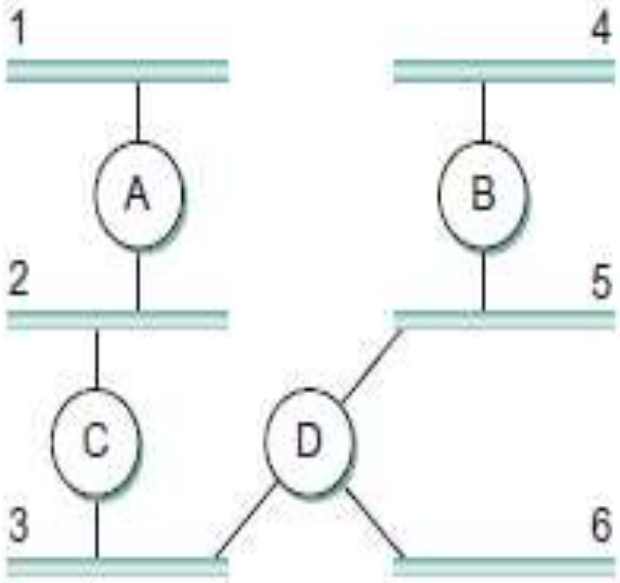


FIGURE 3.30 Example network running RIP.

- One of the more widely used routing protocols in IP networks is the Routing Information Protocol (RIP).
- A straight forward implementation of DVR.
- The goal of the routers is to learn how to forward packets to various *networks.*
- Instead of advertising the cost of reaching other routers, the routers advertise the cost of reaching networks.
- For example, in Figure 3.30, router C would advertise to router A the fact that it can reach networks 2 and 3 (to which it is directly connected) at a cost of 0,
- networks 5 and 6 at cost 1, and
- network 4 at cost 2.

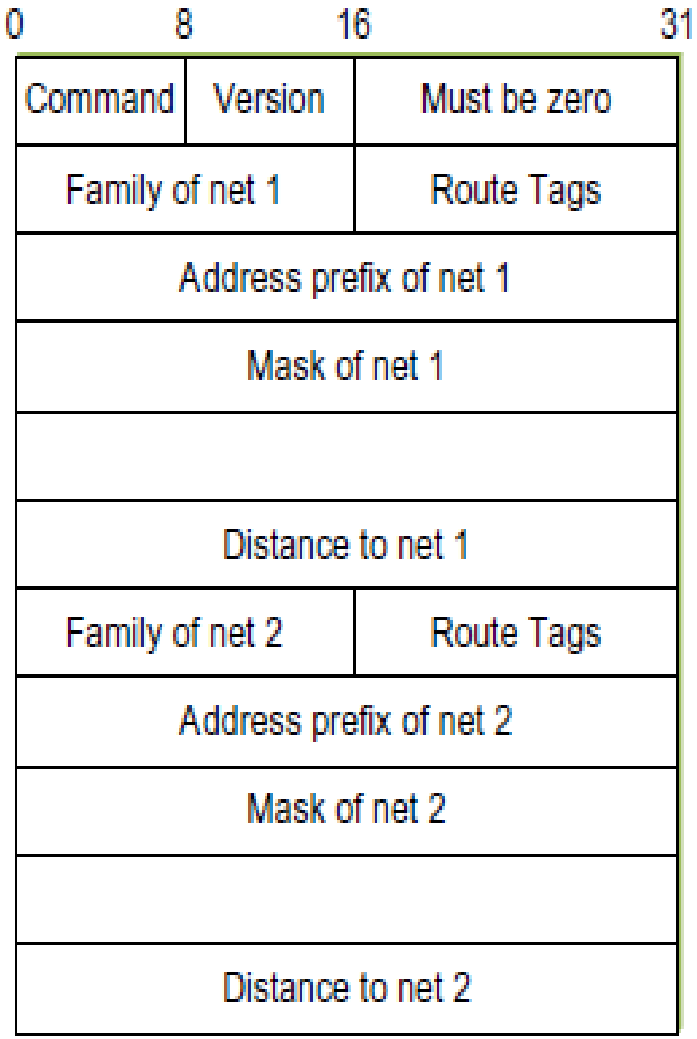


FIGURE 3.31 RIPv2 packet format.

Unit-IV Link state Routing

- Link state routing each router share its knowledge of it neighborhood with all router in the internetwork.
- DVR each router periodically share knowledge about the entire network with its neighbors.
- Each router send neighborhood information to every other router on the network, it does by a process is called Reliable Flooding.

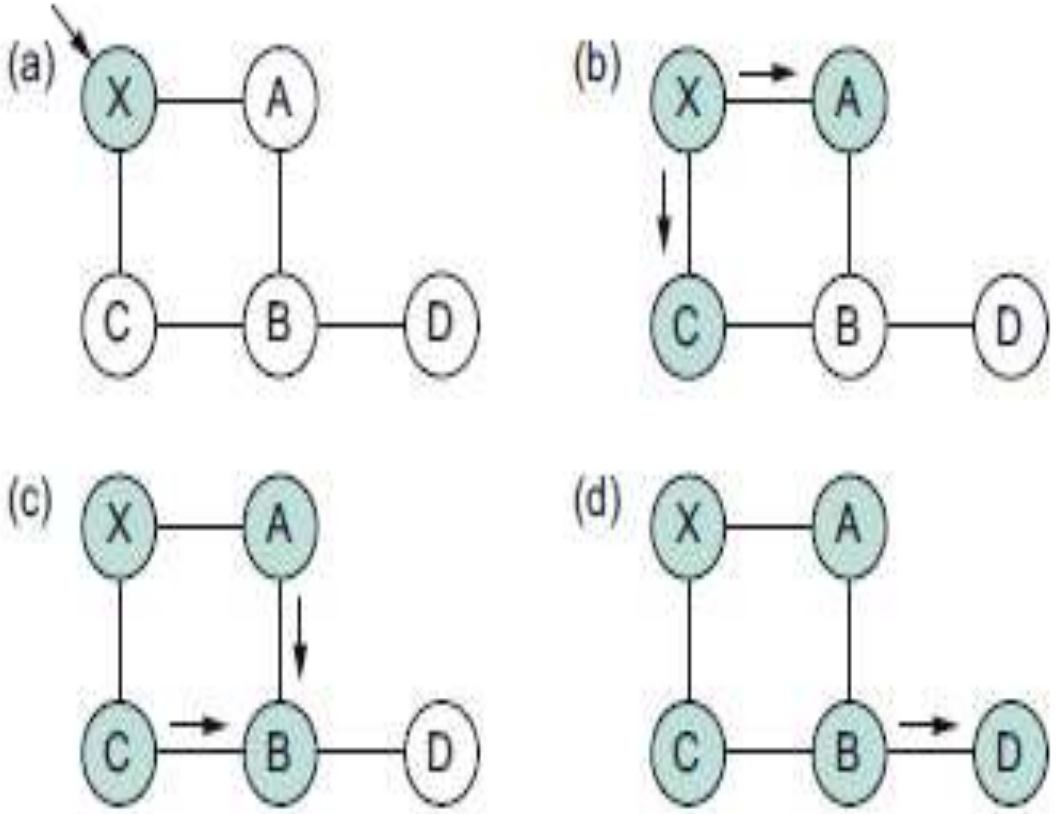
Reliable Flooding:

- *Reliable flooding* is the process of making sure that all the nodes participating in the routing protocol get a copy of the link-state information from all the other nodes.
- *flooding* suggests, the basic idea is for a node to send its link-state information out on all of its directly connected links; each node that receives this information then forwards it out on all of *its* links.
- This process continues until the information has reached all the nodes in the network.

- The each node creates an update packet, also called a *link-state packet* (LSP), which contains the following information:
- The ID of the node that created the LSP
- A list of directly connected neighbors of that node, with the cost of the link to each one
- A sequence number
- A time to live for this packet
- The first two items are needed to enable route calculation; the last two are used to make the process of flooding the packet to all nodes reliable.

Flooding works in the following way:

- Consider a node X that receives a copy of an LSP that originated at some other node Y. Note that Y may be any other router in the same routing domain as X.



■ **FIGURE 3.32** Flooding of link-state packets: (a) LSP arrives at node X; (b) X floods LSP to A and C; (c) A and C flood LSP to B (but not X); (d) flooding is complete.

- X checks to see if it has already stored a copy of an LSP from Y. If not, it stores the LSP. If it already has a copy, it compares the sequence numbers;
- if the new LSP has a larger sequence number, it is assumed to be the more recent, and that LSP is stored, replacing the old one. A smaller or equal sequence number would imply an LSP older so it would be discarded and no further action would be needed.
- If the received LSP was the newer one, X then sends a copy of that LSP to all of its neighbors except the neighbor from which the LSP was just received.
- Figure 3.32 shows an LSP being flooded in a small network. Each node becomes shaded as it stores the new LSP. In Figure 3.32(a) the LSP arrives at node X, which sends it to neighbors A and C in Figure 3.32(b). A and C do not send it back to X, but send it on to B.

- Since B receives two identical copies of the LSP, it will accept whichever arrived first and ignore the second as a duplicate. It then passes the LSP onto D, which has no neighbors to flood it to, and the process is complete.

Route calculation:

- Once a given node has a copy of the LSP from every other node, it is able to compute a complete map for the topology of the network, and from this map it is able to decide the best route to each destination.
- The question, then, is exactly how it calculates routes from this information. The solution is based on a well-known algorithm from graph theory—Dijkstra's shortest-path algorithm.
- In practice, each switch computes its routing table directly from the LSPs it has collected using a realization of Dijkstra's algorithm called the *forward search* algorithm.

- Specifically, each switch maintains two lists,
- Tentative and Confirmed List.
- Each of these lists contains a set of entries of the form (Destination, Cost, NextHop). The algorithm works as follows:

1. Initialize the Confirmed list with an entry for myself; this entry has a cost of 0.

2. For the node just added to the Confirmed list in the previous step, call it node Next and select its LSP.

3. For each neighbor (Neighbor) of Next, calculate the cost (Cost) to reach this Neighbor as the sum of the cost from myself to Next and from Next to Neighbor.

(a) If Neighbor is currently on neither the Confirmed nor the Tentative list, then add (Neighbor, Cost, NextHop) to the Tentative list, where NextHop is the direction I go to reach Next.

(b) If Neighbor is currently on the Tentative list, and the Cost is less than the currently listed cost for Neighbor, then replace the current entry with (Neighbor, Cost, NextHop), where NextHop is the direction I go to reach Next.

- **4.** If the Tentative list is empty, stop. Otherwise, pick the entry from the Tentative list with the lowest cost, move it to the Confirmed list, and return to step 2.

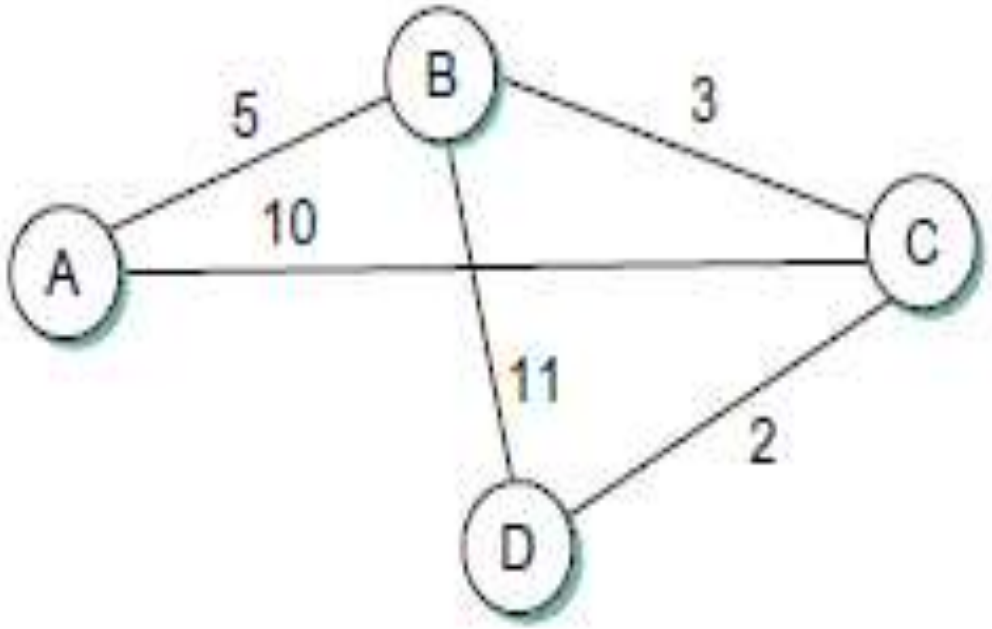


FIGURE 3.33 Link-state routing: an example network.

Table 3.14 Steps for Building Routing Table for Node D (Figure 3.33)

Step	Confirmed	Tentative	Comments
1	(D,0,–)		Since D is the only new member of the confirmed list, look at its LSP.
2	(D,0,–)	(B,11,B) (C,2,C)	D's LSP says we can reach B through B at cost 11, which is better than anything else on either list, so put it on Tentative list; same for C.
3	(D,0,–) (C,2,C)	(B,11,B)	Put lowest-cost member of Tentative (C) onto Confirmed list. Next, examine LSP of newly confirmed member (C).
4	(D,0,–) (C,2,C)	(B,5,C) (A,12,C)	Cost to reach B through C is 5, so replace (B,11,B). C's LSP tells us that we can reach A at cost 12.
5	(D,0,–) (C,2,C) (B,5,C)	(A,12,C)	Move lowest-cost member of Tentative (B) to Confirmed, then look at its LSP.
6	(D,0,–) (C,2,C) (B,5,C)	(A,10,C)	Since we can reach A at cost 5 through B, replace the Tentative entry.
7	(D,0,–) (C,2,C) (B,5,C) (A,10,C)		Move lowest-cost member of Tentative (A) to Confirmed, and we are all done.

Difference between Inter domain and Intra domain Routing protocol

- Inter-domain is any routing protocols that you have setup between two different networks.
- These are usually called Autonomous Systems (AS). The main inter-domain protocol is BGP. Most of the time when talking about inter-domain is routing on the internet.
- Intra-domain(within domain) is any routing on your own network. For that you can use OSPF, RIP routing protocols.

Unit -4 Open shortest path First protocol(OSPF)

- A protocol that implements link-state routing is called *Open Shortest Path First (OSPF)*.
- The first word, “Open,” refers to the fact that it is an open, nonproprietary standard, created under the Internet Engineering Task Force (IETF).
- OSPF adds quite a number of features to the basic link-state algorithm described above, including the following:
- *Authentication of routing messages*
- One feature of distributed routing algorithms is that they disperse information from one node to many other nodes, and the entire network can thus be impacted by bad information from one node. For this reason, it’s a good idea

to be sure that all the nodes taking part in the protocol can be trusted. Authenticating routing messages helps achieve this.

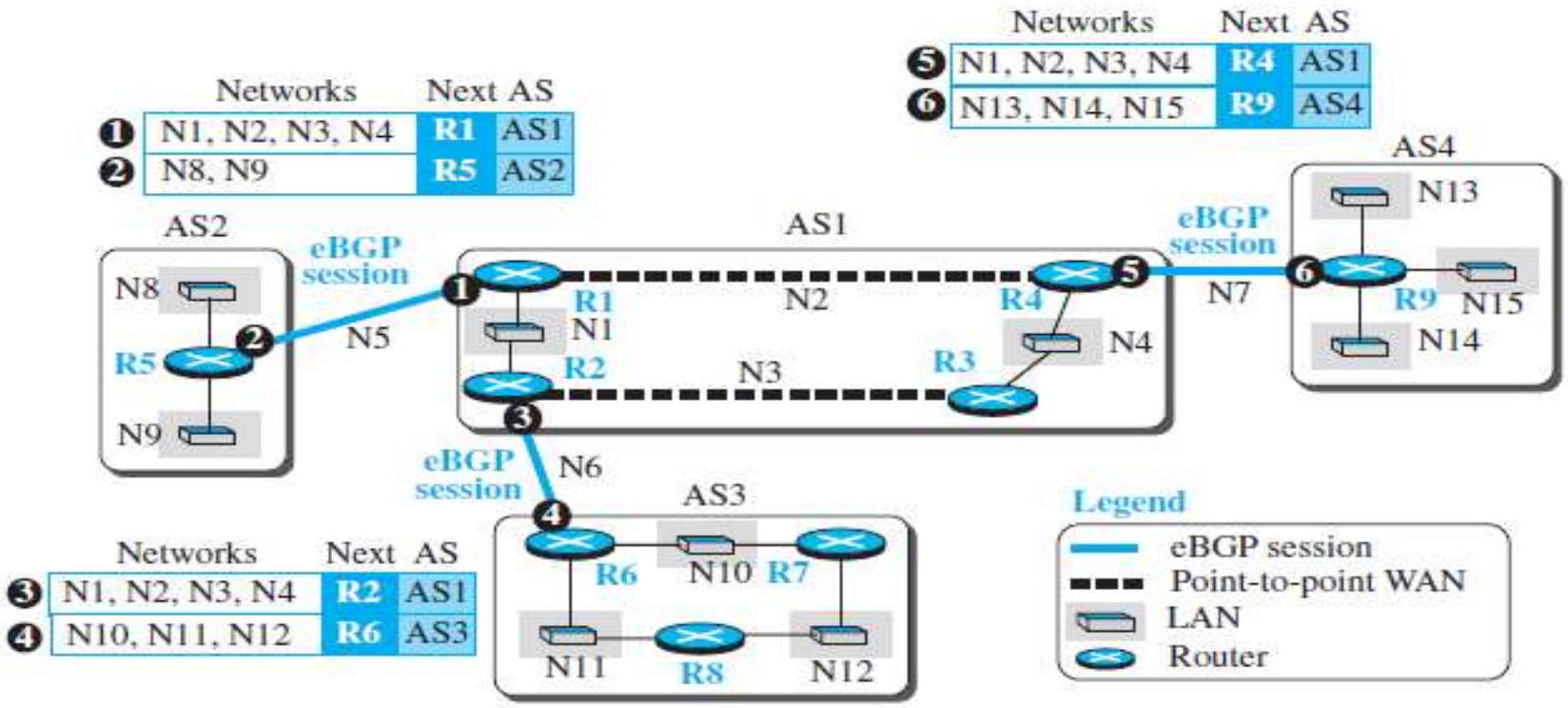
- *Additional hierarchy*
- Hierarchy is one of the fundamental tools used to make systems more scalable.
- OSPF introduces another layer of hierarchy into routing by allowing a domain to be partitioned into *areas*.
- *Load balancing*
- OSPF allows multiple routes to the same place to be assigned the same cost and will cause traffic to be distributed evenly over those routes, thus making better use of available network capacity.

Unit-4 Border Gateway Protocol Version 4 (BGP4)

- The **Border Gateway Protocol version 4 (BGP4)** is the only interdomain routing protocol used in the Internet today.
- BGP4 is based on the path-vector algorithm.

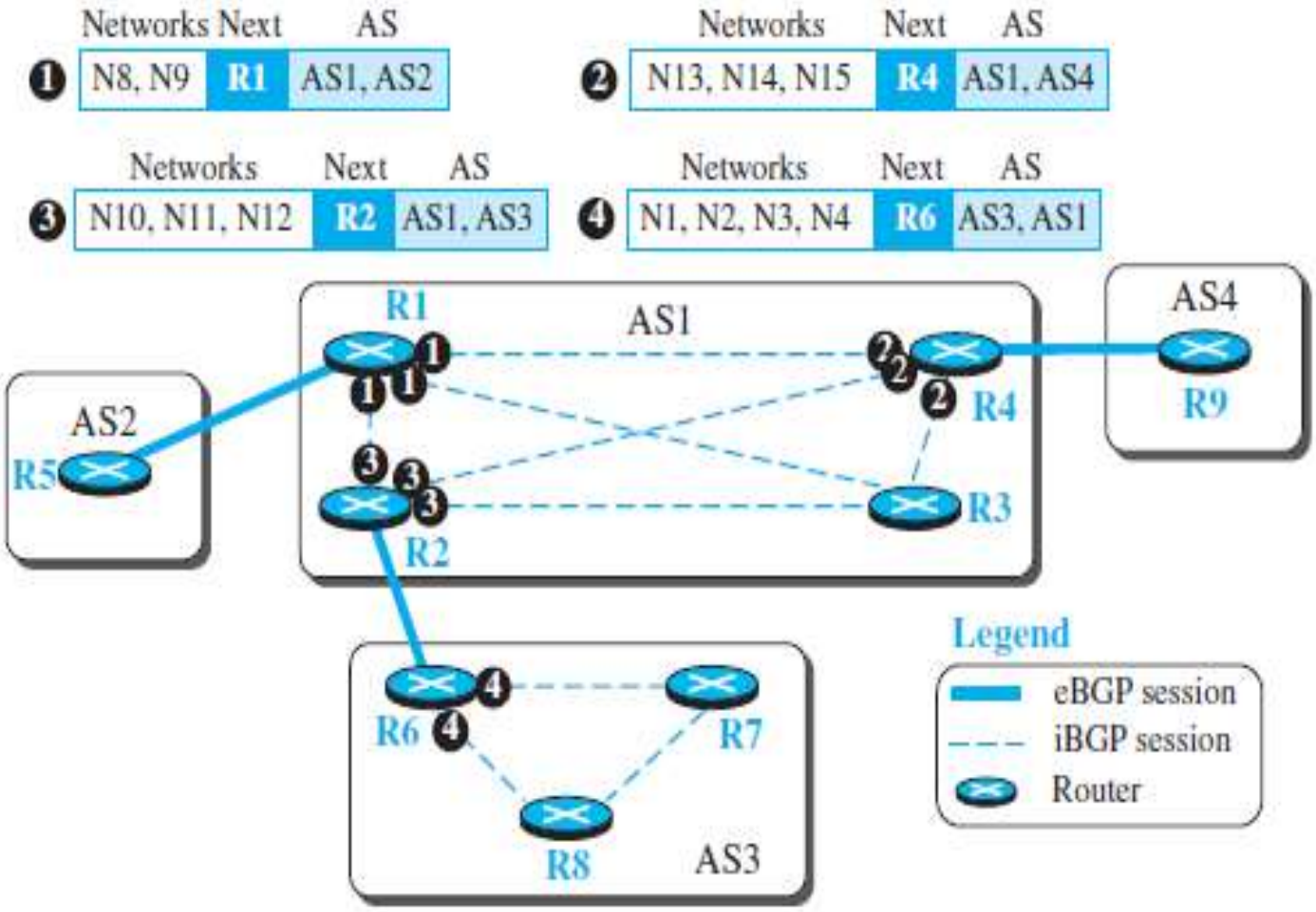
Operation of External BGP (eBGP):

Figure 20.25 *eBGP operation*



- ***Operation of Internal BGP (iBGP)***

Figure 20.26 *Combination of eBGP and iBGP sessions in our internet*



Unit-5 DLC Services

- The **data link control (DLC)** deals with procedures for communication between two adjacent nodes—node-to-node communication.
- Data link control functions include *framing,flow and error control*.

Framing

- Data transmission in the physical layer means moving bits in the form of a signal from the source to the destination.
- *Framing* in the data-link layer separates a message from one source to a destination by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

- This flag can create the same type of problem we saw in the character-oriented protocols.
- That is, if the flag pattern appears in the data, we need to somehow inform the receiver that this is not the end of the frame. We do this by stuffing 1 single bit (instead of 1 byte) to prevent the pattern from looking like a flag. The strategy is called **bit stuffing.**
- In bit stuffing, if a 0 and five consecutive 1 bits are encountered, an extra 0 is added. This extra stuffed bit is eventually removed from the data by the receiver.

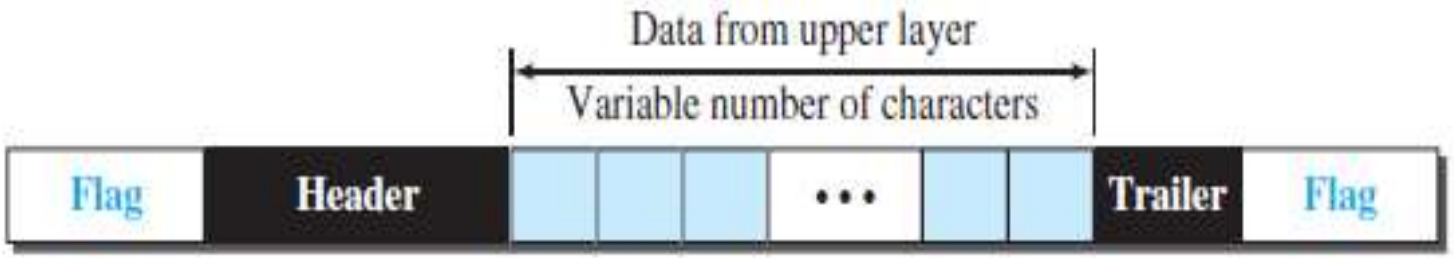
Frame Size

- Frames can be of fixed or variable size. In *fixed-size framing,* there is no need for defining the boundaries of the frames; the size itself can be used as a delimiter.
- An example of this type of framing is the ATM WAN, which uses frames of fixed size called *cells*
- *variable-size framing,* prevalent in local-area networks.
- In variable-size framing, we need a way to define the end of one frame and the beginning of the next. Two approaches were used for this purpose:
- Character-oriented approach
- Bit-oriented approach.

Character-Oriented Framing

- In *character-oriented (or byte-oriented) framing,* data to be carried are 8-bit characters from a coding system such as ASCII
- The header, which normally carries the source and destination addresses and other control information, and the trailer, which carries error detection redundant bits, are also multiples of 8 bits.
- To separate one frame from the next, an 8-bit (1-byte) **flag** is added at the beginning and the end of a frame.

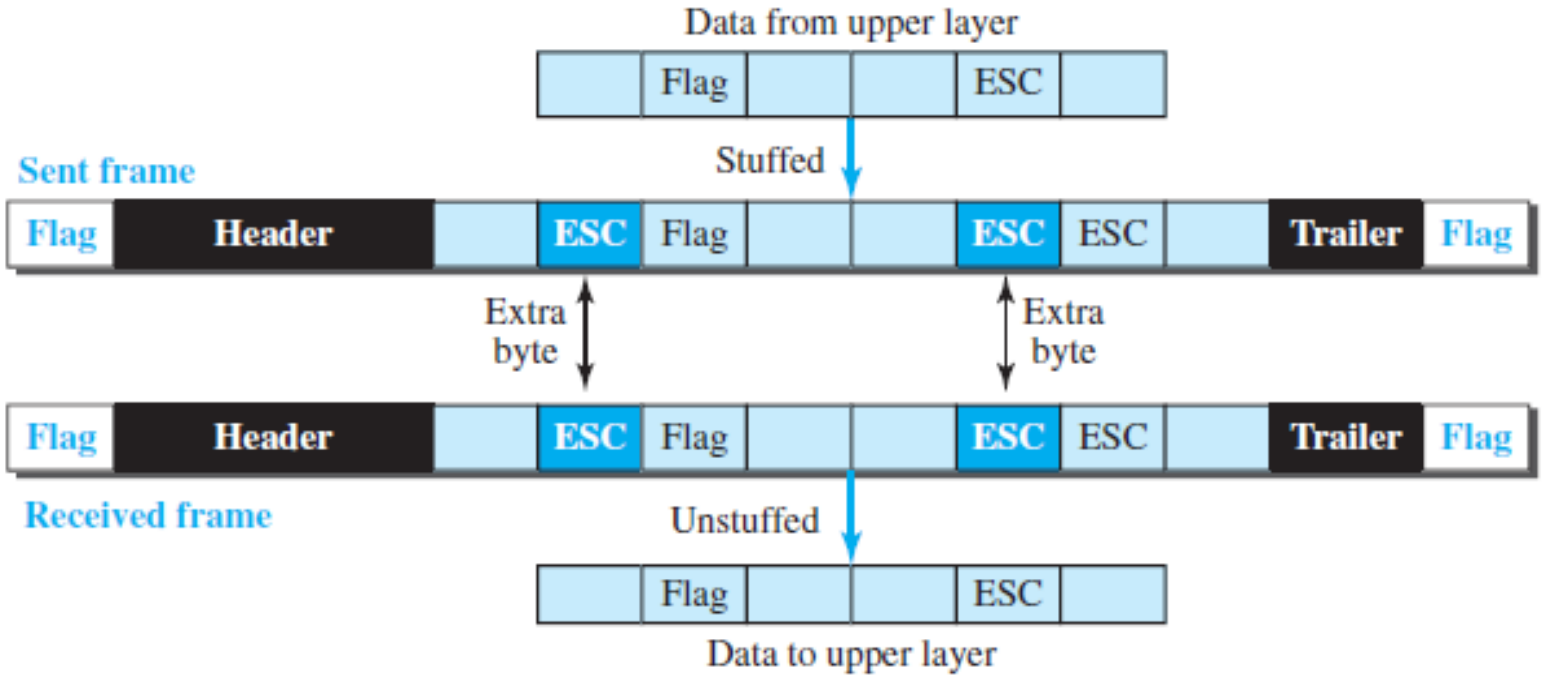
Figure 11.1 *A frame in a character-oriented protocol*



- Character-oriented framing was popular when only text was exchanged by the data-link layers.
- The flag is chosen as a character that is not used for text encoding.
- if the protocol is used for transmitting multimedia messages, there are chances that the pattern of the flag byte is present in the message byte sequence.
- If this happens, the receiver, when it encounters this pattern in the middle of the data, thinks it has reached the end of the frame.
- To fix this problem, a **byte-stuffing** strategy was added to character-oriented framing. In **byte stuffing** (or character stuffing), a special byte is added to the data section of the frame when there is a character with the same pattern as the flag. The data section is stuffed with an extra byte. This byte is usually called the *escape character (ESC)* and has a predefined bit pattern.

- Whenever the receiver encounters the ESC character, it removes it from the data section and treats the next character as data, not as a delimiting flag.

Figure 11.2 *Byte stuffing and unstuffing*



Byte stuffing is the process of adding one extra byte whenever there is a flag or escape character in the text.

Bit-Oriented Framing

- In *bit-oriented framing,* the data section of a frame is a sequence of bits to be interpreted by the upper layer as text, graphic, audio, video, and so on.
- However, in addition to headers (and possible trailers), we still need a delimiter to separate one frame from the other.
- Most protocols use a special 8-bit pattern flag, 01111110, as the delimiter to define the beginning and the end of the frame, as shown in Figure 11.3.

Figure 11.3 *A frame in a bit-oriented protocol*

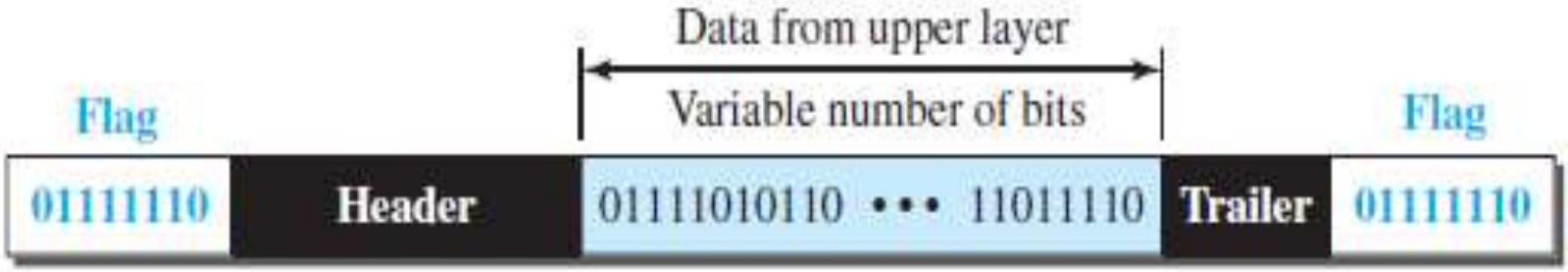
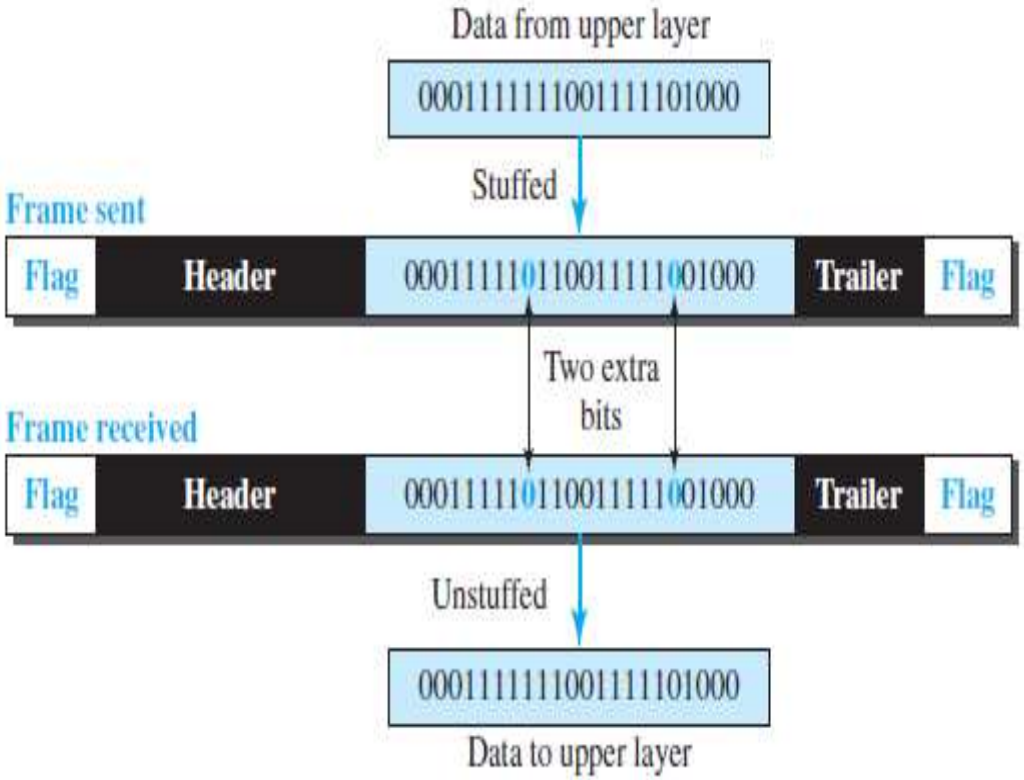


Figure 11.4 *Bit stuffing and unstuffing*



Flow Control

- Flow control coordinates the amount of data that can be sent before receiving acknowledgement
- It is one of the most important functions of data link layer.
- Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- Receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.
- Since the rate of processing is often slower than the rate of transmission, receiver has a block of memory (buffer) for storing incoming data until they are processed.

Error Control

- Error control includes both error detection and error correction.
- It allows the receiver to inform the sender if a frame is lost or damaged during transmission and coordinates the retransmission of those frames by the sender.
- Error control in the data link layer is based on automatic repeat request (ARQ). Whenever an error is detected, specified frames are retransmitted.

Error and Flow Control Mechanisms

- Stop-and-Wait
- Go-Back-N ARQ
- Selective-Repeat ARQ

Connectionless and Connection-Oriented

- A DLC protocol can be either connectionless or connection-oriented.

Connectionless Protocol

- In a connectionless protocol, frames are sent from one node to the next without any relationship between the frames; each frame is independent.
- The frames are not numbered and there is no sense of ordering.
- Most of the data-link protocols for LANs are connectionless protocols.

Connection-Oriented Protocol

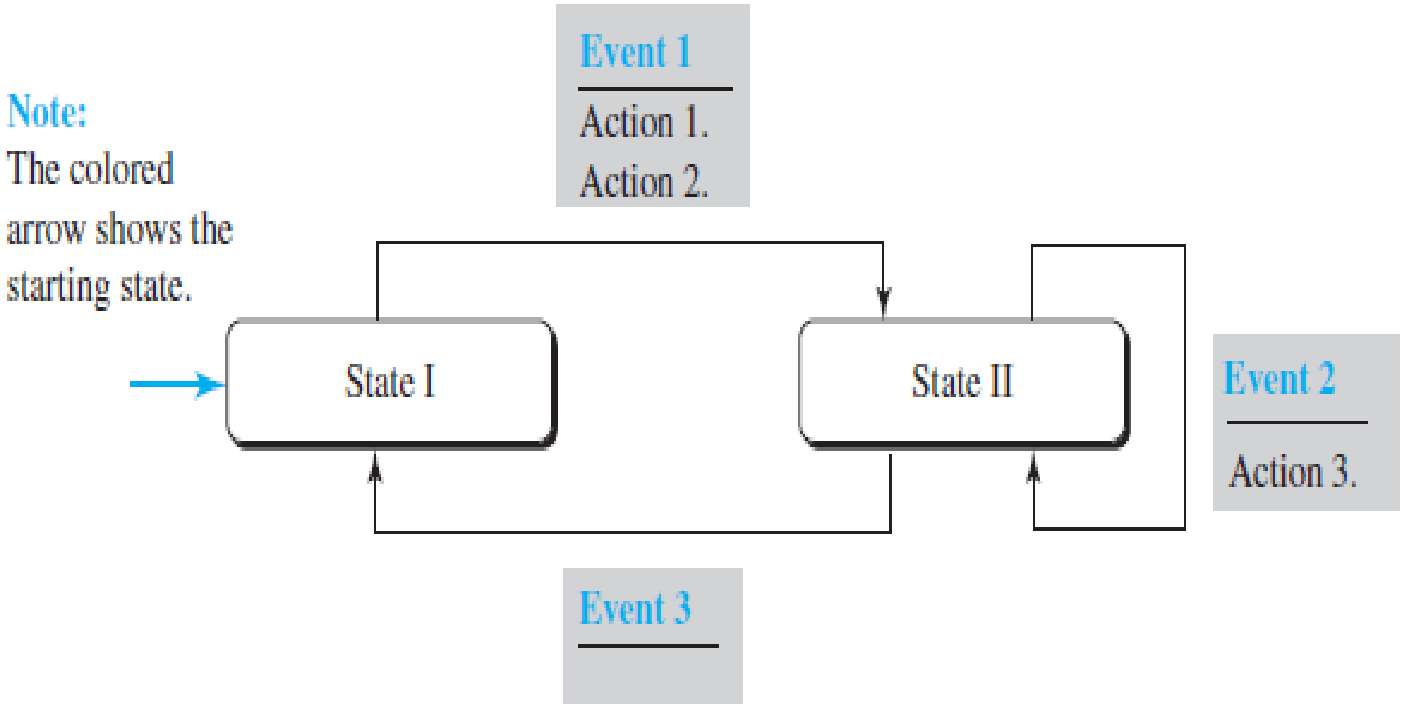
- In a connection-oriented protocol, a logical connection should first be established between the two nodes (setup phase). After all frames that are somehow related to each other are transmitted (transfer phase),
- the logical connection is terminated (teardown phase). In this type of communication, the frames are numbered and sent in order.

Data Link Layer Protocol

- Four protocols have been defined for the data-link layer to deal with flow and error control:
- **Simple,**
- **Stop-and-Wait**
- **Go-Back-N,**
- **Selective-Repeat.**
- Although the first two protocols still are used at the data-link layer, the last two have disappeared.
- The behavior of a data-link-layer protocol can be better shown as a **Finite state machine (FSM).** An FSM is thought of as a machine with a finite number of states.
- The machine is always in one of the states until an *event* occurs.

- Each event is associated with two reactions: defining the list (possibly empty) of actions to be performed and determining the next state (which can be the same as the current state).
- In Figure 11.6, we show an example of a machine using FSM. We have used rounded-corner rectangles to show states, colored text to show events, and regular black text to show actions.
- A horizontal line is used to separate the event from the actions, although later we replace the horizontal line with a slash. The arrow shows the movement to the next state.

Figure 11.6 *Connectionless and connection-oriented service represented as FSMs*



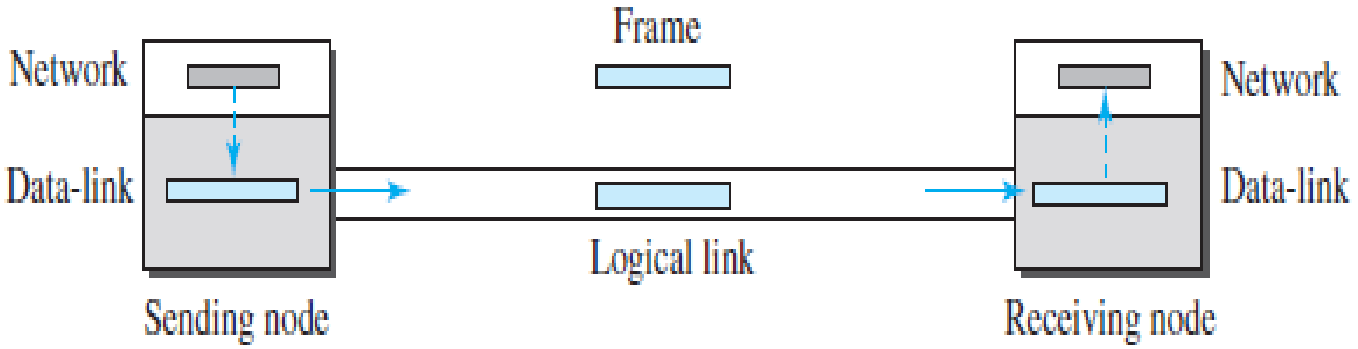
- The figure shows a machine with three states. There are only three possible events and three possible actions. The machine starts in state I.
- If event 1 occurs, the machine performs actions 1 and 2 and moves to state II.
- When the machine is in state II, two events may occur. If event 1 occurs, the machine performs action 3 and remains in the same state, state II.
- If event 3 occurs, the machine performs no action, but move to state I.

Simple Protocol

- Our first protocol is a **simple protocol** with neither flow nor error control. We assume that the receiver can immediately handle any frame it receives. In other words, the receiver can never be overwhelmed with incoming frames. Figure 11.7 shows the layout for this protocol.

- The data-link layer at the sender gets a packet from its network layer, makes a frame out of it, and sends the frame. The data-link layer at the receiver receives a frame from the link, extracts the packet from the frame, and delivers the packet to its network layer. The data-link layers of the sender and receiver provide transmission services for their network layers.

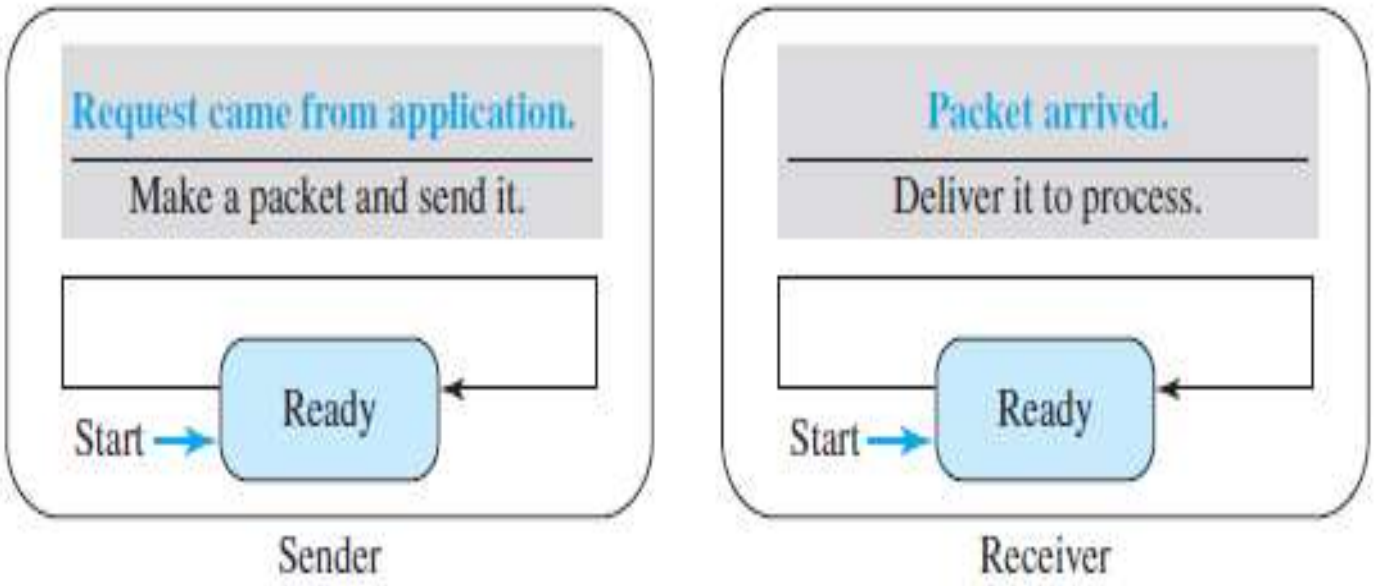
Figure 11.7 *Simple protocol*



FSMs

- The sender site should not send a packet until its application layer has a message to send.
- The receiver site cannot deliver a message to its application layer until a packet arrives.
- Each FSM has only one state, the *ready state*. The sending machine remains in the ready state until a request comes from the process in the application layer.
- When this event occurs, the sending machine encapsulates the message in a packet and sends it to the receiving machine.
- The receiving machine remains in the ready state until a packet arrives from the sending machine. When this event occurs, the receiving machine decapsulates the message out of the packet and delivers it to the process at the application layer

Figure 23.18 *FSMs for the simple protocol*



Stop-and-Wait Protocol

- Our second protocol is a connection-oriented protocol called the **Stop-and-Wait protocol,** which uses both flow and error control.
- Both the sender and the receiver use a sliding window of size 1.
- The sender sends one packet at a time and waits for an acknowledgment before sending the next one
- To detect corrupted packets, we need to add a checksum to each data packet. When a packet arrives at the receiver site, it is checked. If its checksum is incorrect, the packet is corrupted and silently discarded.
- The silence of the receiver is a signal for the sender that a packet was either corrupted or lost.

- Every time the sender sends a packet, it starts a timer. If an acknowledgment arrives before the timer expires, the timer is stopped and the sender sends the next packet (if it has one to send).
- If the timer expires, the sender resends the previous packet, assuming that the packet was either lost or corrupted.
- This means that the sender needs to keep a copy of the packet until its acknowledgment arrives.

Figure 23.20 *Stop-and-Wait protocol*

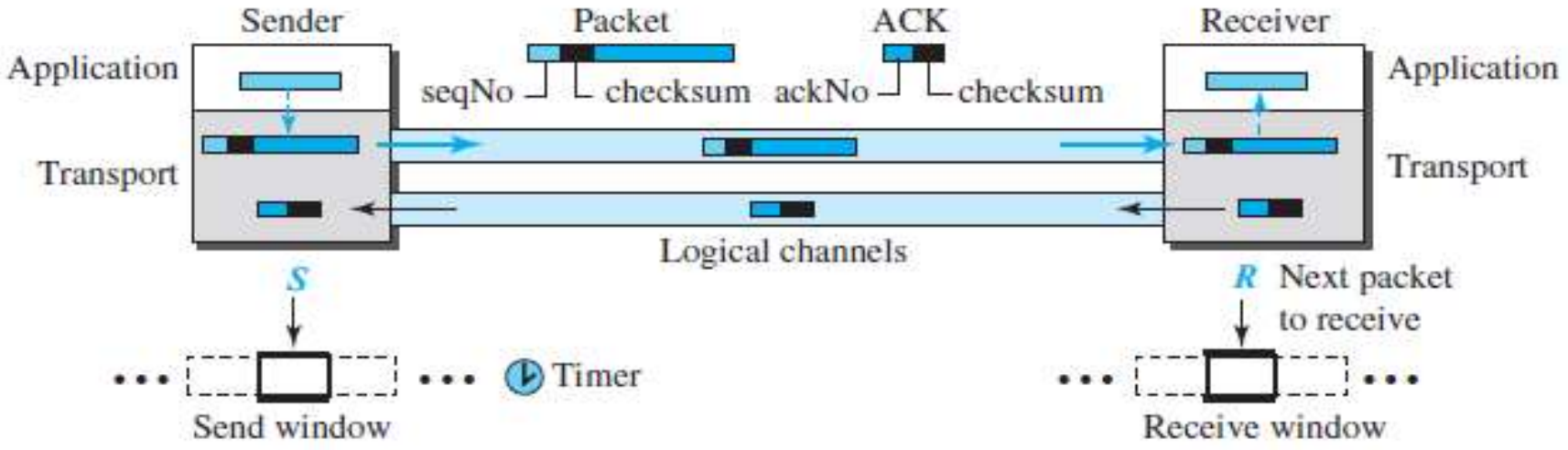
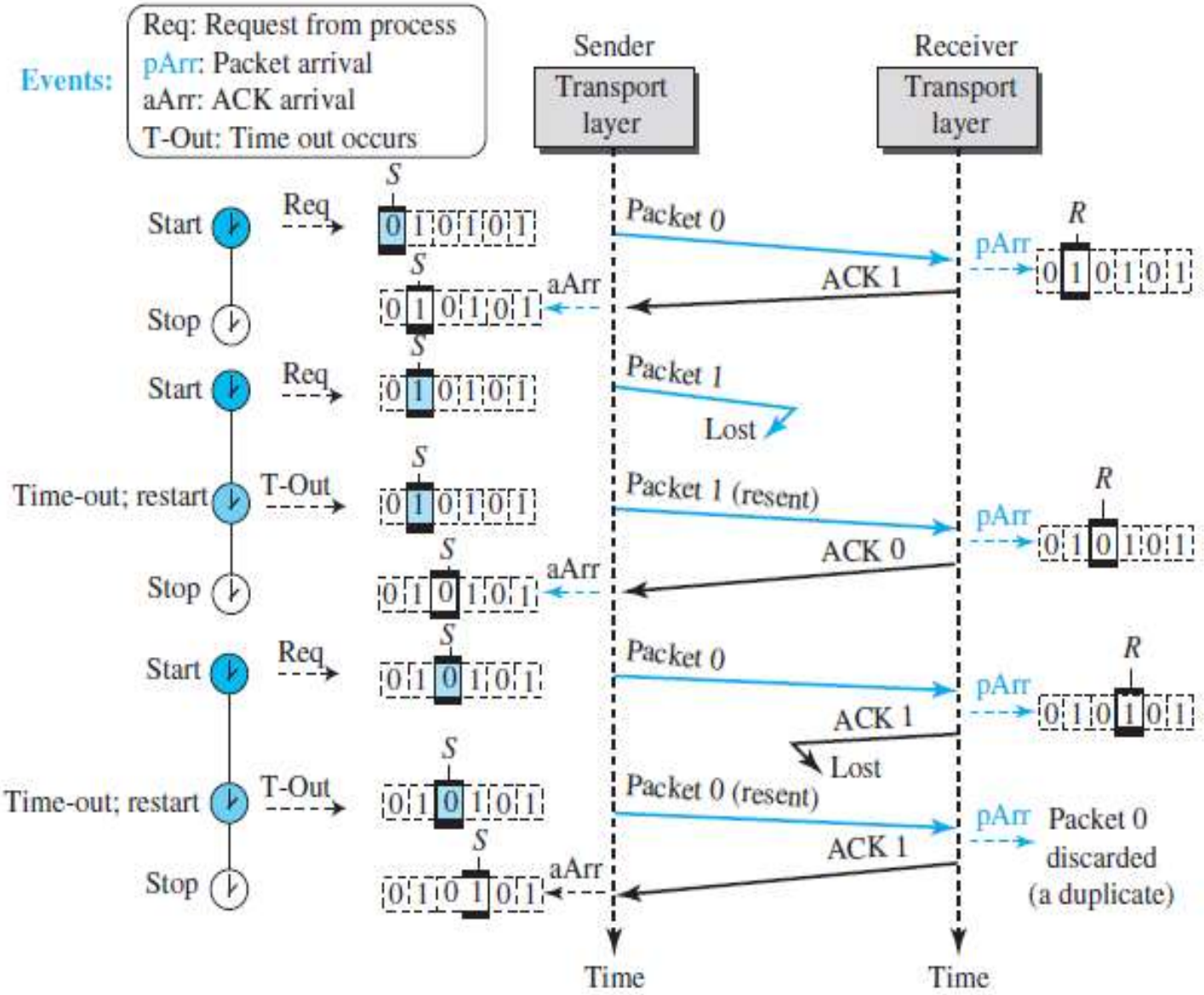
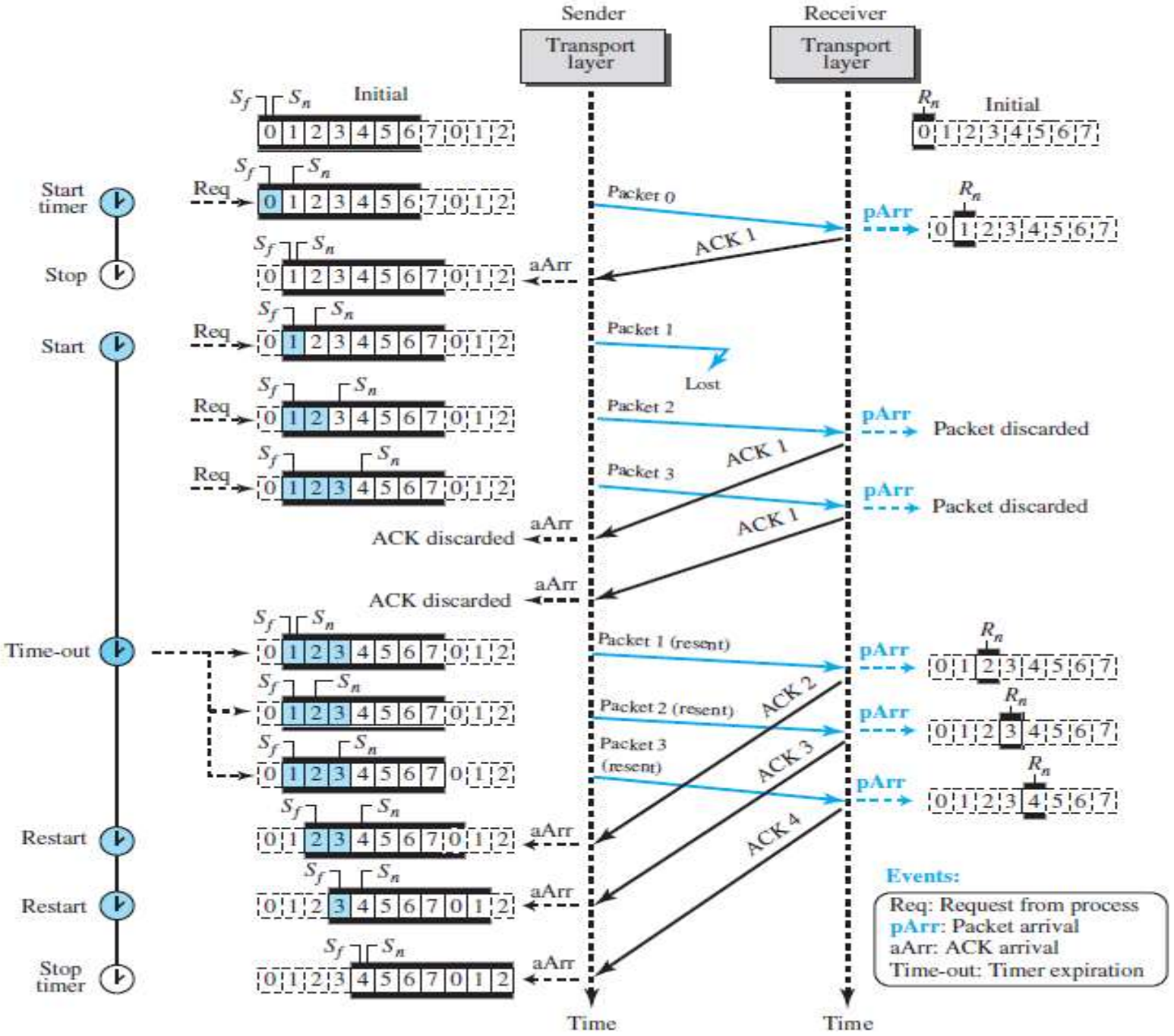


Figure 23.22 *Flow diagram for Example 23.4*




Sender
Receiver
Transport layer
Transport layer
Initial
Start timer
Stop
Start
Time-out
Restart
Restart
Stop timer
Req
Packet 0
ACK 1
aArr
pArr
Packet 1
Lost
Packet 2
Packet discarded
Packet 3
ACK discarded
Packet 1 (resent)
ACK 2
Packet 2 (resent)
ACK 3
Packet 3 (resent)
ACK 4
Events:
Req: Request from process
pArr: Packet arrival
aArr: ACK arrival
Time-out: Timer expiration
Time
Time

Piggybacking

- In which data is flowing only in one direction although the acknowledgment may travel in the other direction.
- Protocols have been designed in the past to allow data to flow in both directions.
- However, to make the communication more efficient, the data in one direction is piggybacked with the acknowledgment in the other direction.
- In other words, when node A is sending data to node B, Node A also acknowledges the data received from node B.

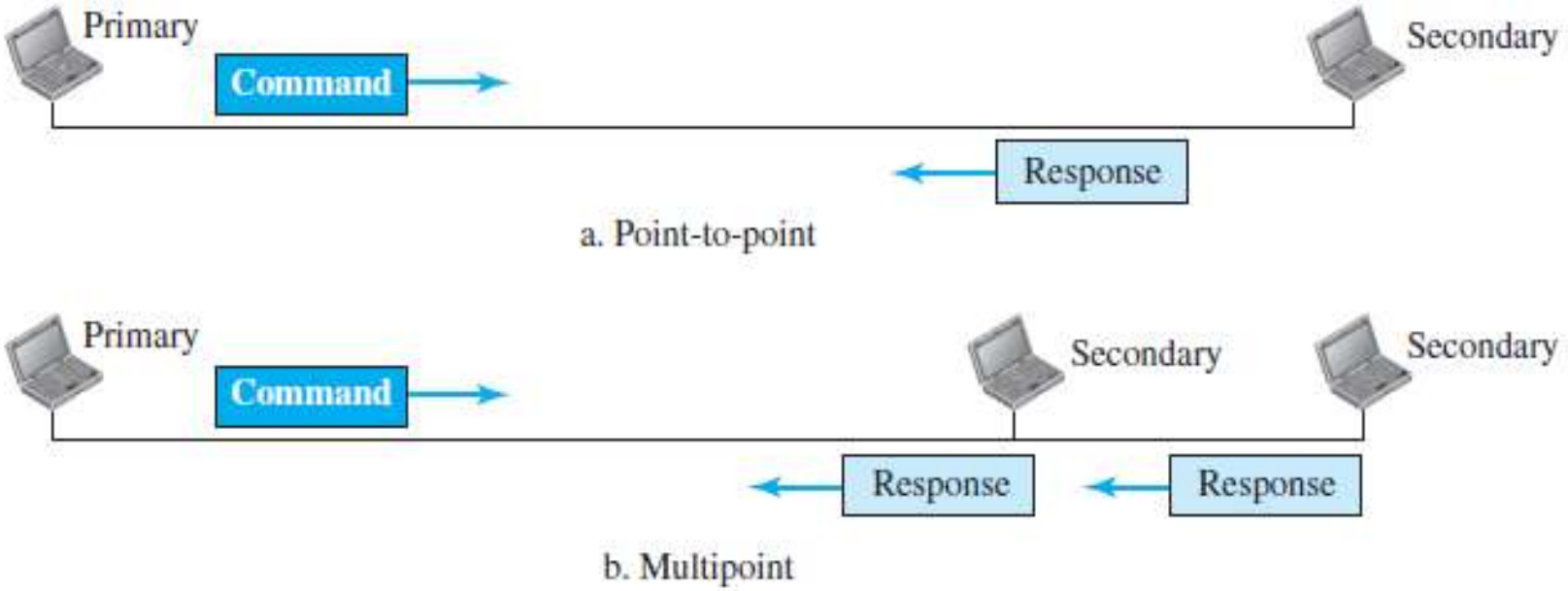
UNIT-5 HDLC

- **High-level Data Link Control (HDLC)** is a bit-oriented protocol for communication over point-to-point and multipoint links.
- It implements the Stop-and-Wait protocol.

Configurations and Transfer Modes

- HDLC provides two common transfer modes that can be used in different configurations:
- *normal response mode (NRM)*
- *asynchronous balanced mode (ABM)*.
- In *normal response mode (NRM)*, the station configuration is unbalanced. We have one primary station and multiple secondary stations.
- A *primary station* can send commands; a *secondary station* can only respond.
- The NRM is used for both point-to-point and multipoint links, as shown in Figure 11.14.

Figure 11.14 *Normal response mode*



- In ABM, the configuration is balanced. The link is point-to-point, and each station can function as a primary and a secondary (acting as peers), as shown in Figure 11.15.
- This is the common mode today.
- FCS-Frame Check Sequence

Figure 11.15 *Asynchronous balanced mode*



Framing

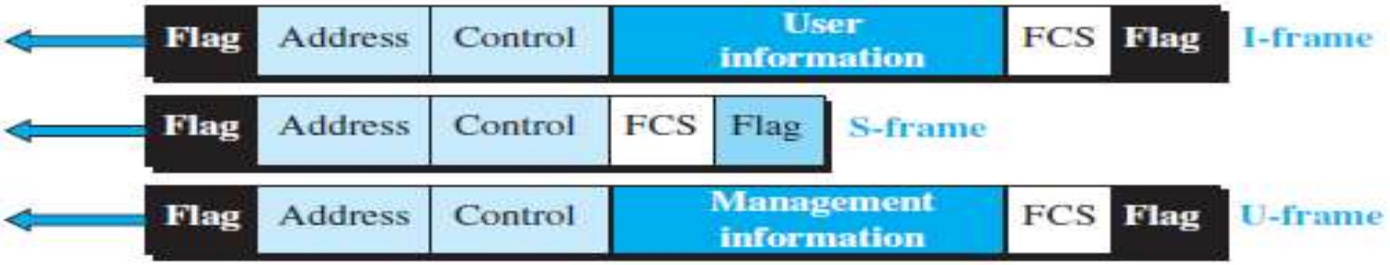
HDLC defines three types of frames:

- *information frames (I-frames),*
- *supervisory frames (S-frames)*
- *unnumbered frames (U-frames)*

Each type of frame serves as an envelope for the transmission of a different type of message

- I-frames are used to data-link user data and control information relating to user data (piggybacking).
- S-frames are used only to transport control information.
- U-frames are reserved for system management.

Figure 11.16 *HDLC frames*



Point-Point Protocol:

- One of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP).
- PPP uses only one type of frame, but allows multiplexing of different payloads to achieve a kind of connection-oriented service authentication.
- Encapsulating different packets in a frame allows PPP to move to different states to provide necessary services.

Services Provided by PPP

- PPP defines the format of the frame to be exchanged between devices.
- It also defines how two devices can negotiate the establishment of the link and the exchange of data.
- PPP is designed to accept payloads from several network layers (not only IP).

- Authentication is also provided in the protocol, but it is optional.
- The new version of PPP, called *Multilink PPP,* provides connections over multiple links. One interesting feature of PPP is that it provides network address configuration.

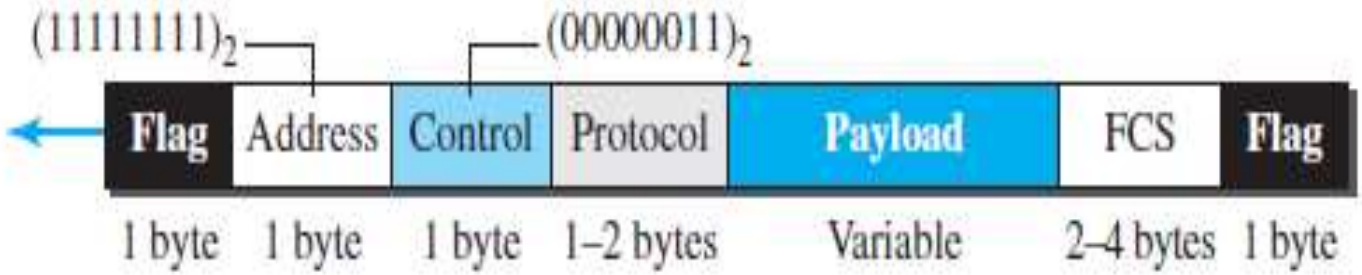
Services Not Provided by PPP

- PPP does not provide flow control. A sender can send several frames one after another with no concern about overwhelming the receiver. PPP has a very simple mechanism for error control.
- A CRC field is used to detect errors. If the frame is corrupted, it is silently discarded; the upper-layer protocol needs to take care of the problem.

Framing

- PPP uses a character-oriented (or byte-oriented) frame.
- Figure 11.20 shows the format of a PPP frame. The description of each field follows:

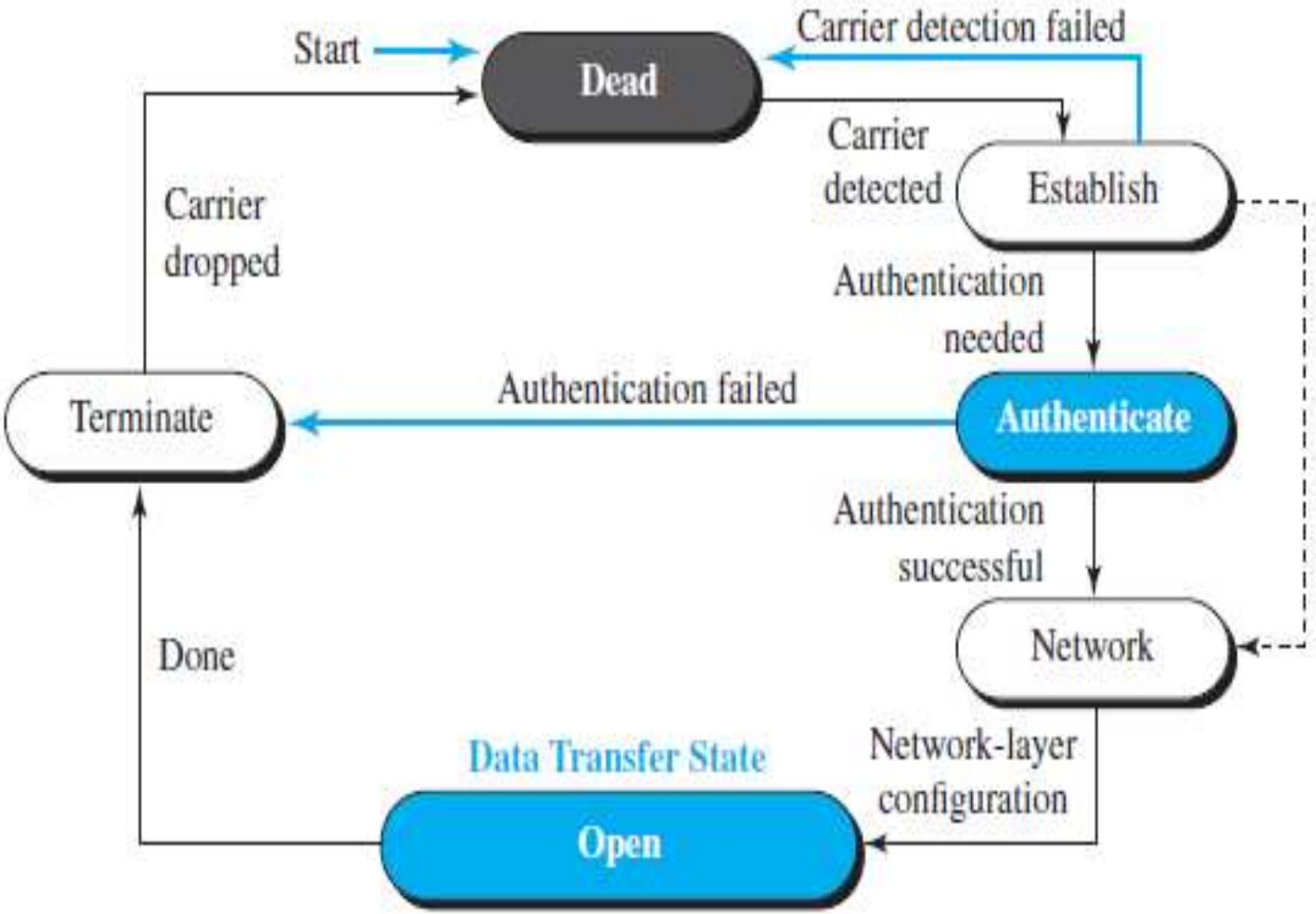
Figure 11.20 *PPP frame format*



Transition Phases

- A PPP connection goes through phases which can be shown in a *transition phase* diagram (see Figure 11.21).
- The transition diagram, which is an FSM, starts with the *dead* state. In this state, there is no active carrier (at the physical layer) and the line is quiet.
- When one of the two nodes starts the communication, the connection goes into the *establish* state

Figure 11.21 *Transition phases*



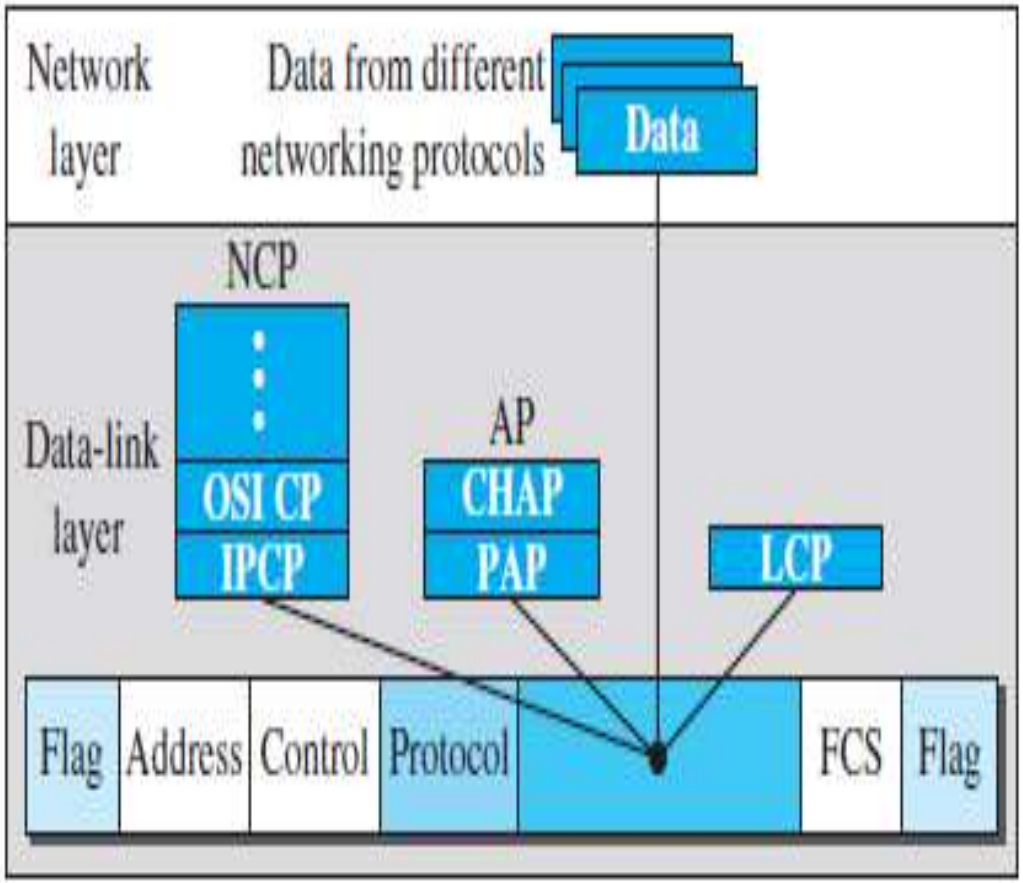
Multiplexing

- PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate the parties involved, and carry the network-layer data.
- Three sets of protocols are defined to make PPP powerful: the Link Control Protocol (LCP), two Authentication Protocols (APs), and several Network Control Protocols (NCPs).
- At any moment, a PPP packet can carry data from one of these protocols in its data field, as shown in Figure 11.22. Note that there are one LCP, two APs, and several NCPs.
- Data may also come from several different network layers

Figure 11.22 *Multiplexing in PPP*

Legend

LCP: Link control protocol
AP: Authentication protocol
NCP: Network control protocol



Protocol values:
LCP: 0xC021
AP: 0xC023 and 0xC223
NCP: 0x8021 and ....
Data: 0x0021 and ....

Unit –5 802.11 Wireless LAN

- A client is always associated with one AP and when the client moves closer to another AP, it associates with the new AP (Hand-Off)
- Three flavors:
  - 802.11b
  - 802.11a
  - 802.11g
- The most popular 802.11b standard currently in deployment.
- Supports 1, 2, 5.5 and 11 Mbps data rates in the 2.4 GHz ISM (Industrial-Scientific-Medical) band

- **802.11a**
- **Operates in the 5 GHz UNII (Unlicensed National Information Infrastructure) band**
- **Incompatible with devices operating in 2.4GHz**
- **Supports Data rates up to 54 Mbps.**

- **802.11g**
- **Supports data rates as high as 54 Mbps on the 2.4 GHz band**

- A family of wireless LAN (WLAN) specifications developed by a working group at the Institute of Electrical and Electronic Engineers (IEEE)
- Defines standard for WLANs using the following four technologies
  - Frequency Hopping Spread Spectrum (FHSS)
  - Direct Sequence Spread Spectrum (DSSS)
  - Infrared (IR)
  - Orthogonal Frequency Division Multiplexing (OFDM)
- Versions: 802.11a, 802.11b, 802.11g

802.11a Vs. 802.11b

802.11a vs. 802.11b	802.11a	802.11b
Raw data rates	Up to 54 Mbps (54, 48, 36, 24,18, 12 and 6 Mbps)	Up to 11 Mbps (11, 5.5, 2, and 1 Mbps)
Range	50 Meters	100 Meters
Bandwidth	UNII and ISM (5 GHz range)	ISM (2.4000—2.4835 GHz range)
Modulation	OFDM technology	DSSS technology

- A **wireless ad hoc network (WANET)** is a decentralized type of wireless network.[1][2]
- The network is ad hoc because it does not rely on a pre existing infrastructure, such as routers in wired networks or access points in managed (infrastructure) wireless networks.
- Instead, each node participates in routing by forwarding data for other nodes, so the determination of which nodes forward data is made dynamically on the basis of network connectivity[2]. In addition to the classic routing, ad hoc networks can use flooding for forwarding data.

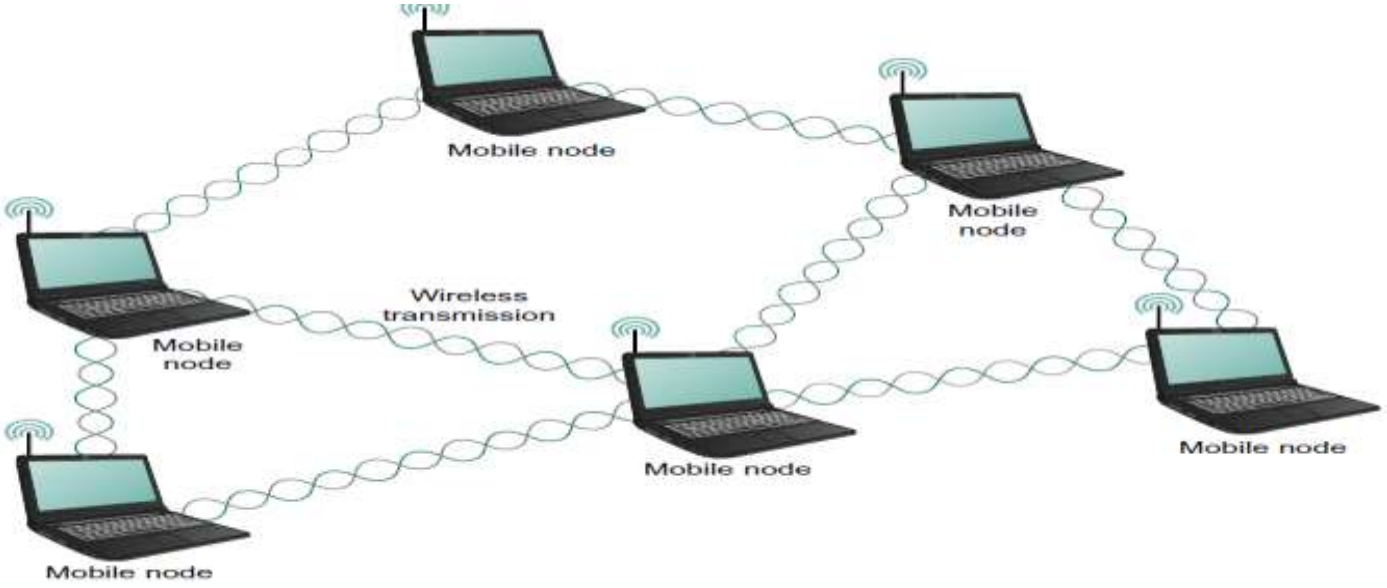


FIGURE 2.29 A wireless *ad hoc* or mesh network.

802.11/WiFi:

- Most readers will have used a wireless network based on the IEEE 802.11 standards, often referred to as *Wi-Fi*.
- Designed for use in a limited geographical area (homes, office buildings, campuses), and its primary challenge is to mediate access to a shared communication medium—in this case, signals propagating through space.

Physical Properties

- 802.11 defines a number of different physical layers that operate in various frequency bands and provide a range of different data rates.
- The original 802.11 standard defined two radio-based physical layers standards, one using frequency hopping and the other using direct sequence spread spectrum Both provided data rates in the 2 Mbps range.

802.11b provides up to 11 Mbps.

- Then came 802.11a, which delivers up to 54 Mbps using a variant of FDM called *orthogonal frequency division multiplexing (OFDM)*

Collision Avoidance:

Hidden node problem:

- Consider the situation depicted in Figure 2.30 where A and C are both within range of B but not each other.
- Suppose both A and C want to communicate with B and so they each send it a frame.
- A and C are unaware of each other since their signals do not carry that far. These two frames collide with each other at B, but unlike an Ethernet, neither A nor C is aware of this collision. A and C are said to be ***hidden nodes*** with respect to each other

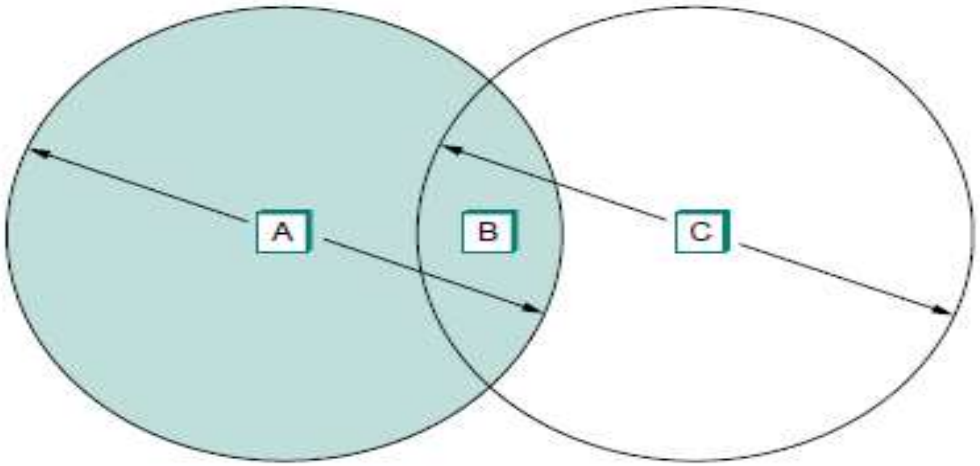
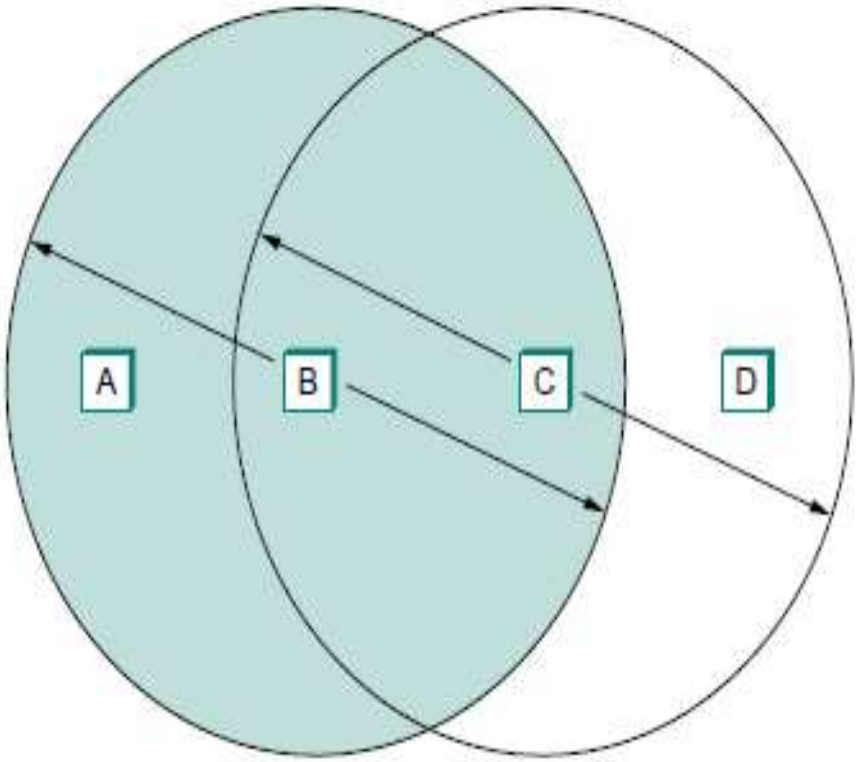


FIGURE 2.30 The hidden node problem. Although A and C are hidden from each other, their signals can collide at B. (B's reach is not shown.)

Exposed Node problem:

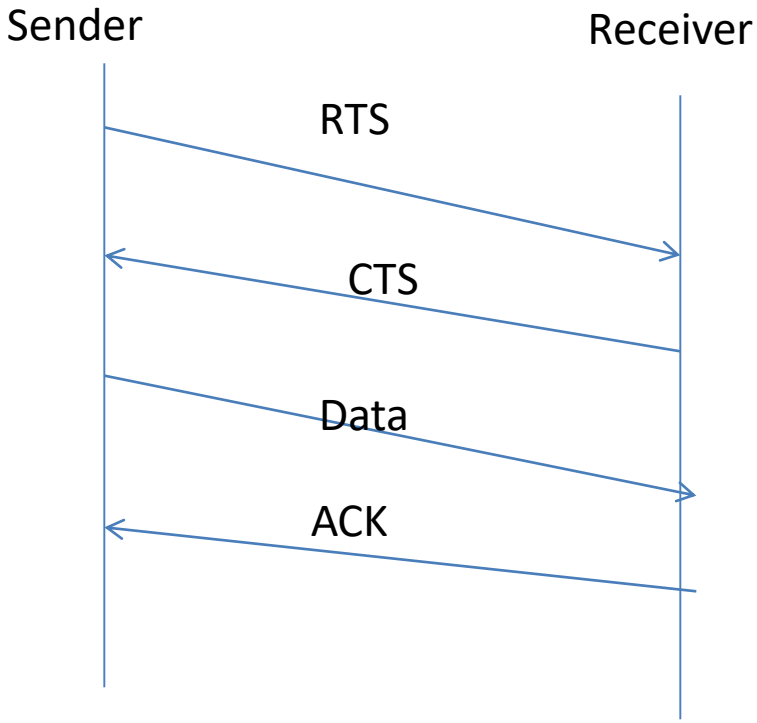
- A related problem, called the *exposed node problem*, occurs under the circumstances illustrated in Figure 2.31(P.No:138)
- where each of the four nodes is able to send and receive signals that reach just the nodes to its immediate left and right.



■ **FIGURE 2.31** The exposed node problem. Although B and C are exposed to each other's signals, there is no interference if B transmits to A while C transmits to D. (A and D's reaches are not shown.)

- For example, B can exchange frames with A and C but it cannot reach D, while C can reach B and D but not A.
- Suppose B is sending to A. Node C is aware of this communication because it hears B's transmission.
- It would be a mistake, however, for C to conclude that it cannot transmit to anyone just because it can hear B's transmission. For example, suppose C wants to transmit to node D.
- This is not a problem since C's transmission to D will not interfere with A's ability to receive from B.
- 802.11 addresses these problems by using CSMA/CA, where the CA stands for collision *avoidance*, in contrast to the collision *detection* of CSMA/CD used on Ethernets
- The Carrier Sense part seems simple enough: Before sending a packet, the transmitter checks if it can hear any other transmissions; if not, it sends

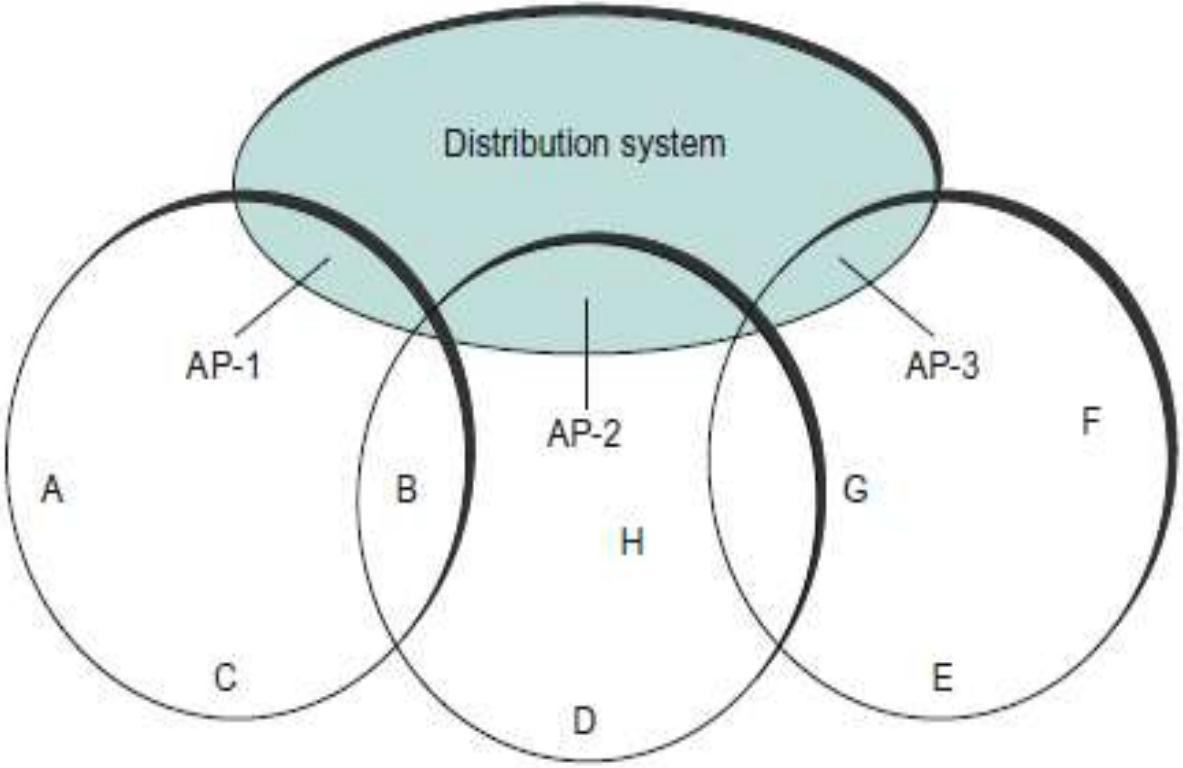
- 802.11 adds an mechanism called RTS-CTS (Ready to Send-Clear to Send).
- This goes some way toward addressing the hidden terminal problem.



- The sender sends an RTS— a short packet—to the intended receiver, and if that packet is received successfully the receiver responds with another short packet, the CTS.
- Two nodes might detect an idle link and try to transmit an RTS frame at the same time, causing their RTS frames to collide with each other.
- The senders realize the collision has happened when they do not receive the CTS frame after a period of time, in which case they each wait a random amount of time before trying again
- After a successful RTS-CTS exchange, the sender sends its data packet and, if all goes well, receives an ACK for that packet.
- In the absence of a timely ACK, the sender will try again to request usage of the channel again, using the same process described above.

Distribution System:

- Instead of all nodes being created equal, some nodes are allowed to roam (e.g., your laptop) and some are connected to a wired network infrastructure. 802.11 calls these base stations *access points* (APs), and they are connected to each other by a so-called *distribution system*.
- Figure 2.32 illustrates a distribution system that connects three access points, each of which services the nodes in some region.
- Each access point operates on some channel in the appropriate frequency range.
- The only important point is that the distribution network operates at the link layer, the same protocol layer as the wireless links. In other words, it does not depend on any higher-level protocols (such as the network layer).



■ **FIGURE 2.32** Access points connected to a distribution system.

- Two nodes can communicate directly with each other if they

are within reach of each other, the idea behind this configuration is that each node associates itself with one access point.

- For node A to communicate with node E, for example, A first sends a frame to its access point (AP-1), which forwards the frame across the distribution system to AP-3, which finally transmits the frame to E.
- What 802.11 does specify is how nodes select their access points and how this algorithm works in light of nodes moving from one cell to another.
- The technique for selecting an AP is called *scanning* and involves the following four steps:
- **1.** The node sends a Probe frame.
- **2.** All APs within reach reply with a Probe Response frame.
- **3**.The node selects one of the access points and sends that AP an Association Request frame.
- **4.** The AP replies with an Association Response frame

Frame Format:

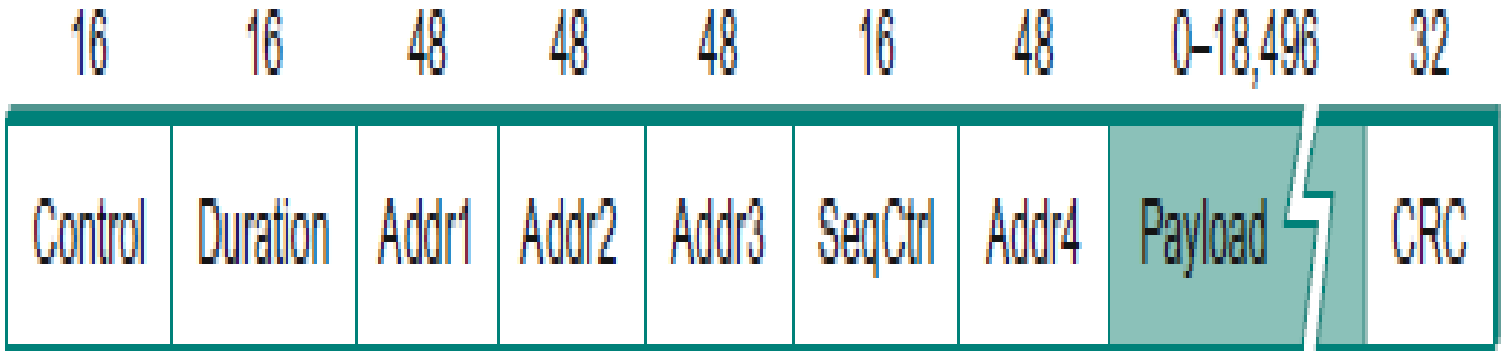
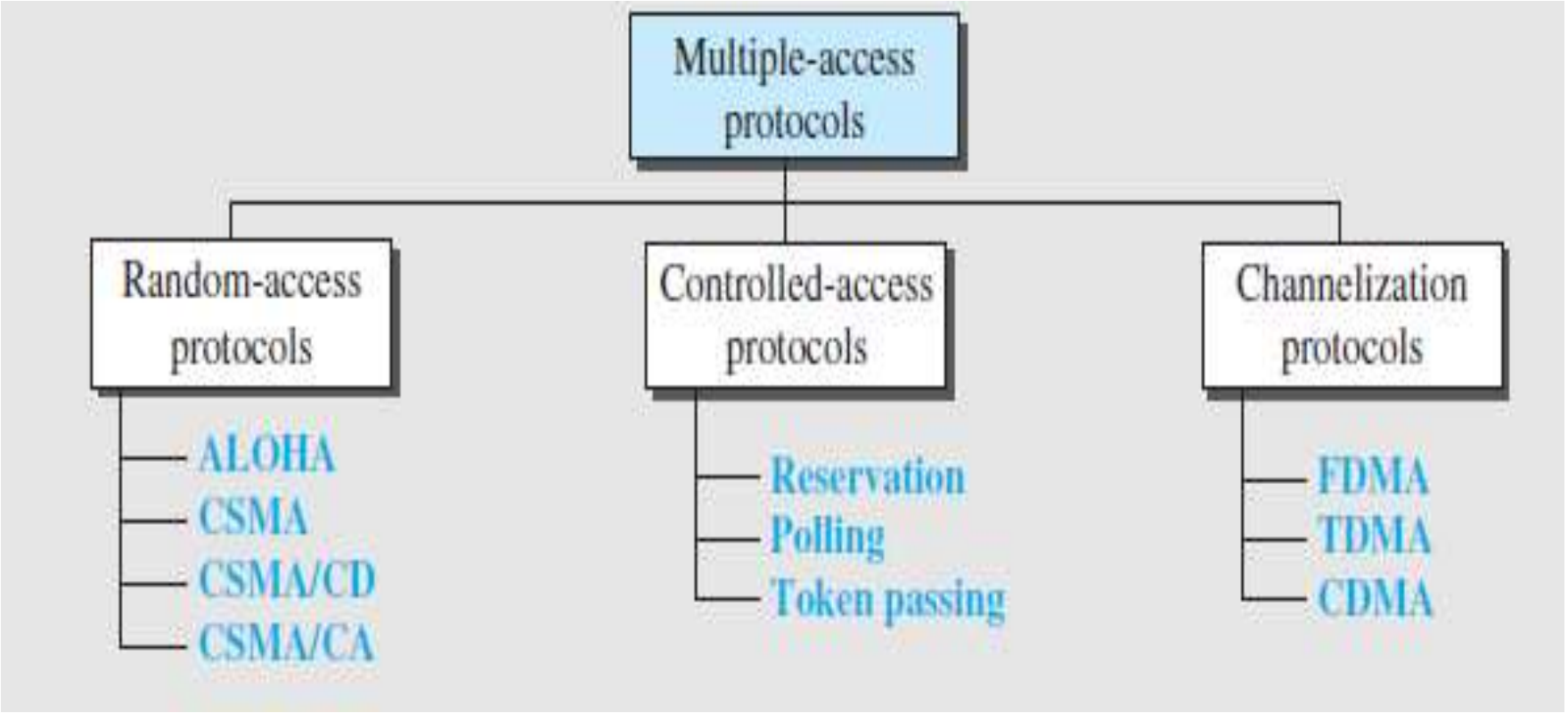


FIGURE 2.34 802.11 frame format.

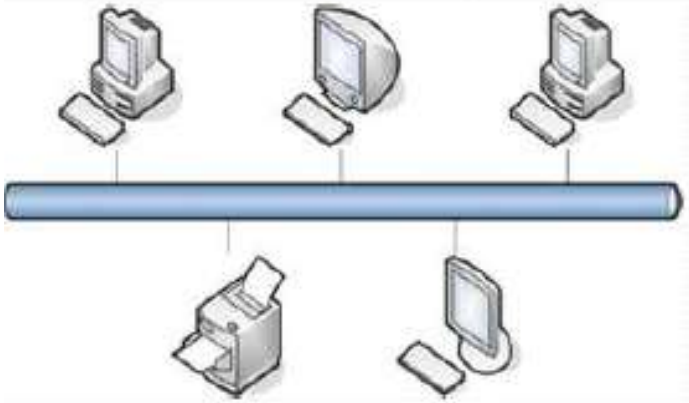
Unit-5 Media Access Control

Media Access Control

When nodes or stations are connected and use a common link, called a *multipoint* or *broadcast link,* we need a multiple-access protocol to coordinate access to the link.



Multiple access links





- **There is ‘collision if more than one node try to sends the data at same time**

1- RANDOM ACCESS

In random access or contention methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. This decision depends on the state of the medium(idle or busy)

Random Access Methods:

- **ALOHA**
- **Carrier Sense Multiple Access ; CSMA**
- **Carrier Sense Multiple Access with Collision Detection (CSMA/CD)**
- **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)**

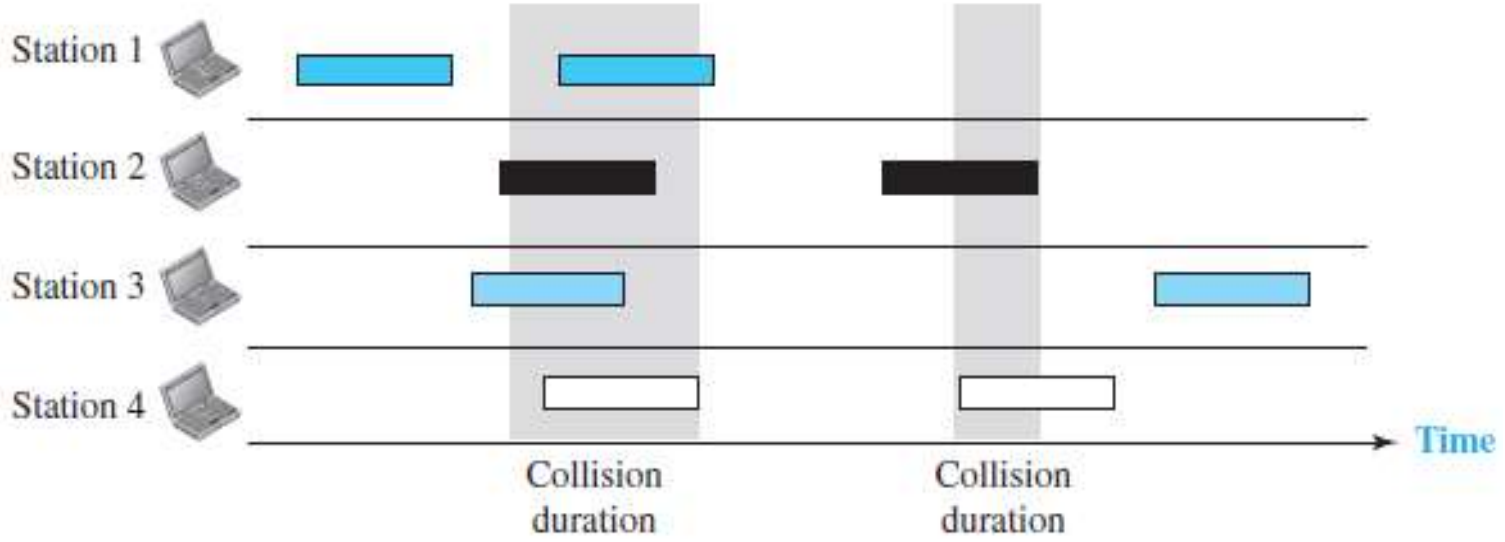
ALOHA

- **ALOHA,** the earliest random access method, was developed at the University of Hawaii in early 1970. It was designed for a radio (wireless) LAN, but it can be used on any shared medium

Pure ALOHA

- The original ALOHA protocol is called ***pure ALOHA.*** The idea is that each station sends a frame whenever it has a frame to send (multiple access). However, since there is only one channel to share, there is the possibility of collision between frames from different stations.

Figure 12.2 *Frames in a pure ALOHA network*



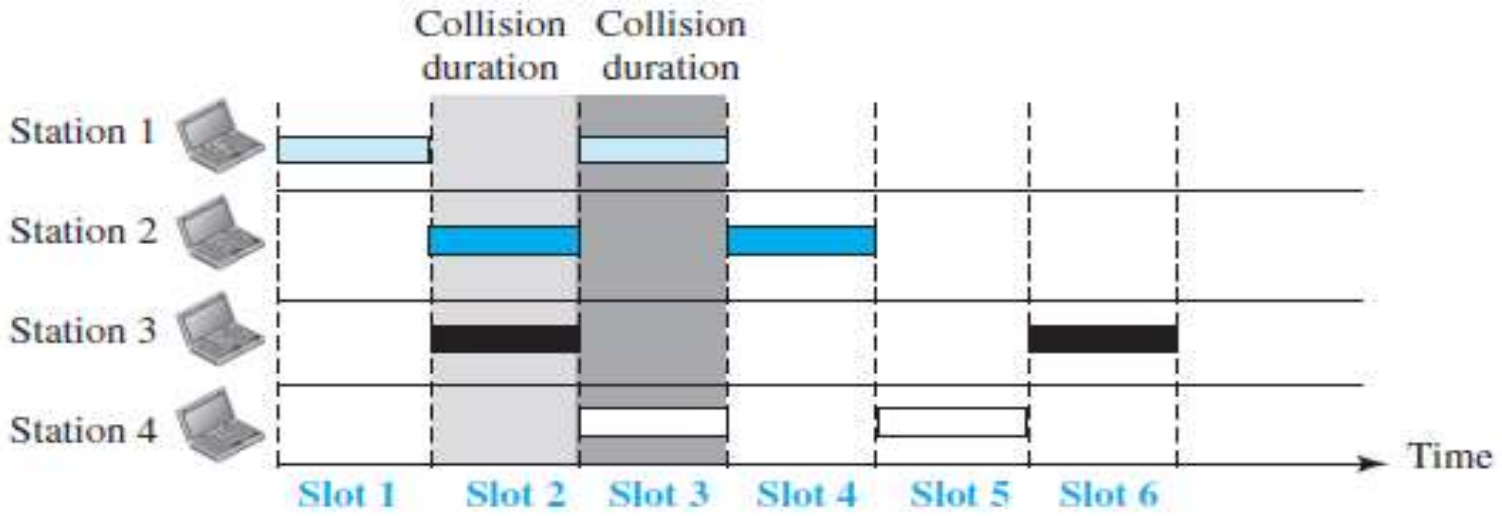
- The figure shows that each station sends two frames; there are a total of eight frames on the shared medium.
- Some of these frames collide because multiple frames are in contention for the shared channel.

- Figure 12.2 shows that only two frames survive: one frame from station 1 and one frame from station 3.
- The pure ALOHA protocol relies on acknowledgments from the receiver. When a station sends a frame, it expects the receiver to send an acknowledgment.
- If the acknowledgment does not arrive after a time-out period, the station resends the frame.

Slotted ALOHA

- In **slotted ALOHA** we divide the time into slots of seconds and force the station to send only at the beginning of the time slot.

Figure 12.5 *Frames in a slotted ALOHA network*



- Because a station is allowed to send only at the beginning of the synchronized time slot, if a station misses this moment, it must wait until the beginning of the next time slot.

- This means that the station which started at the beginning of this slot has already finished sending its frame.
- Of course, there is still the possibility of collision if two stations try to send at the beginning of the same time slot.

CSMA

Carrier sense multiple access (CSMA)

- requires that each station first listen to the medium (or check the state of the medium) before sending.
- In other words, CSMA is based on the principle "sense before transmit" or "listen before talk.
- CSMA can reduce the possibility of collision, but it cannot eliminate it.
- ***1-Persistent***
- ***Non persistent***
- **p-*Persistent***

1-persistent

- *method* is simple, after the station finds the line idle, it sends its frame immediately (with probability 1).
- This method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately.

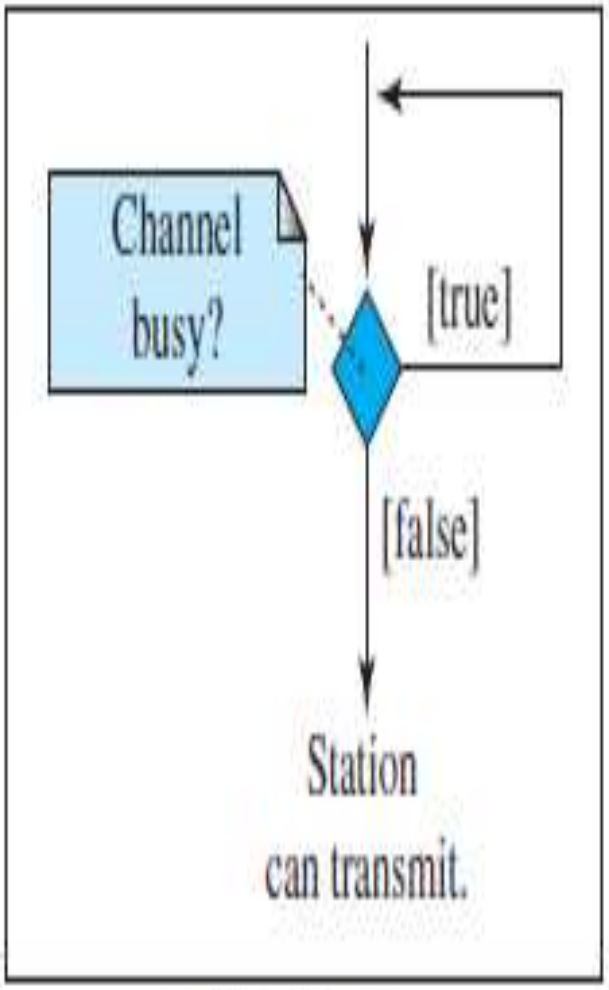
Nonpersistent

- In the *nonpersistent method,* a station that has a frame to send senses the line.
- If the line is idle, it sends immediately. If the line is not idle, it waits a **random amount of time** and then senses the line again.

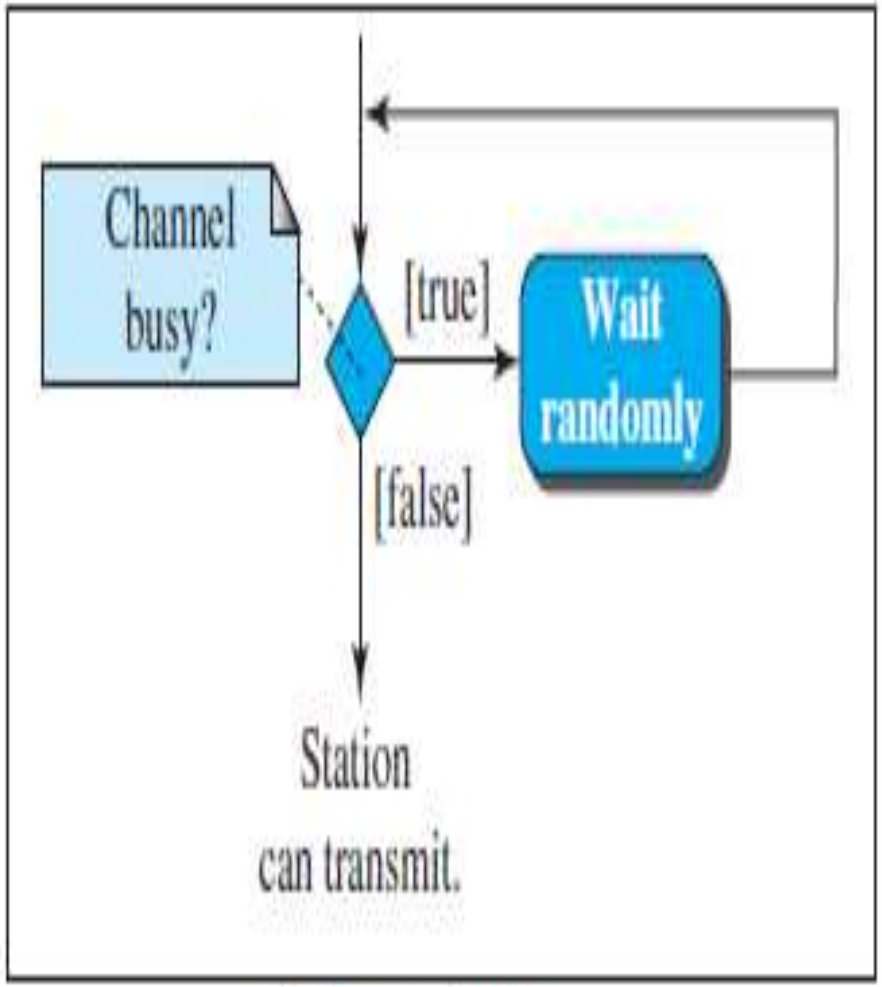
- The nonpersistent approach reduces the chance of collision because it is unlikely that two or more stations will wait the same amount of time and retry to send simultaneously.

p-*Persistent*

- The p-*persistent method* is used if the channel has time slots with a slot duration equal to or greater than the maximum propagation time.
- The *p*-persistent approach combines the advantages of the other two strategies. It reduces the chance of collision and improves efficiency.



a. 1-Persistent



b. Nonpersistent

CSMA/CD

- CSMA/CD mainly used in Ethernet**.**
- **Carrier sense multiple access with collision detection (CSMA/CD)** augments the algorithm to handle the collision.
- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however, there is a collision, the frame is sent again.

Carrier Sense



Is anyone transmitting data?

Yes — Do not transmit

No — Transmit Data

Multiple Access



Multiple devices are connected...

The Perfect environment for collisions!

Collision Detect





Backoff Algorithm

Computer A has transmission priority

How to CSMA/CD Protocol works?

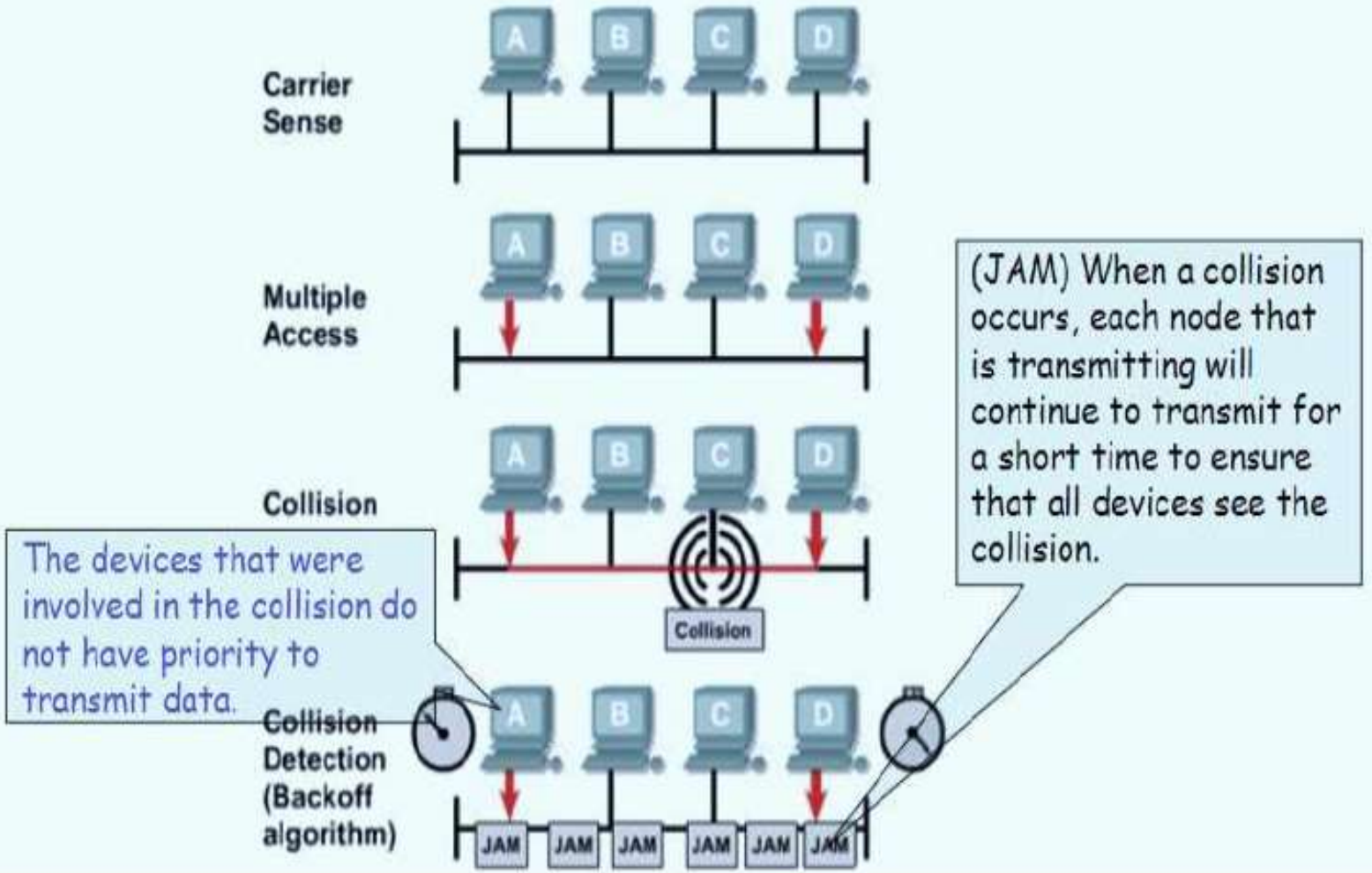
1. **If Medium IDLE**
   - TRANSMIT,
   - Otherwise step 2

2. **If Medium BUSY**
   - WAIT until idle,
   - Then, TRANSMISSION with p=1

3. **If Collision DETECTED**
   - Transmit brief "**JAMMING SIGNAL**"
   - ABORT Transmission

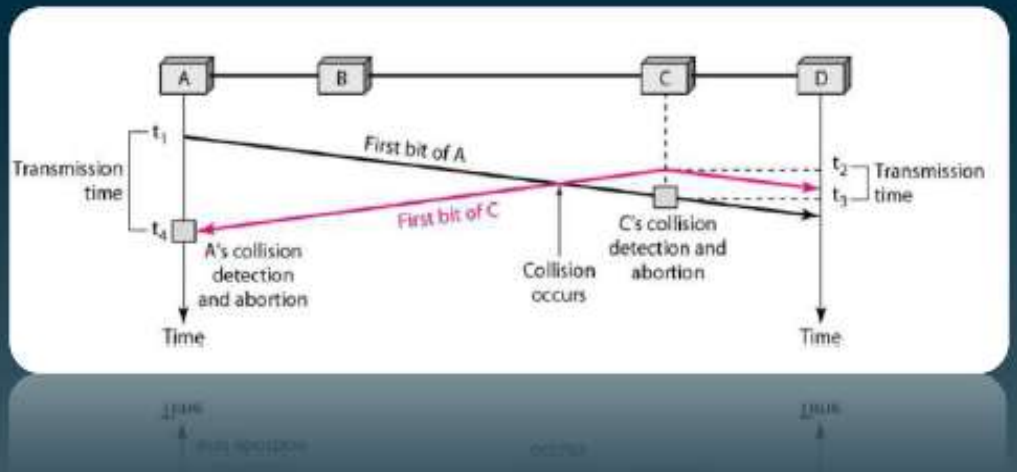
After Aborting- WAIT RANDOM Time, Try again

COLLISION DETECTION PROCEDURE



Collision of the First Bit in CSMA/CD

Each Station continues to send bits in the frame until it detects the Collision.



CSMA/CA

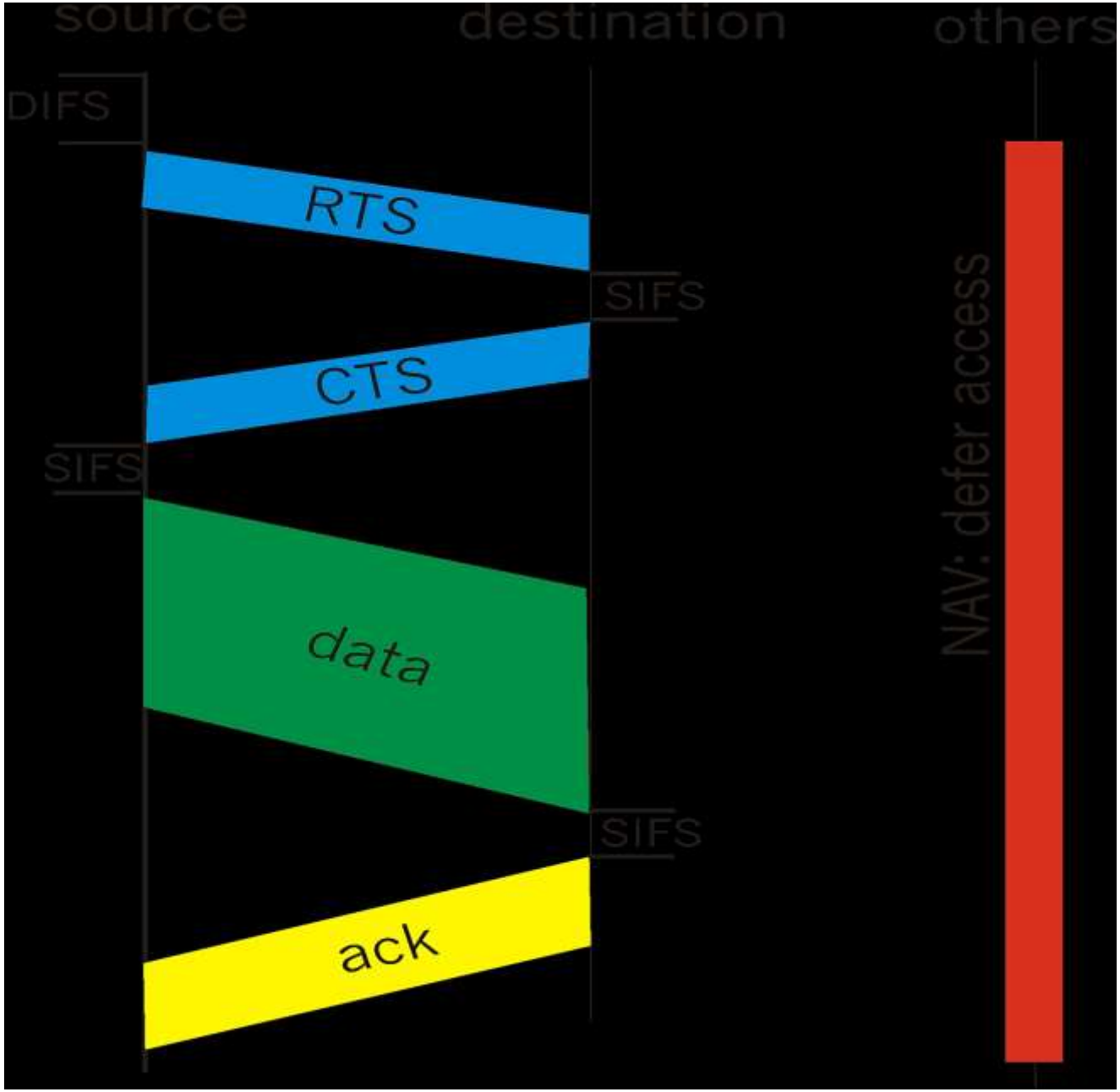
Carrier sense multiple access with collision avoidance

- It was invented for wireless networks
- It avoid from collision
- In this protocol the station, before sending the data senses that, is there any other transmissions from other stations for a particular **PERIOD.**
- These periods are measured by a special timers called as **Network Allocation Vector(NAV)**
- **•The operation remains as same as the CSMA–CD except the waiting time. There are 2 timers involved in this process.**
- **DIFS Distributed Coordinated Function Inter Frame Space.**
- **SIFS–Short Inter Frame Space.**

Important Timers in CSMA -CA

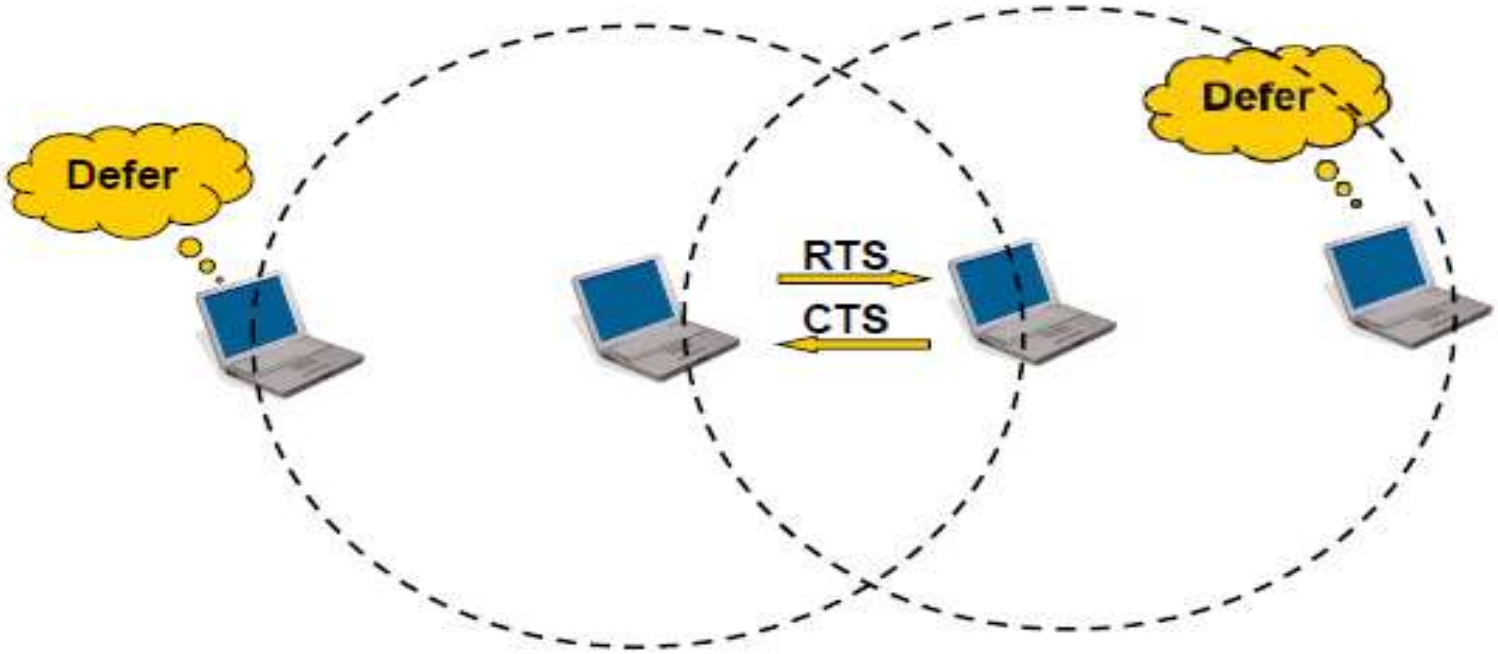
- If station finds that the medium is continuously idle for **Distributed Coordinated Function Inter Frame Space(DIFS)** duration, it is then permitted to transmit a frame . If the channel is found busy during the DIFS interval, the station should defer(postpone) its transmission.

- **Short Inter Frame Space** (**SIFS**) ,is the amount of **time** in micro seconds **required for a wire less interface to process a received frame and to respond** with are response frame


source
destination
others
DIFS
RTS
SIFS
CTS
SIFS
data
SIFS
ack
NAV: defer access

- Before sending a frame the source station senses the channel for a period of DIFS.
- If the channel is found idle for the entire period of DIFS then the sender sends the control frame RTS “Request To Send”
- The receiver takes some time for processing the received frame and to respond this time is referred to as SIFS. Then the receiver responds with the control frame CTS“ Clear To Send”
- Other stations are not allowed to sense the channel during this period

CTS and RTS??



CONTROLLED ACCESS

- In **controlled access,** the stations consult one another to find which station has the right to send.
- A station cannot send data unless it has been authorized by other stations.
- We discuss three controlled-access methods.

1.Reservation

2.Polling

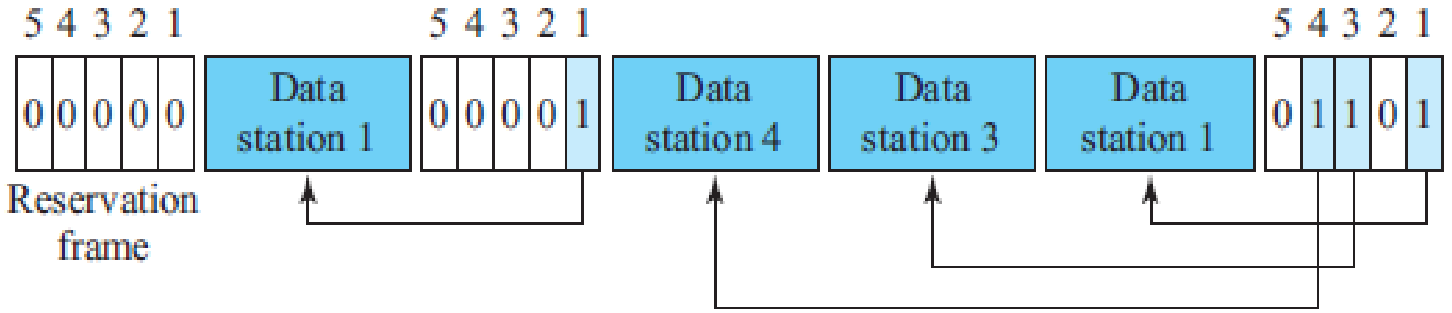
3.Token passing

Reservation

- In the **reservation** method, a station needs to make a reservation before sending data.

- Time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.

- If there are *N* stations in the system, there are exactly *N* reservation mini slots in the reservation frame. Each mini slot belongs to a station.

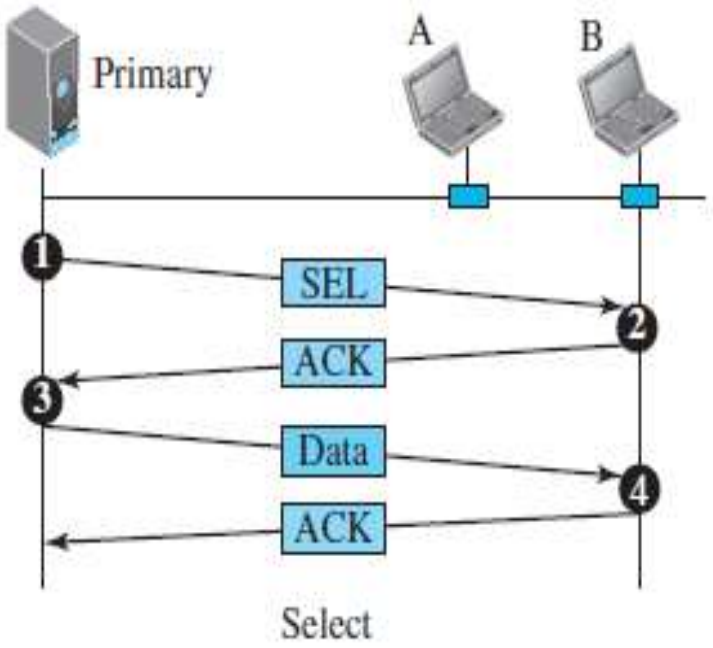


- Above Figure shows a situation with five stations and a five-mini slot reservation frame.
- In the first interval, only stations 1, 3, and 4 have made reservations.
- In the second interval, only station 1 has made a reservation

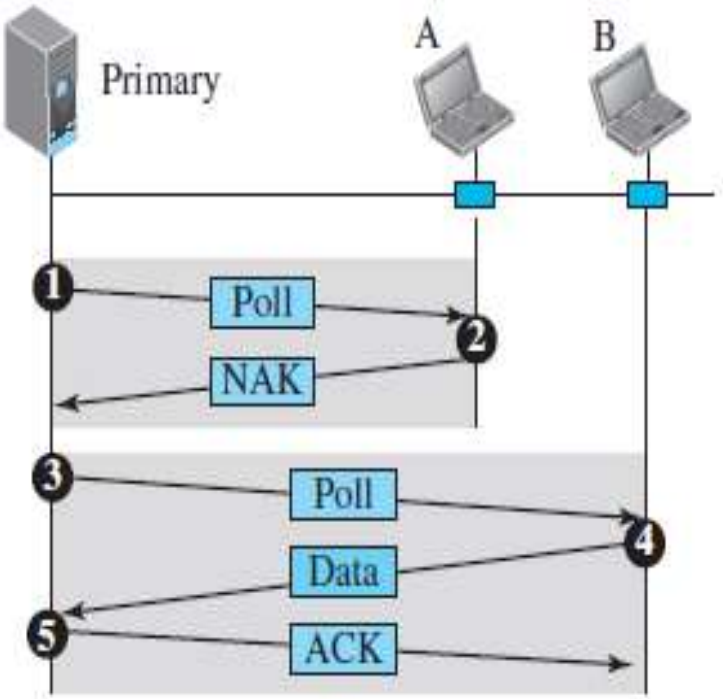
Polling

- **Polling** works with topologies in which one device is designated as a ***primary station*** and the other devices are ***secondary stations*****.**
- All data exchanges must be made through the primary device even when the ultimate destination is a secondary device.
- The primary device controls the link; the secondary devices follow its instructions.
- Primary device to determine which device is allowed to use the channel at a given time.

- The primary device, therefore, is always the initiator of a session (see Figure 12.19).
- This method uses **poll** and **select** functions to prevent collisions.



Select



Poll

Select

- The *select* function is used whenever the primary device has something to send.
- Remember that the primary controls the link. If the primary is neither sending nor receiving data, it knows the link is available.
- The primary must alert the secondary to the upcoming transmission and wait for an acknowledgment of the secondary's ready status. Before sending data, the primary creates and transmits a select (SEL) frame, one field of which includes the address of the intended secondary.

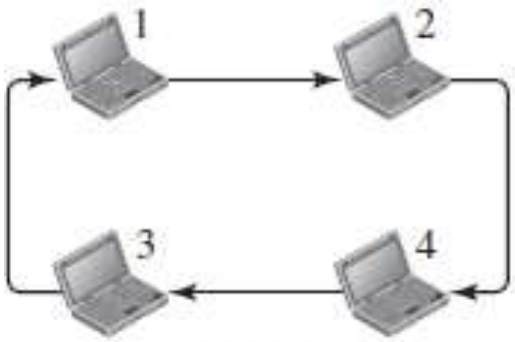
Poll

- . When the primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send.
- When the first secondary is approached, it responds either with a NAK frame if it has nothing to send or with data (in the form of a data frame) if it does.
- If the response is negative (a NAK frame), then the primary polls the next secondary in the same manner until it finds one with data to send.
- When the response is positive (a data frame), the primary reads the frame and returns an acknowledgment (ACK frame), verifying its receipt.

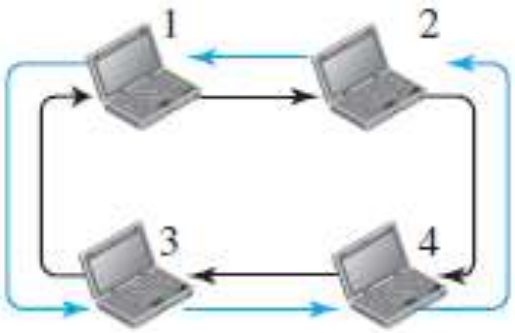
Token Passing

- In the **token-passing** method, the stations in a network are organized in a logical ring.
- In other words, for each station, there is a *predecessor* and a *successor*.
- In this method, a special packet called a ***token*** circulates through the ring.
- The possession of the token gives the station the right to access the channel and send its data.
- When a station has some data to send, it waits until it receives the token from its predecessor.

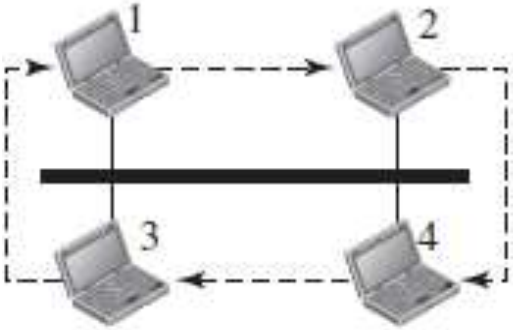
Logical ring and physical topology in token-passing access method



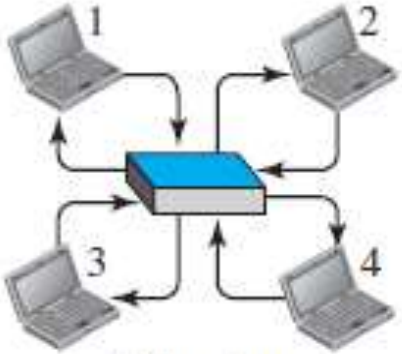
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

- It then holds the token and sends its data. When the station has no more data to send, it releases the token, passing it to the next logical station in the ring.
- The station cannot send data until it receives the token again in the next round. In this process, when a station receives the token and has no data to send, it just passes the data to the next station.

CHANNELIZATION

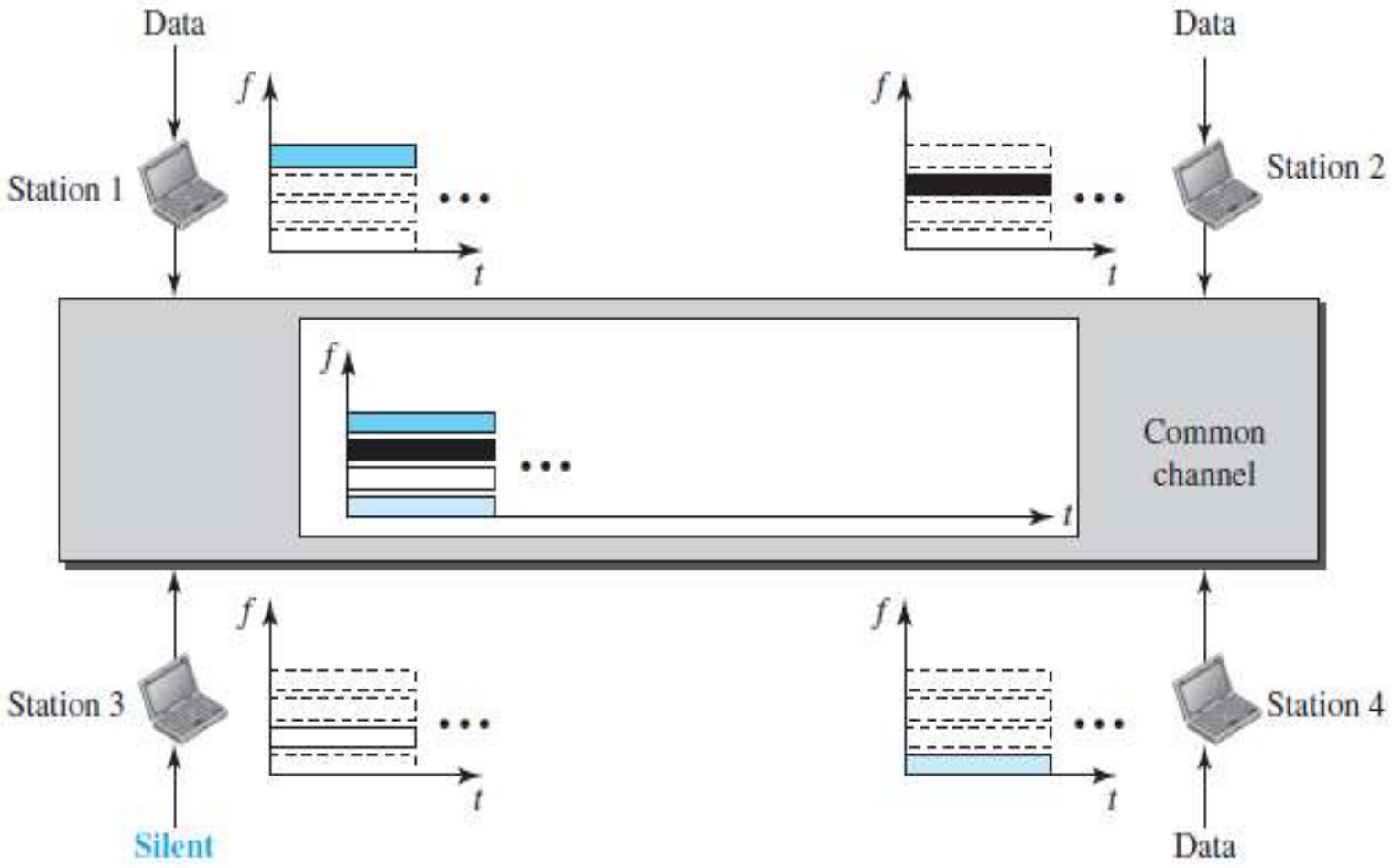
- **Channelization** (or *channel partition,* as it is sometimes called) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, among different stations.
- In this section, we discuss three channelization protocols:
- FDMA
- TDMA
- CDMA.

FDMA

- In **frequency-division multiple access (FDMA),** the available bandwidth is divided into frequency bands.
- Each station is allocated a band to send its data.
- In other words, each band is reserved for a specific station, and it belongs to the station all the time.
- FDMA specifies a predetermined frequency band for the entire period of communication.

In real time example of FDMA is

- AM or FM radio broadcasting
- Walkie talkies for closed group user



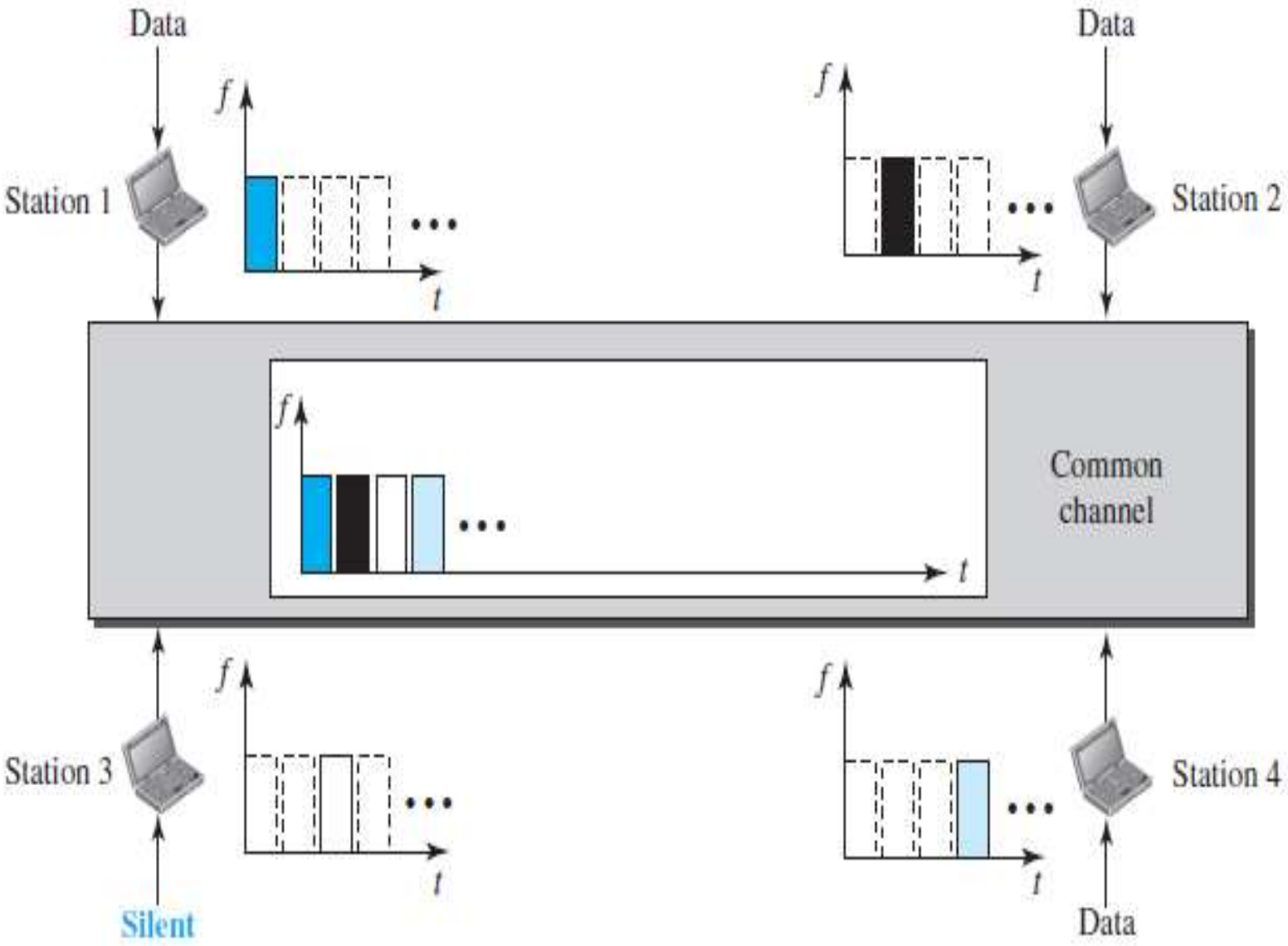
In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

TDMA

- In **Time-division multiple access (TDMA),** the stations share the bandwidth of the channel in time.
- Each station is allocated a time slot during which it can send data.
- Each station transmits its data in its assigned time slot.
- TDMA allows a mobile station's radio component to listen and broadcast only in its assigned time slot.

Example of TDMA

- Personal Digital Cellular(PDC)
- 2nd Generation (2G) Global system of Mobile Communication


Data
Station 1
f
t
Data
Station 2
f
t
Common
channel
f
t
Station 3
Silent
f
t
Station 4
Data
f
t

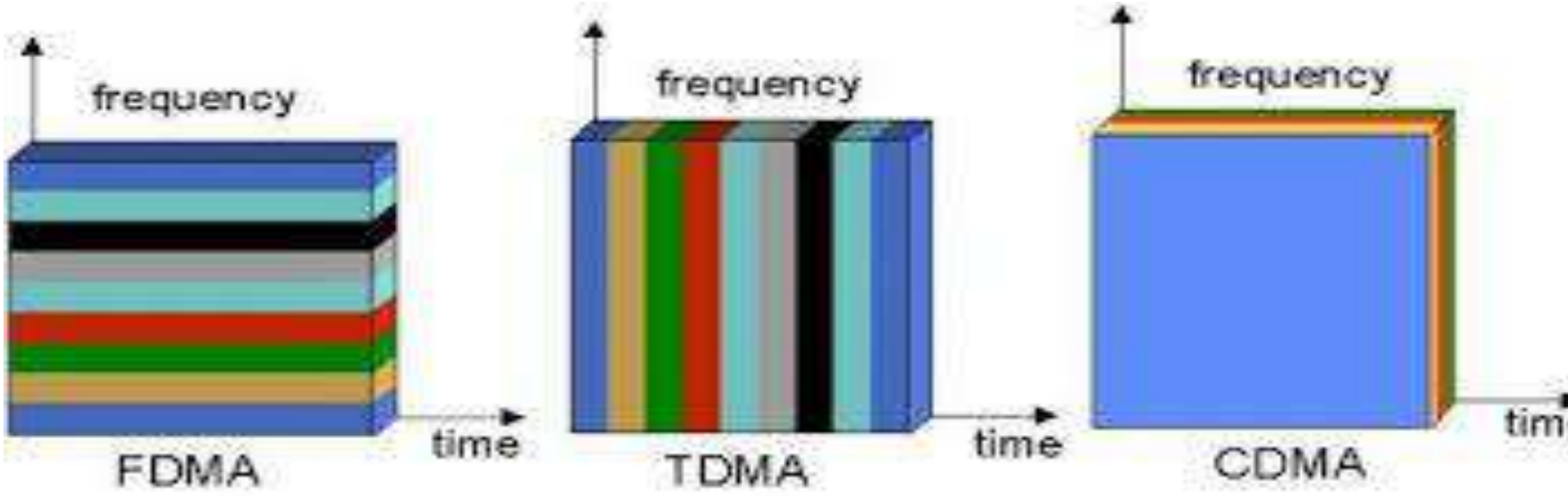
CDMA

- **Code-division multiple access (CDMA)**
- CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link.
- It differs from TDMA in that all stations can send data simultaneously; there is no timesharing.

Chips

- CDMA is based on coding theory. Each station is assigned a code, which is a sequence of numbers called *chips,*

- **CDMA** networks allow for a greater number of users, meaning their capacity is greater than that of **GSM** networks.
- Moreover, **CDMA** is the infrastructure on which all 3G networks are based — for both **GSM** and **CDMA** carriers.



Physical Layer:

Major function of physical layer is move data in the form of electromagnetic signals across the transmission medium.

UNIT-5

Performance:

important issue in networking is the performance of the network—how good is it?

Bandwidth

- To measures network performance in terms of bandwidth. The term can be used in two different contexts with two different measuring values:
- bandwidth in hertz and
- bandwidth in bits per second.

Bandwidth in Hertz

- Bandwidth in hertz is the range of frequencies contained in a composite signal or the range of frequencies a channel can pass.
- For example, we can say the bandwidth of a subscriber telephone line is 4 kHz.

Bandwidth in Bits per Seconds

- The term *bandwidth* can also refer to the number of bits per second that a channel, a link, or even a network can transmit.
- For example, one can say the bandwidth of a Fast Ethernet network is a maximum of 100 Mbps. This means that this network can send 100 Mbps.

Throughput

- The **throughput** is a measure of how fast we can actually send data through a network.
- Bandwidth in bits per second and throughput seem the same, but they are different.
- For example, we may have a link with a bandwidth of 1 Mbps, but the devices connected to the end of the link may handle only 200 kbps. This means that we cannot send more than 200 kbps through this link.

Latency (Delay)

- The **latency** or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

- We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.

Latency =propagation time + transmission time + queuing time + processing delay

Bandwidth-Delay Product

- Bandwidth and delay are two performance metrics of a link. what is very important in data communications is the product of the two, the bandwidth-delay product.

Figure 3.32 *Filling the link with bits for case 1*

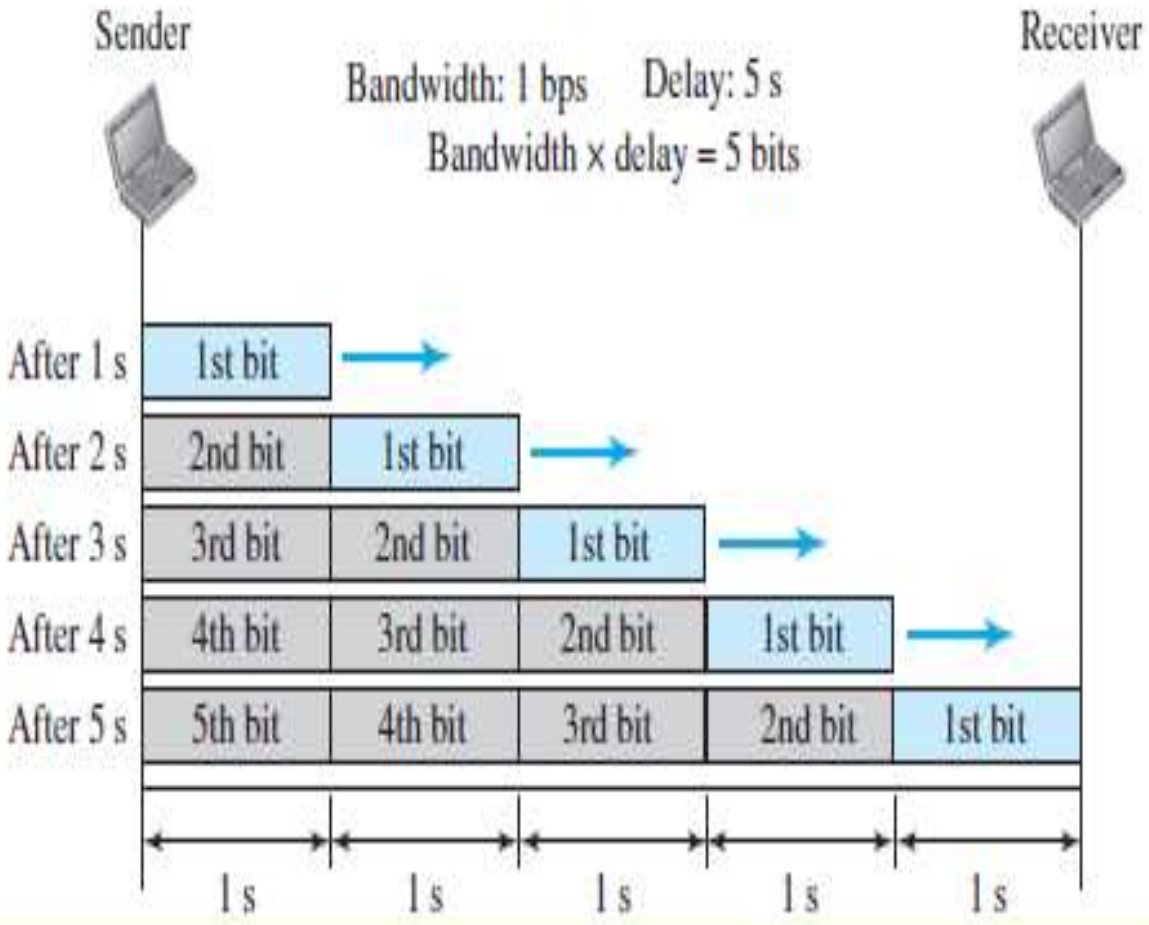
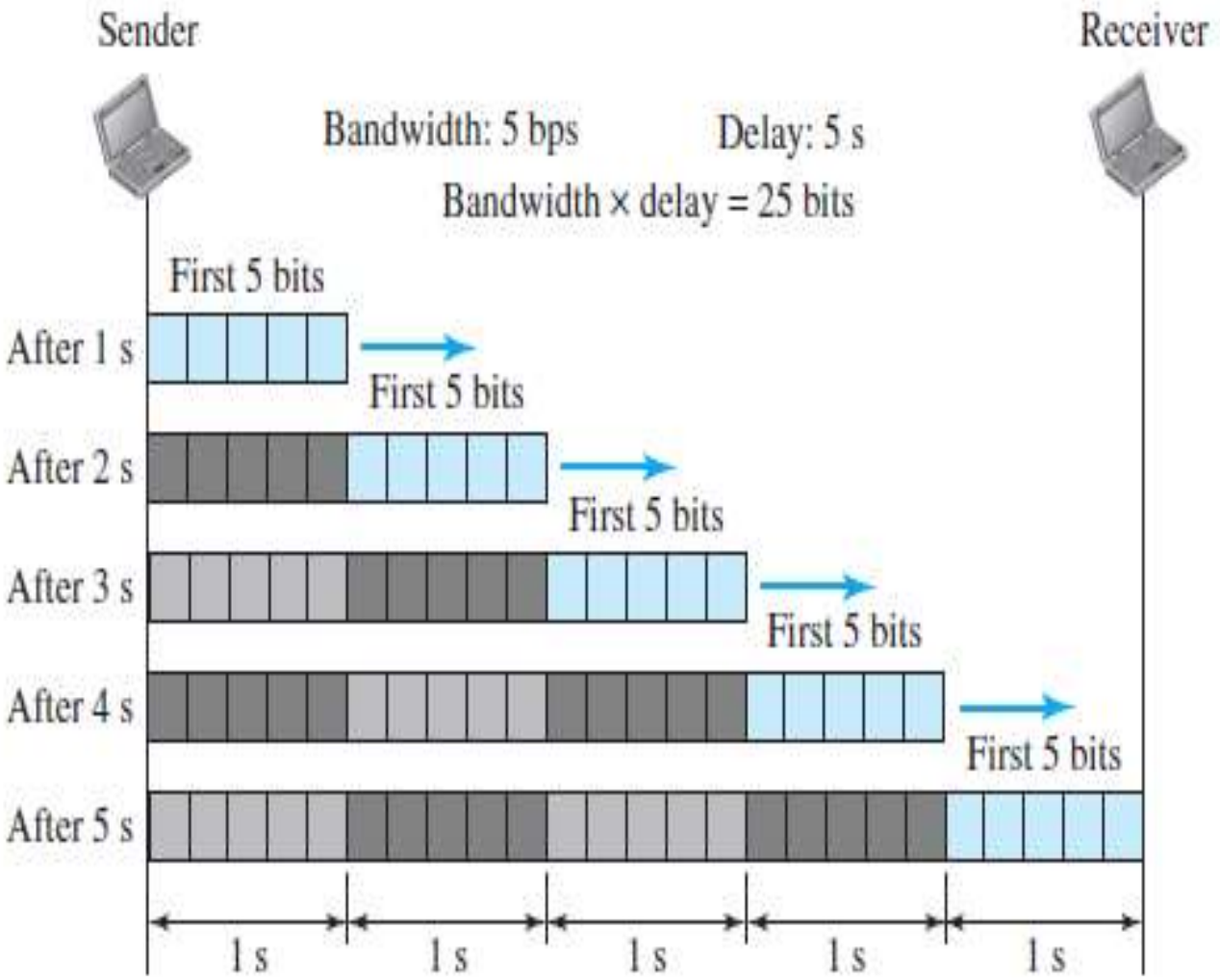


Figure 3.33 *Filling the link with bits in case 2*



Jitter:

Another performance issue that is related to delay is **jitter.** We can roughly say that

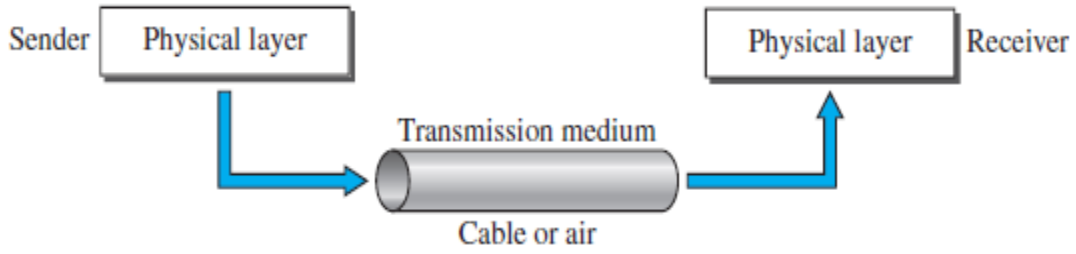
jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example).

- If the delay for the first packet is 20 ms, for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter.

UNIT-5- Transmission Media(Medium)

- Transmission media are actually located below the physical layer and are directly controlled by the physical layer
- A **transmission medium** can be broadly defined as anything that can carry information from a source to a destination.
- For example, the transmission medium for two people having a dinner conversation is the air.

Figure 7.1 *Transmission medium and physical layer*

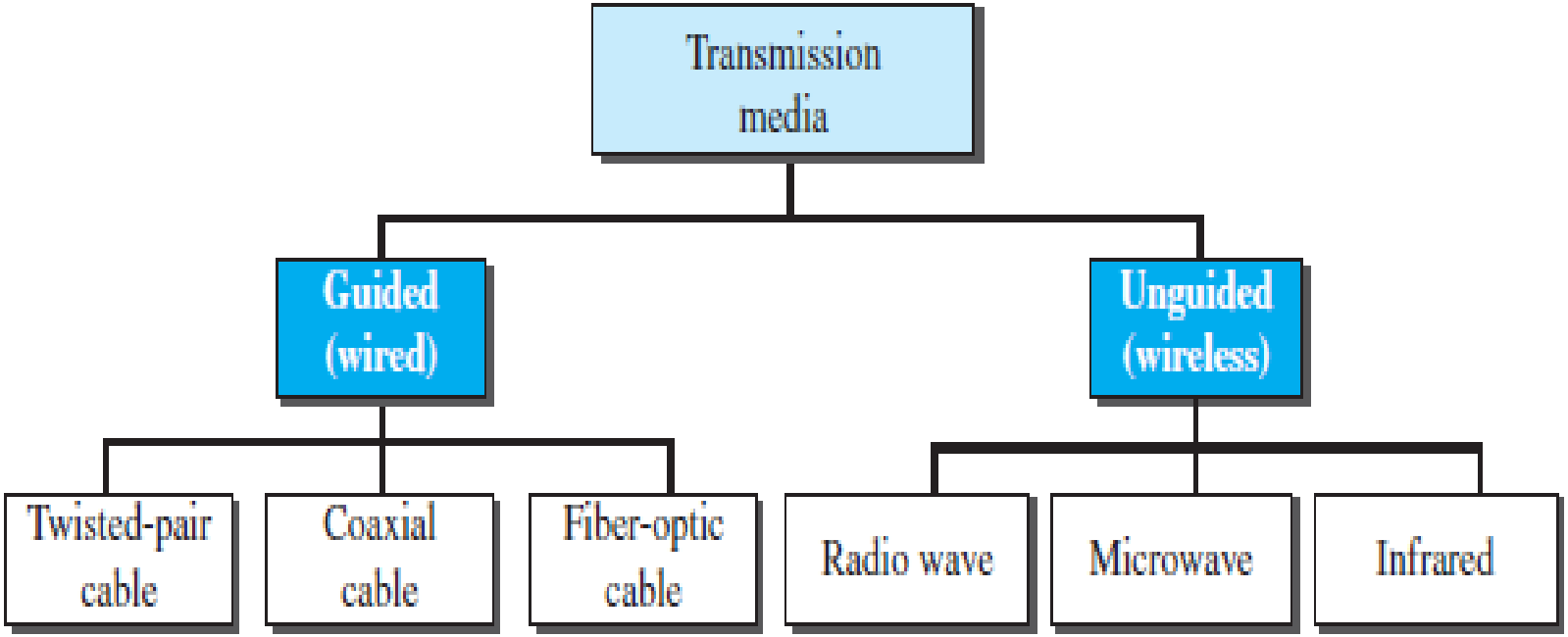


In telecommunications, transmission media can be divided into two broad categories:

1.Guided

2.Unguided

Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable. Unguided medium is free space.



Guided Media:

- **Guided media,** which are those that provide a conduit from one device to another,
- Include **twisted-pair cable, coaxial cable,** and **fiber-optic cable.**

Twisted pair cable:

A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.

- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.

- Twisted pair cable uses electrical signal for transmission.
- TP cable comes in two forms

1.Unshielded TP(UTP)

2.Shielded TP(STP)

Figure 7.3 *Twisted-pair cable*

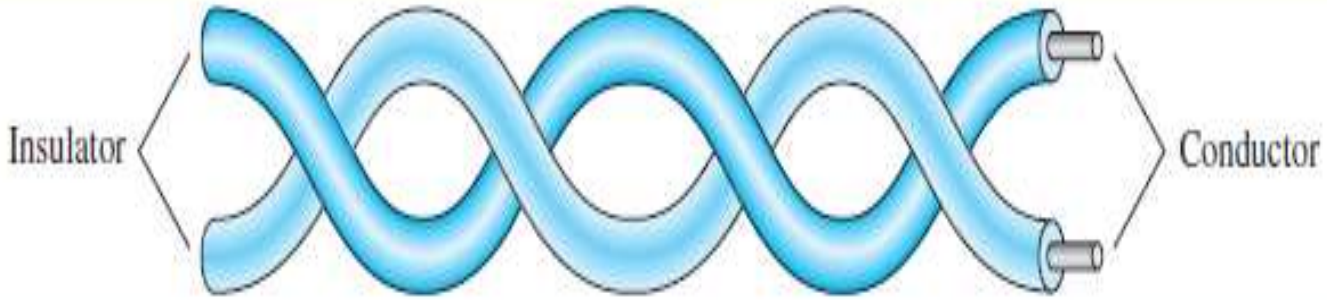
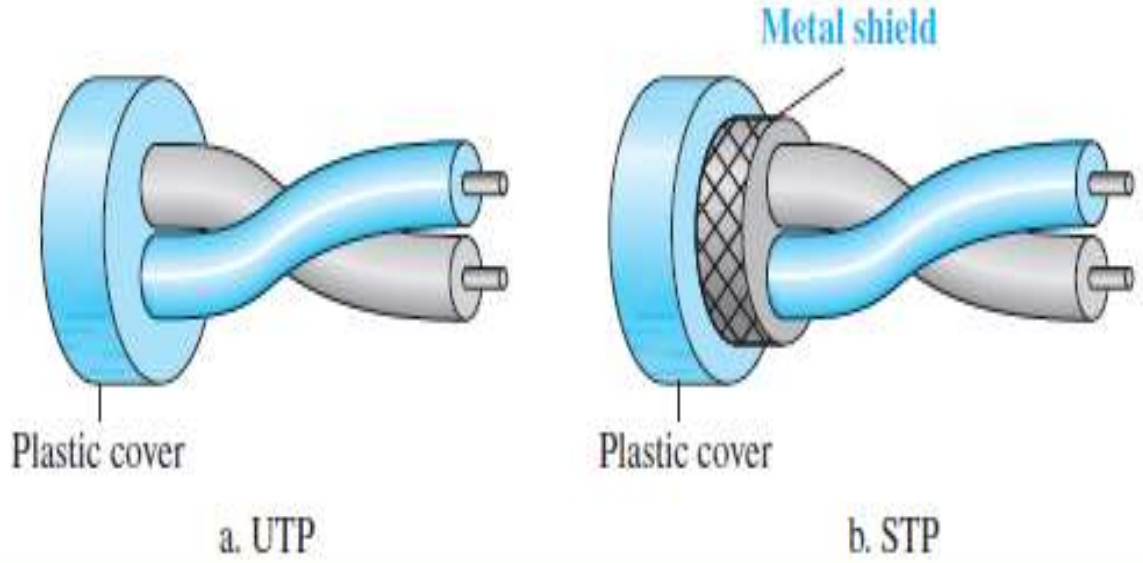


Figure 7.4 *UTP and STP cables*



Unshielded Twisted Pair(UTP):

- The most common twisted-pair cable used in communications is referred to as ***unshielded twisted-pair*** **(UTP).**
- UTP is an ordinary telephone wire.
- UTP is a set of twisted pairs of cable within plastic

Cover

Least expensive

Commonly used in LAN

Simple to install

Categories

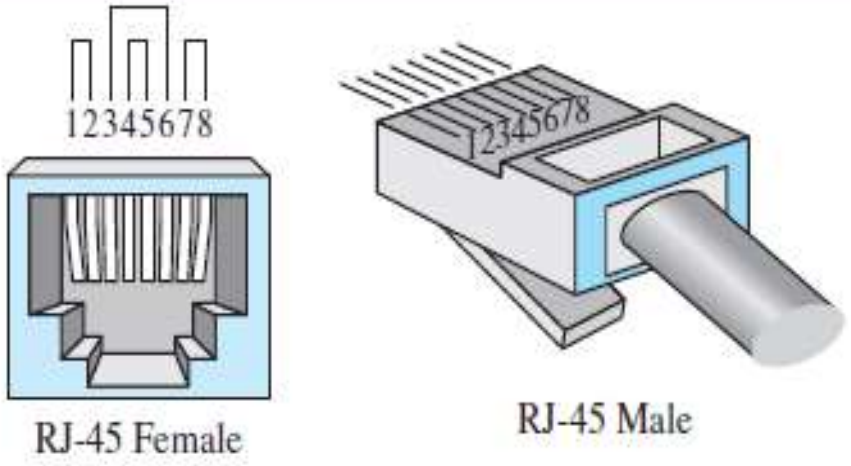
- The Electronic Industries Association (EIA) has developed standards to classify unshielded twisted-pair cable into seven categories. Categories are determined by cable

Category	Specification	Data Rate (Mbps)	Use
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called *SSTP (shielded screen twisted-pair)*. Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Connectors

- The most common UTP connector is **RJ45** (RJ stands for registered jack), The RJ45 is a keyed connector, meaning the connector can be inserted in only one way.

Figure 7.5 *UTP connector*



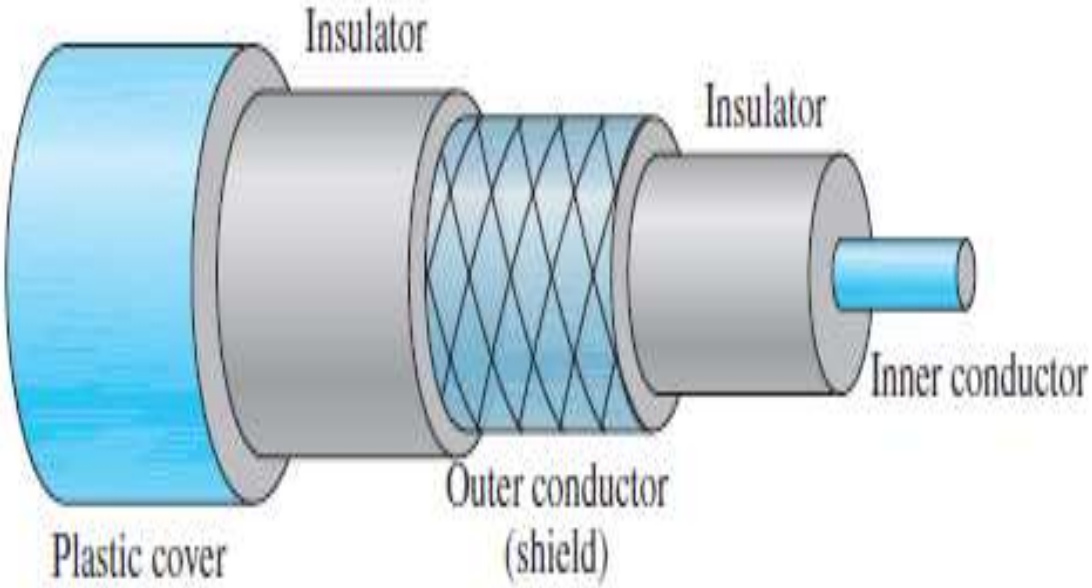
Shielded Twisted Pair:

- IBM has also produced a version of twisted-pair cable for its use, called ***shielded twisted-pair*** **(STP).**
- Metal shield improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive.

Coaxial Cable:

- Coaxial cable (or *coax*) carries signals of higher frequency ranges than those in twisted-pair cable, in part because the two media are constructed quite differently.
- Coaxial cable used to transmit both analog and digital signal

Figure 7.7 *Coaxial cable*



- It is made up of two conductors that share a common axis.

Coaxial cable standard:

- Coaxial cables are categorized by their **Radio Government (RG)** ratings.
- Each RG number denotes a unique set of physical specifications

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coaxial cable model:

Base band mode

Broad band mode

- In base band mode the cable bandwidth denoted to a single stream of data.
- Mostly used in LAN

In Broad band mode the bandwidth is divided into ranges.

It allows transmission of multiple data streams over the same cable simultaneously.

Advantages of Coaxial cable:

Used for both analog and digital data transmission

It has a higher bandwidth

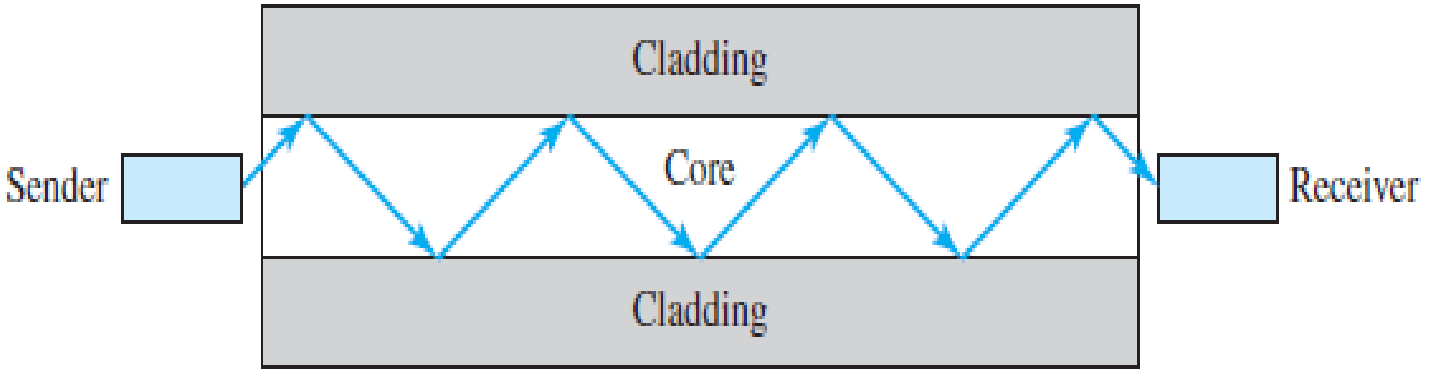
- Used for longer distance and higher data rates.
- Excellent noise immunity
- Used in LAN and Television distribution

Fiber-Optic Cable:

- A fiber-optic cable is made of glass or plastic and transmits data signals in the form of light.
- To understand optical fiber, we first need to explore several aspects of the nature of light.
- Light travels in a straight line as long as it is moving through a single uniform substance.
- If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction.

- Optical fibers use reflection to guide light through a channel.
- A glass or plastic **core** is surrounded by a **cladding** of less dense glass or plastic.

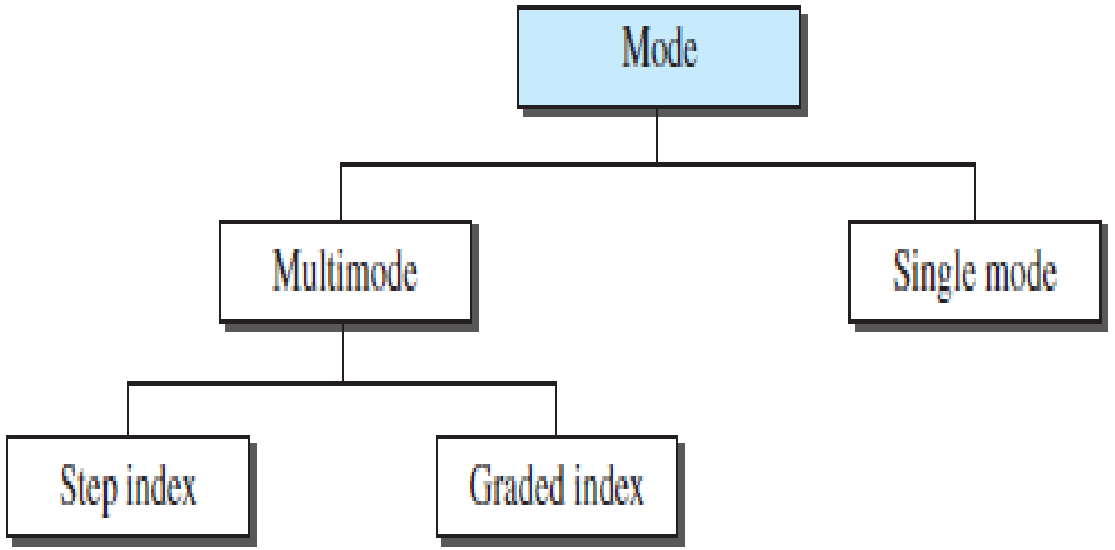
Figure 7.11 *Optical fiber*



Propagation Modes

- Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics. Multimode can be implemented in two forms: step-index or graded-index

Figure 7.12 *Propagation modes*



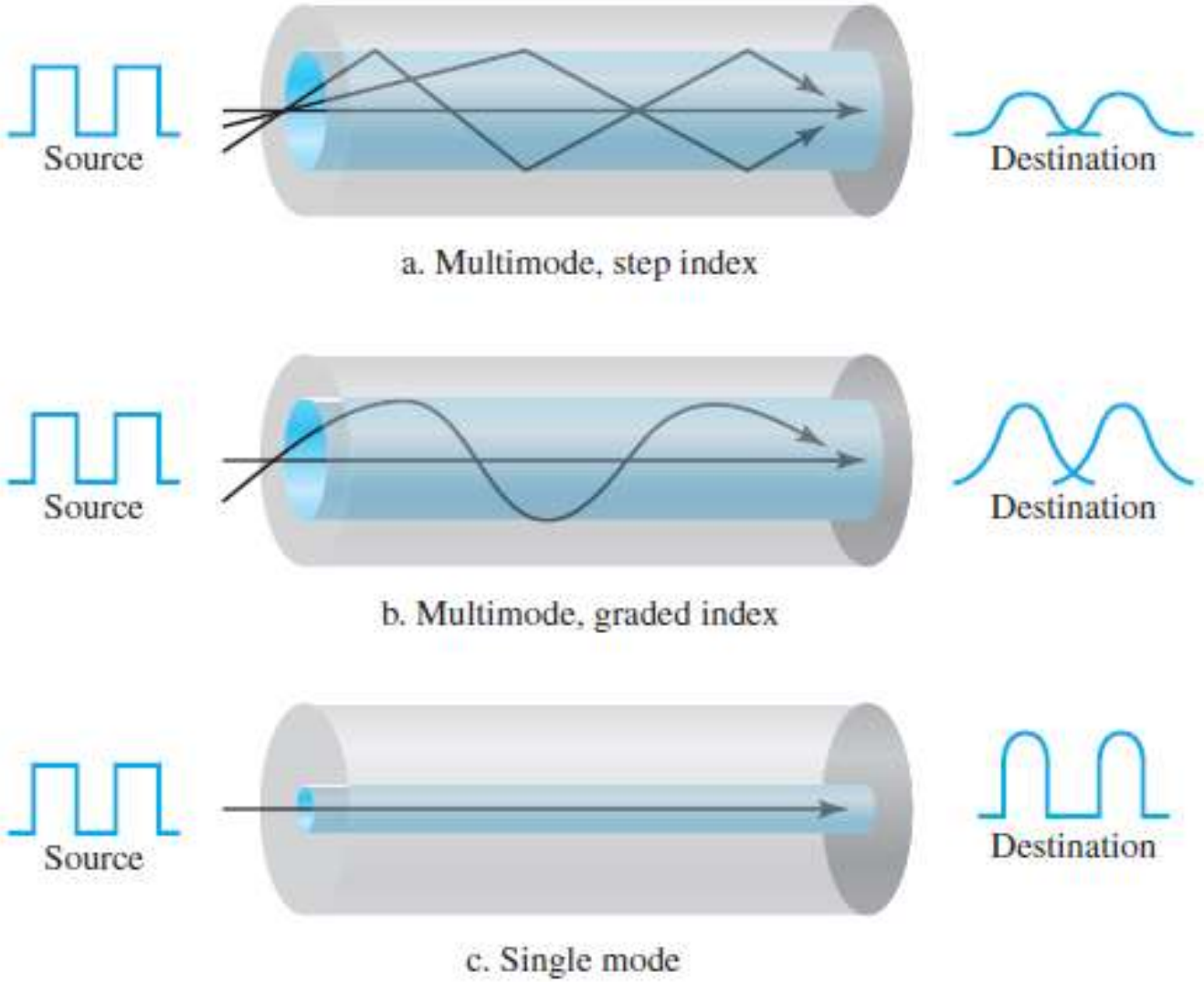
Multimode

- Multimode is so named because multiple beams from a light source move through the core in different paths.

Single-Mode

- Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.

Figure 7.13 *Modes*



Advantages

- Fiber-optic cable has several advantages over metallic cable (twisted-pair or coaxial).
- **Higher bandwidth.**
- **Less signal attenuation.** Fiber-optic transmission distance is significantly greater than that of other guided media.
- **Immunity to electromagnetic interference.** Electromagnetic noise cannot affect fiber-optic cables.
- **Light weight.** Fiber-optic cables are much lighter than copper cables.

Disadvantages

- **Installation and maintenance.** Fiber-optic cable is a relatively new technology. Its installation and maintenance require expertise that is not yet available everywhere.
- **Unidirectional light propagation.** Propagation of light is unidirectional. If we need bidirectional communication, two fibers are needed.
- **Cost is expensive**

UNGUIDED MEDIA: WIRELESS

- **Unguided medium** transport electromagnetic waves without using a physical conductor.
- This type of communication is often referred to as ***wireless communication*.**
- Signals are normally broadcast through free space and signals are available to anyone who has a device capable of receiving them.

Figure 7.17 *Electromagnetic spectrum for wireless communication*

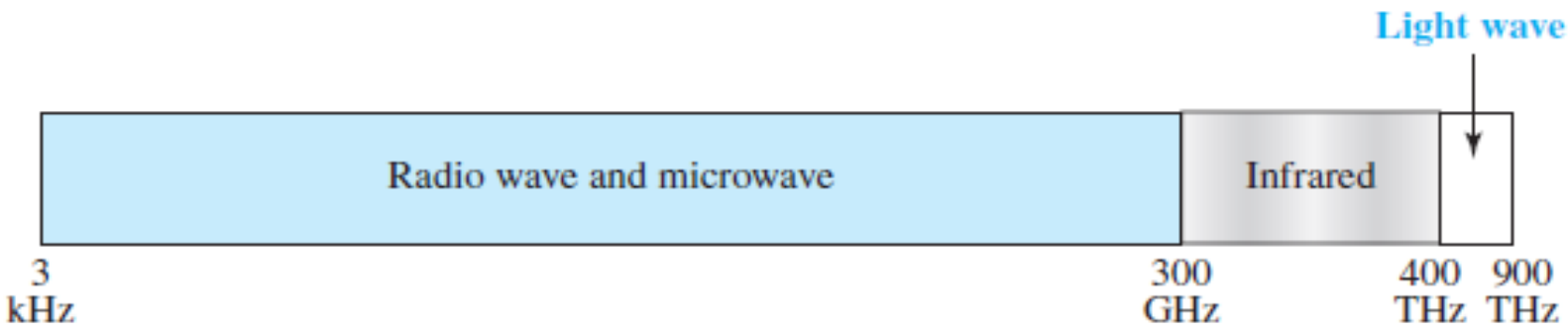
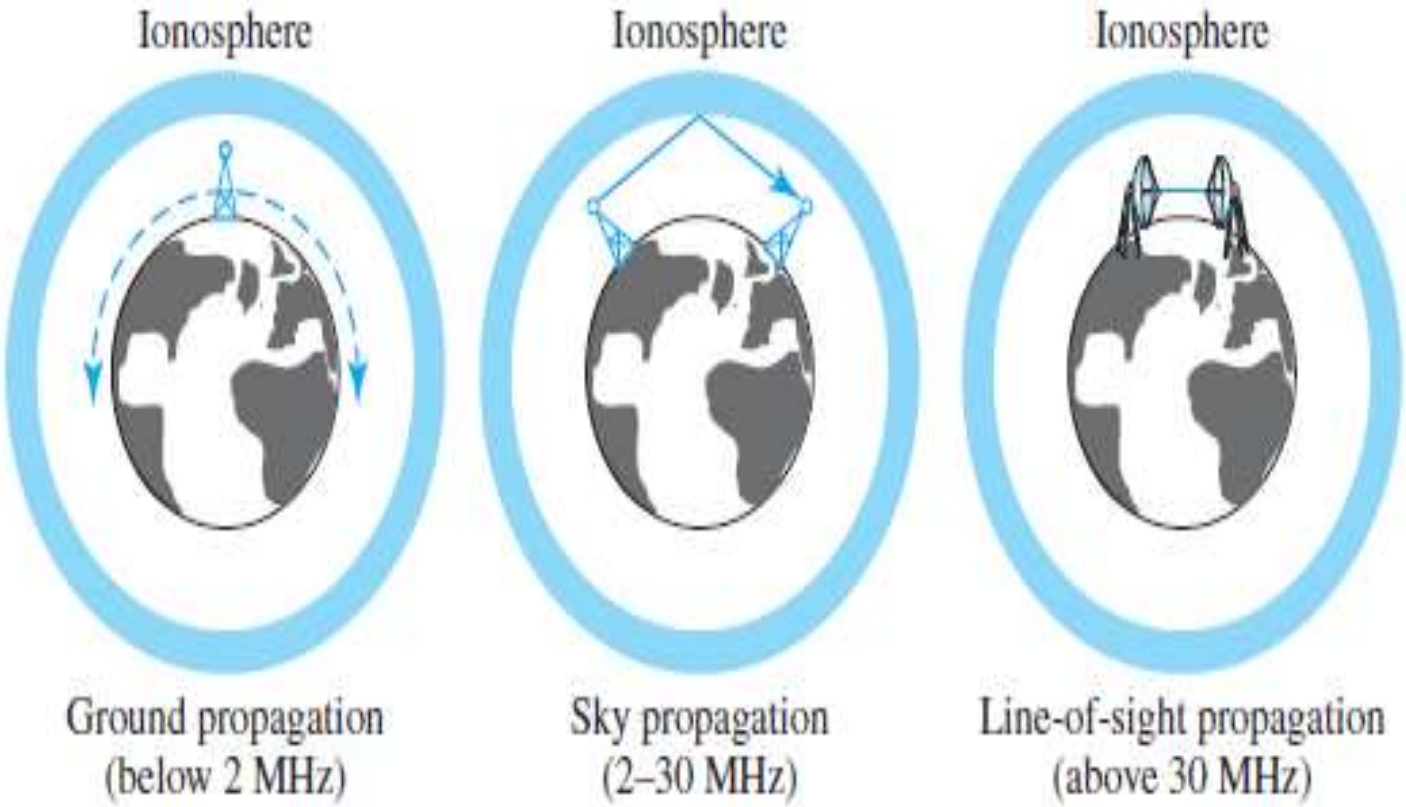


Figure 7.18 *Propagation methods*



- In **ground propagation,** radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- Distance depends on the amount of power in the signal The greater the power, the greater the distance.

Frequency Band	Range	Application
very low frequency (VLF)	3–30 kHz	Long-range radio Navigation
low frequency (LF) Radio beacons and navigational locators	30–300 kHz	Radio beacons and navigational locators

- In **sky propagation,** higher-frequency radio waves radiate upward into the ionosphere where they are reflected back to earth.

Frequency band	Range	Application
middle frequency (MF)	300 kHz–3 MHz	AM radio
high frequency (HF)	3–30 MHz Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF) VHF	30–300 MHz Sky and line-of-sight	TV, FM radio

- In **line-of-sight propagation,** very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth.

Frequency Band	Range	Propagation	Application
ultrahigh frequency (UHF) UHF	300 MHz–3 GHz	Line-of-sight	TV, cellular phones, paging, satellite
Super high frequency (SF)	3–30 GHz	Line-of-sight	Satellite
Extremely High Frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite

Radio Waves

- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called **radio waves**
- waves ranging in frequencies between 1 and 300 GHz are called **microwaves.**
- Radio waves, for the most part, are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. A sending antenna sends waves that can be received by any receiving antenna.
- Radio waves, particularly those waves that propagate in the sky mode, can travel long distances. This makes radio waves a good candidate for long-distance broadcasting such as AM radio

Omnidirectional Antenna

- Radio waves use **omnidirectional antennas** that send out signals in all directions.

Applications:

- The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers.
- AM and FM radio, television,
- cordless phones, and paging are examples of multicasting.



Microwaves:

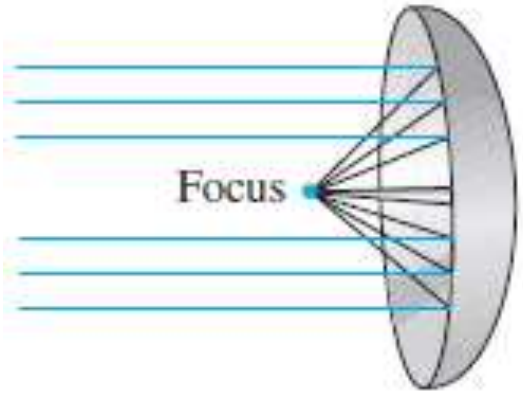
- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves.
- Microwaves are unidirectional. When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.

Unidirectional Antenna:

- Microwaves need **unidirectional antennas** that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish
- and the horn antenna

Applications:

- Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.



a. Parabolic dish antenna



b. Horn antenna

Infrared:

- **Infrared waves,** with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication.
- Infrared waves, having high frequencies, cannot penetrate walls
- This advantages short-range communication system in one room cannot be affected by another system in the next room
- When we use our infrared remote control, we do not interfere with the use of the remote by our neighbours.

Applications:

- The infrared band, almost 400 THz, has an excellent potential for data transmission.
- Such a wide bandwidth can be used to transmit digital data with a very high data rate.
- For example, some manufacturers provide a special port called the **IrDA port** that allows a wireless keyboard to communicate with a PC.
- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

(Not in present syllabus)
Bluetooth[802.15.1]

- Bluetooth used for very short range communication between mobile phones, PDAs, notebook computers, and other personal or peripheral devices.

For example, Bluetooth can be used to connect a mobile phone to a headset or a notebook computer to a keyboard.

- The Bluetooth radios can use quite low power transmission.
- Bluetooth operates in the license-exempt band at 2.45 GHz. Bluetooth links have typical bandwidths around 1 to 3 Mbps and a range of about 10 m.
- For this reason, and because the communicating devices typically belong to one individual or group, Bluetooth is sometimes categorized as a Personal Area Network (PAN)

- The basic Bluetooth network configuration, called a ***piconet***, consists of a master device and up to six slave devices, as shown in Figure 2.35.

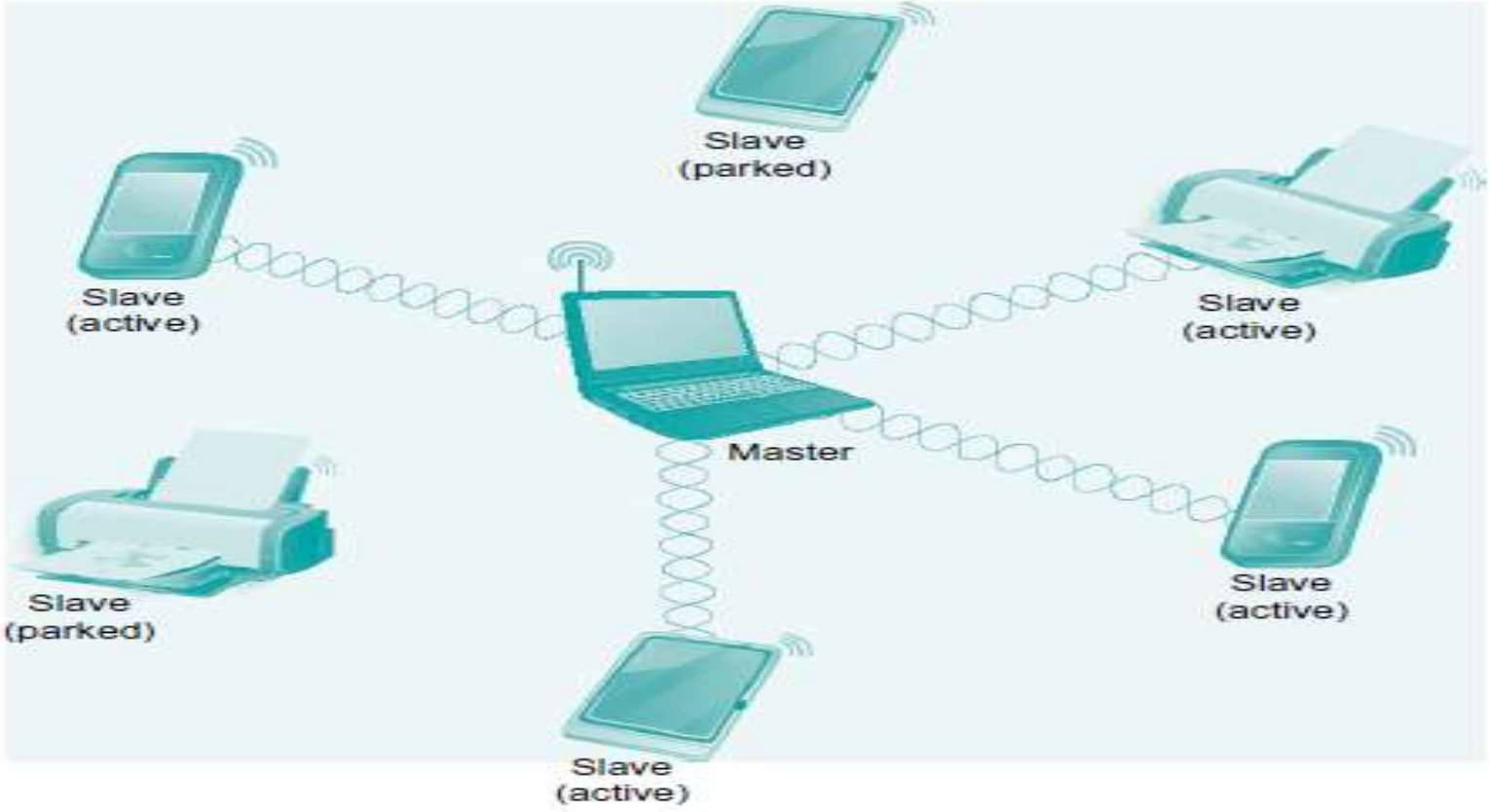


FIGURE 2.35 A Bluetooth piconet.

Any communication is between the master and a slave; the slaves do not communicate directly with each other.

Because slaves have a simpler role, their Bluetooth hardware and software can be simpler and cheaper.

- Bluetooth to use for synchronous time division multiplexing. A frame takes up 1, 3, or 5 consecutive time slots.
- Only the master can start to transmit in odd-numbered slots. A slave can start to transmit in an even-numbered slot.
- A slave device can be *parked* that is, it is set to an inactive, low-power state.
- A parked device cannot communicate on the piconet; it can only be reactivated by the master.
- A piconet can have up to 255 parked devices in addition to its active slave devices.